NAMIBIA UNIVERSITY OF SCIENCE AND TECHNOLOGY

Artificial Intelligence

Lecture 2: Uniformed Search (Demo version)

Search Problems

agent

entity that perceives its environment
and acts upon that environment

state

a configuration of the agent and its environment

2	4	5	7
8	3	1	11
14	6		10
9	13	15	12

12	9	4	2
8	7	3	14
	1	6	11
5	13	10	15

15	4	10	3
13	1	11	12
9	5	14	7
6	8		2

initial state

the state in which the agent begins

initial state

2	4	5	7
8	3	1	11
14	6		10
9	13	15	12

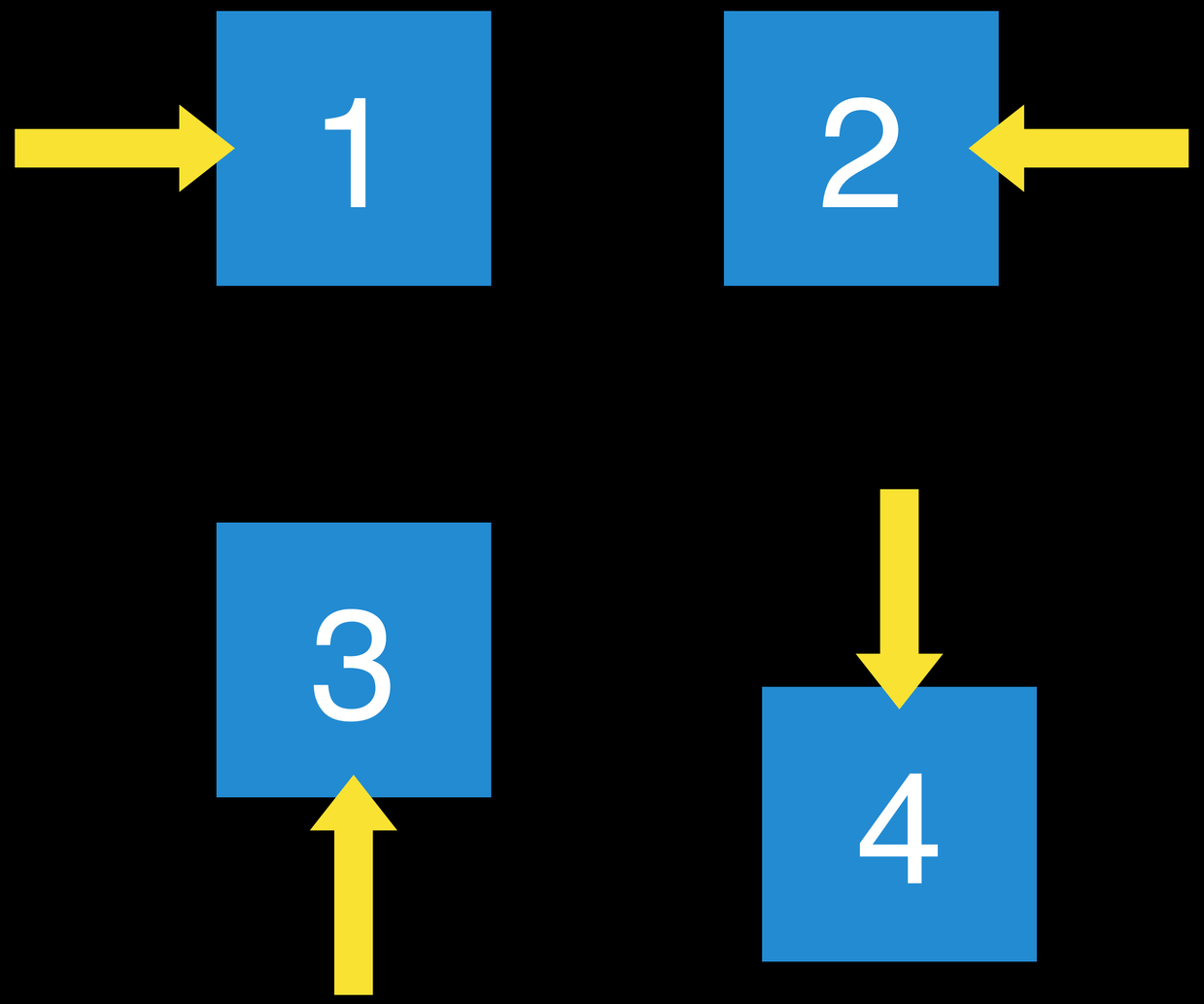
actions

choices that can be made in a state

actions

$\textsc{Actions}(s)$ returns the set of actions that can be executed in state s

actions



transition model

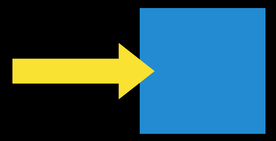
a description of what state results from performing any applicable action in any state

transition model

$\textsc{Result}(s, a)$ returns the state resulting from performing action a in state s

RESULT(

2	4	5	7
8	3	1	11
14	6	10	12
9	13	15	

, ) =

2	4	5	7
8	3	1	11
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2	4	5	7
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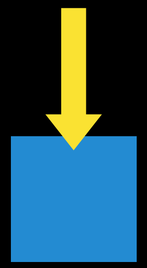
, ) =

2	4	5	7
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transition model

RESULT(

2	4	5	7
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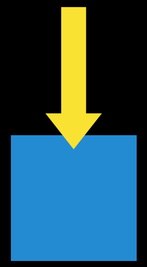

) =

2	4	5	7
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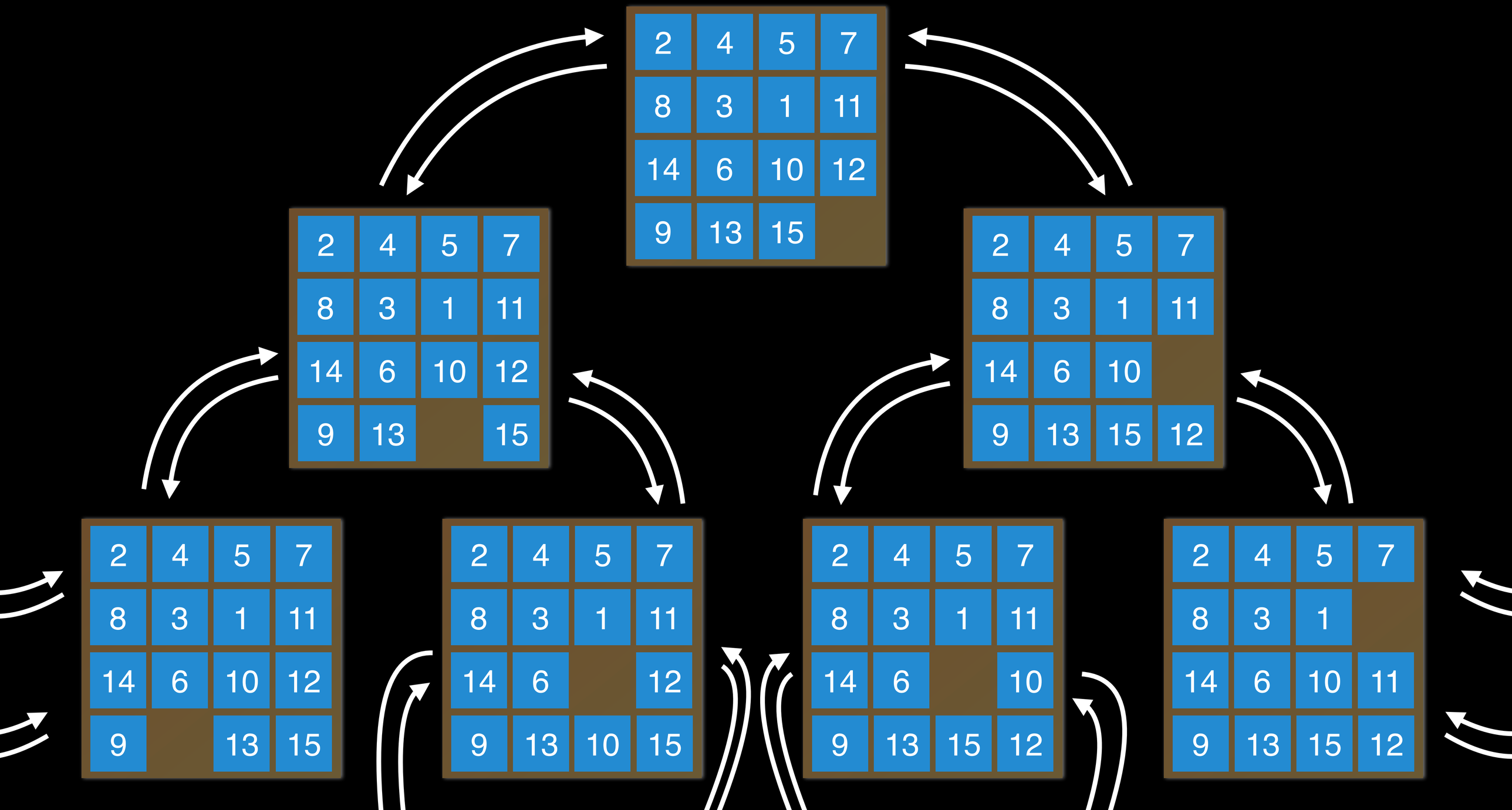
9	13	15	12

state space

the set of all states reachable from the initial state by any sequence of actions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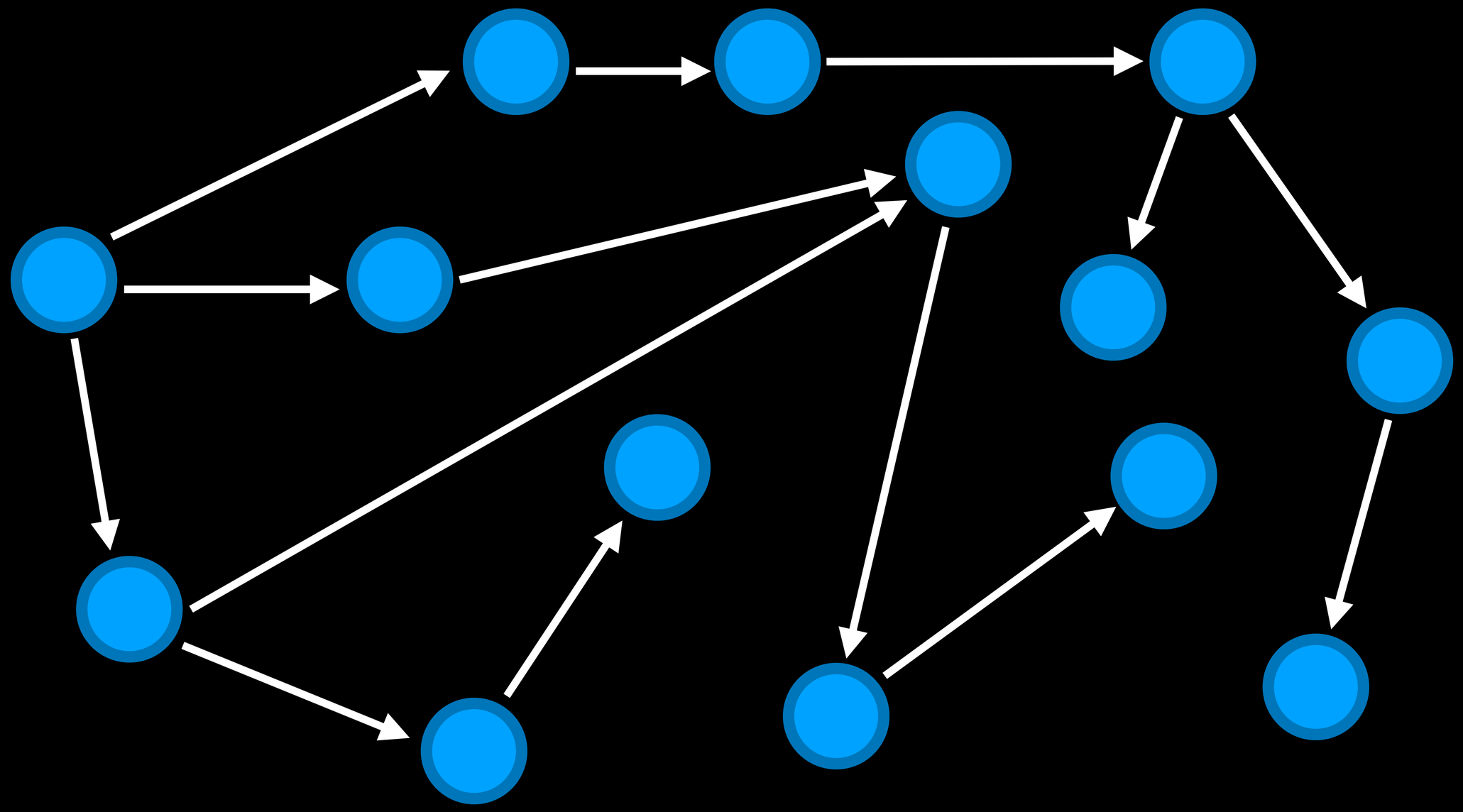
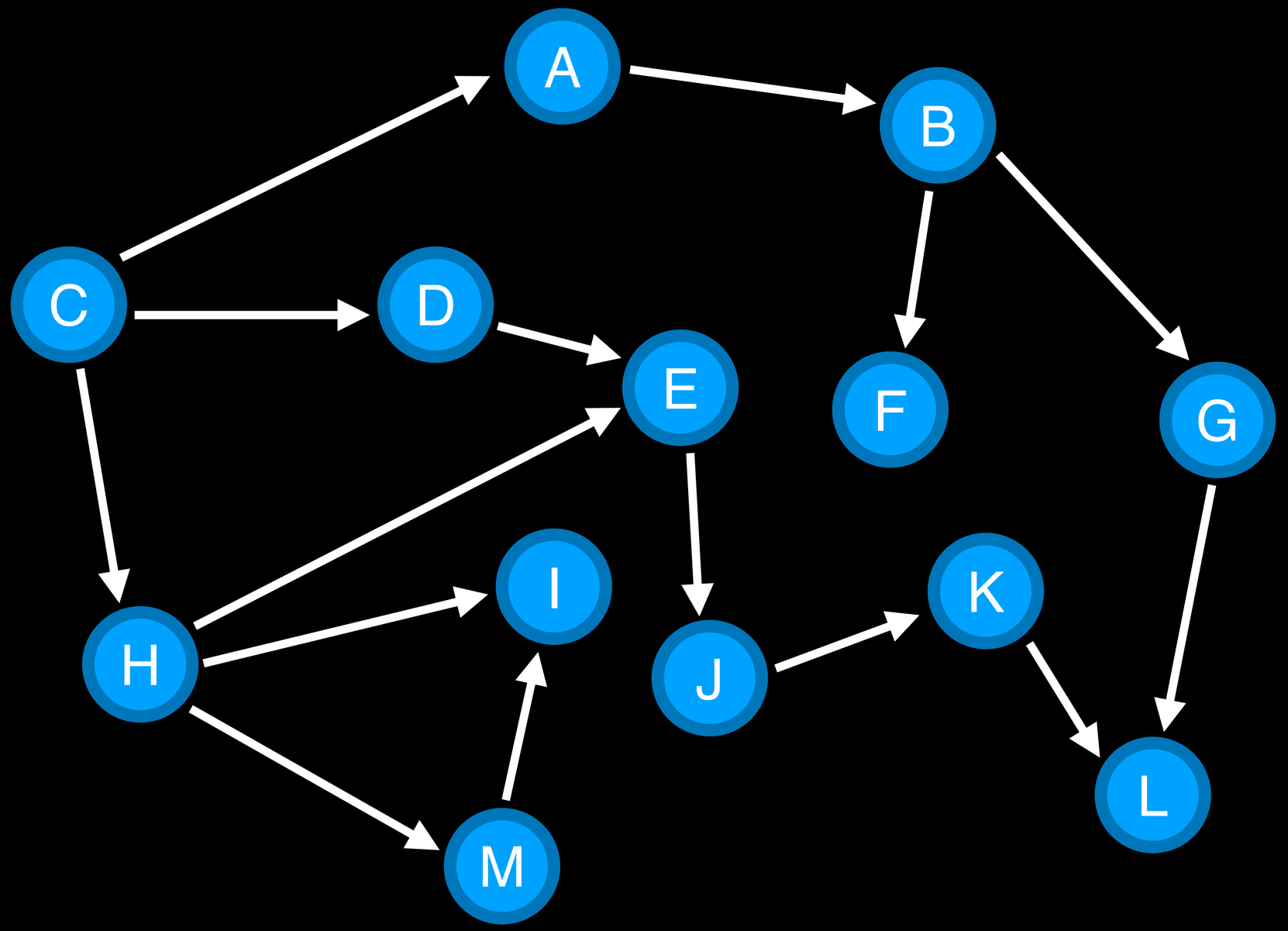


goal test

way to determine whether a given state is a goal state

path cost

numerical cost associated with a given path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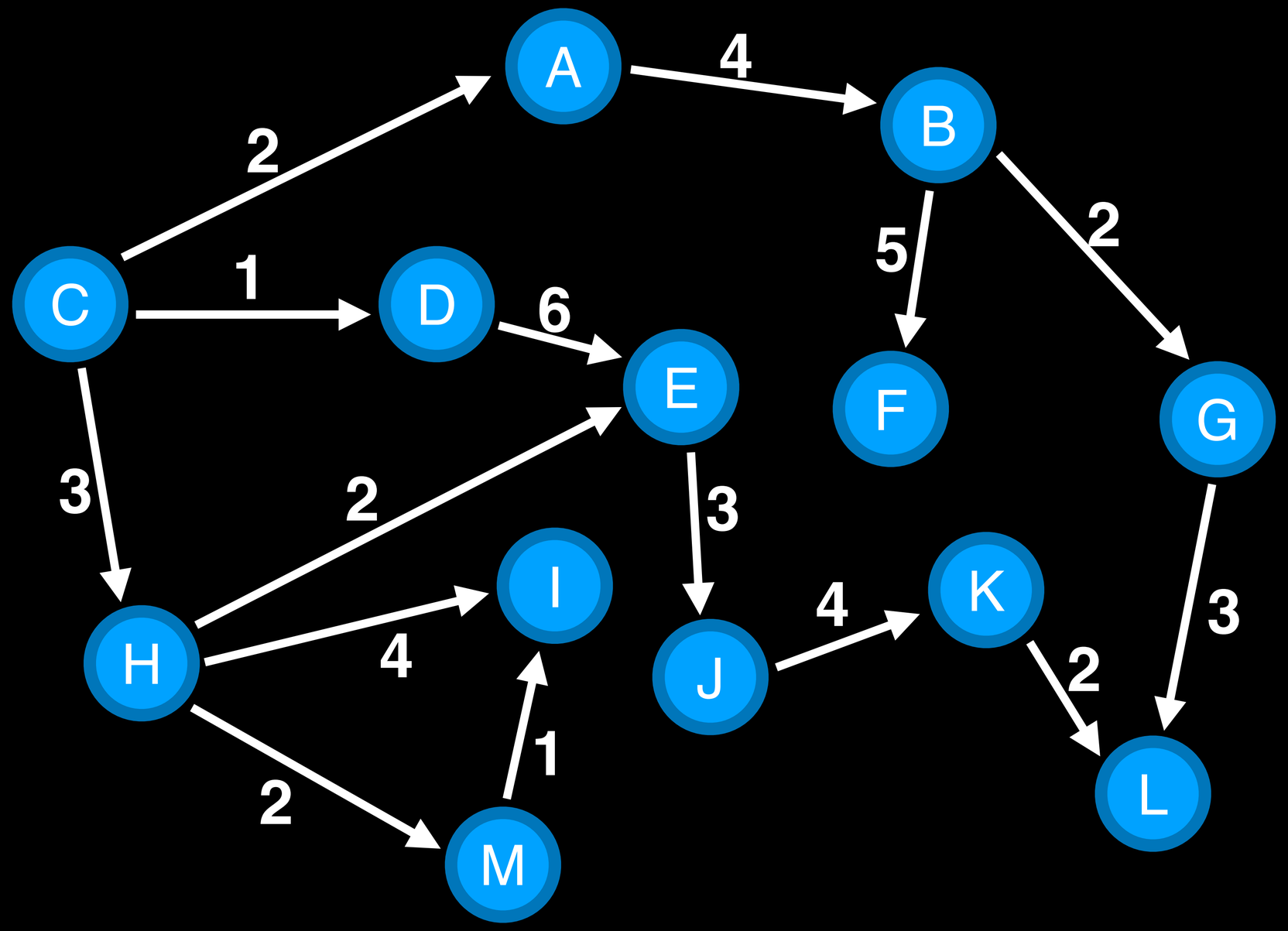
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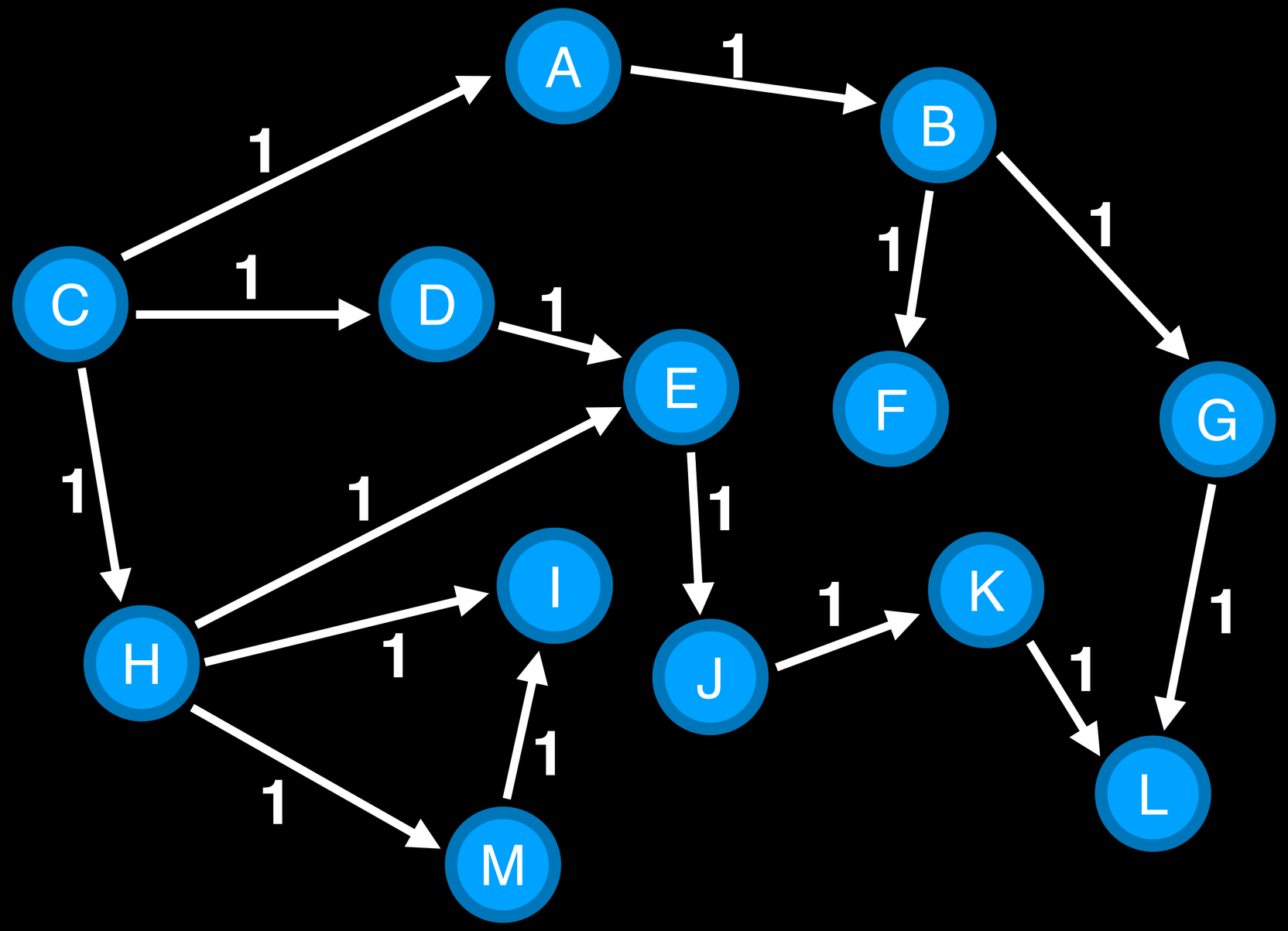
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Search Problems

- initial state
- actions
- transition model
- goal test
- path cost function

solution

a sequence of actions that leads from the initial state to a goal state

optimal solution

a solution that has the lowest path cost among all solutions

node

a data structure that keeps track of

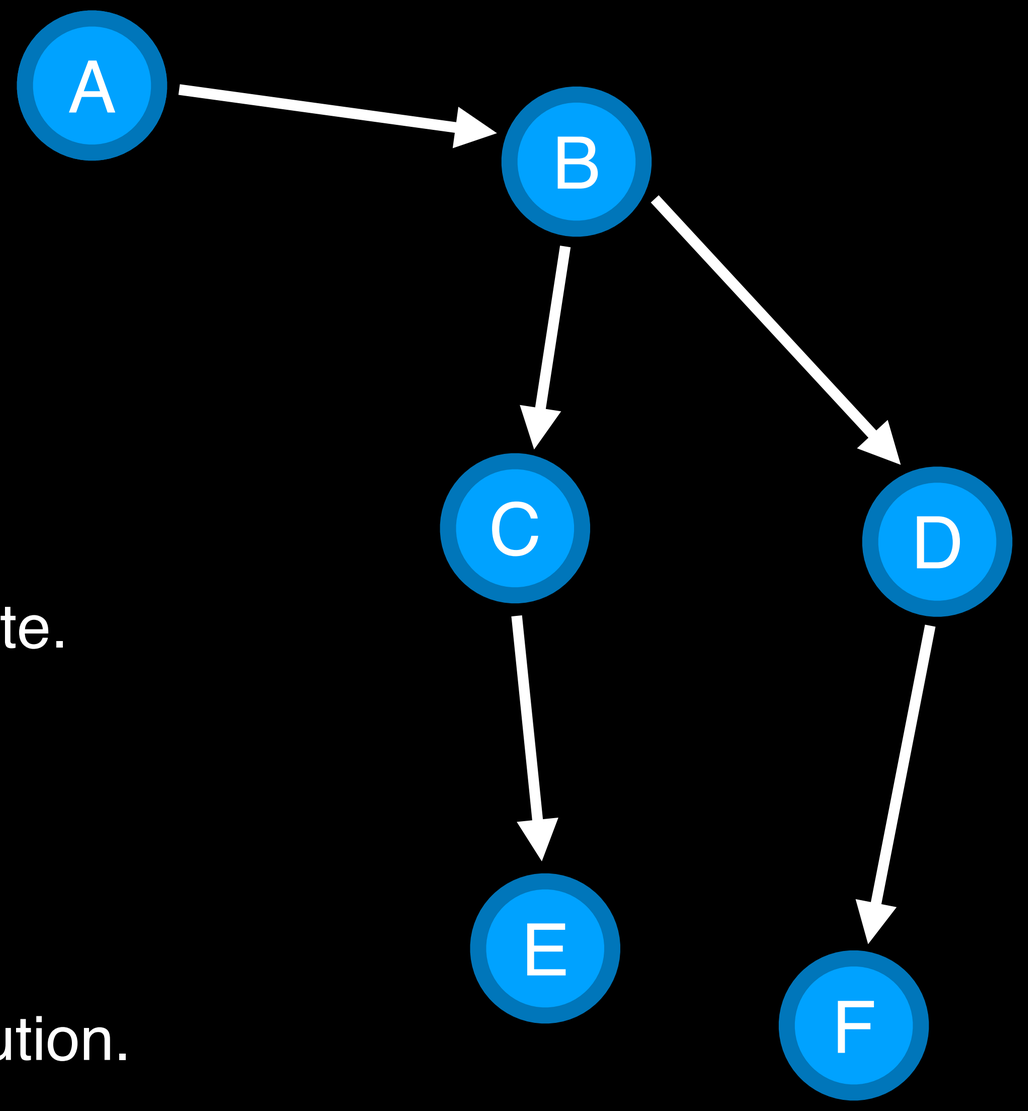
- a **state**
- a **parent** (node that generated this node)
- an **action** (action applied to parent to get node)
- a **path cost** (from initial state to node)

Approach

- Start with a **frontier** that contains the initial state.
- Repeat:
  - If the frontier is empty, then no solution.
  - Remove a node from the frontier.
  - If node contains goal state, return the solution.
  - **Expand** node, add resulting nodes to the frontier.

Find a path from A to E.

Frontier



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Frontier

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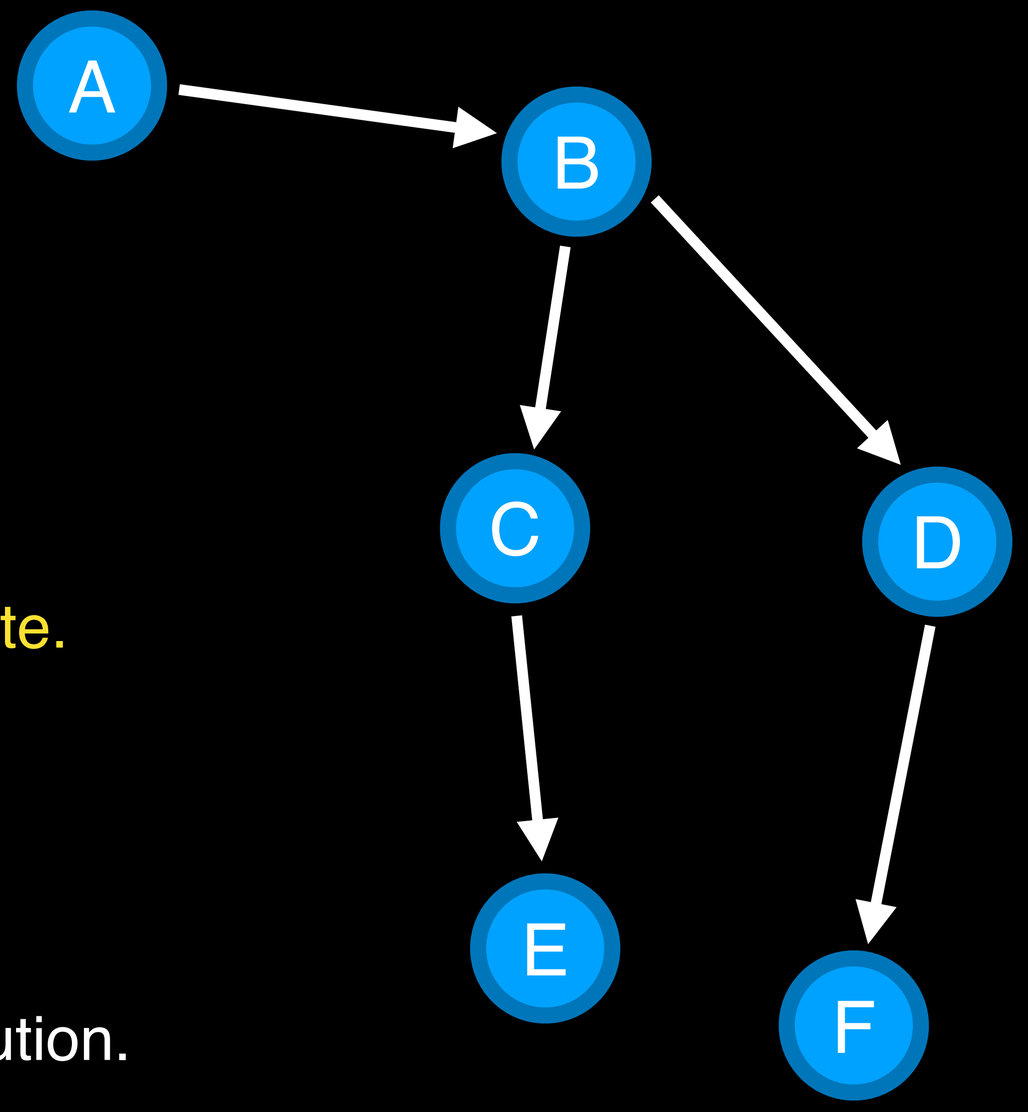

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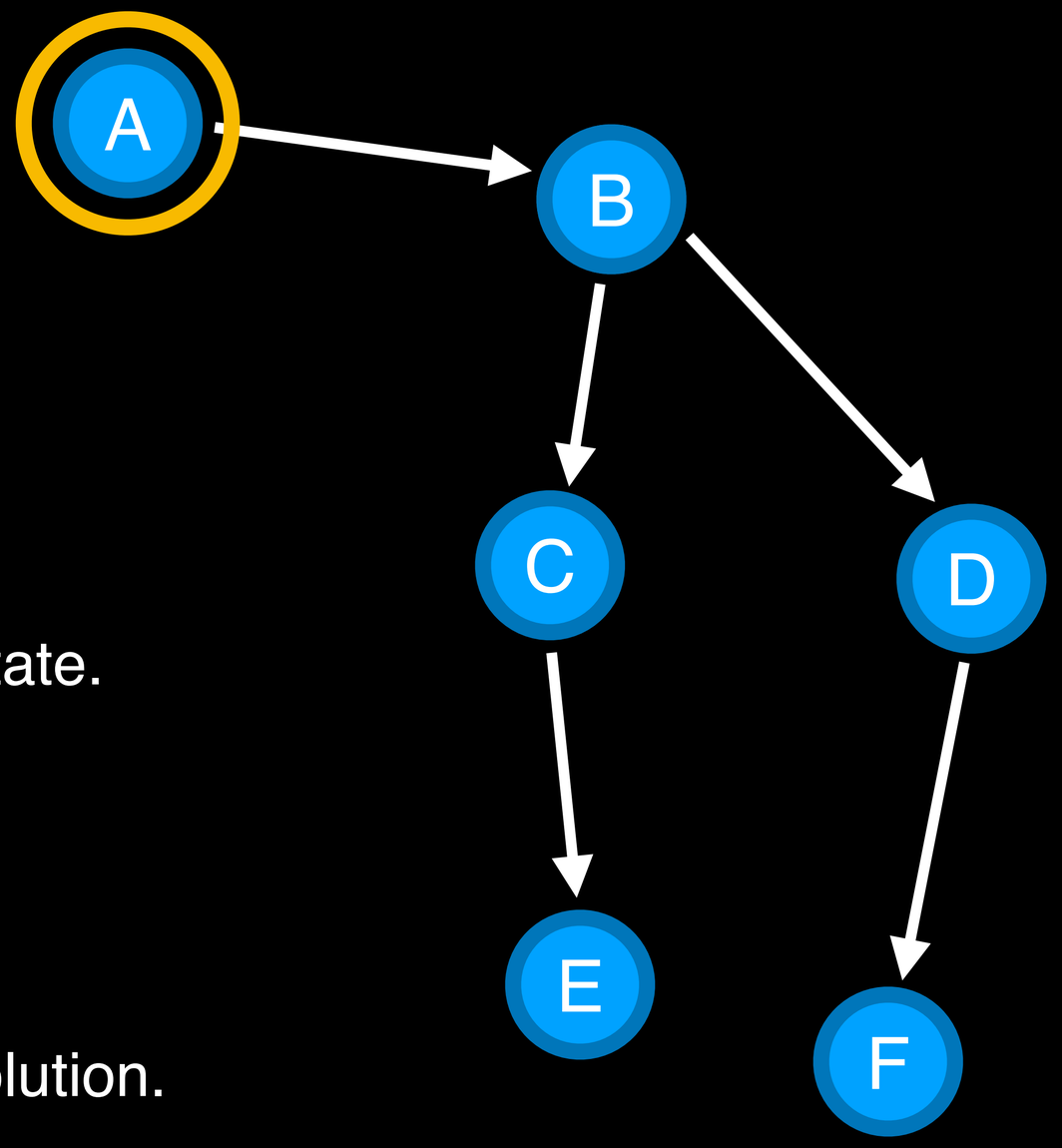

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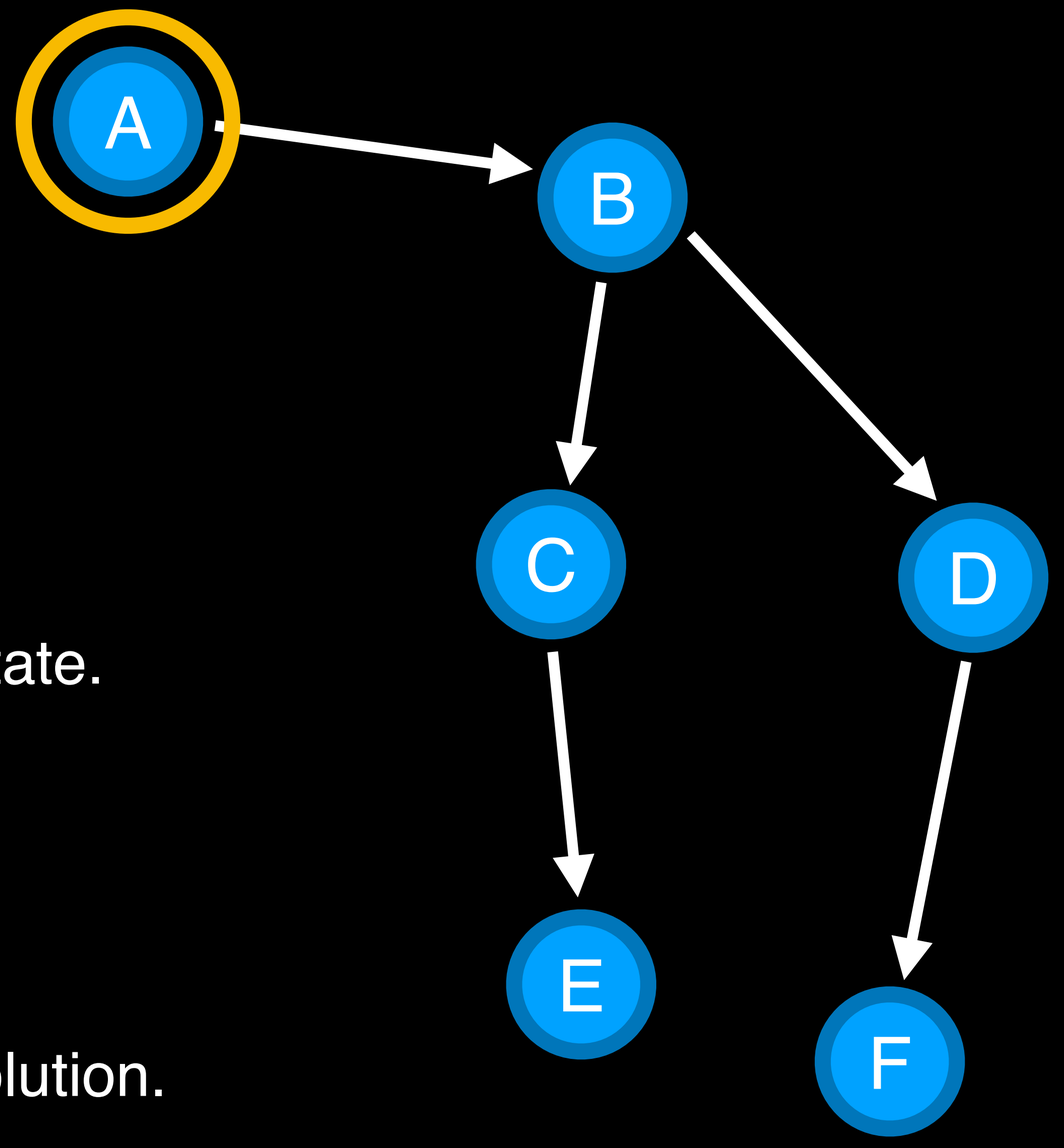
Find a path from A to E.

Frontier

B



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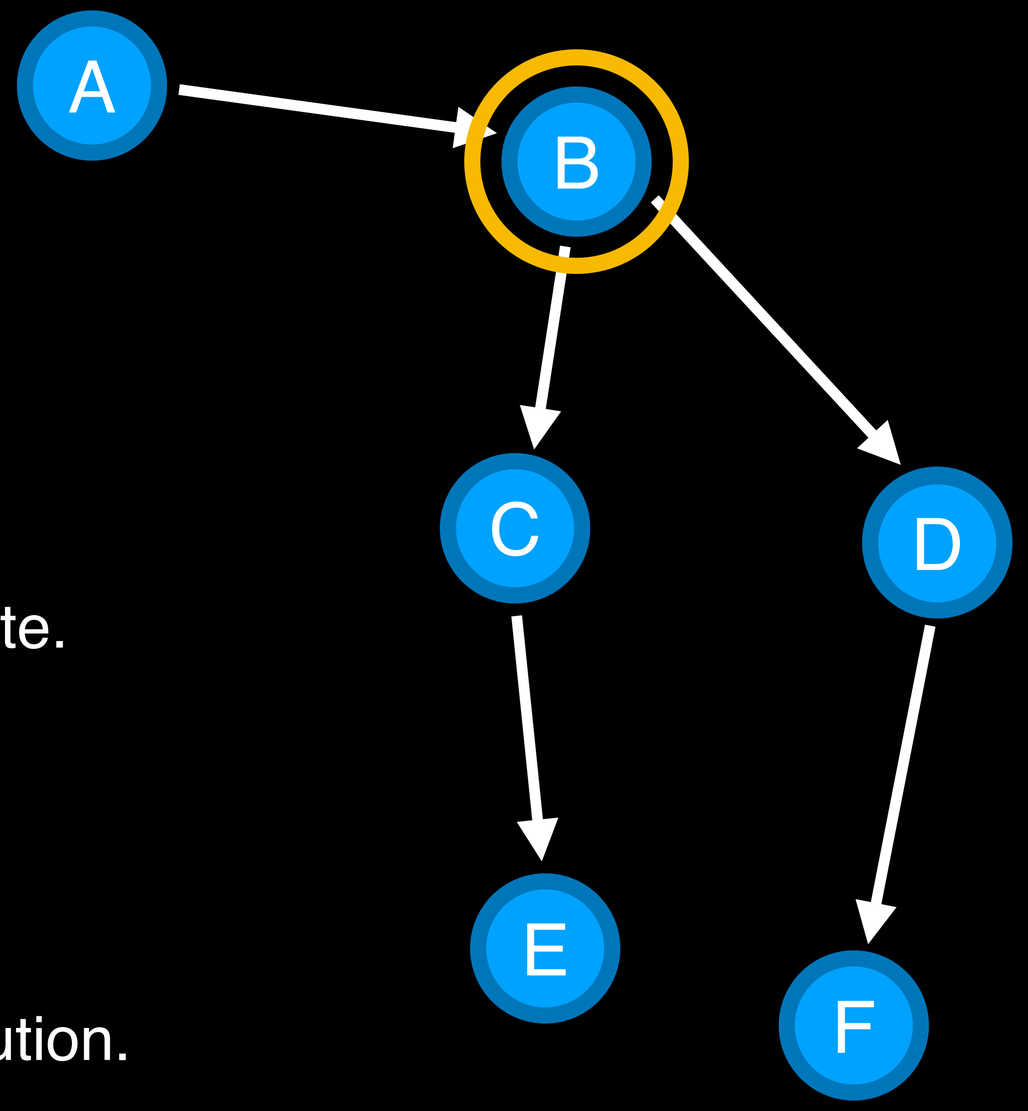
Frontier



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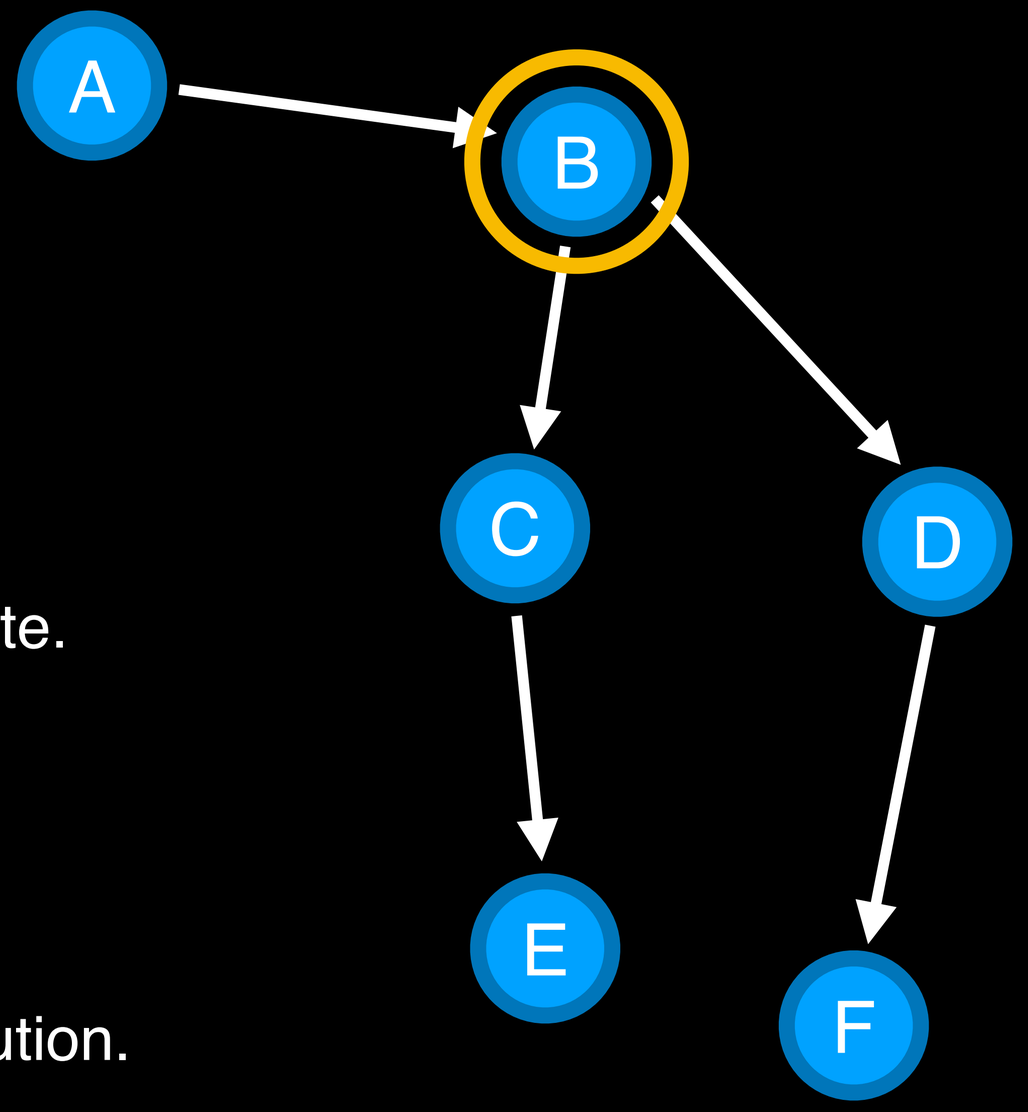
Find a path from A to E.

Frontier

C D



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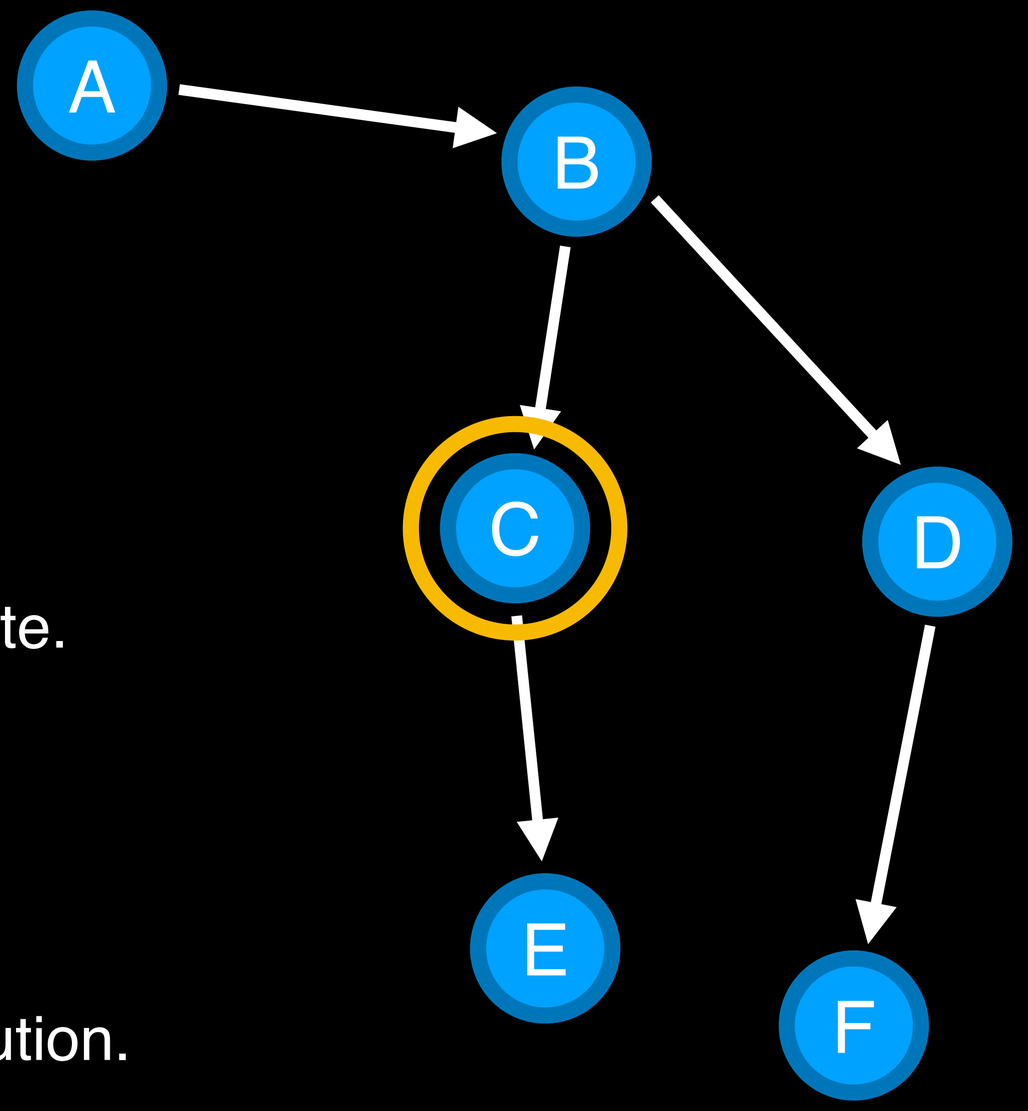
Find a path from A to E.

Frontier

D



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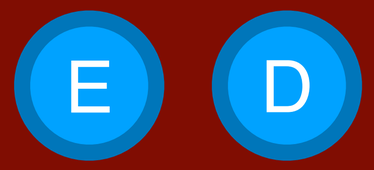
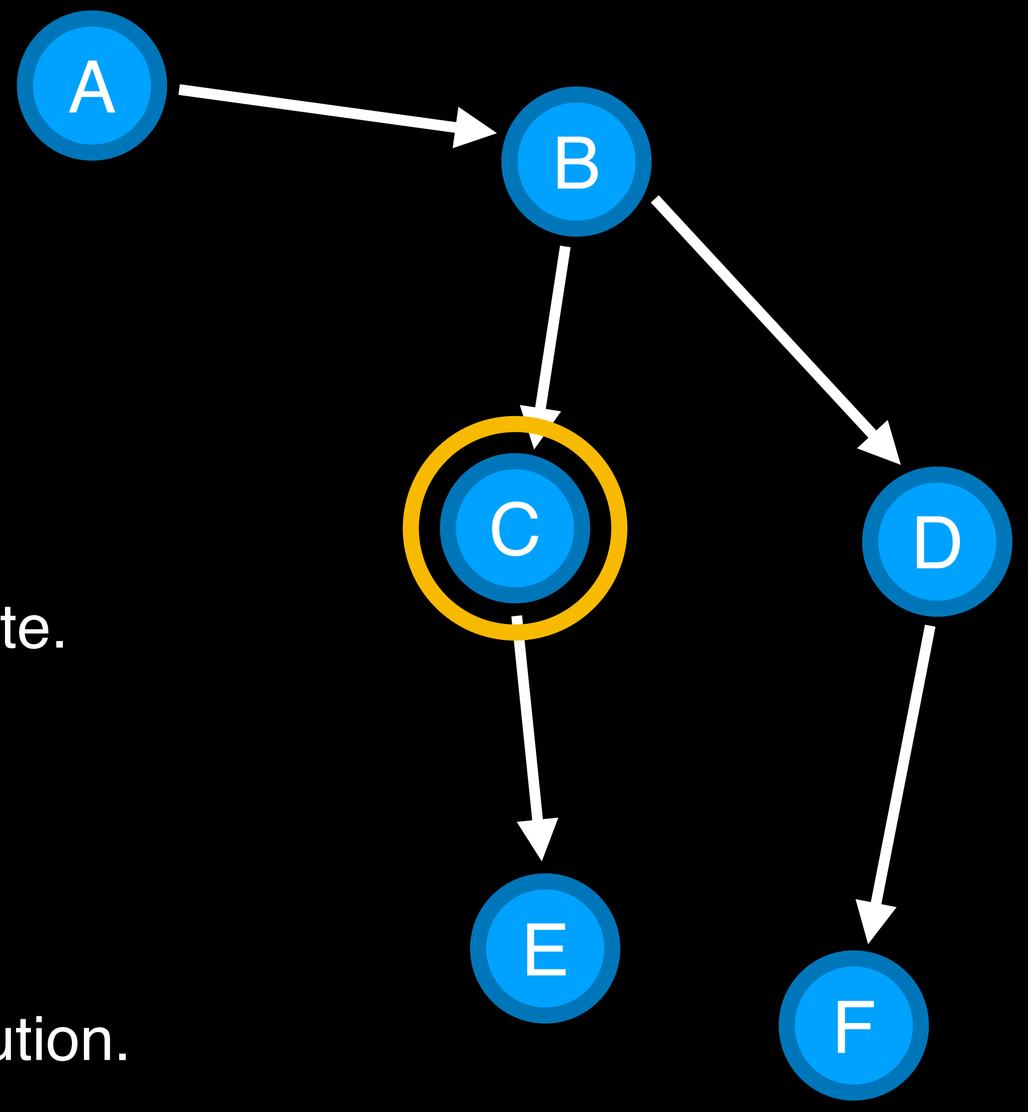
Find a path from A to E.

Frontier





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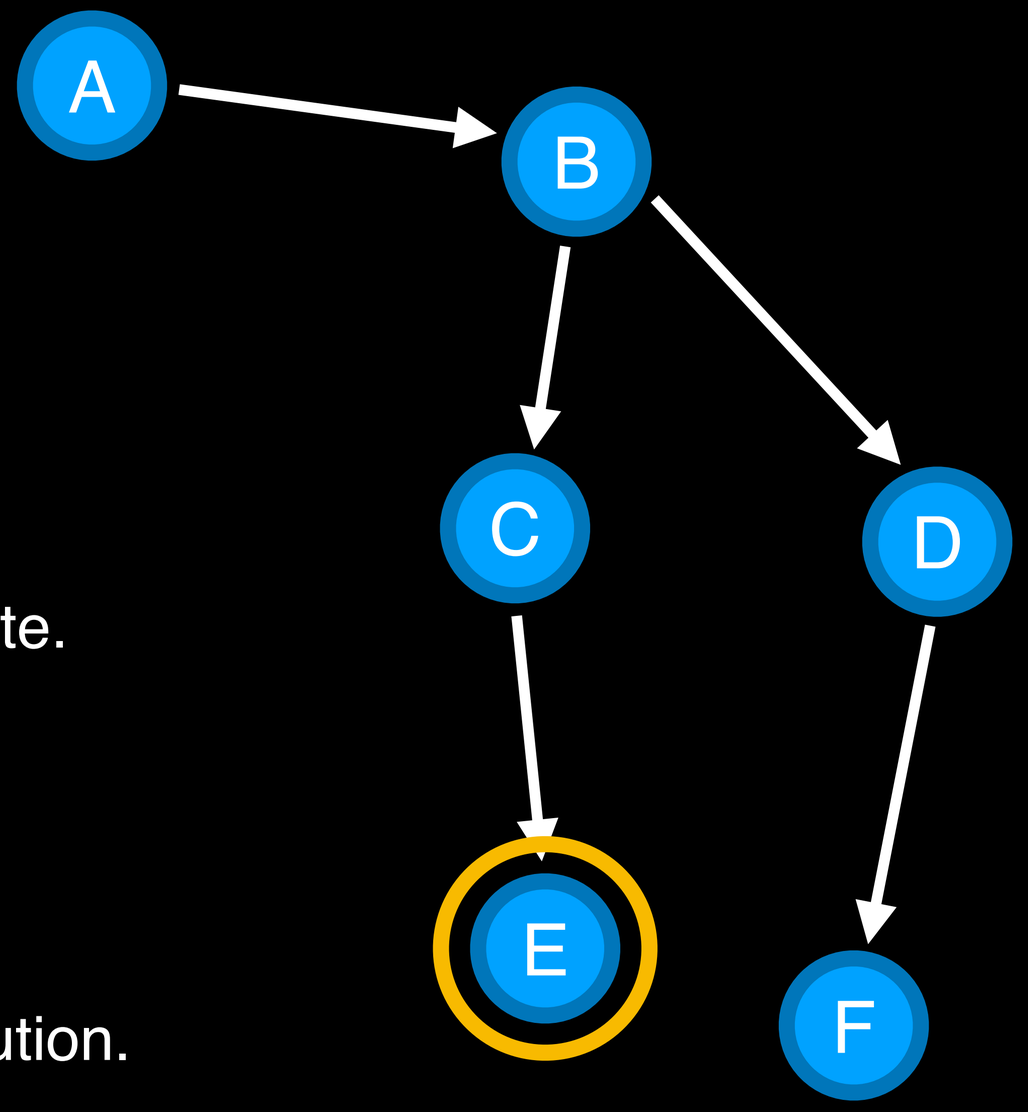

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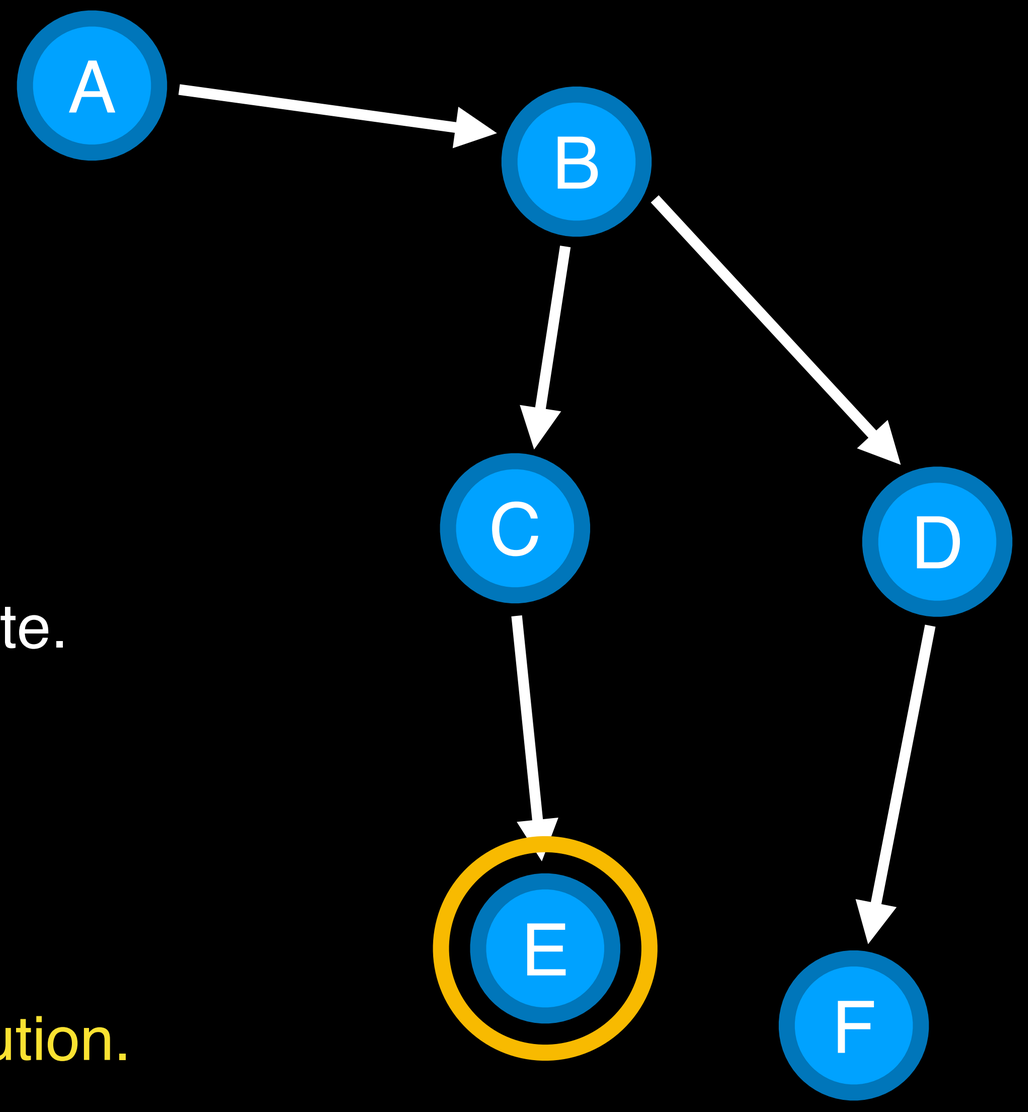
Find a path from A to E.

Frontier

D



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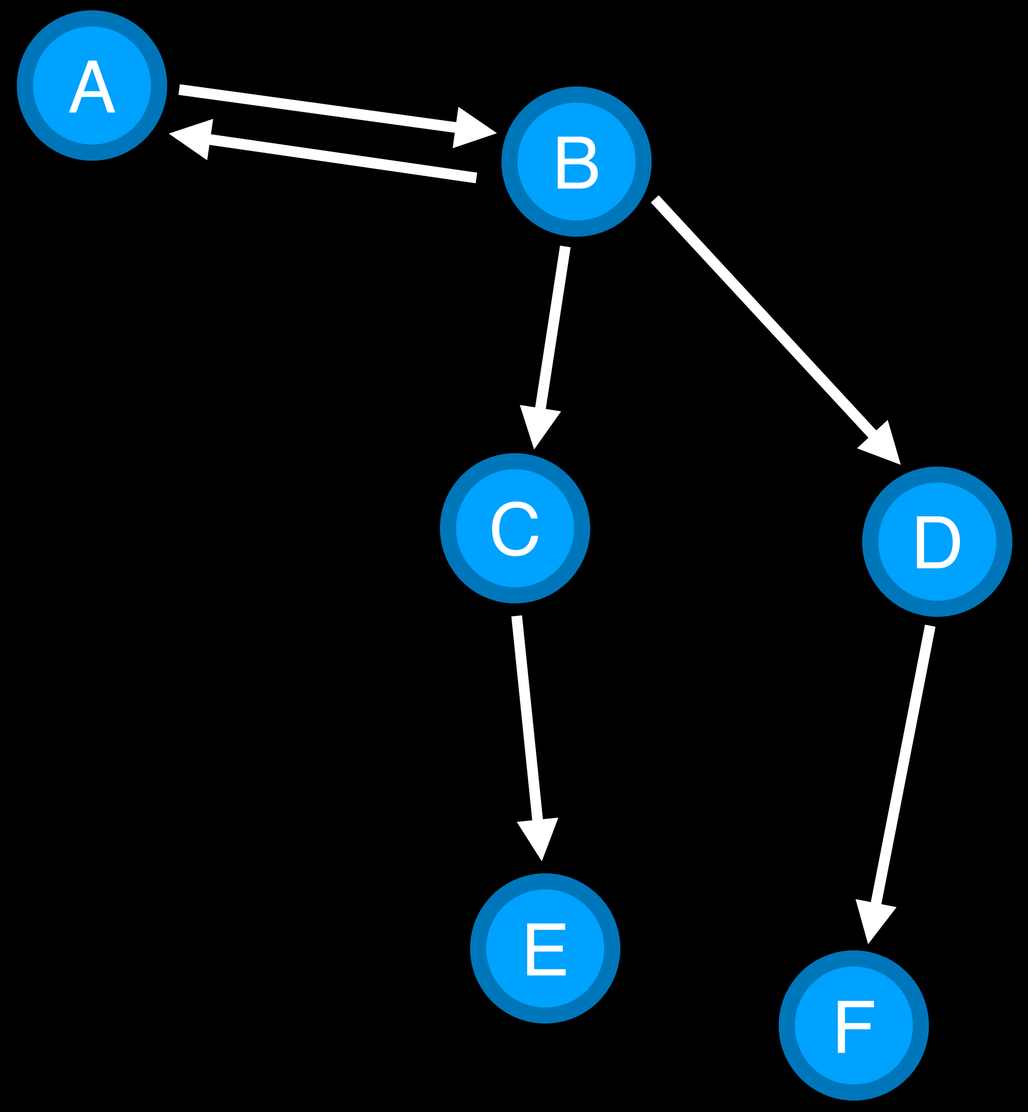
What could go wrong?

Find a path from A to E.

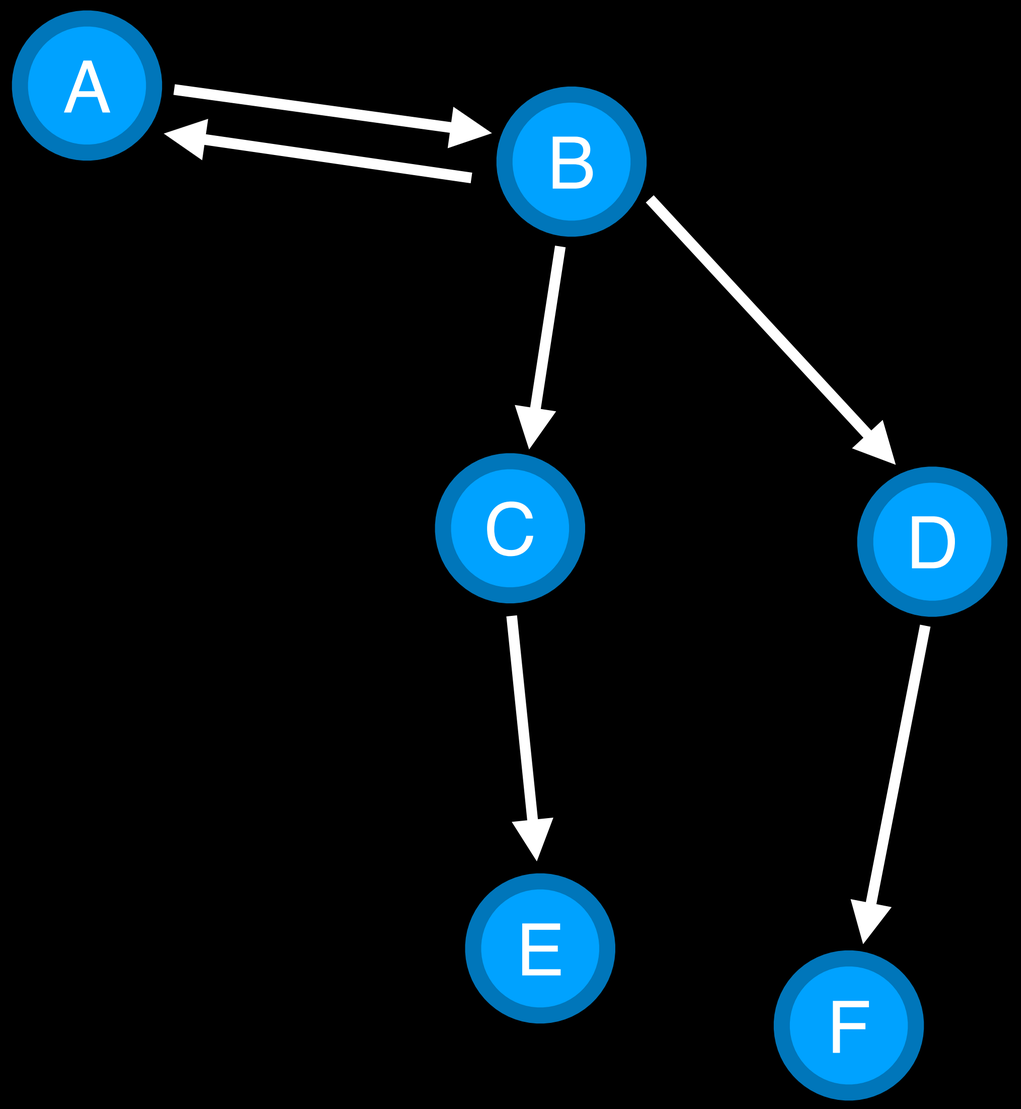
Frontier



Find a path from A to E.

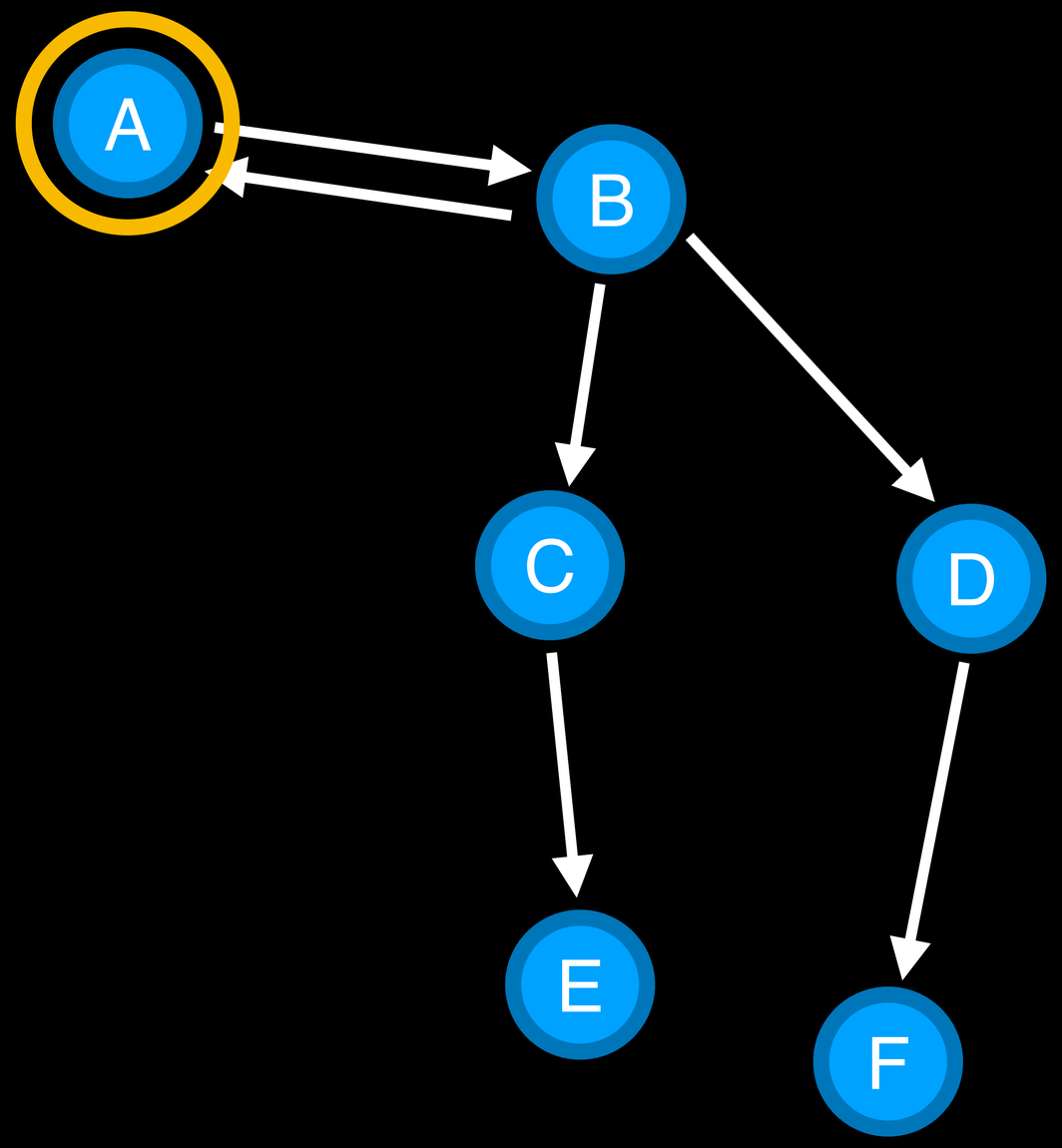
Frontier

A



Find a path from A to E.

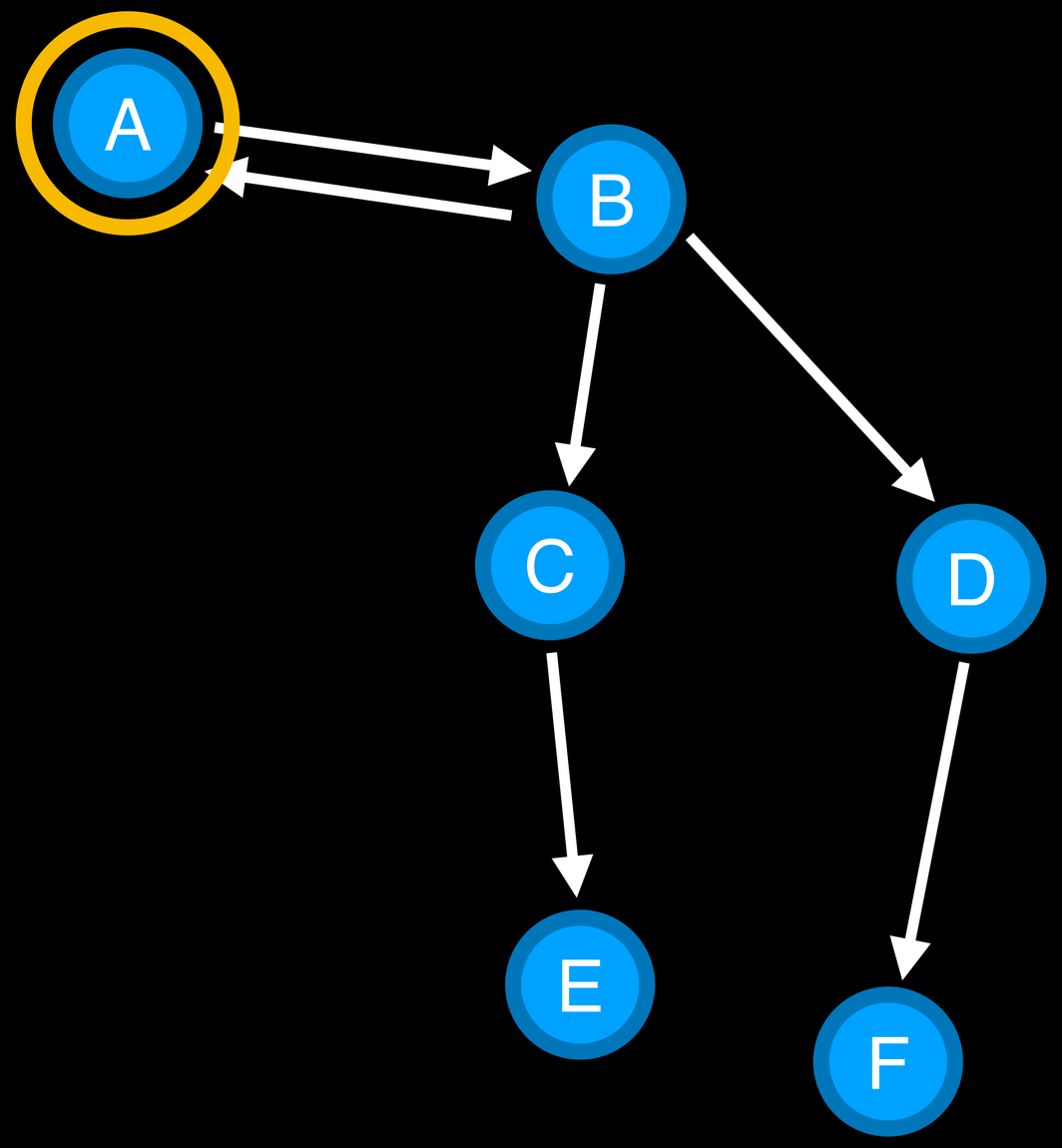
Frontier



Find a path from A to E.

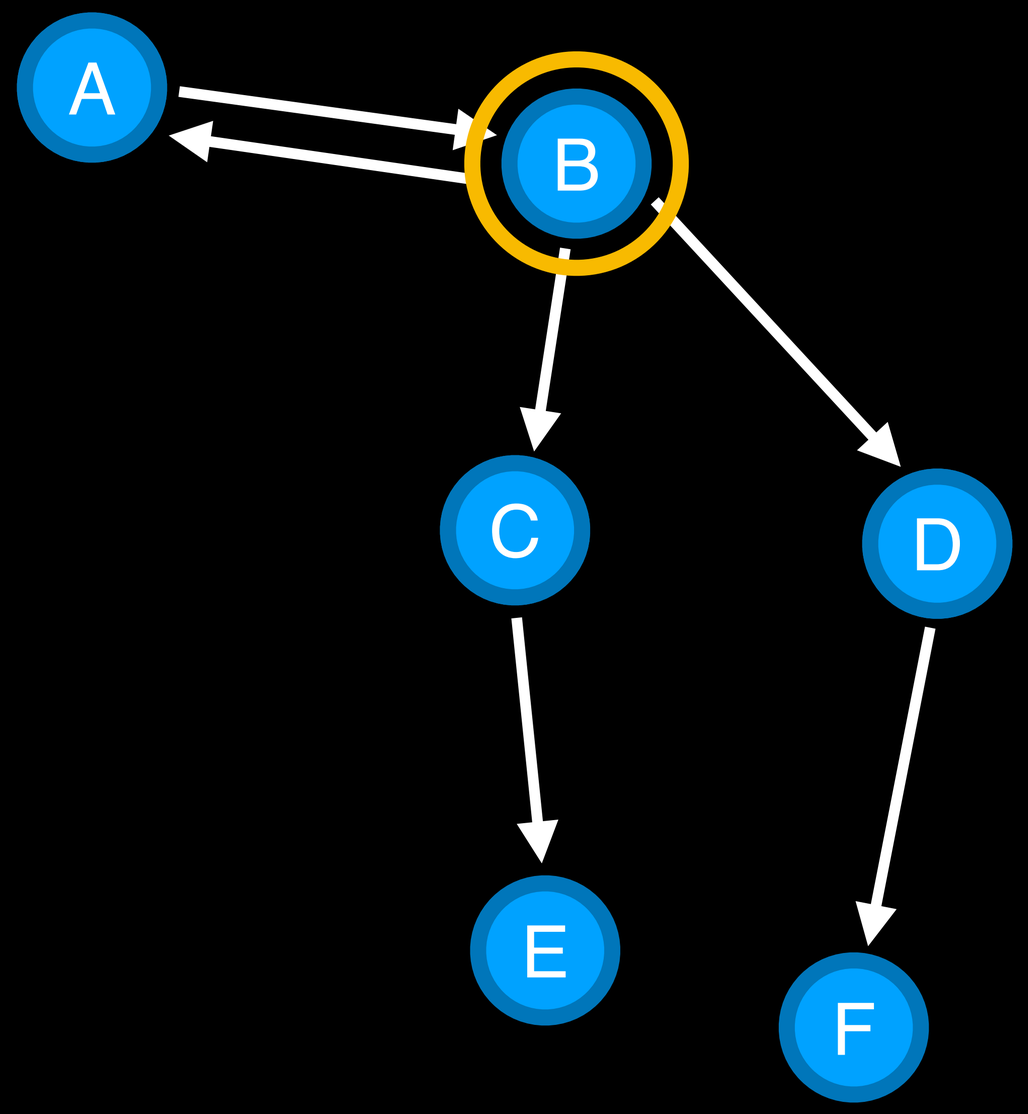
Frontier

B



Find a path from A to E.

Frontier



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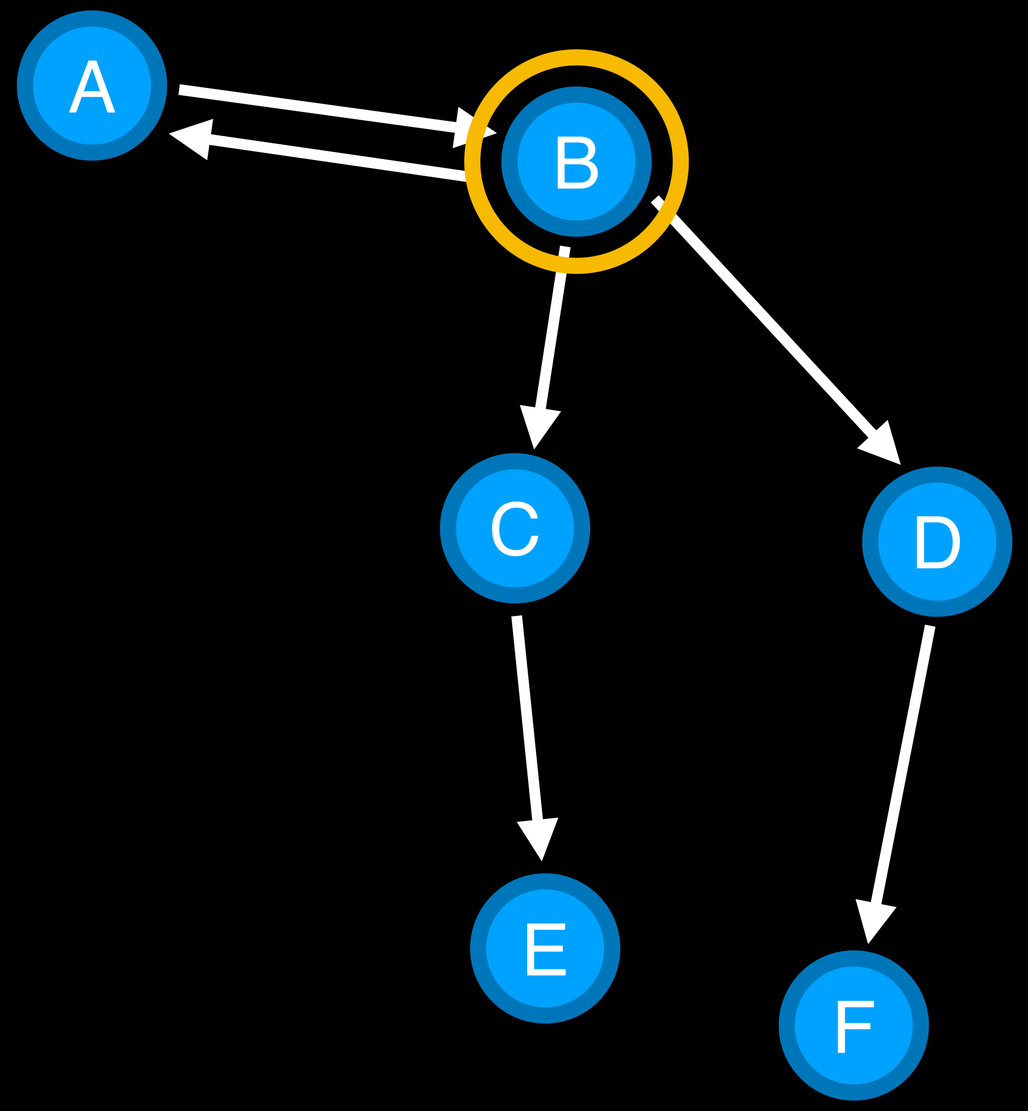
Frontier



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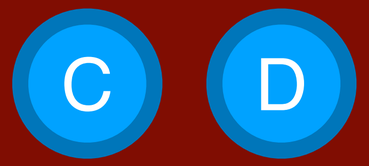
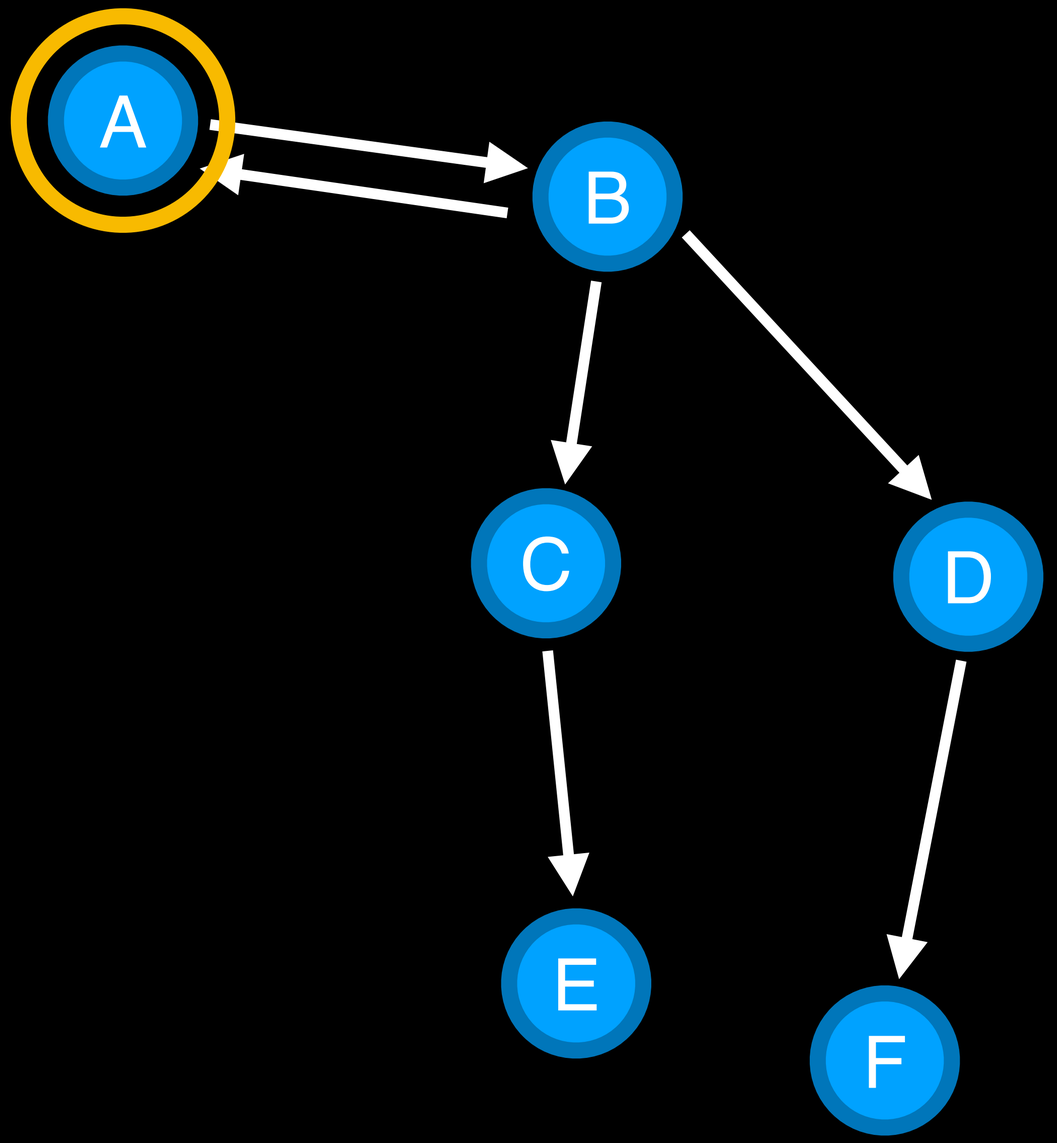
Frontier





Revised Approach

- Start with a **frontier** that contains the initial state.
- Start with an empty **explored set**.
- Repeat:
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Revised Approach

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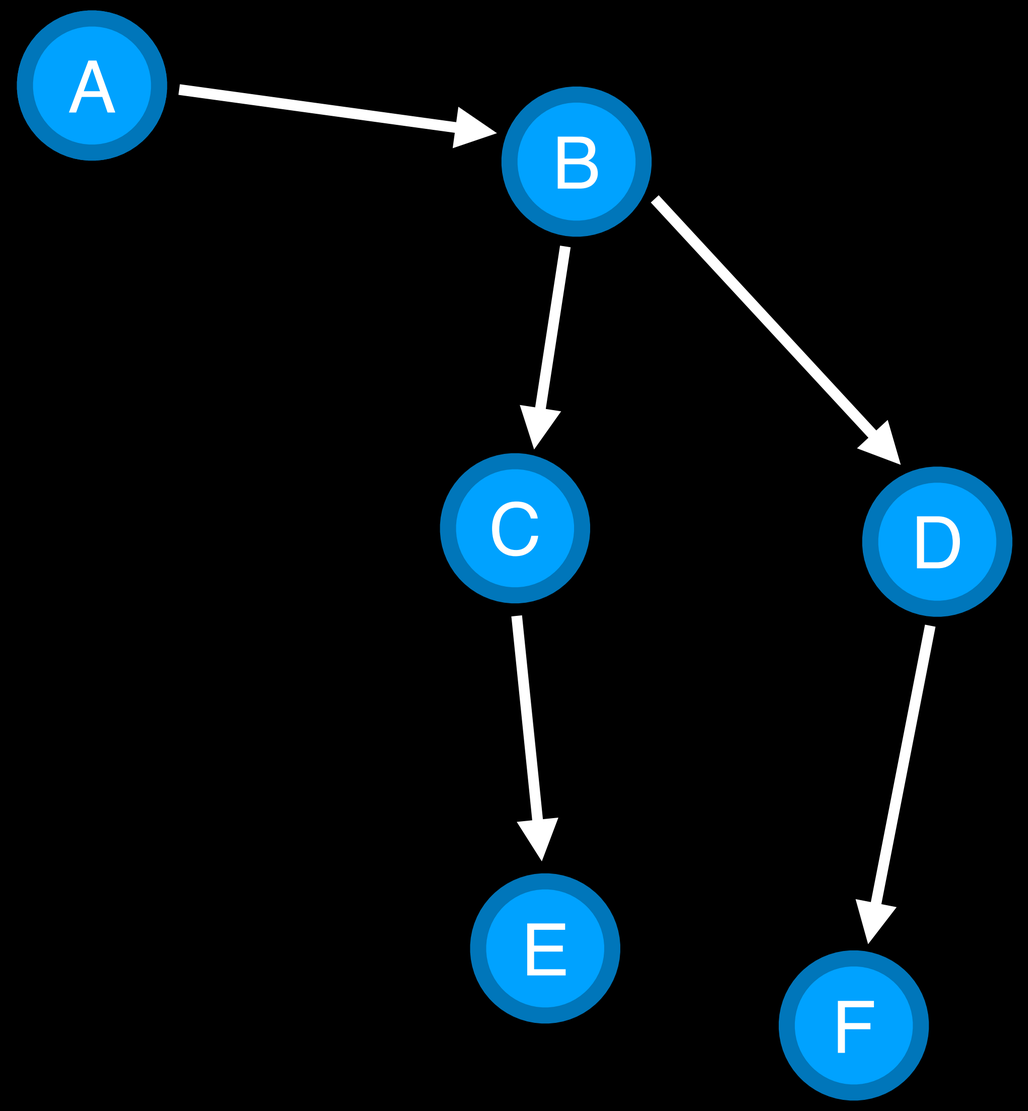
stack

last-in first-out data type

Find a path from A to E.

Frontier

Explored Set

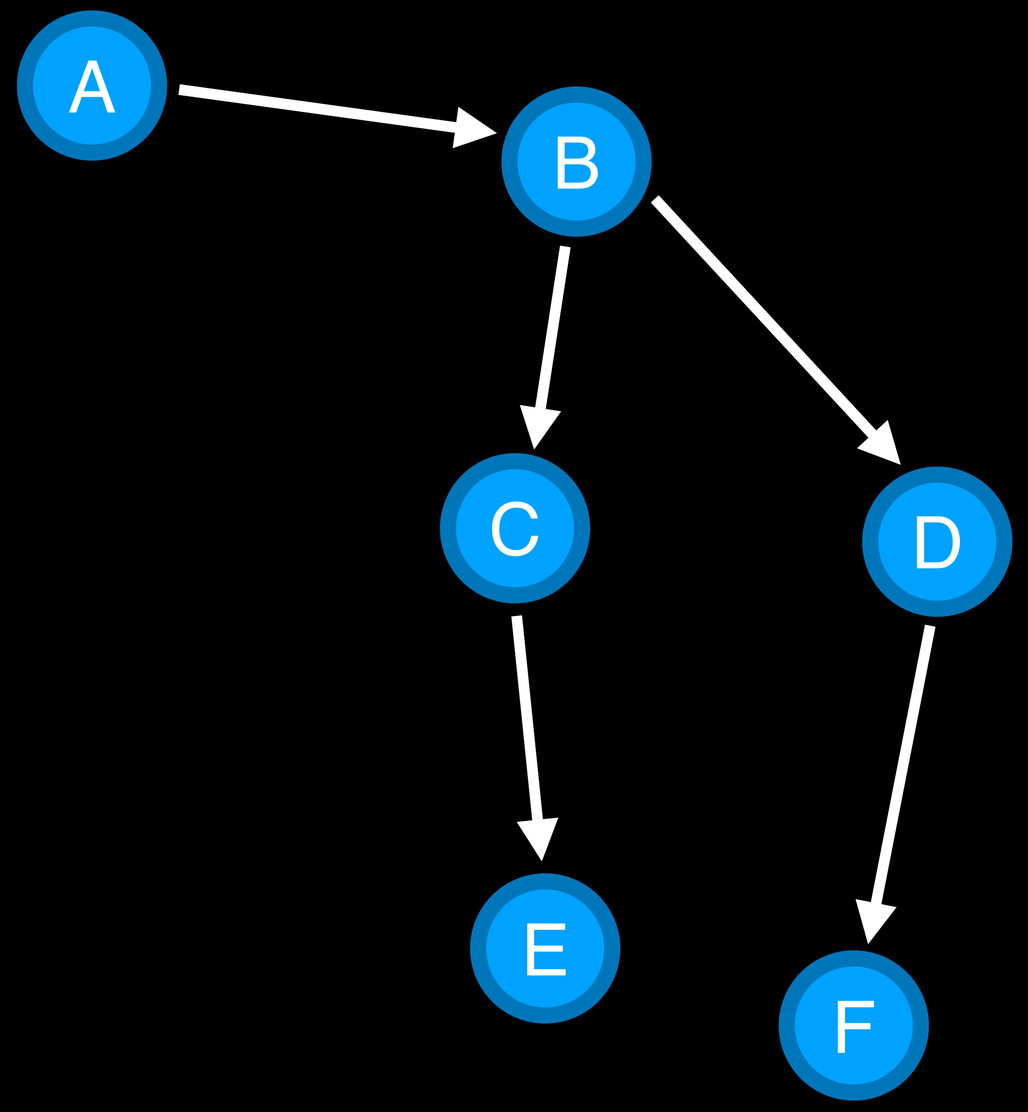


Find a path from A to E.

Frontier

A

Explored Set

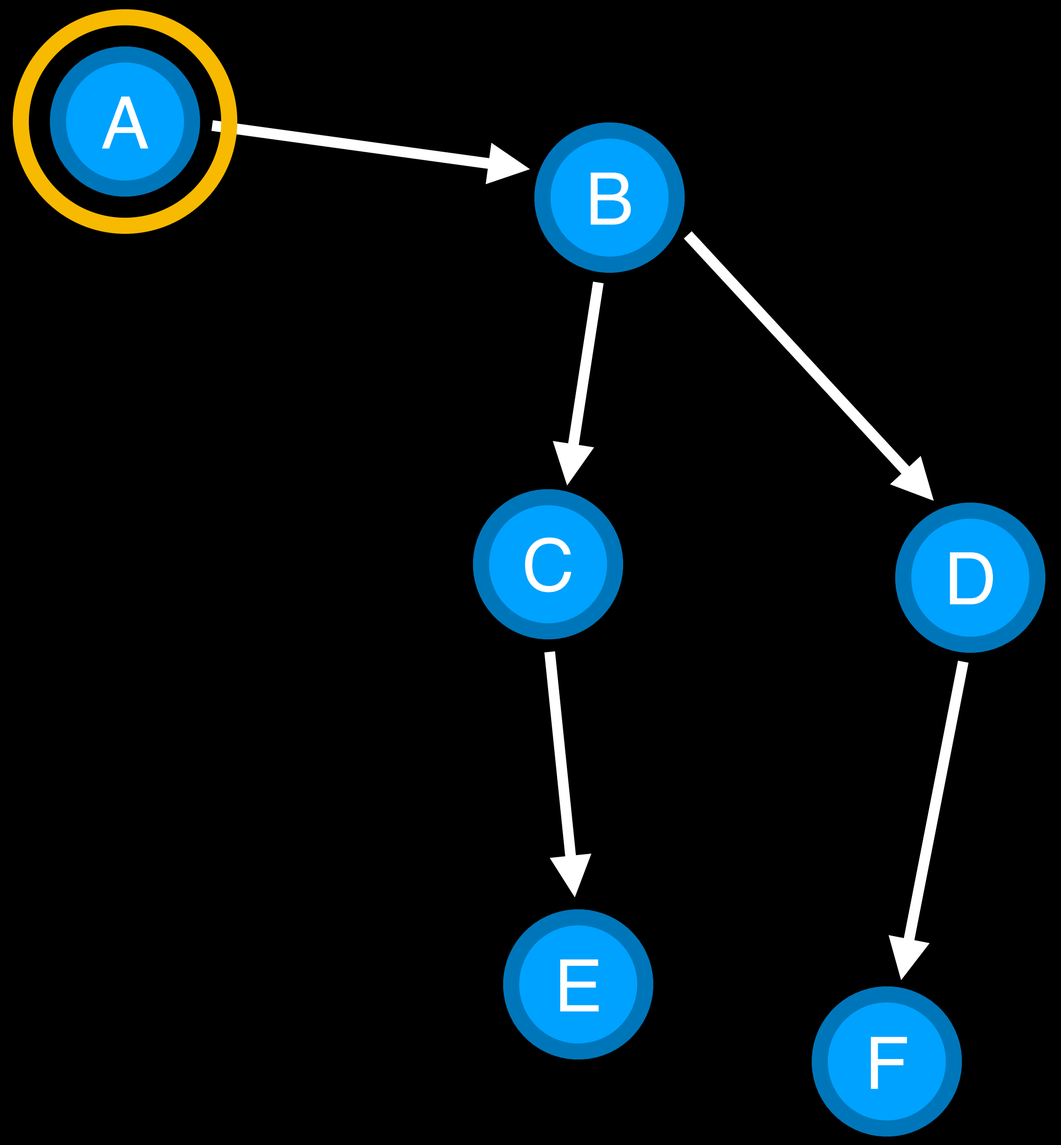


Find a path from A to E.

Frontier

Explored Set

A



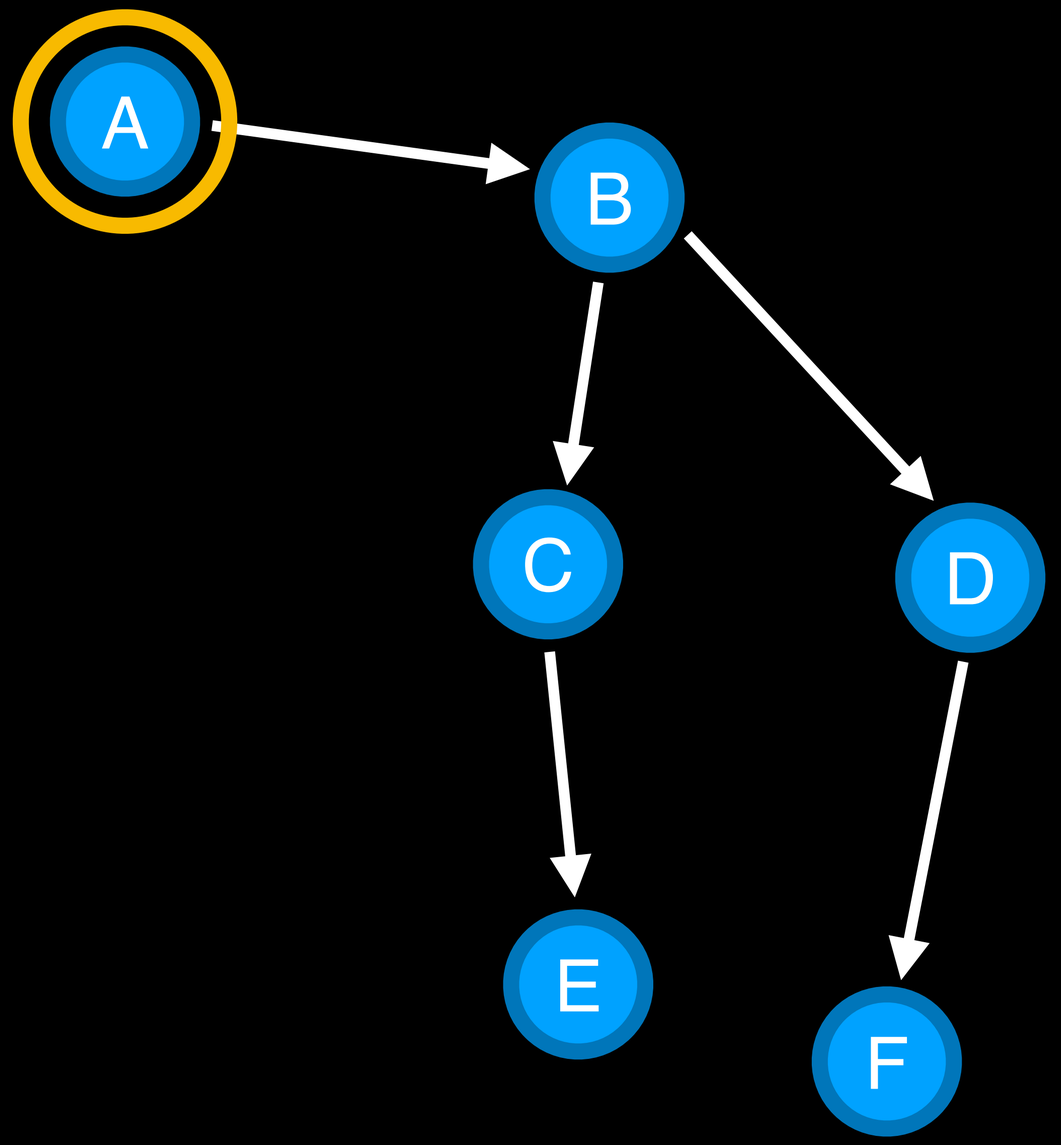
Find a path from A to E.

Frontier

B

Explored Set

A

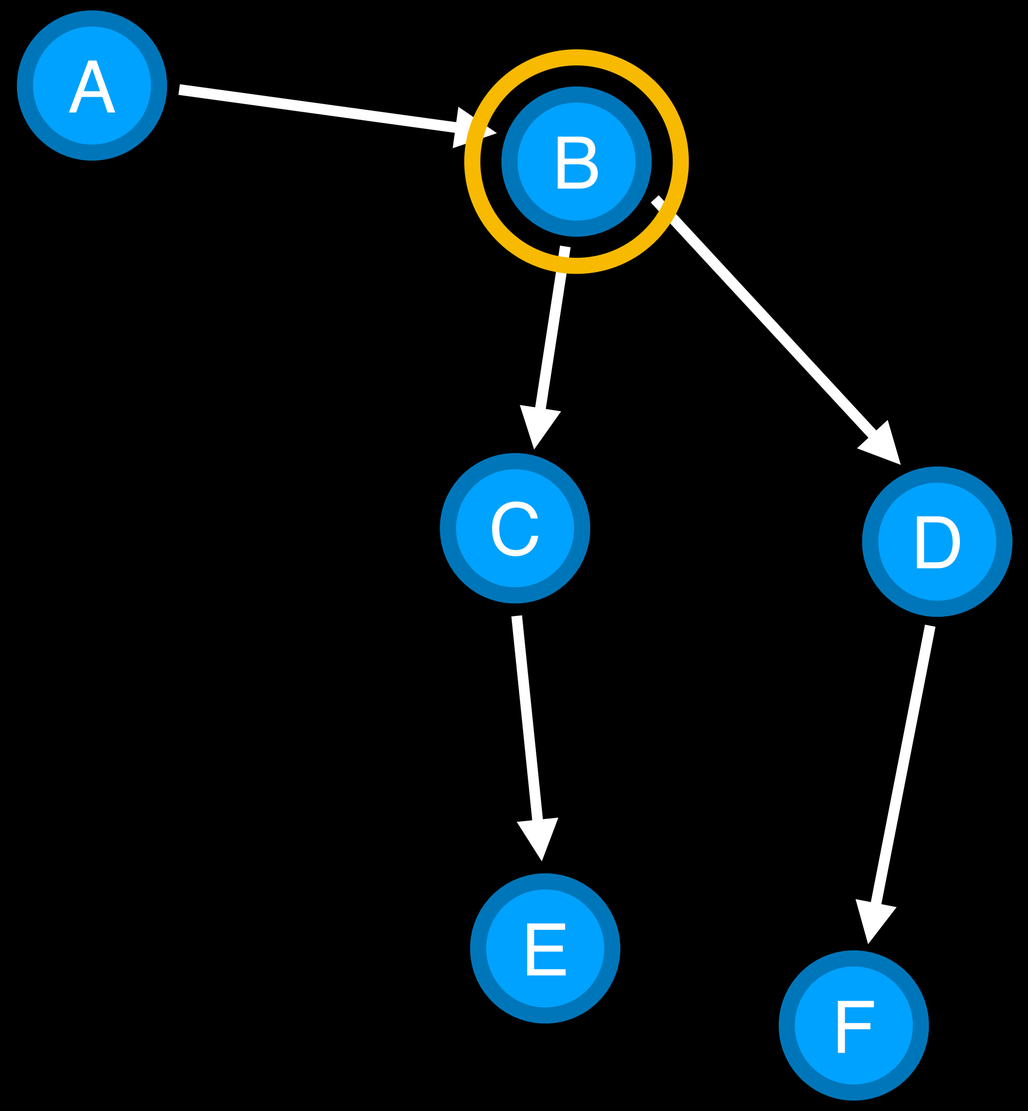


Find a path from A to E.

Frontier

Explored Set

A B



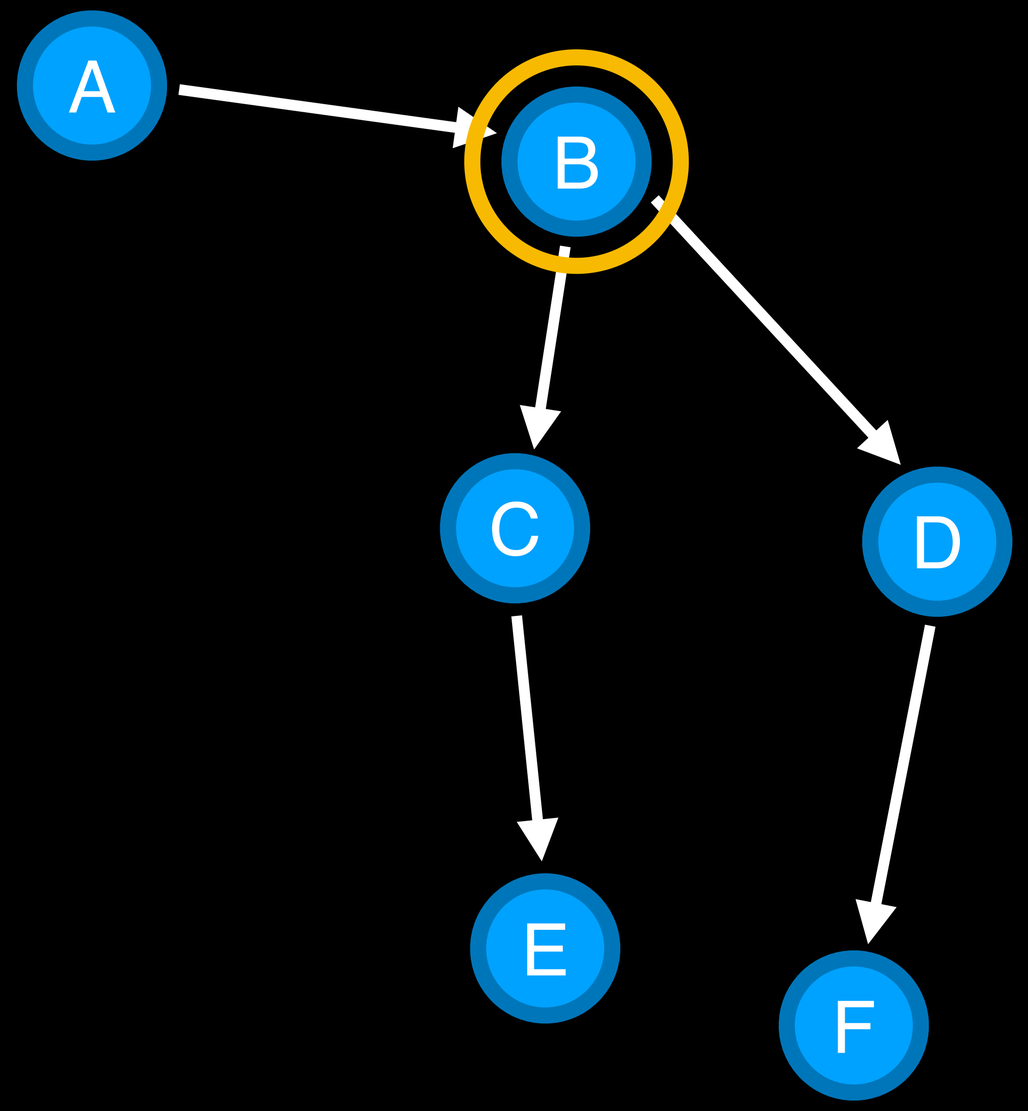
Find a path from A to E.

Frontier

C D

Explored Set

A B



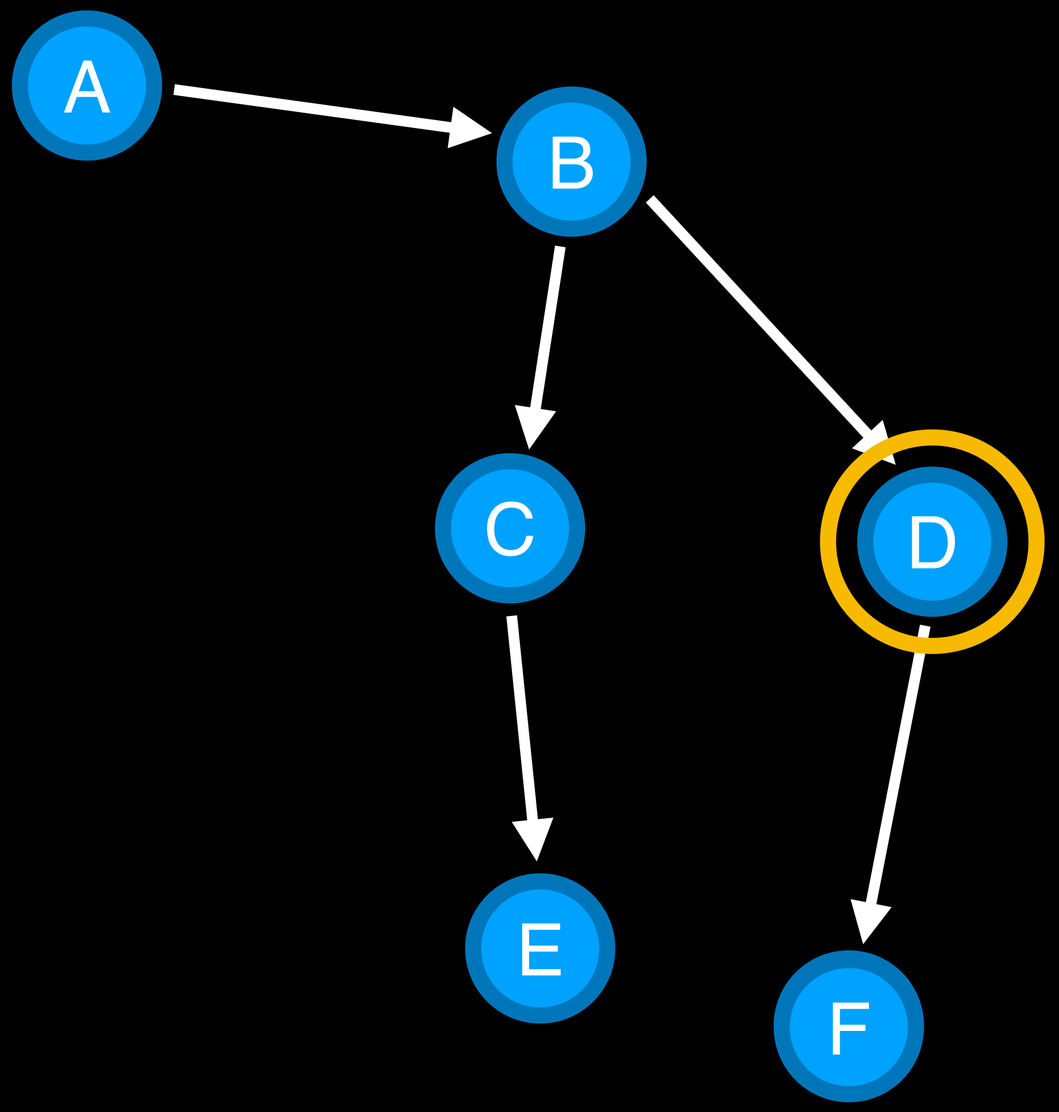
Find a path from A to E.

Frontier

C

Explored Set

A B D



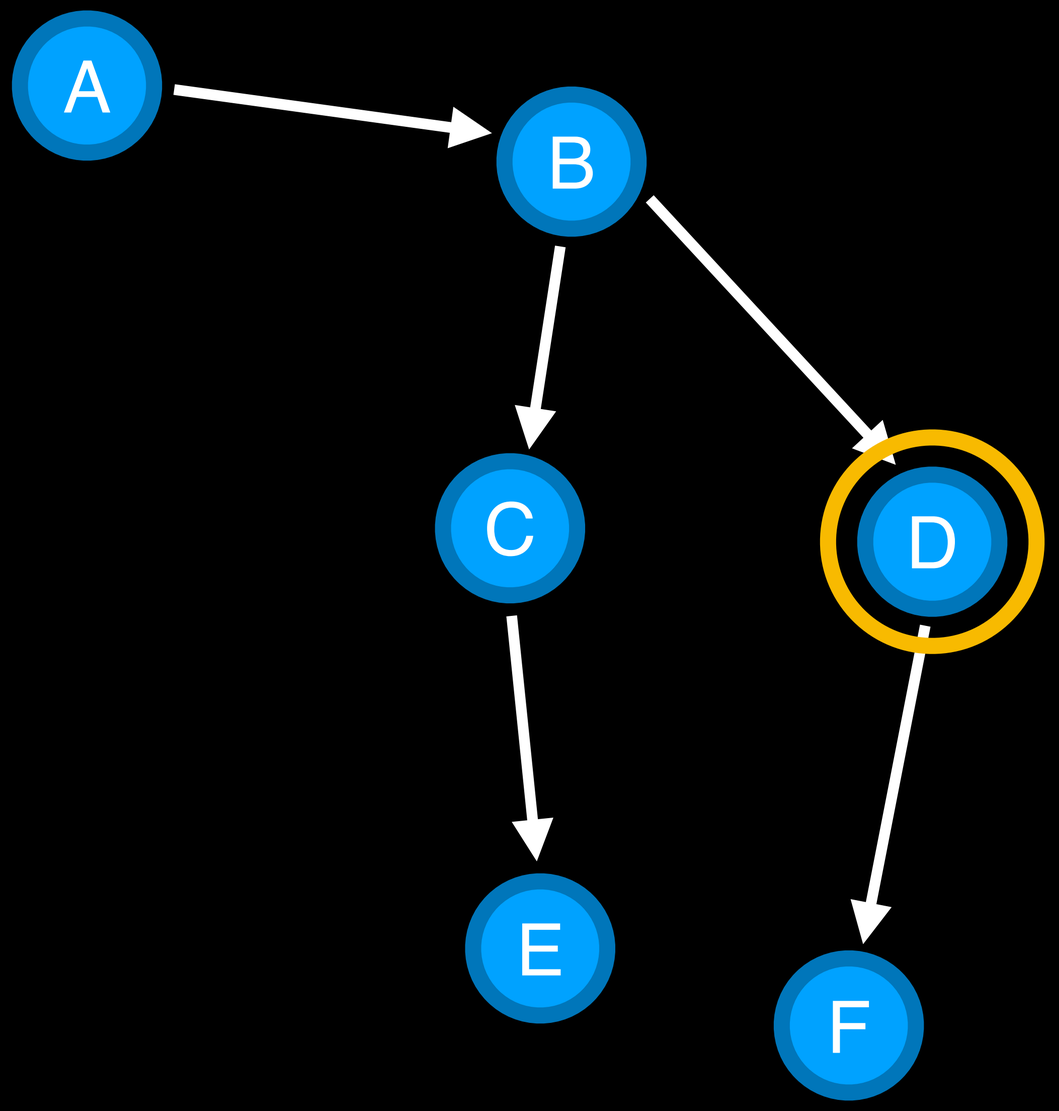
Find a path from A to E.

Frontier

C F

Explored Set

A B D



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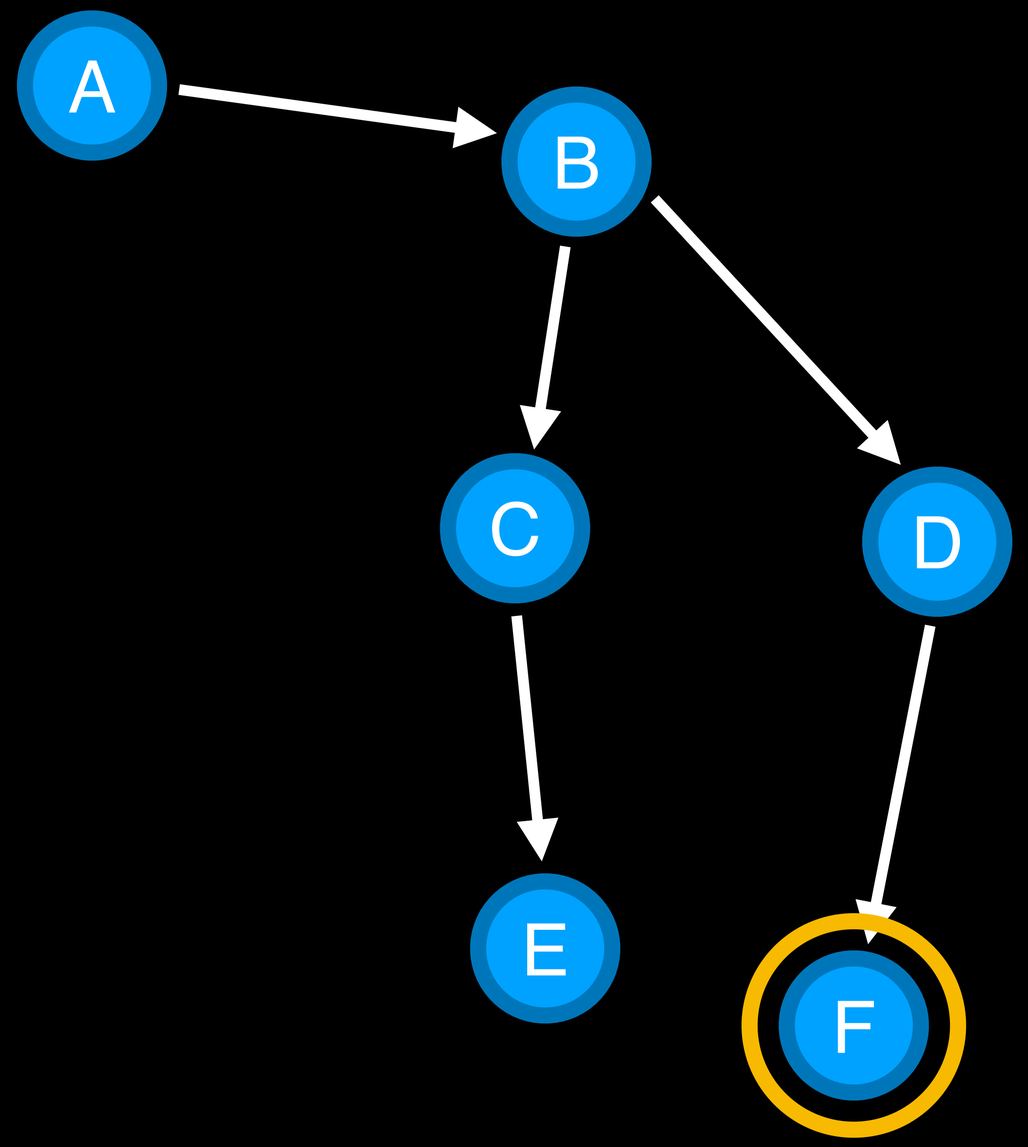
Frontier

C

Explored Set

A B D F

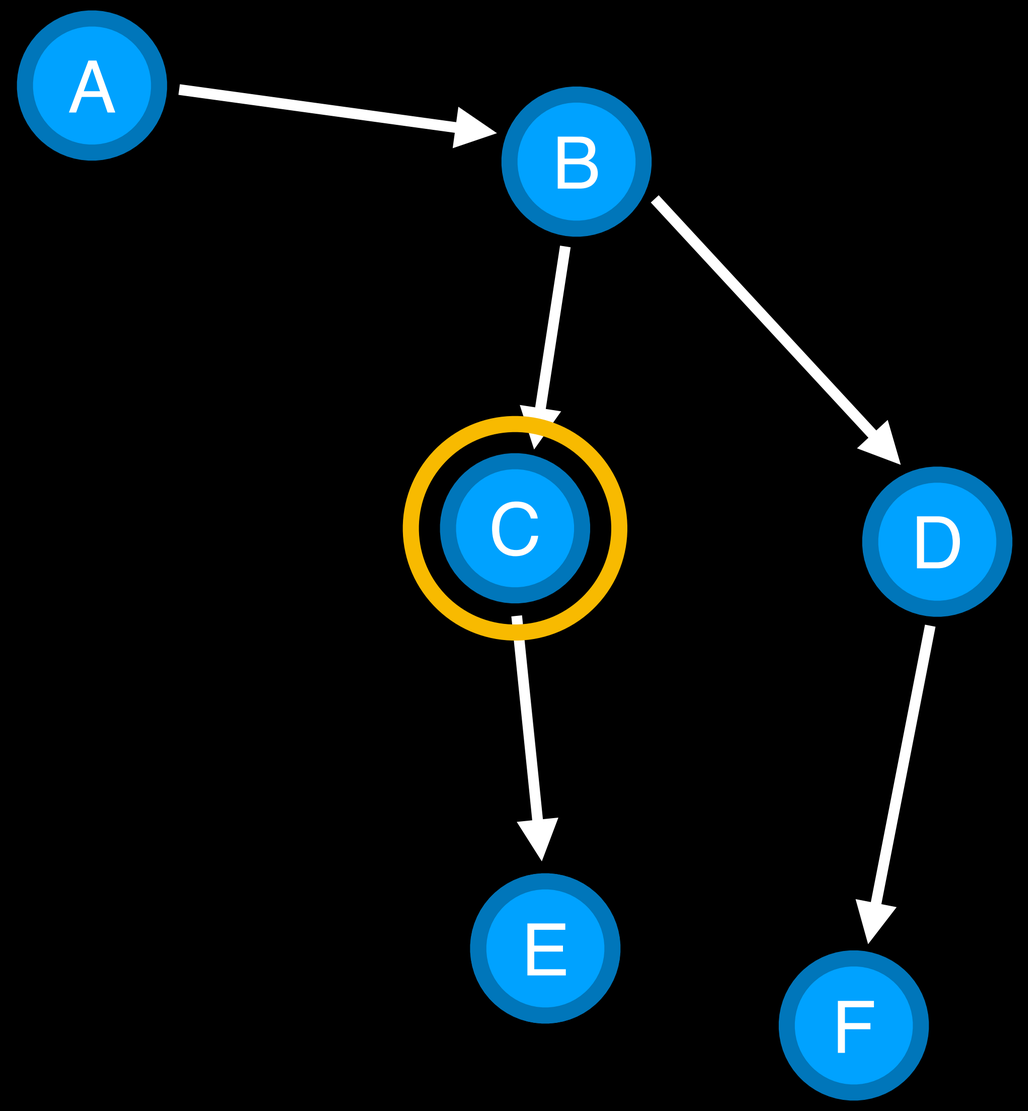


Find a path from A to E.

Frontier

Explored Set

A B D F C



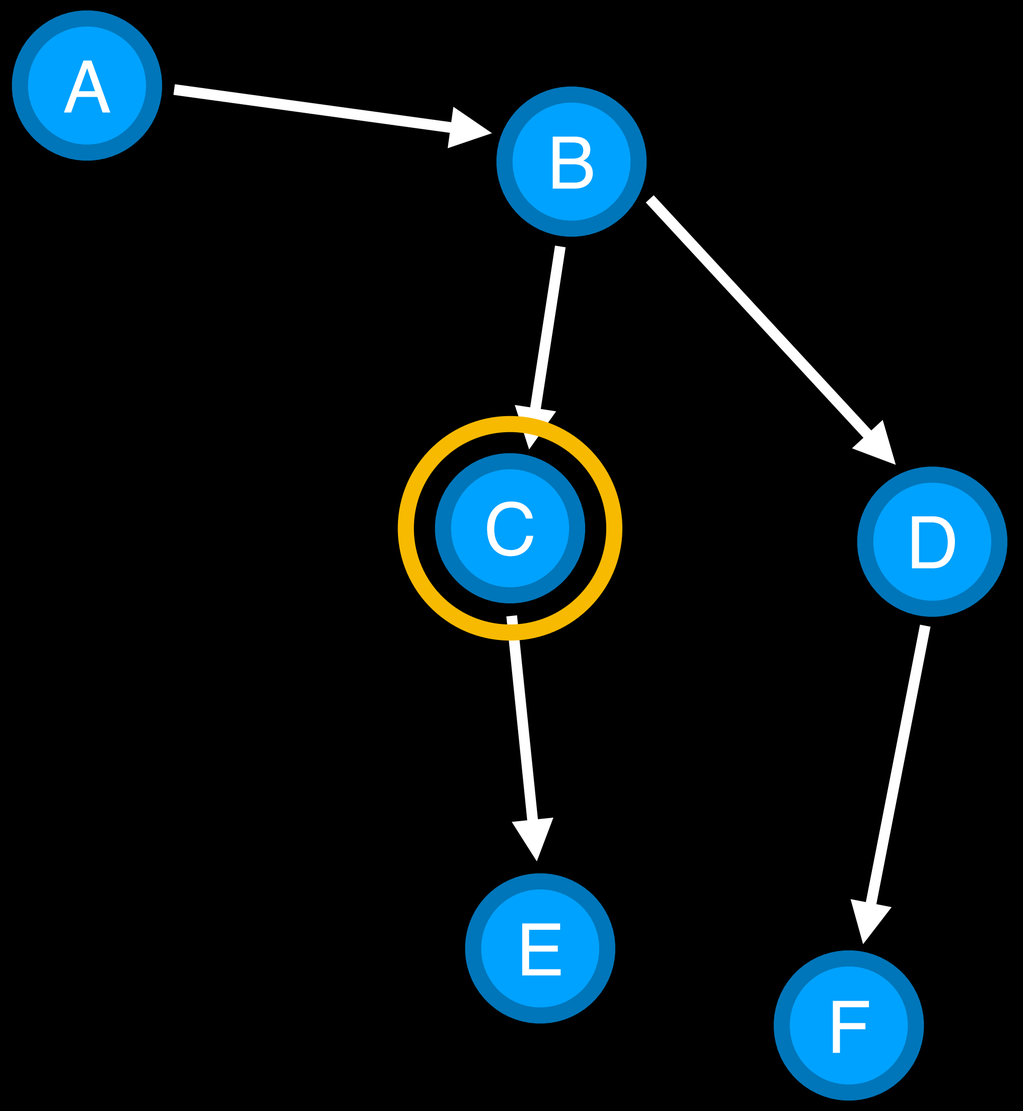
Find a path from A to E.

Frontier

E

Explored Set

A B D F C

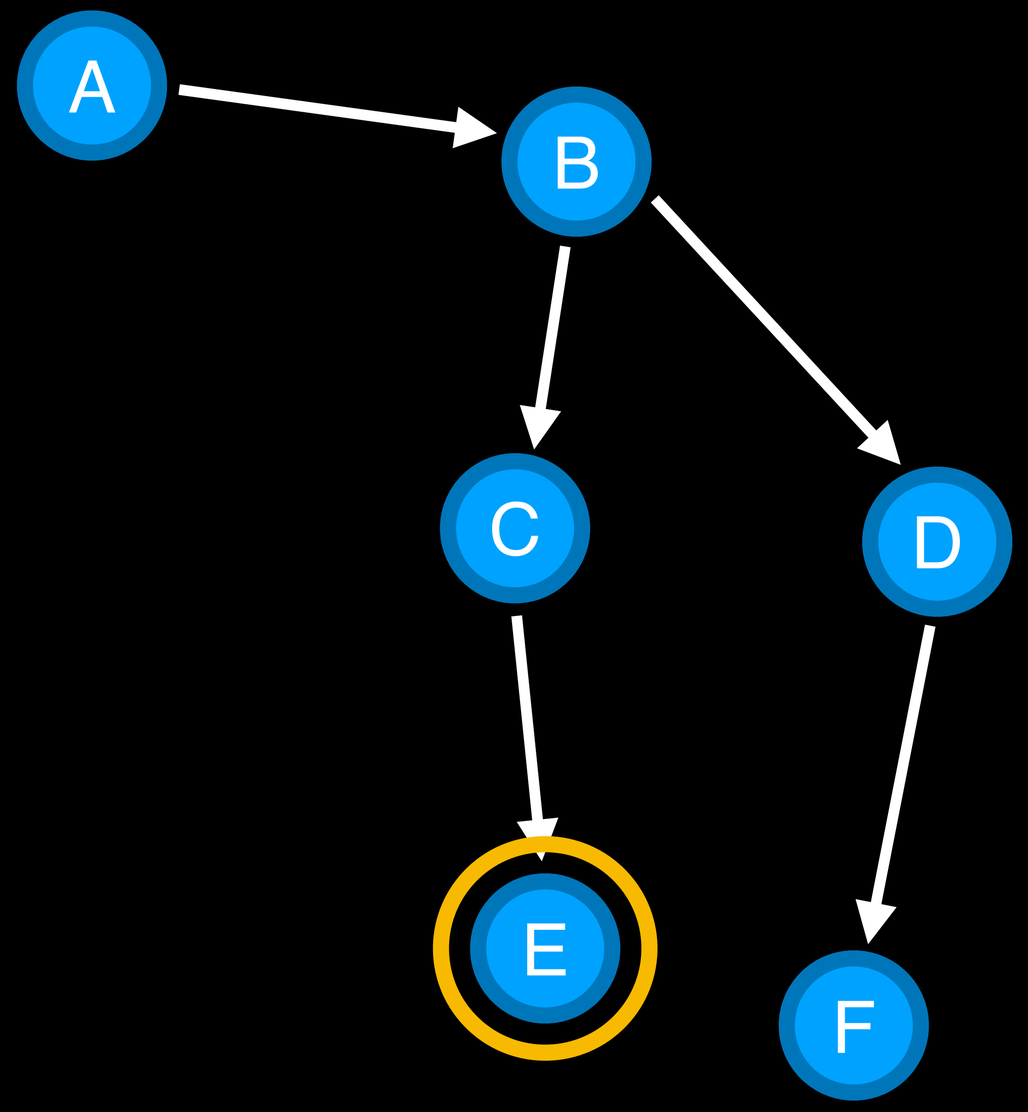


Find a path from A to E.

Frontier

Explored Set

A B D F C



Depth-First Search

depth-first search

search algorithm that always expands the deepest node in the frontier

Breadth-First Search

breadth-first search

search algorithm that always expands the shallowest node in the frontier

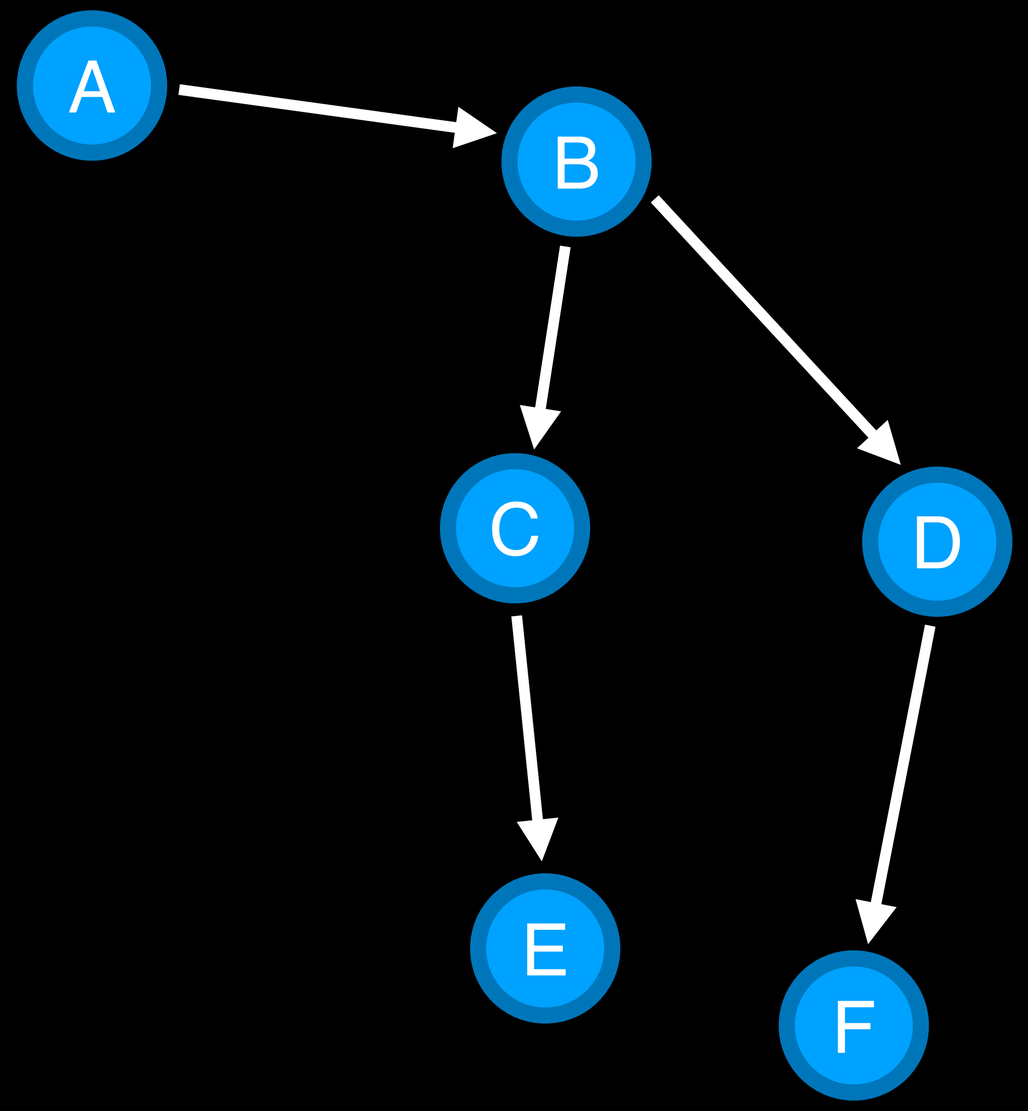
queue

first-in first-out data type

Find a path from A to E.

Frontier

Explored Set

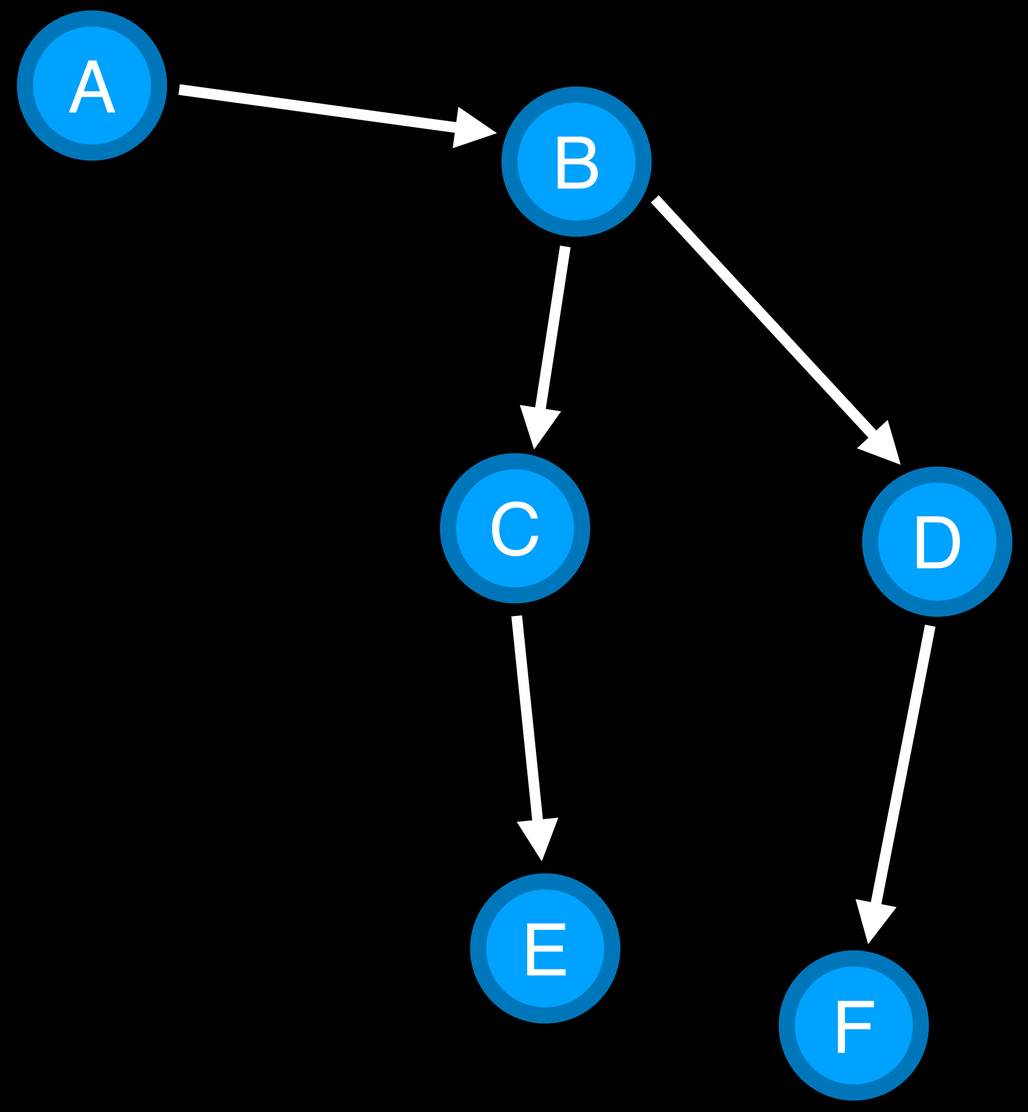


Find a path from A to E.

Frontier

A

Explored Set

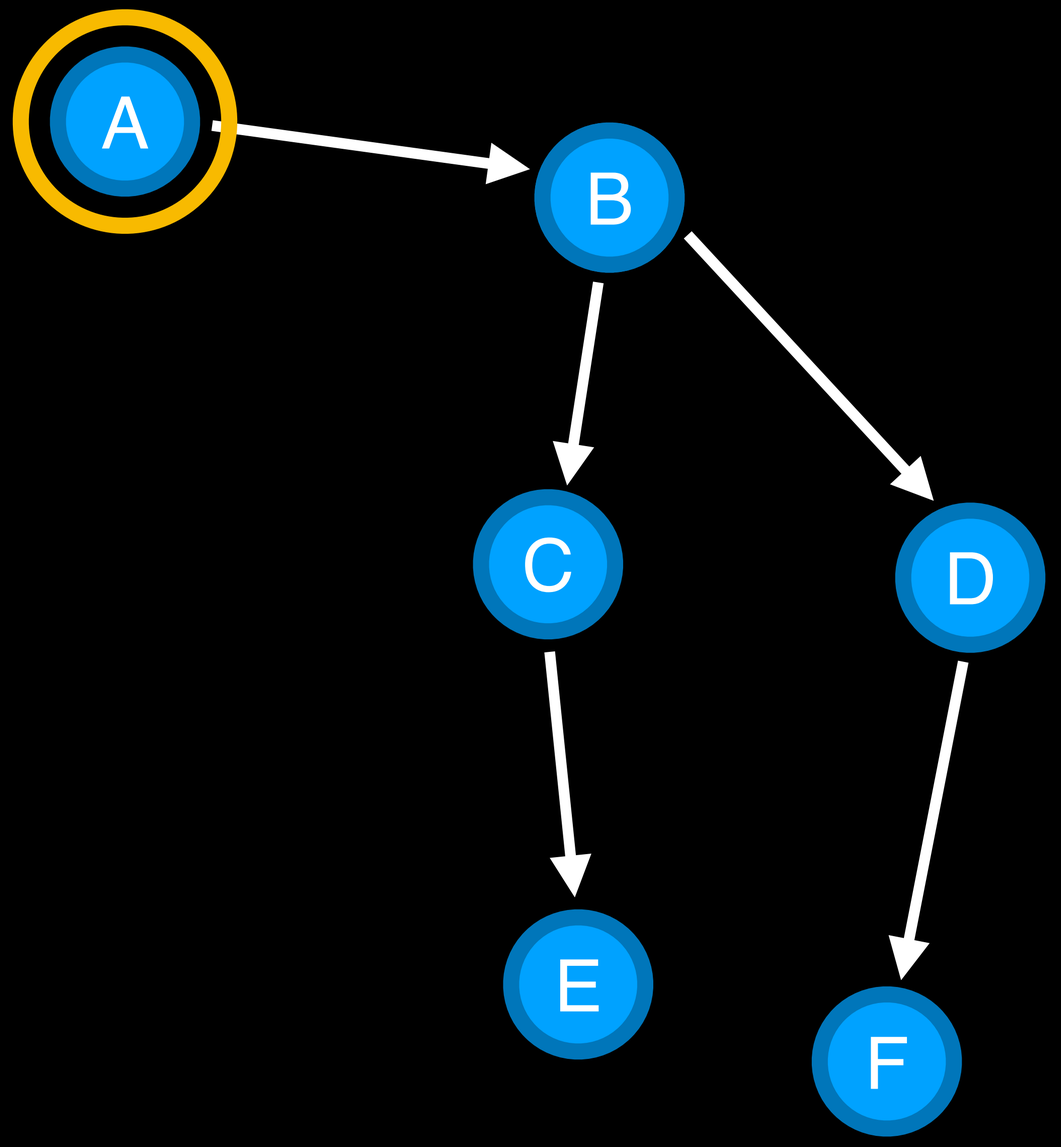


Find a path from A to E.

Frontier

Explored Set

A



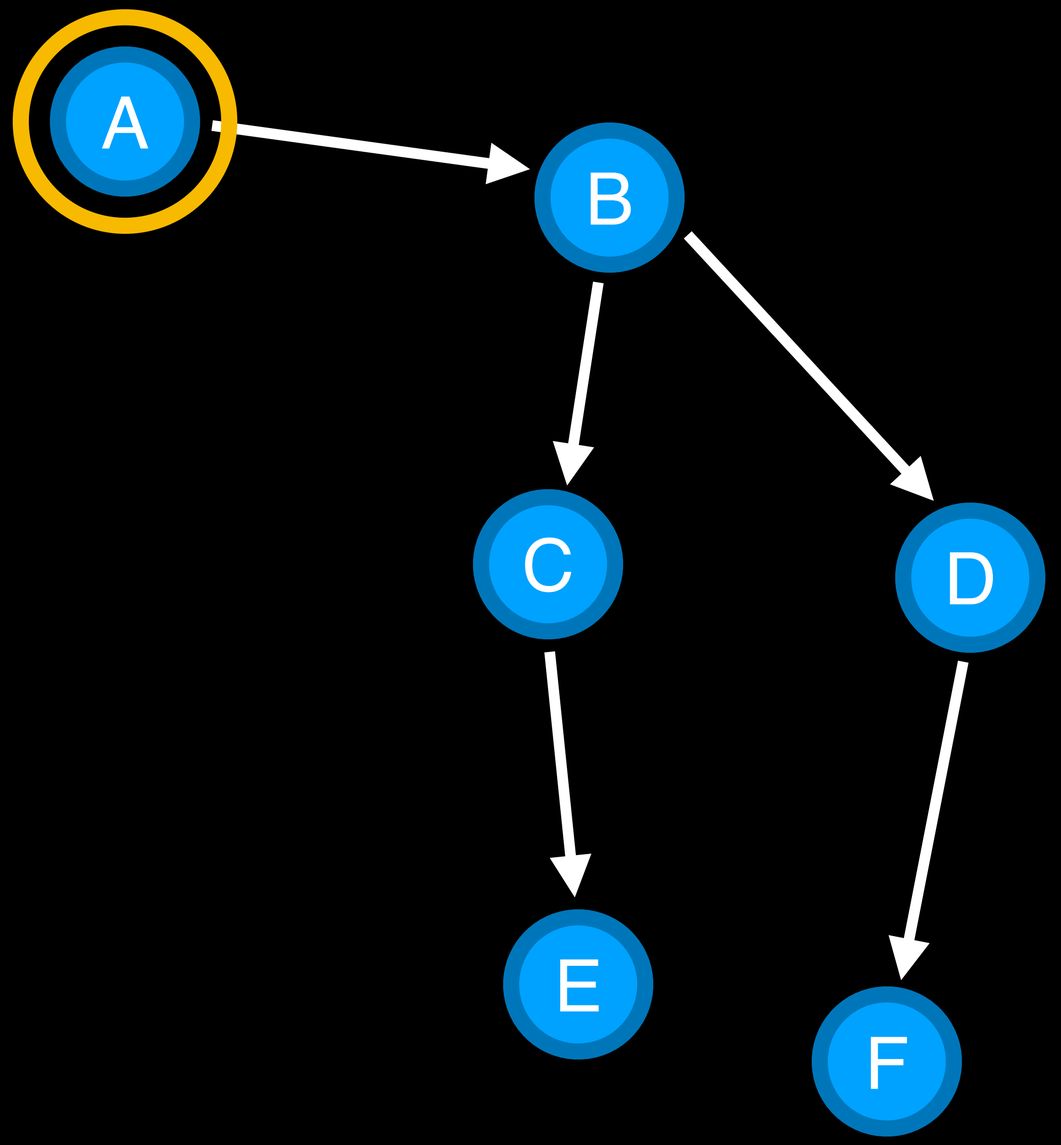
Find a path from A to E.

Frontier

B

Explored Set

A

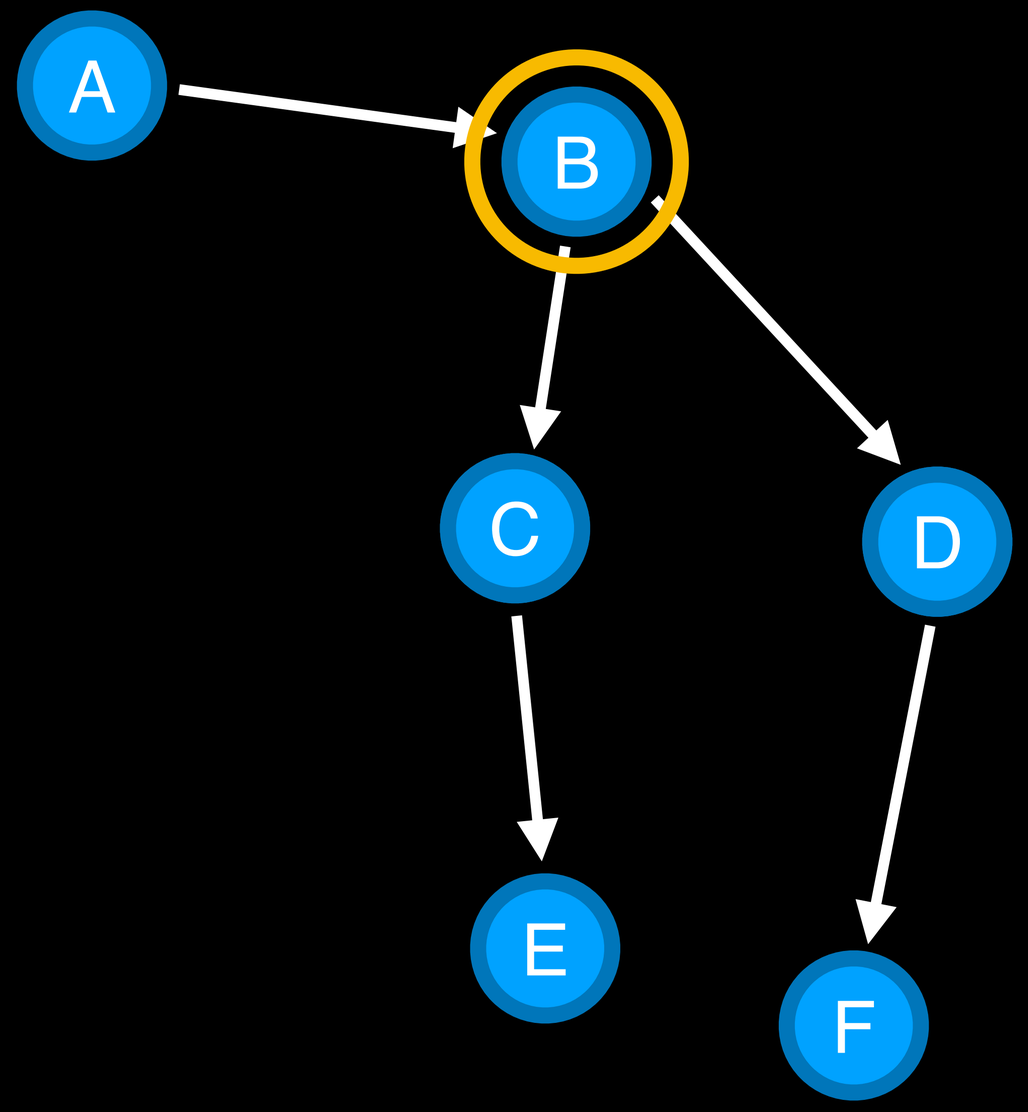


Find a path from A to E.

Frontier

Explored Set

A B



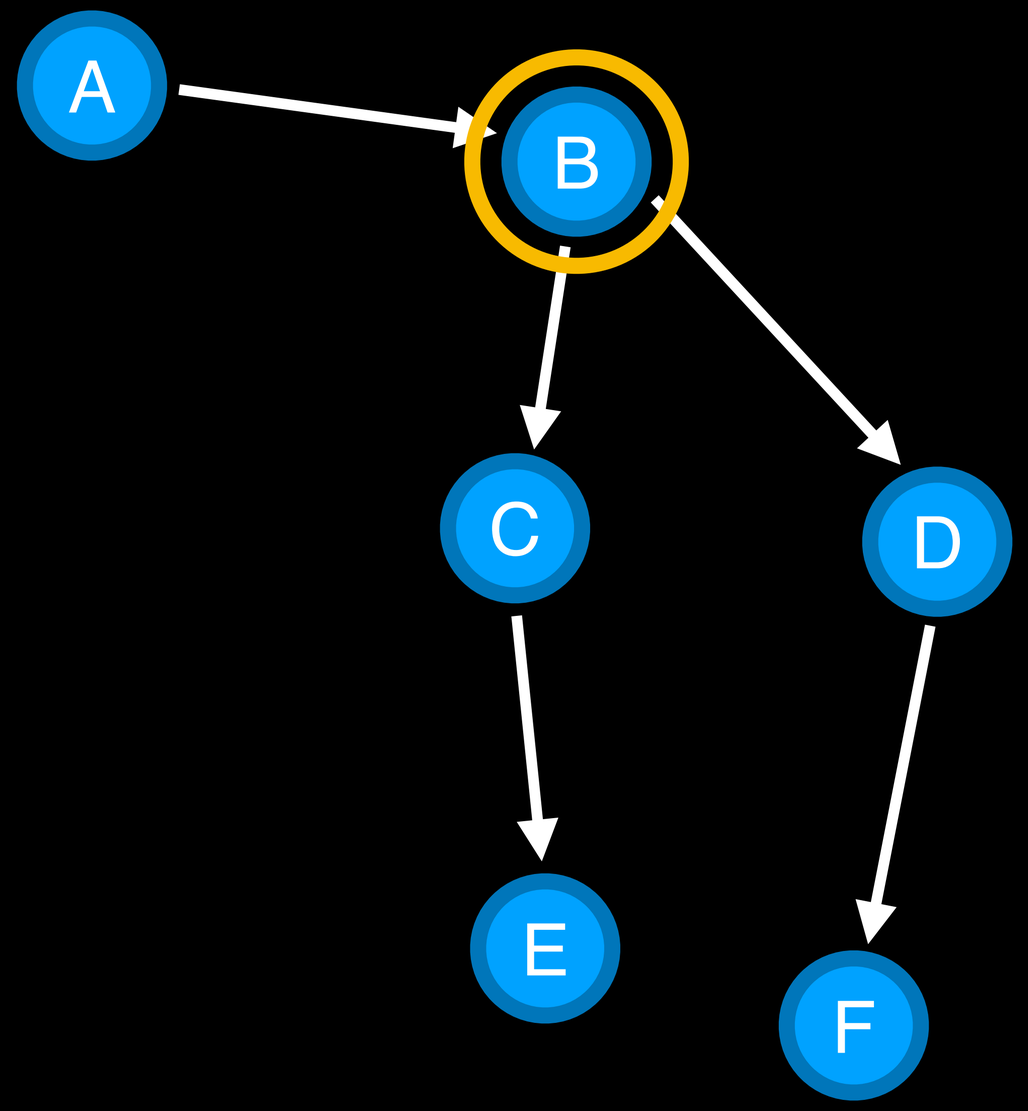
Find a path from A to E.

Frontier

C D

Explored Set

A B



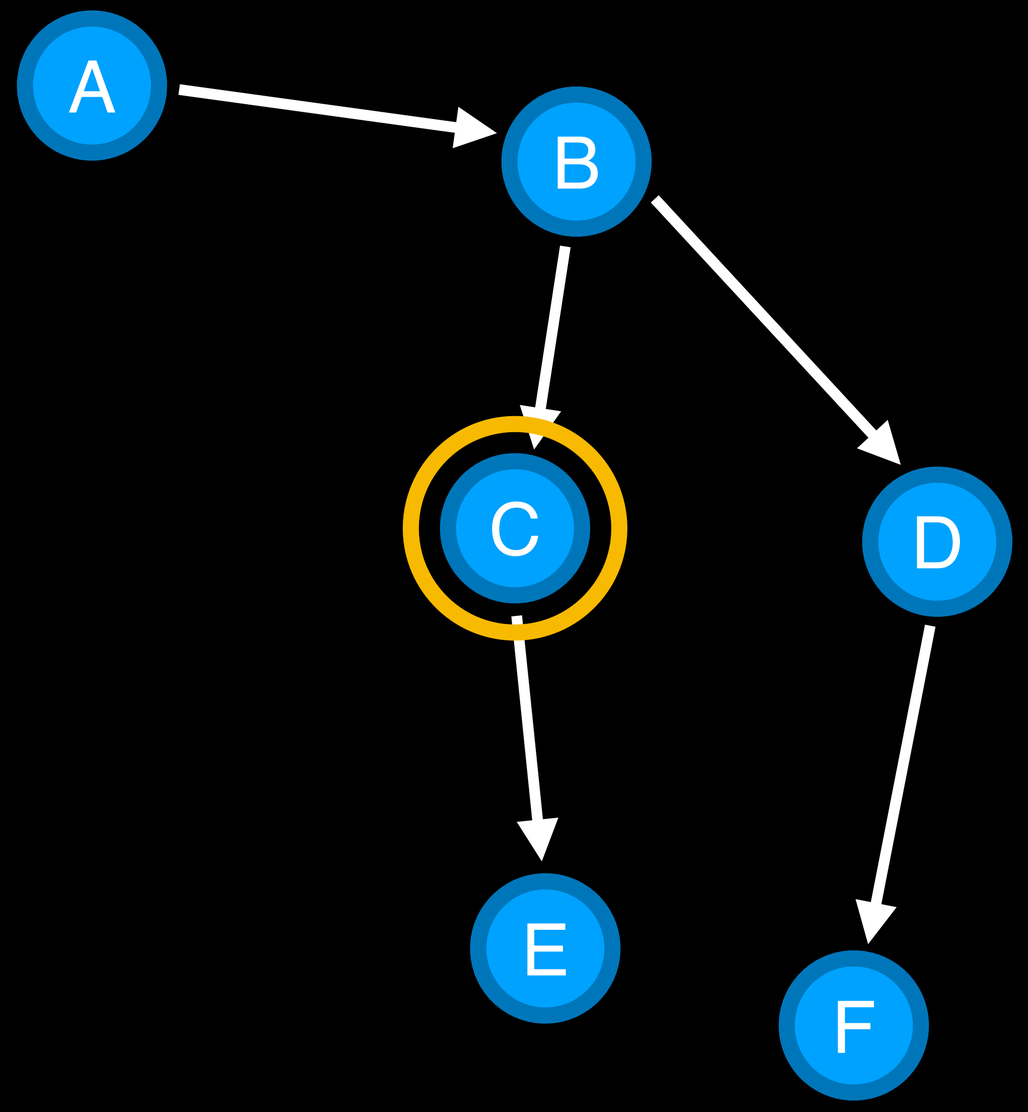
Find a path from A to E.

Frontier

D

Explored Set

A B C



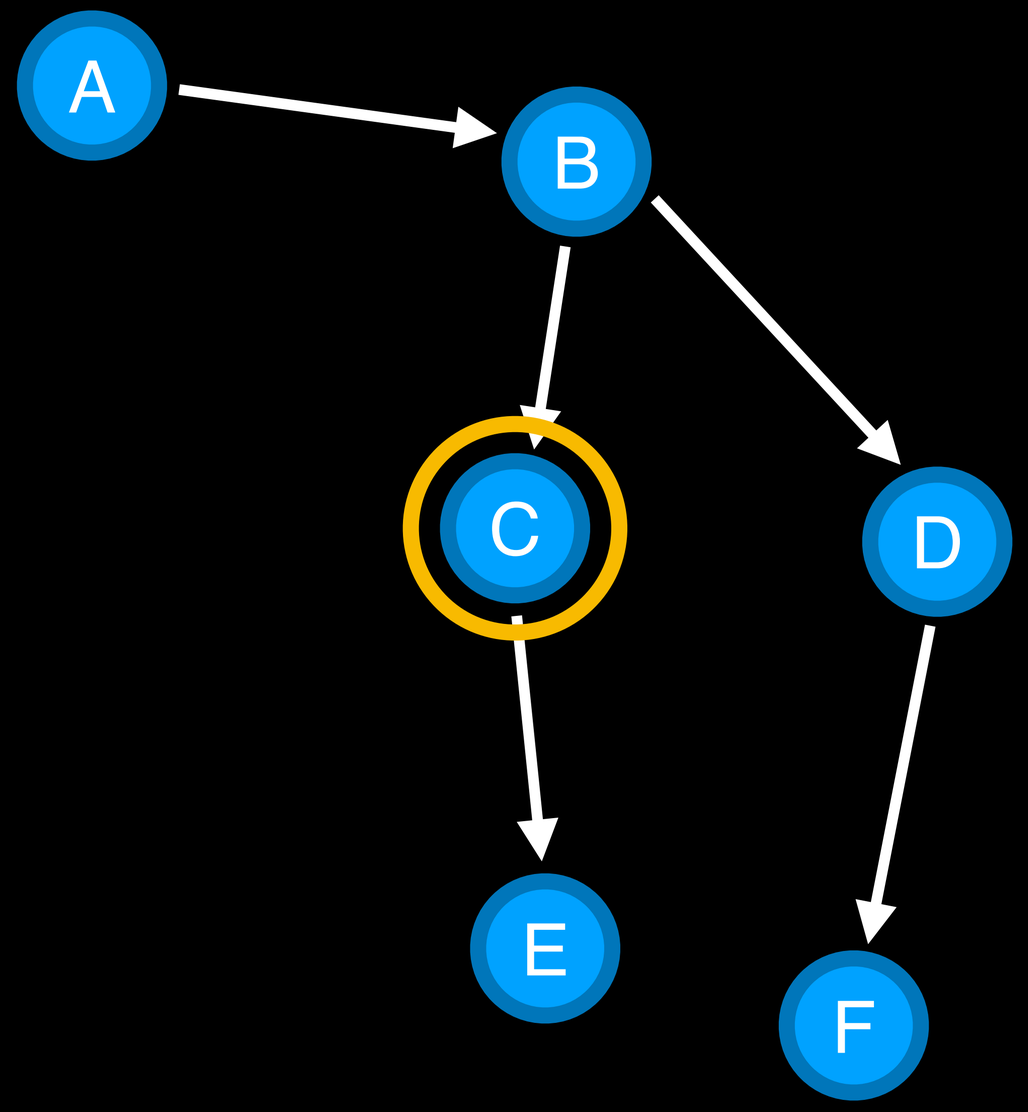
Find a path from A to E.

Frontier

D E

Explored Set

A B C



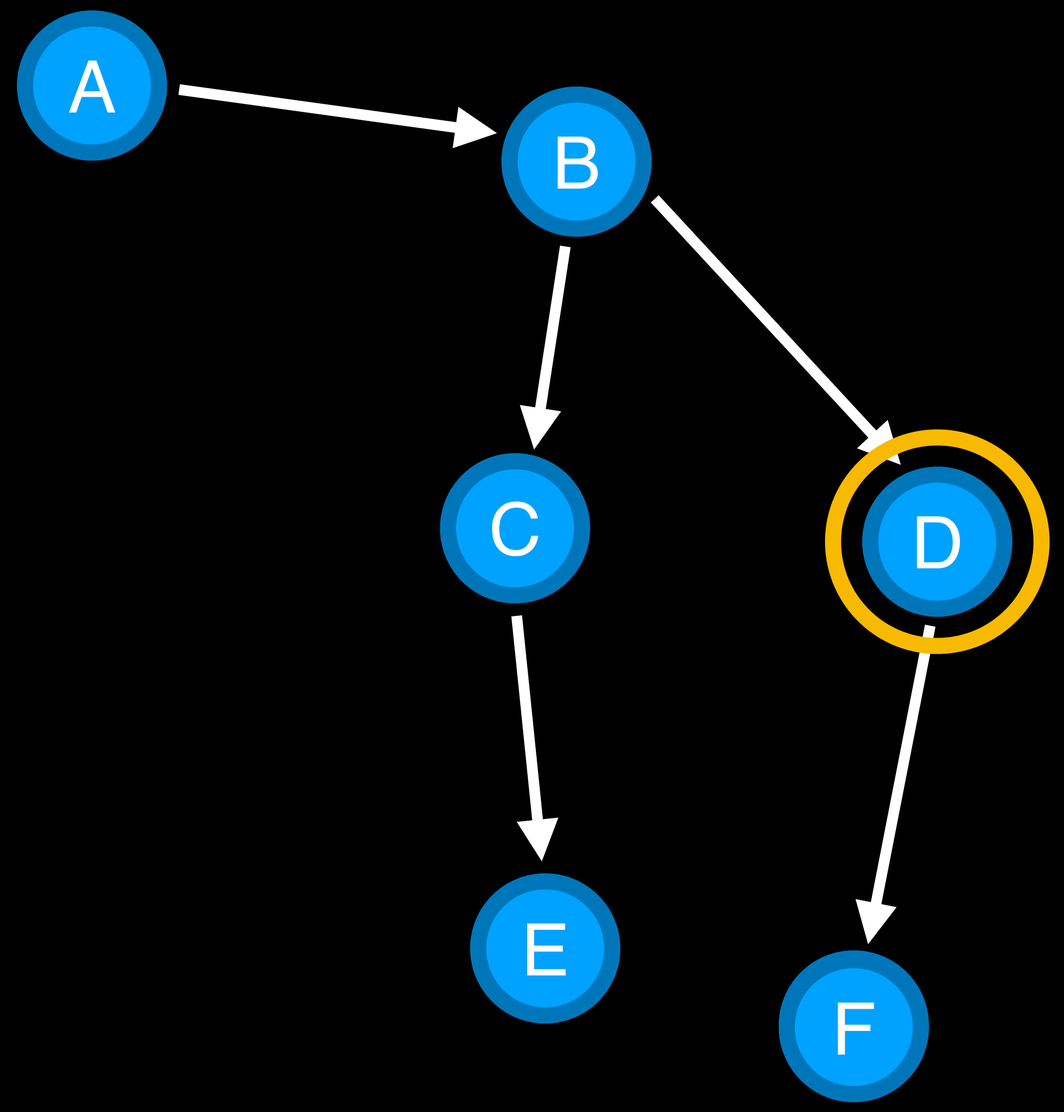
Find a path from A to E.

Frontier

E

Explored Set

A B C D



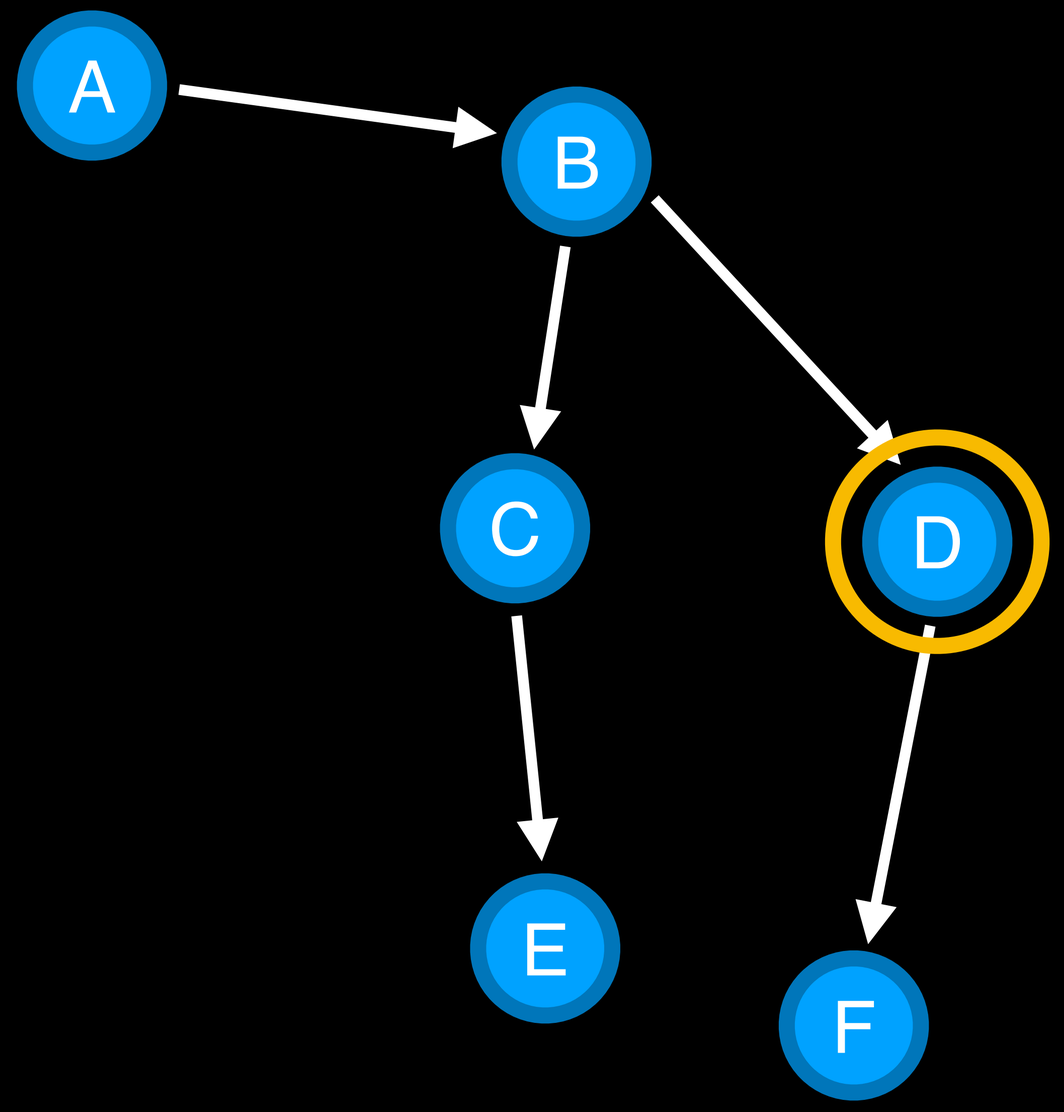
Find a path from A to E.

Frontier

E F

Explored Set

A B C D



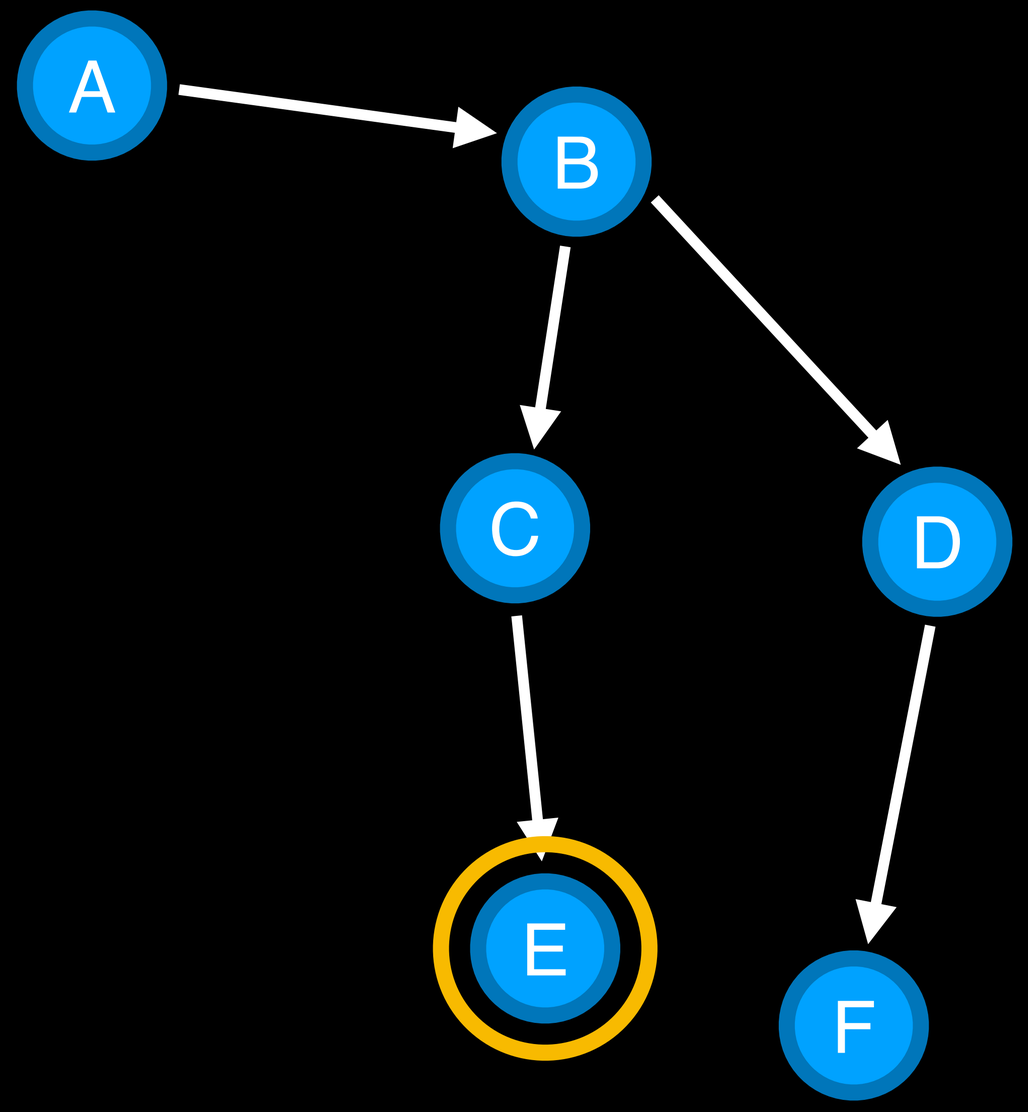
Find a path from A to E.

Frontier

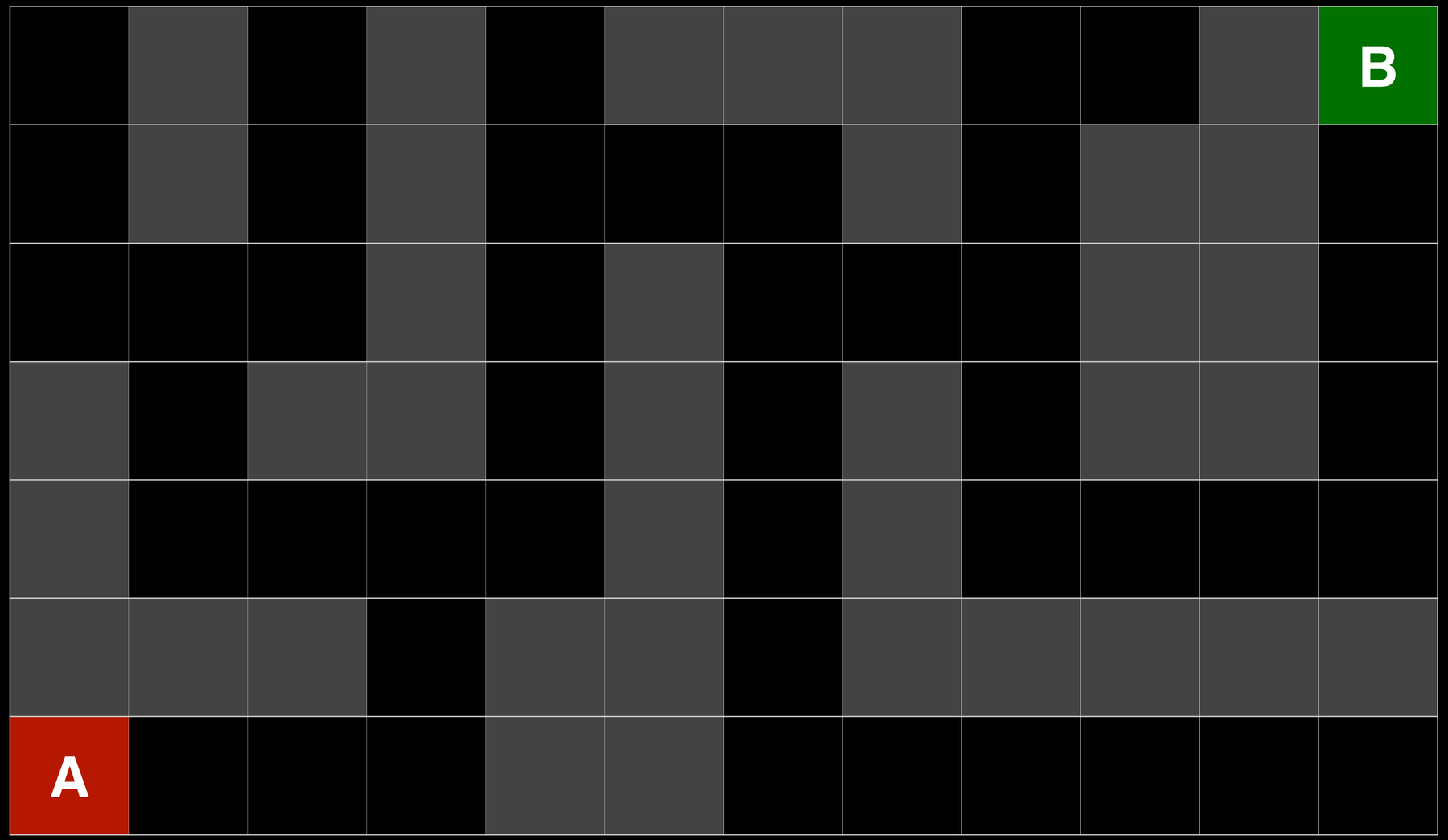
F

Explored Set

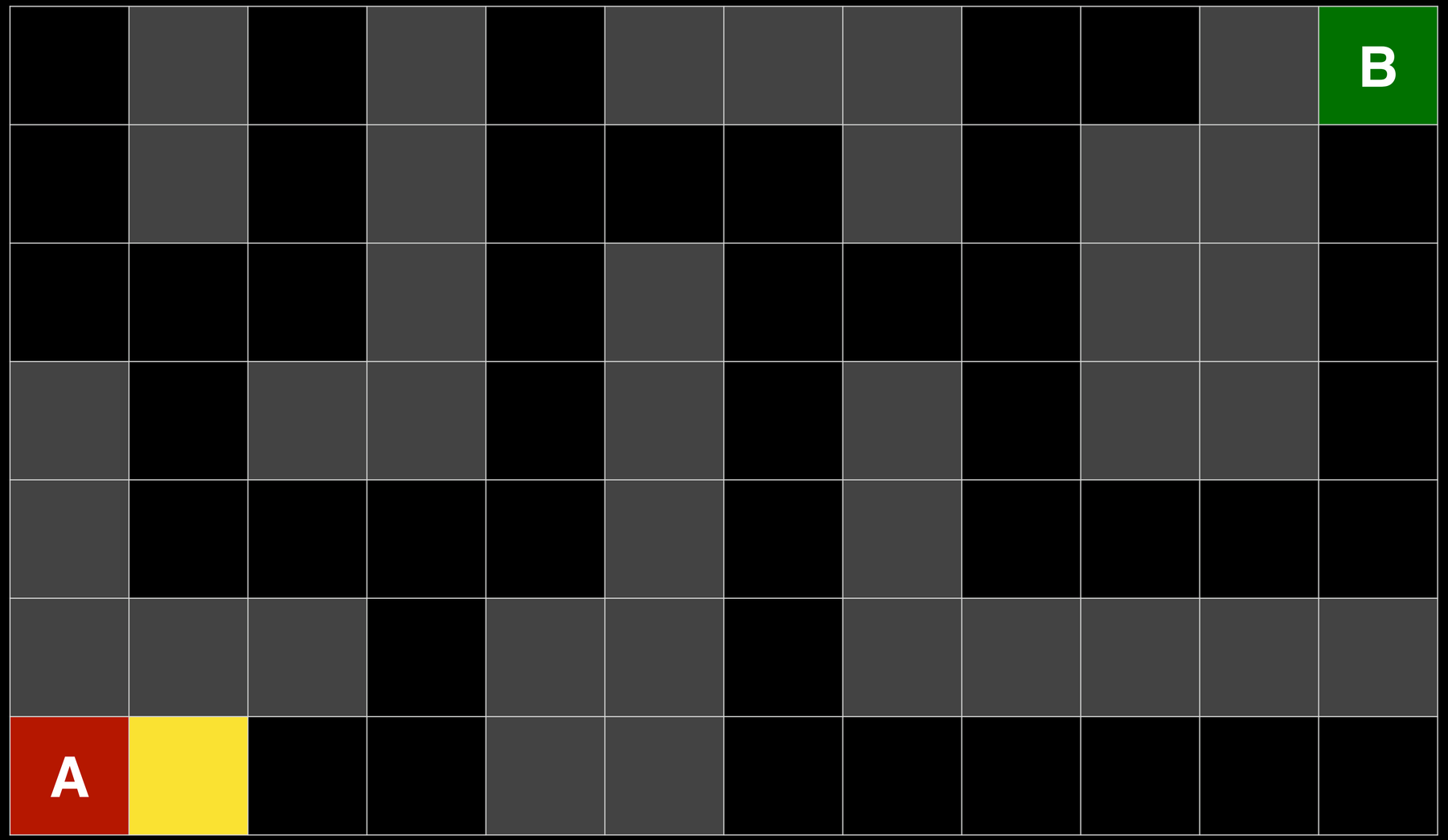
A B C D



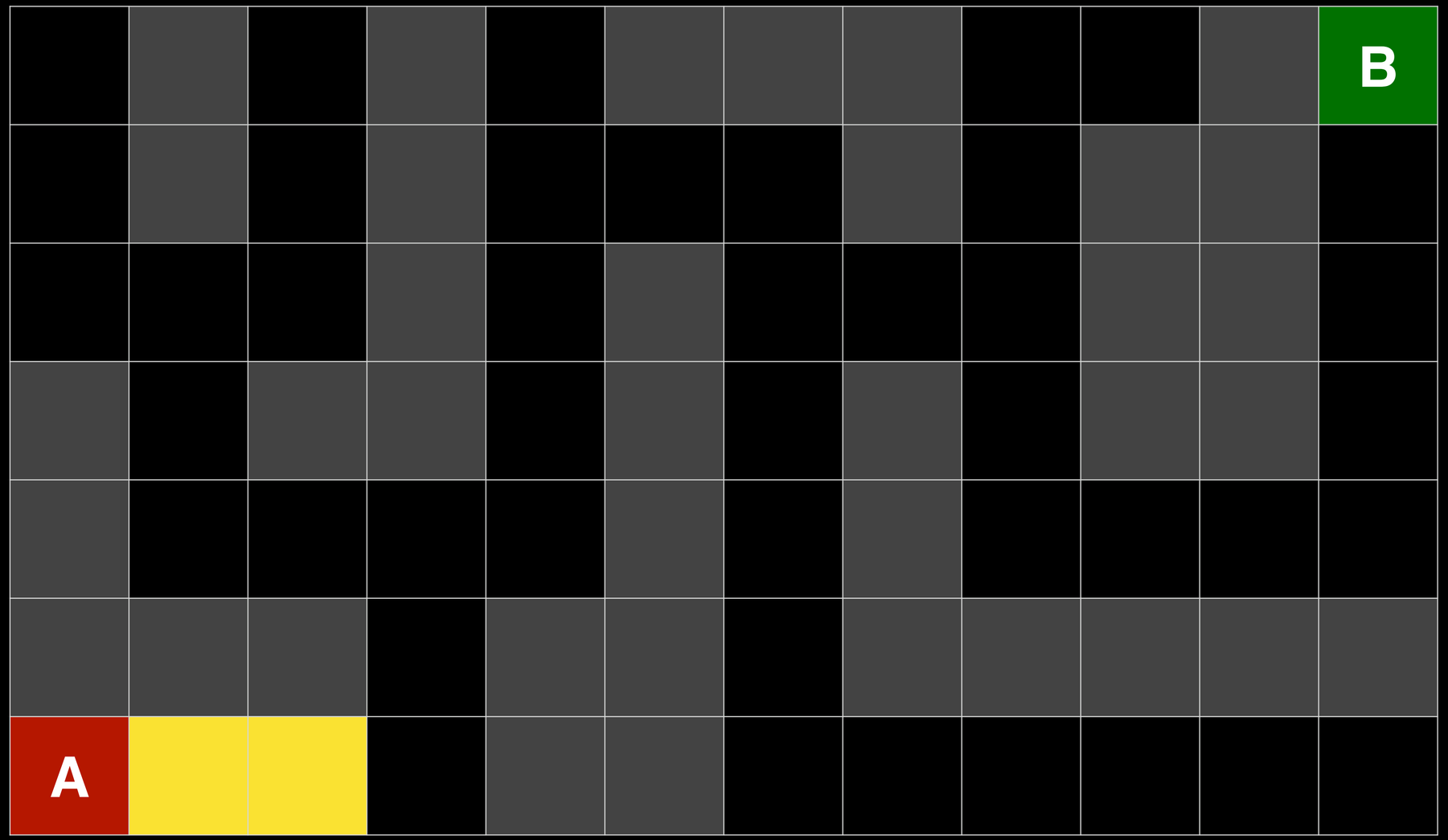
Depth-First Search



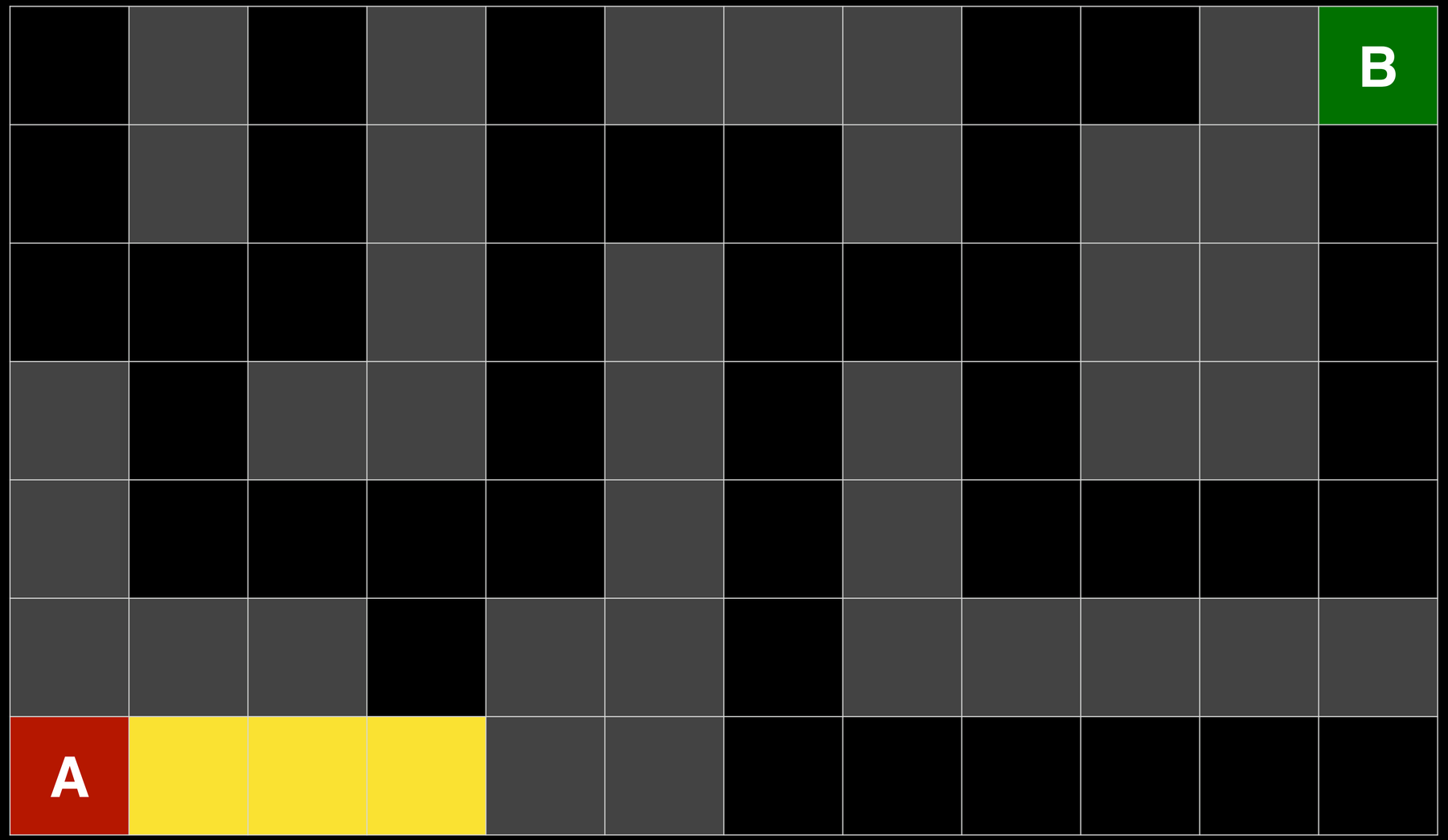
Depth-First Search



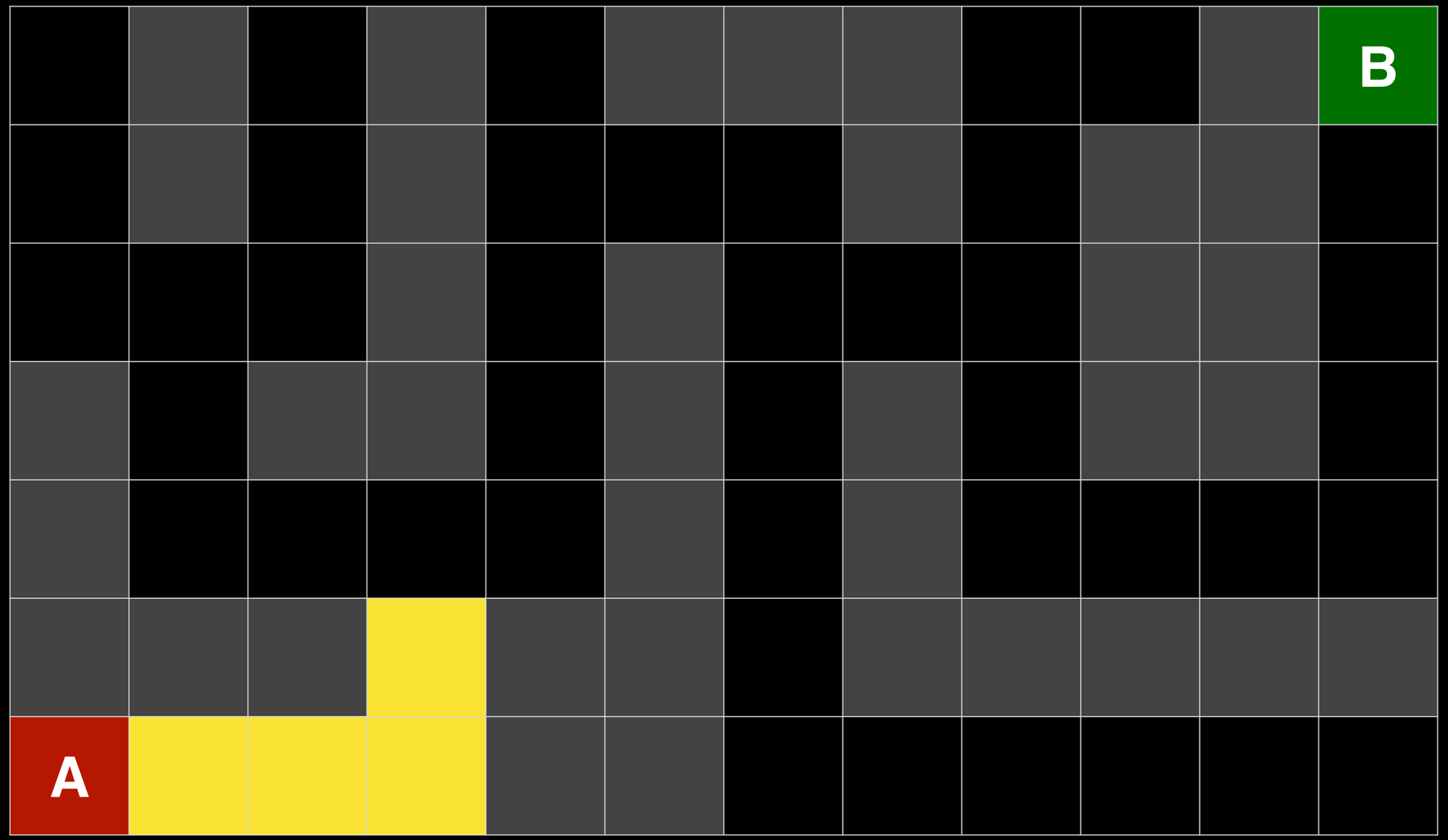
Depth-First Search



Depth-First Search



Depth-First Search



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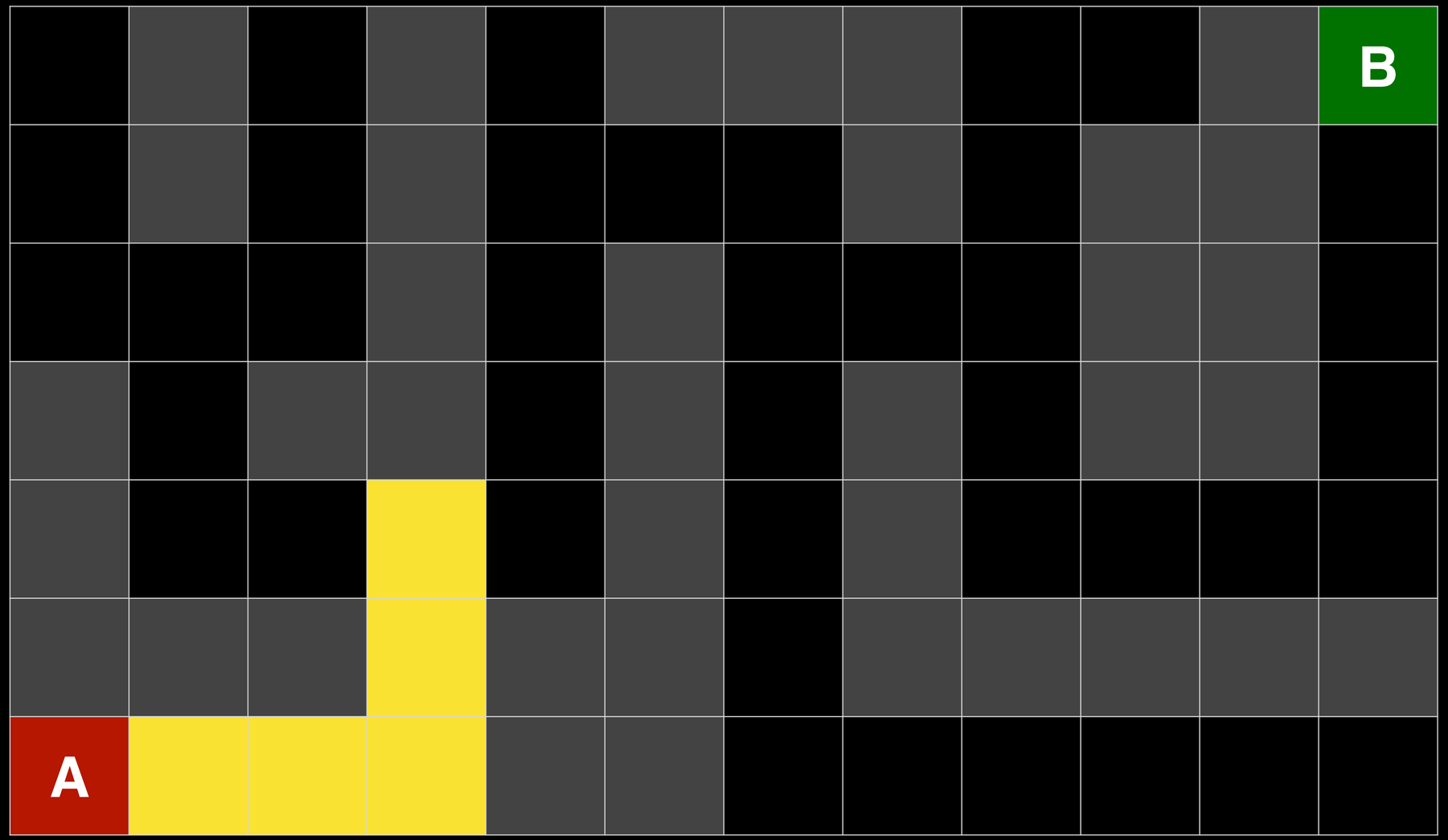

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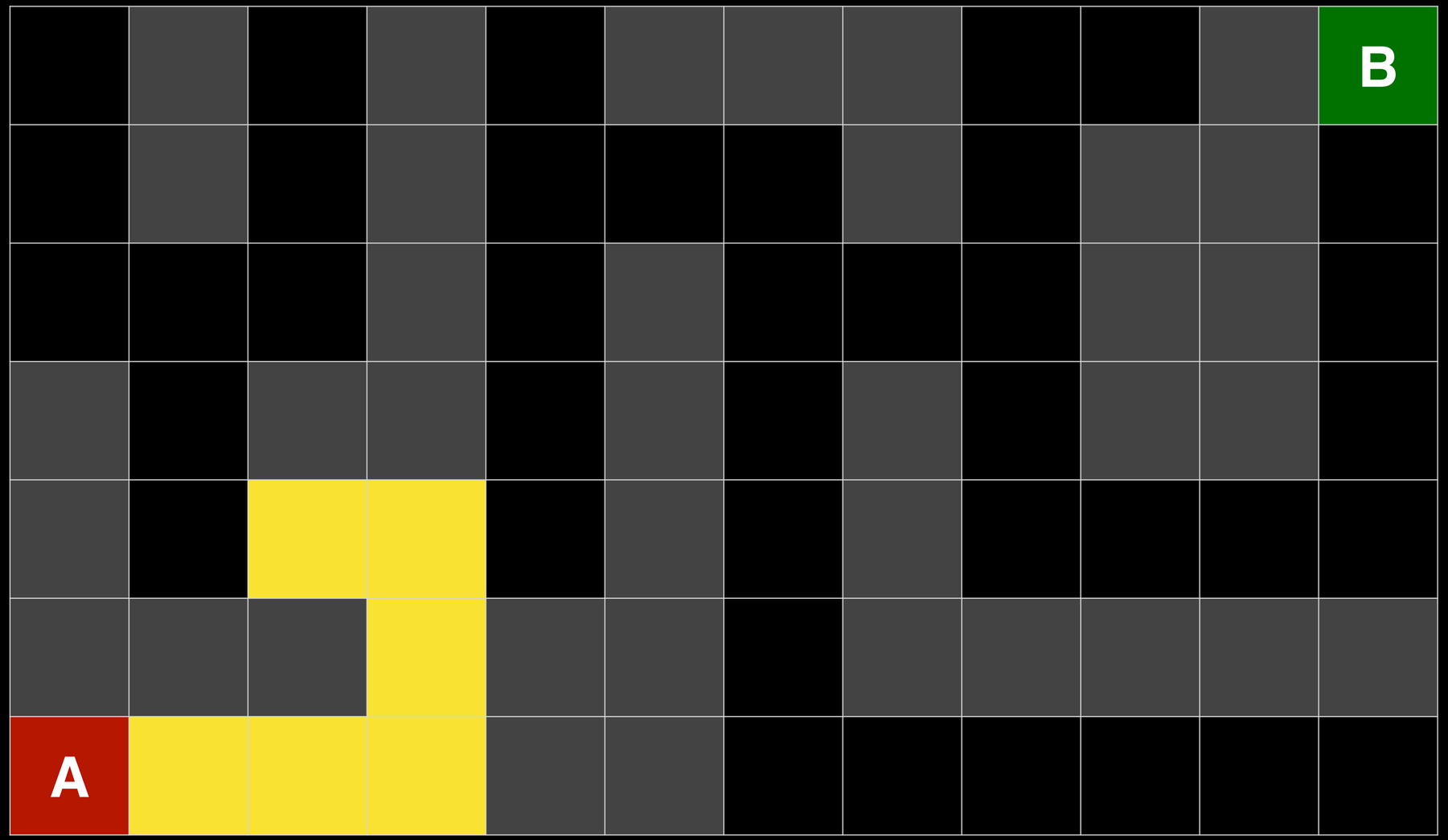
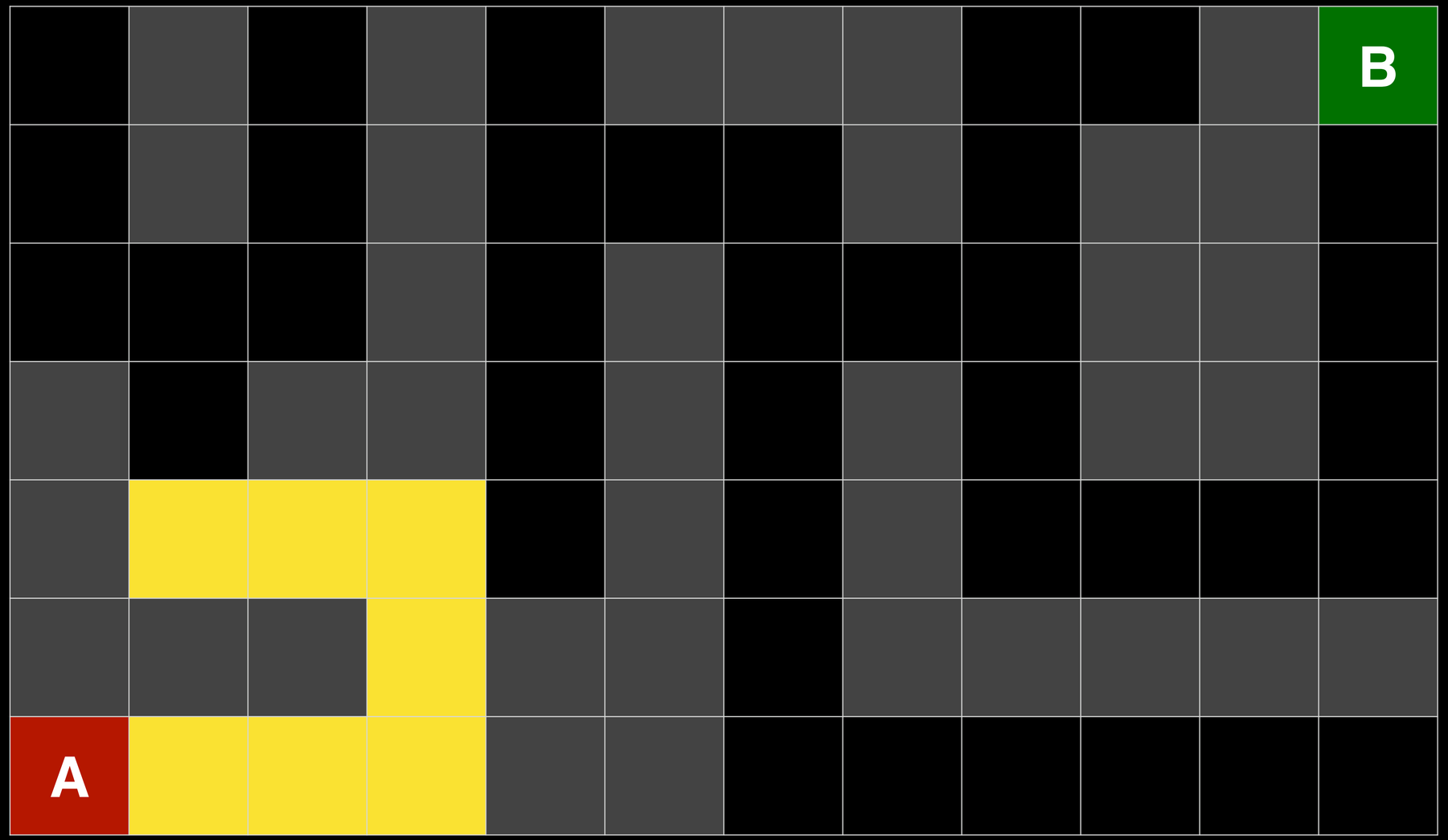
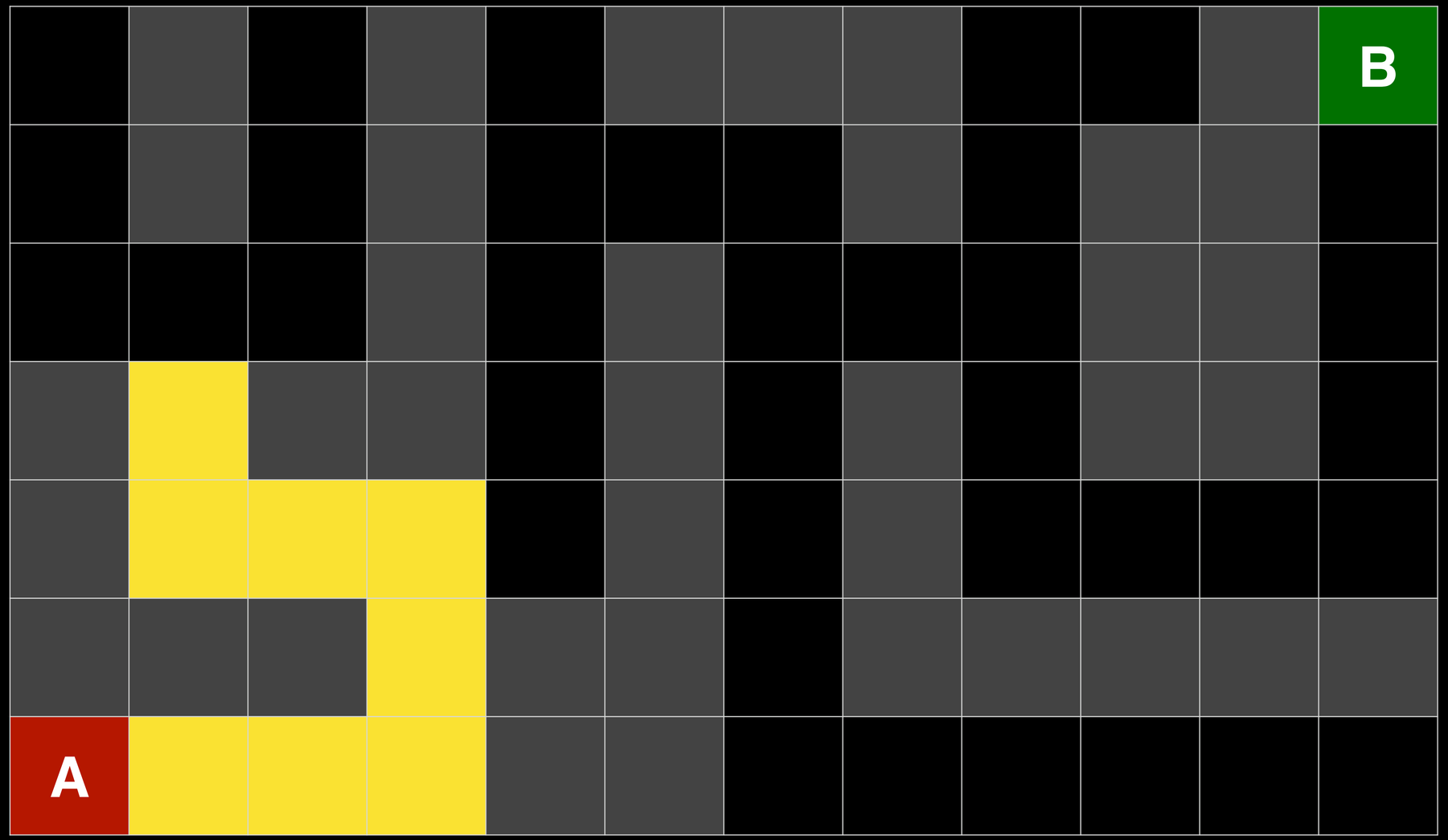


Depth-First Search



Depth-First Search



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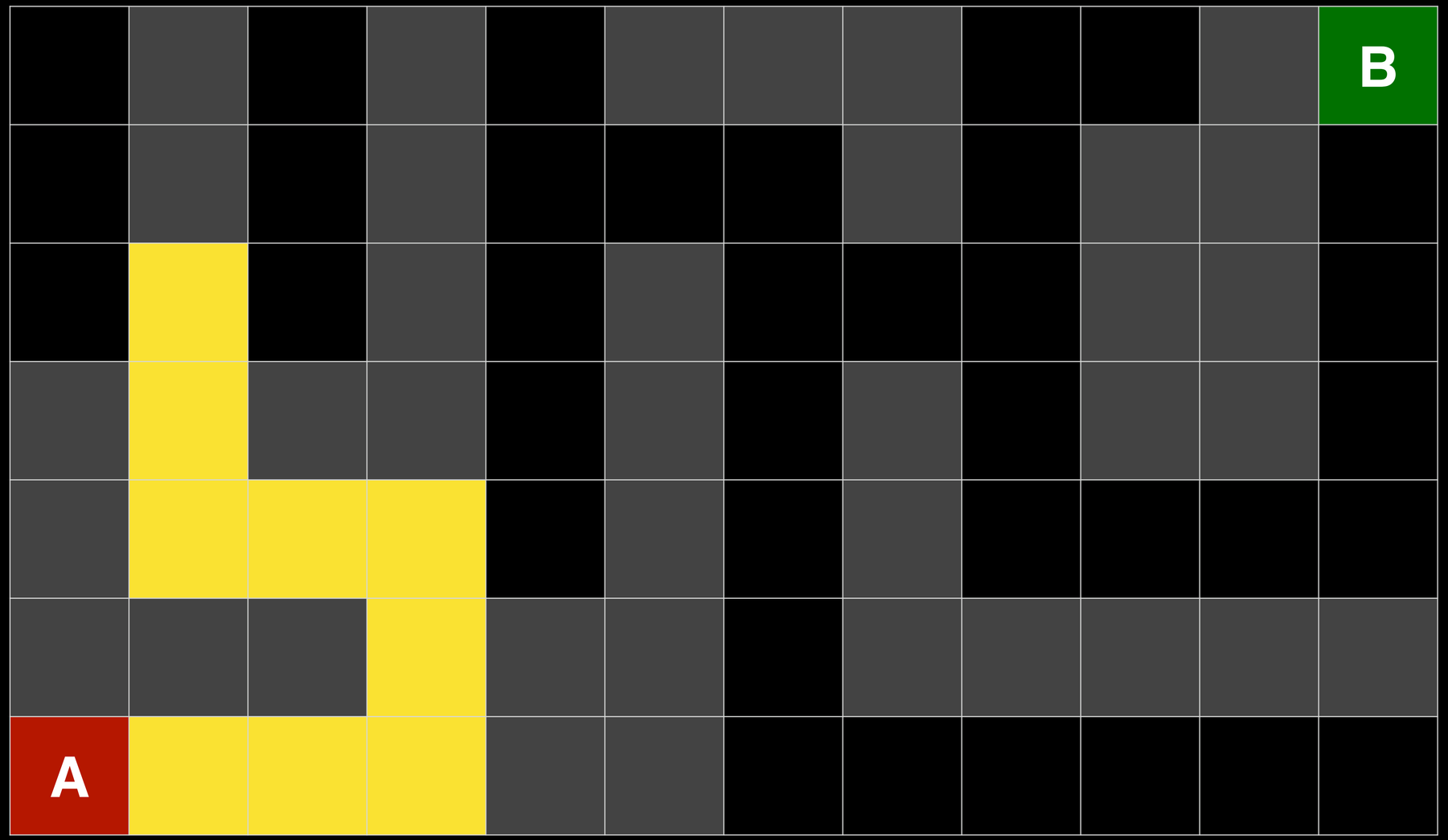

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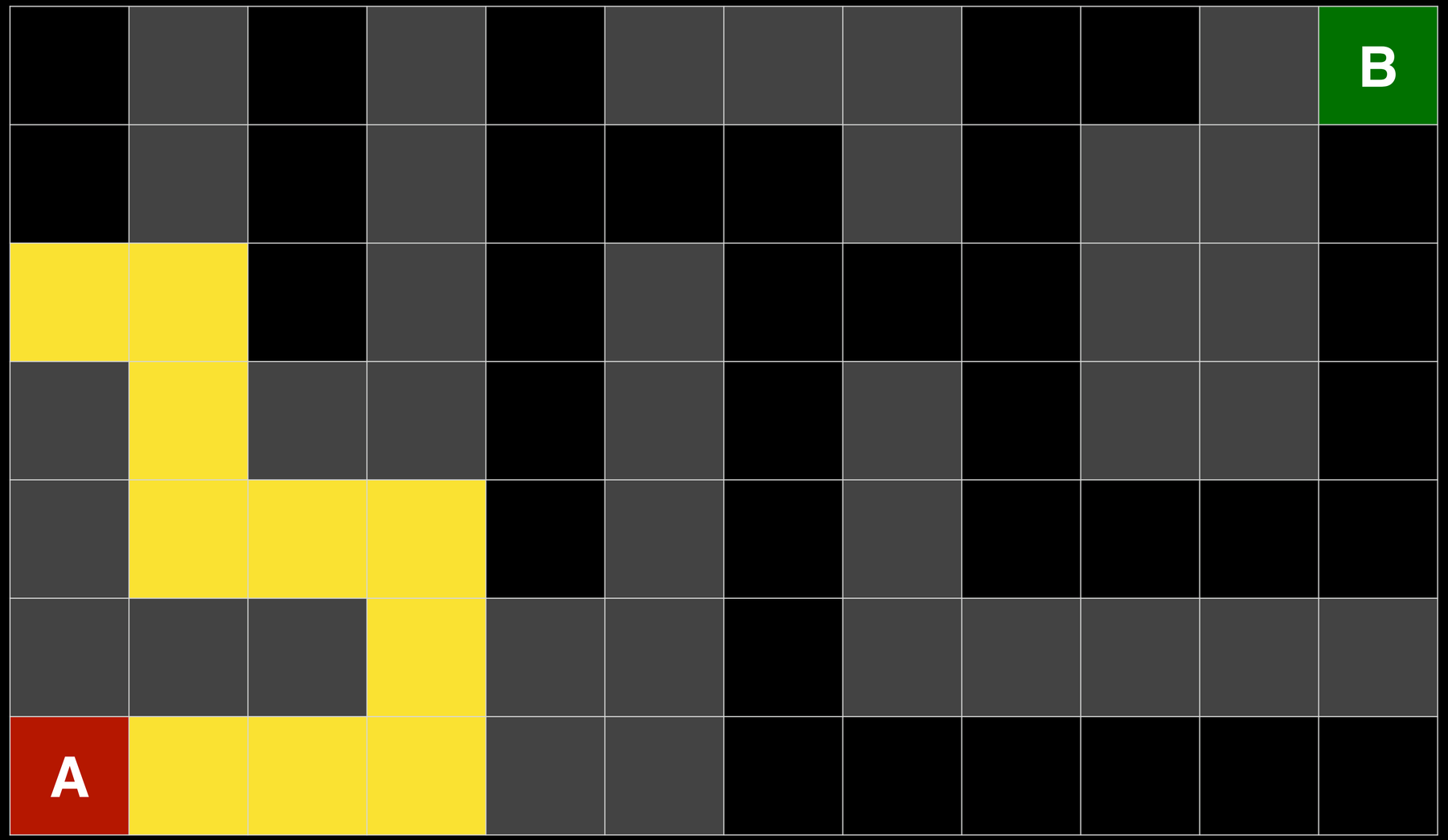
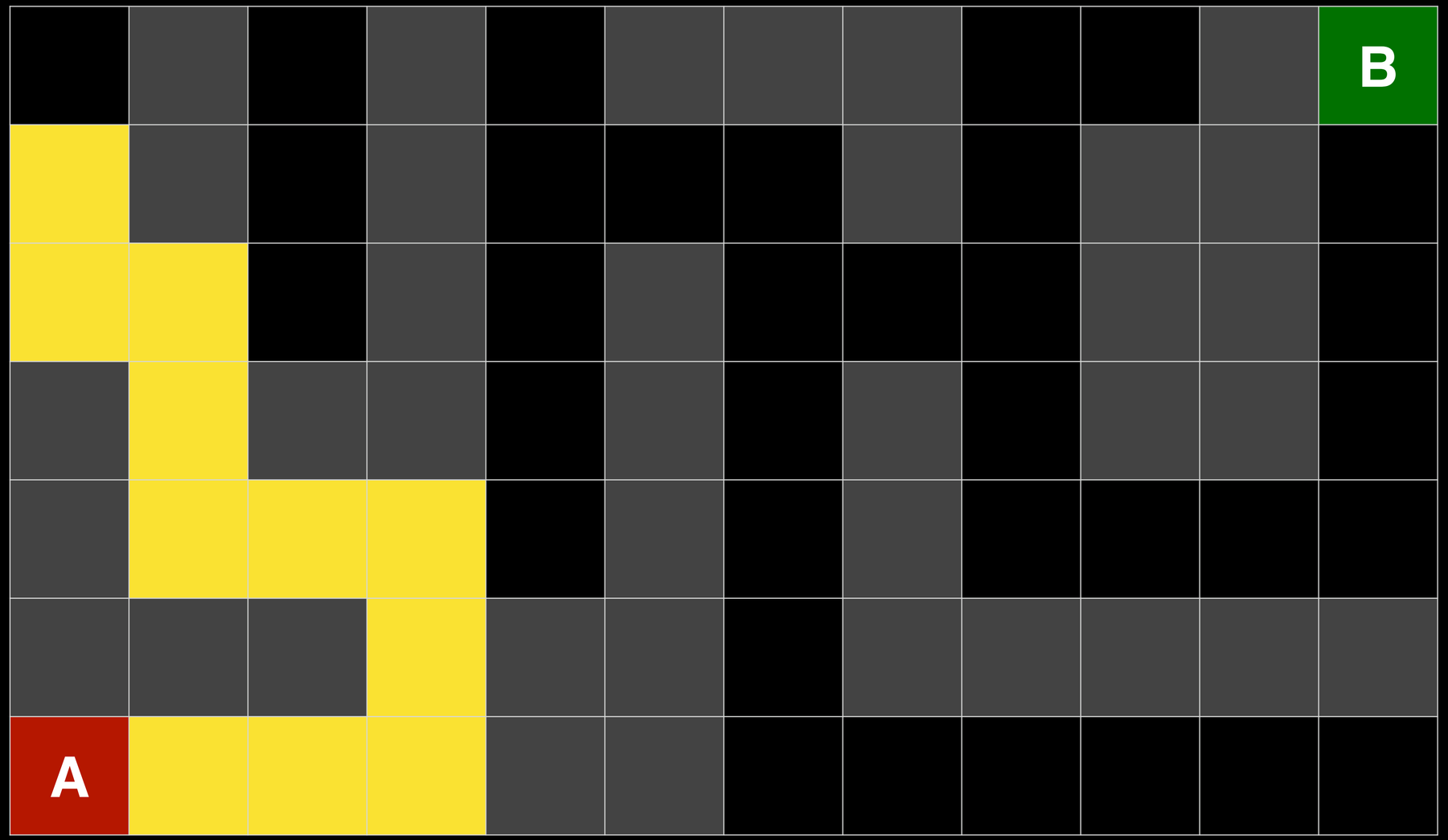


Depth-First Search



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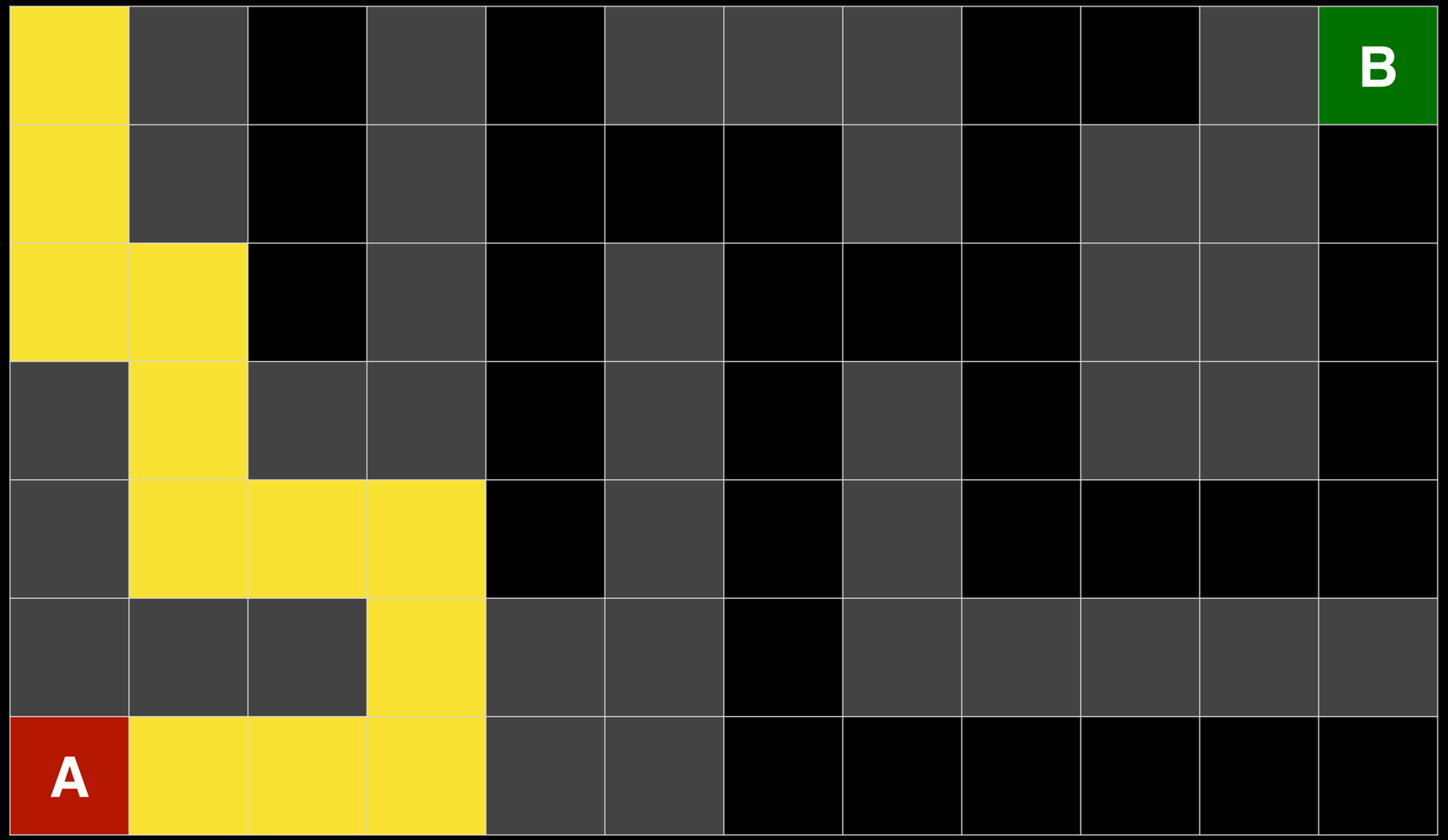
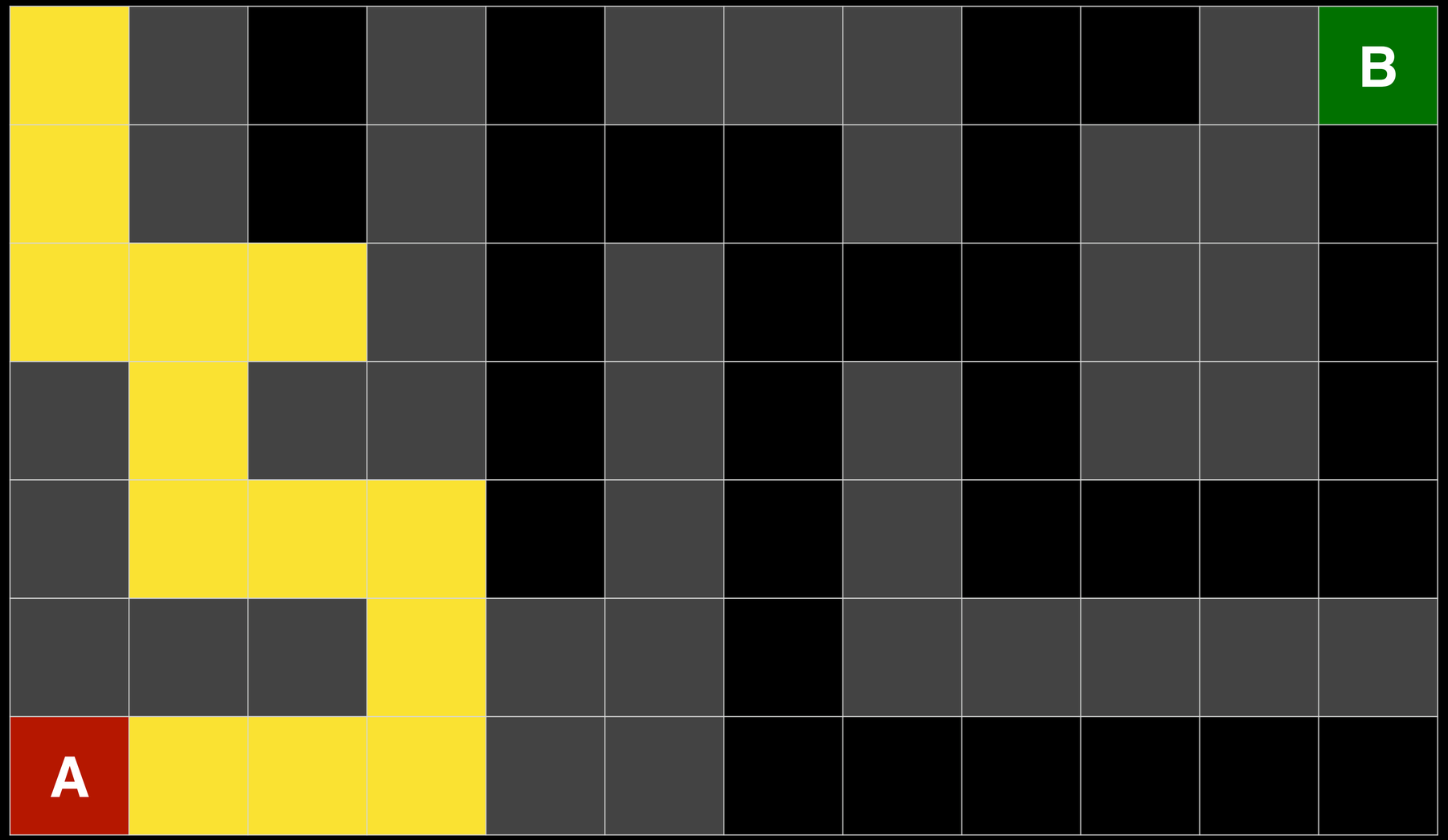


Depth-First Search



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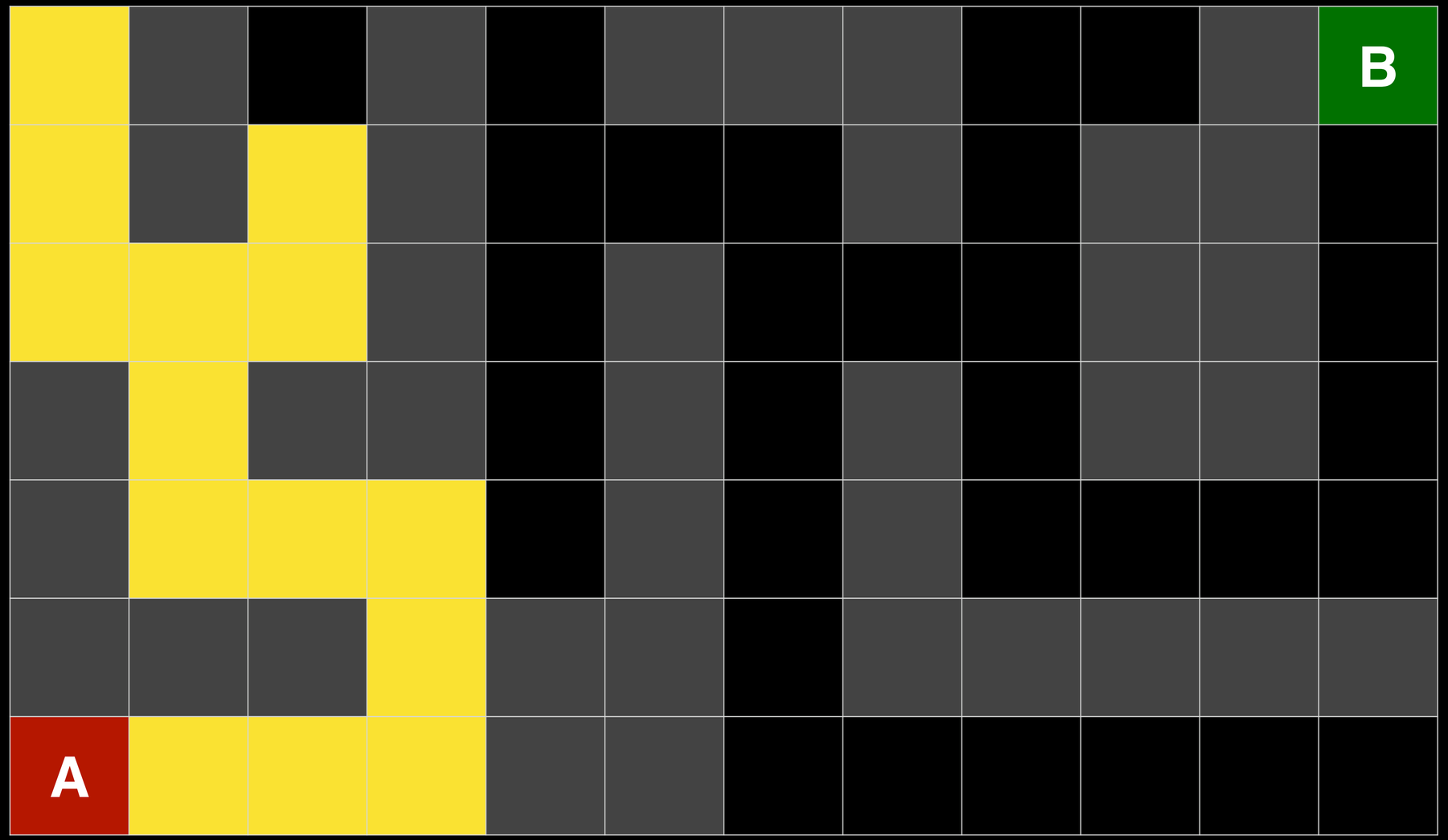
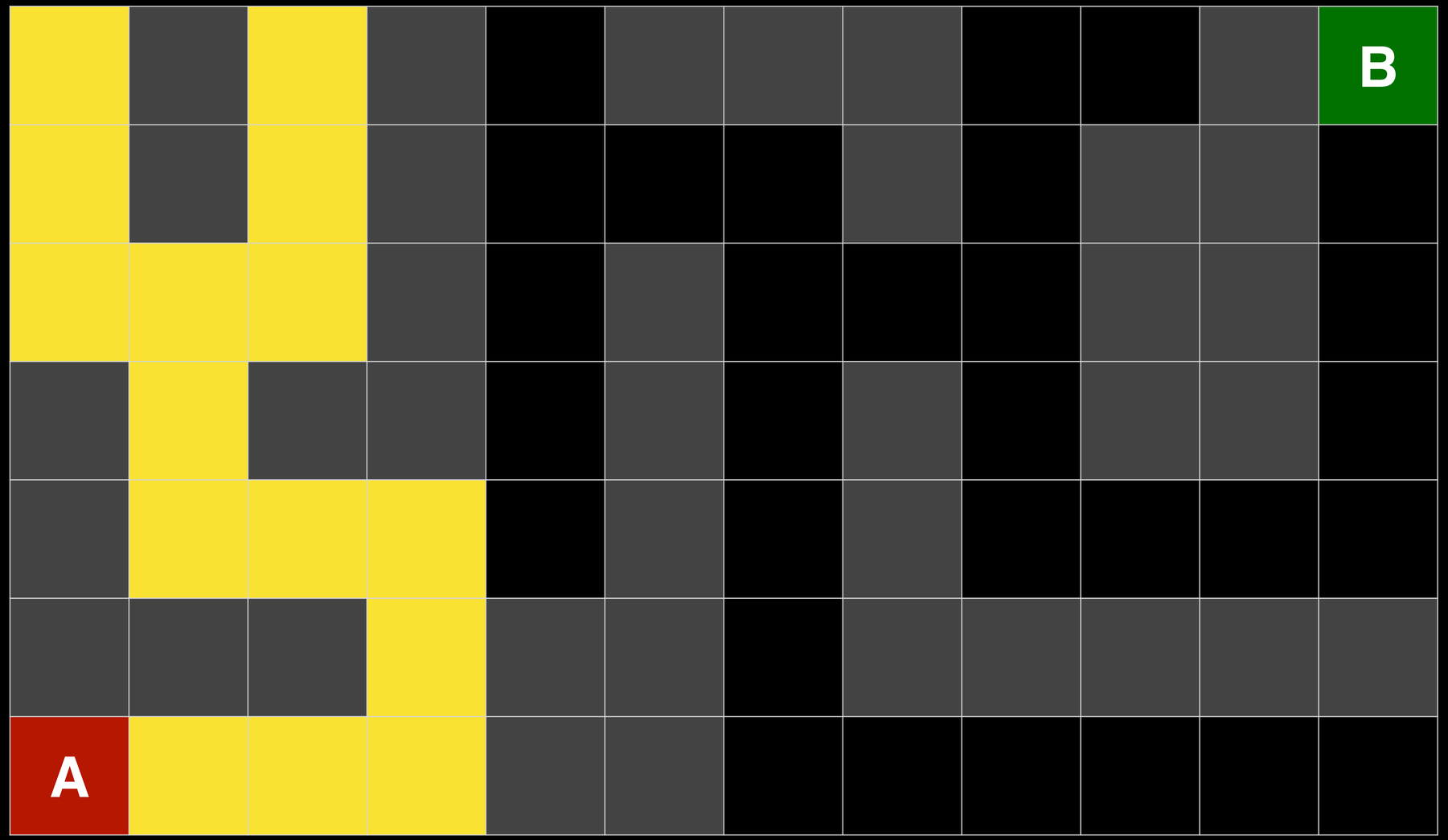


Depth-First Search



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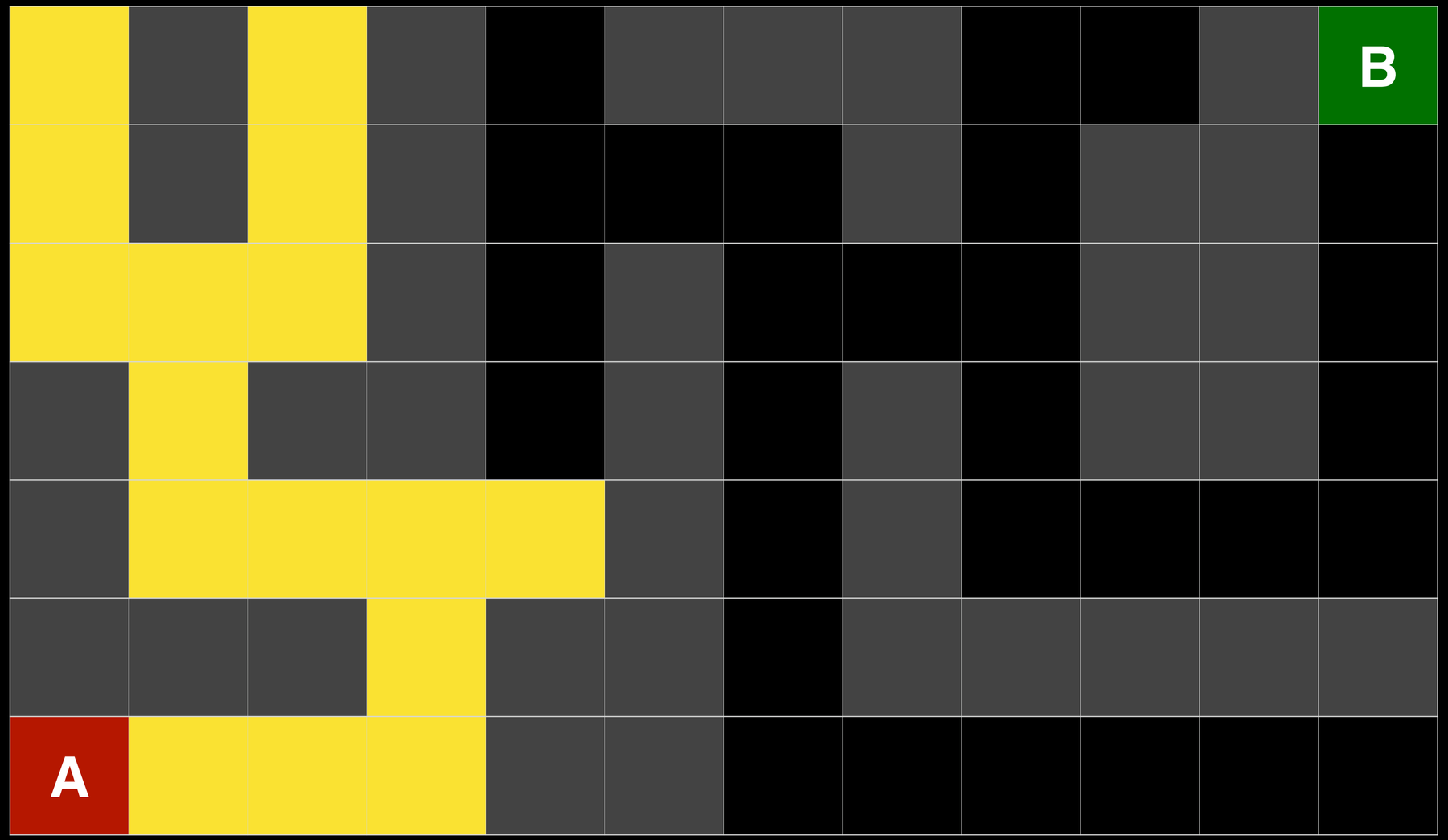
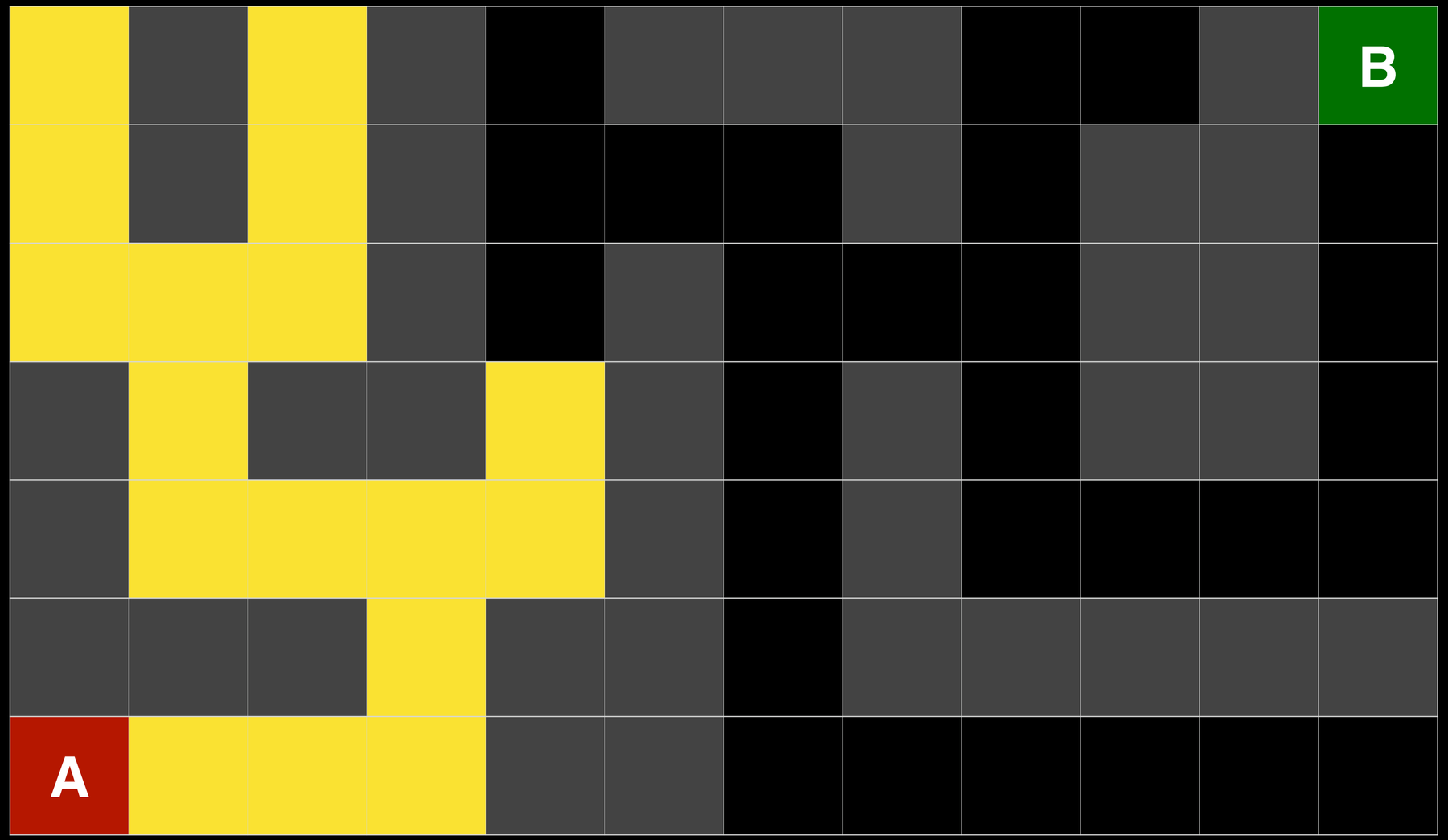
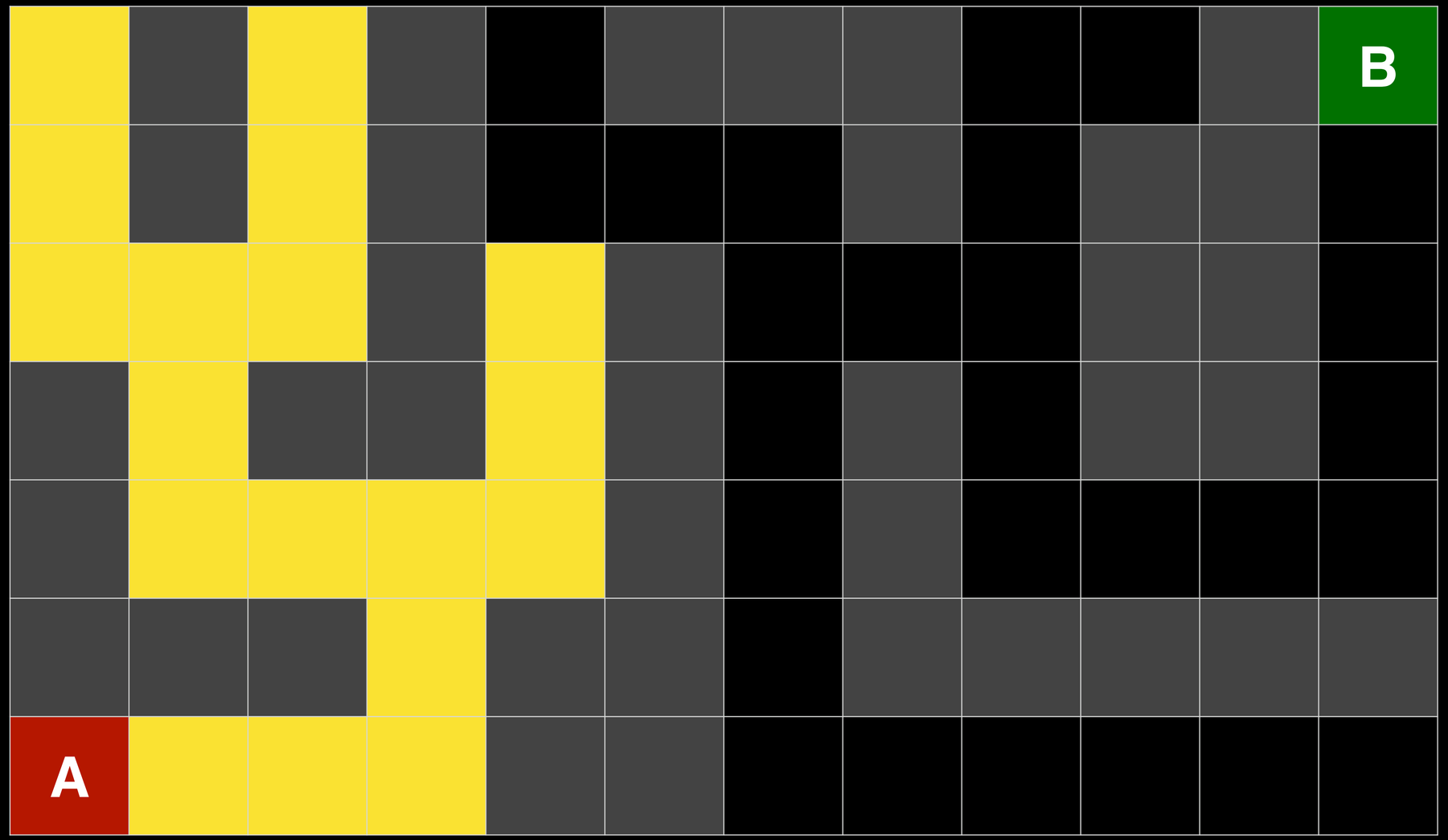


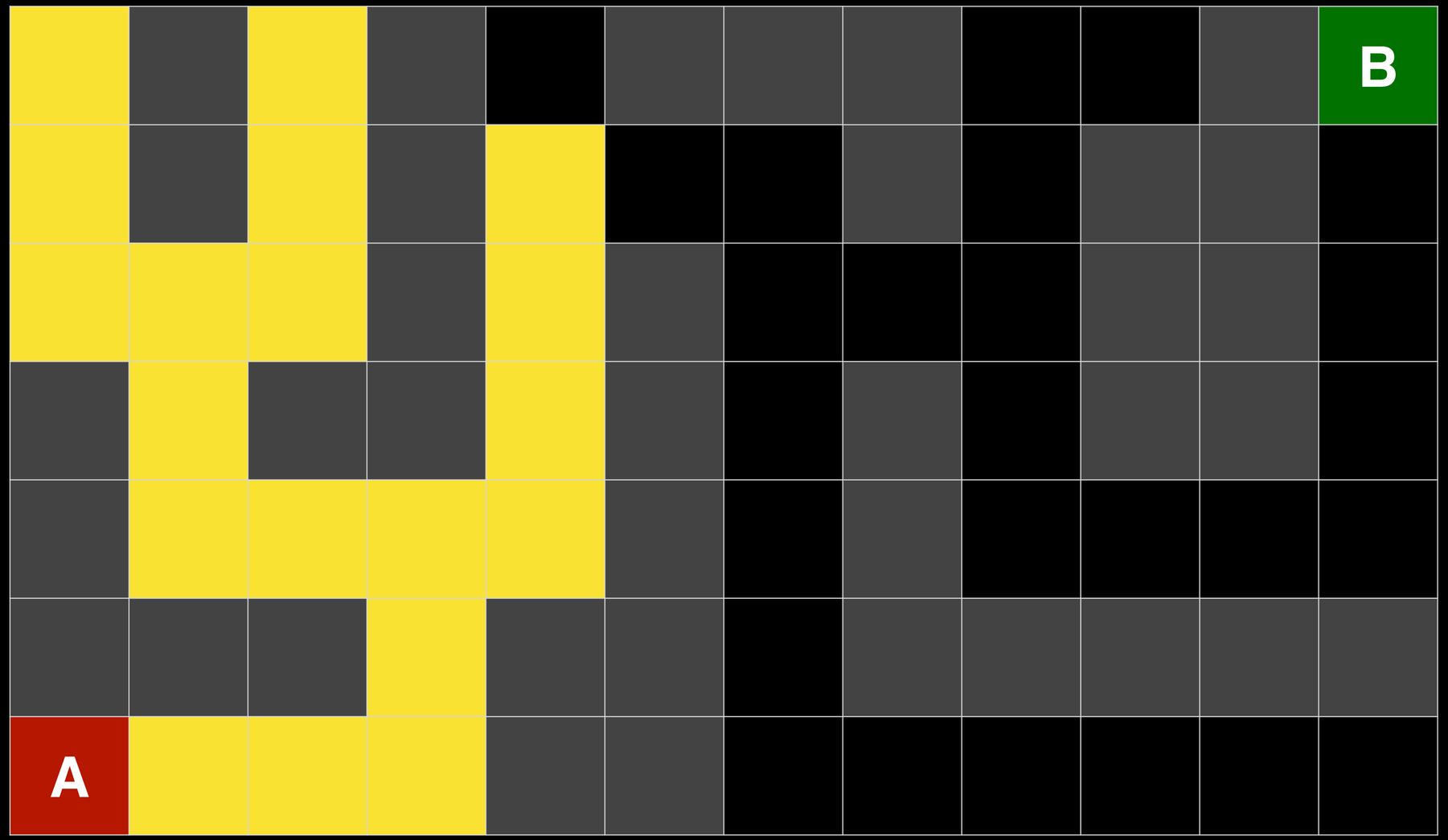
Depth-First Search



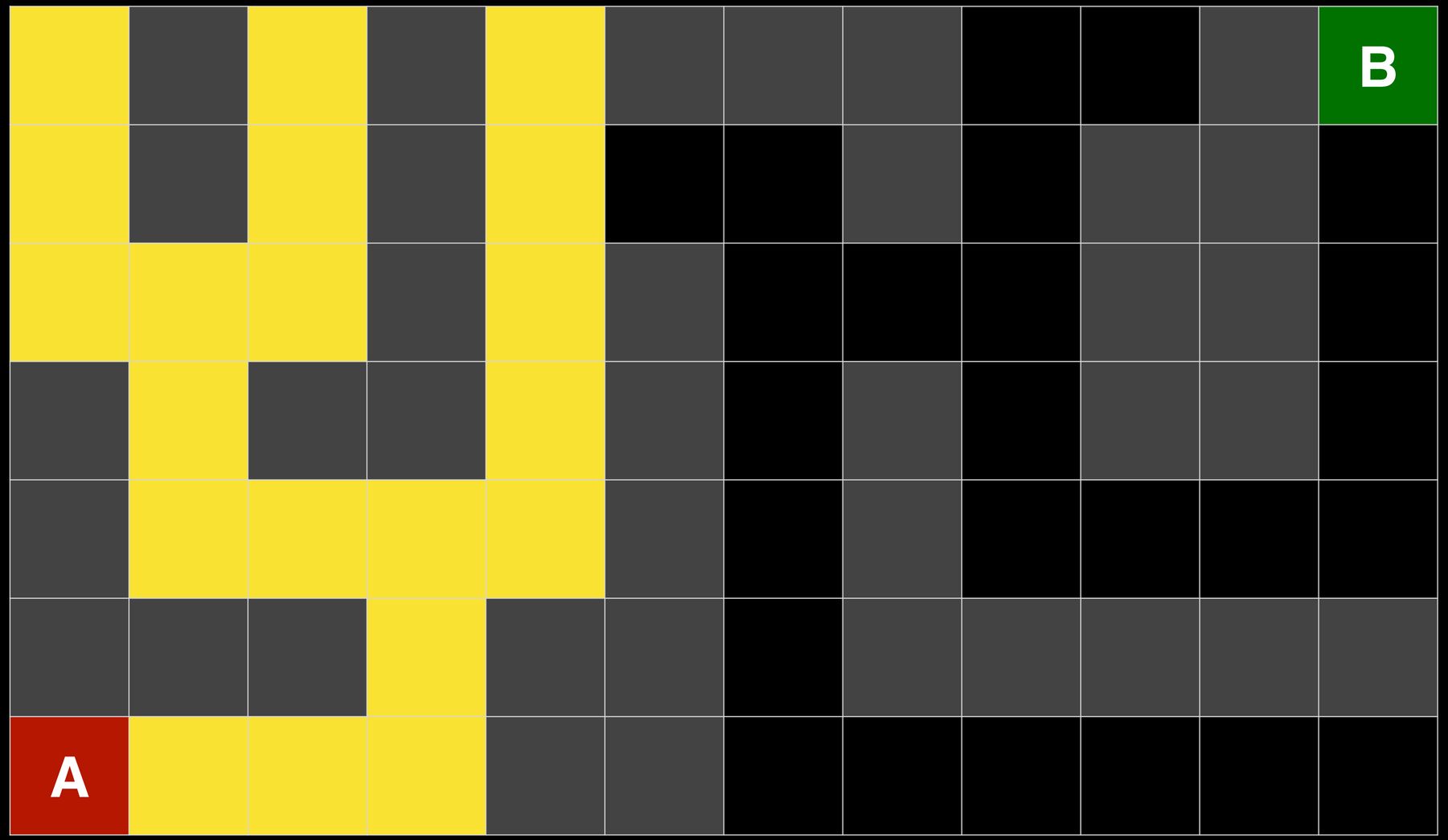
Depth-First Search



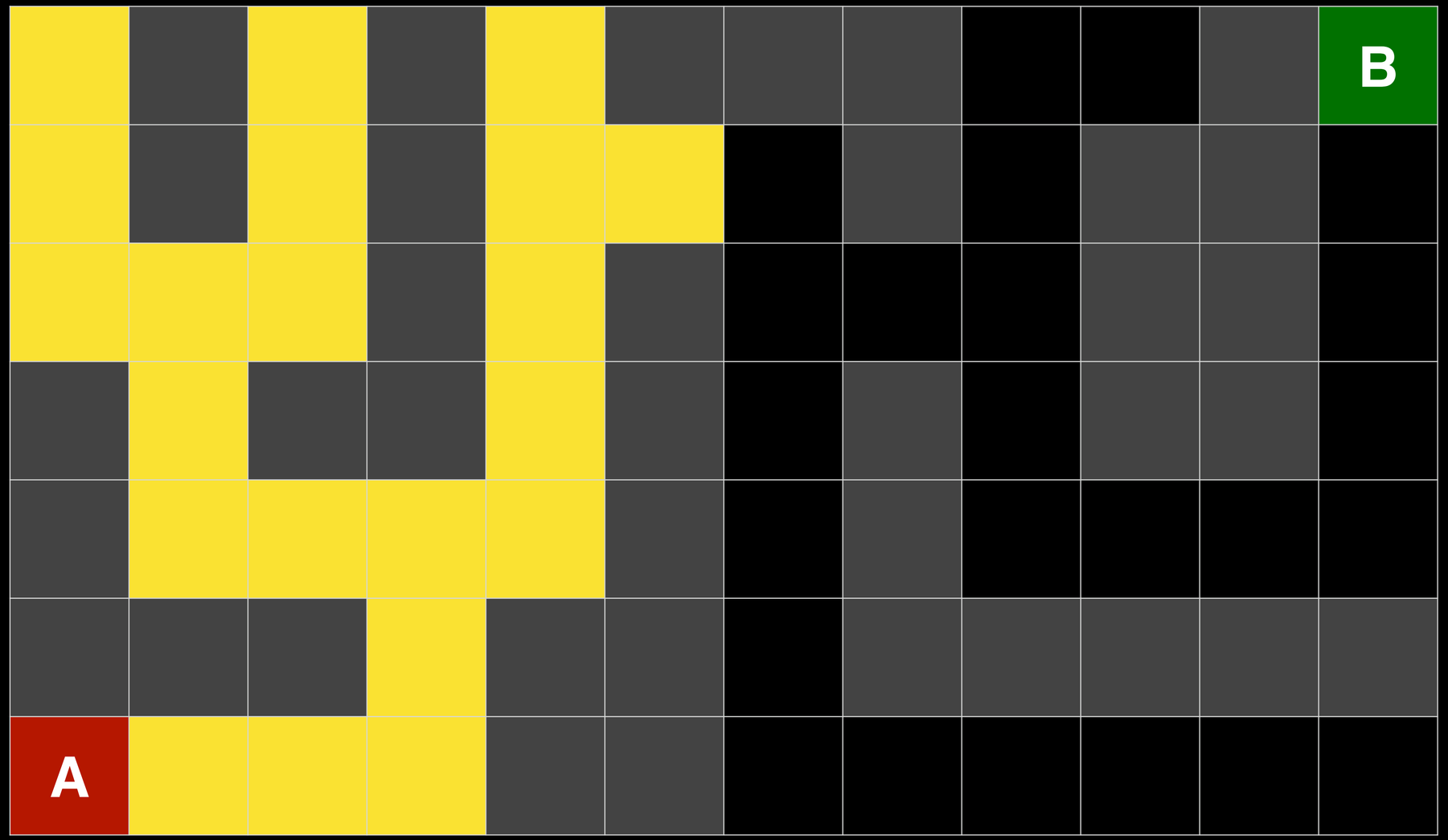
Depth-First Search



Depth-First Search



Depth-First Search



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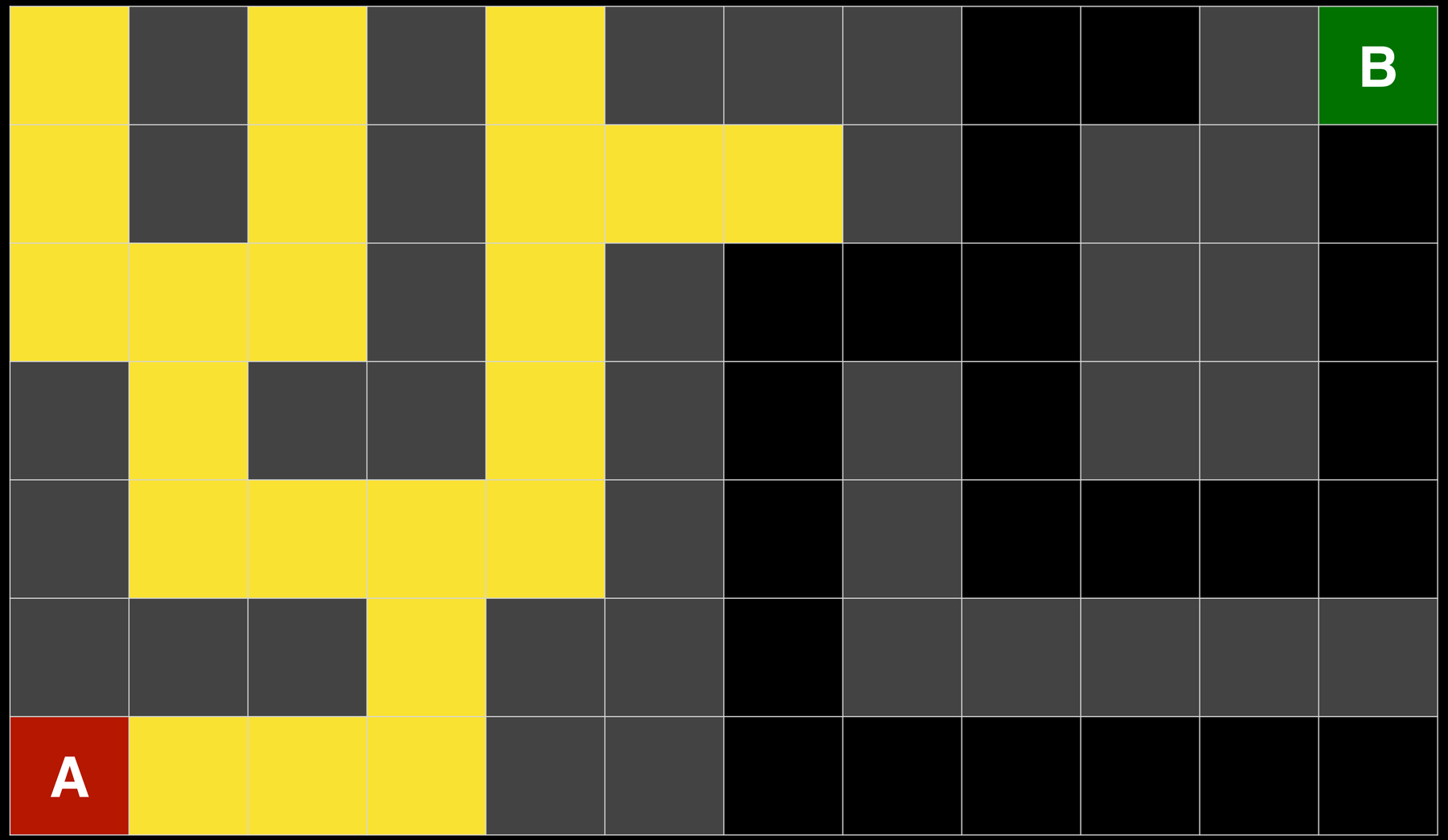
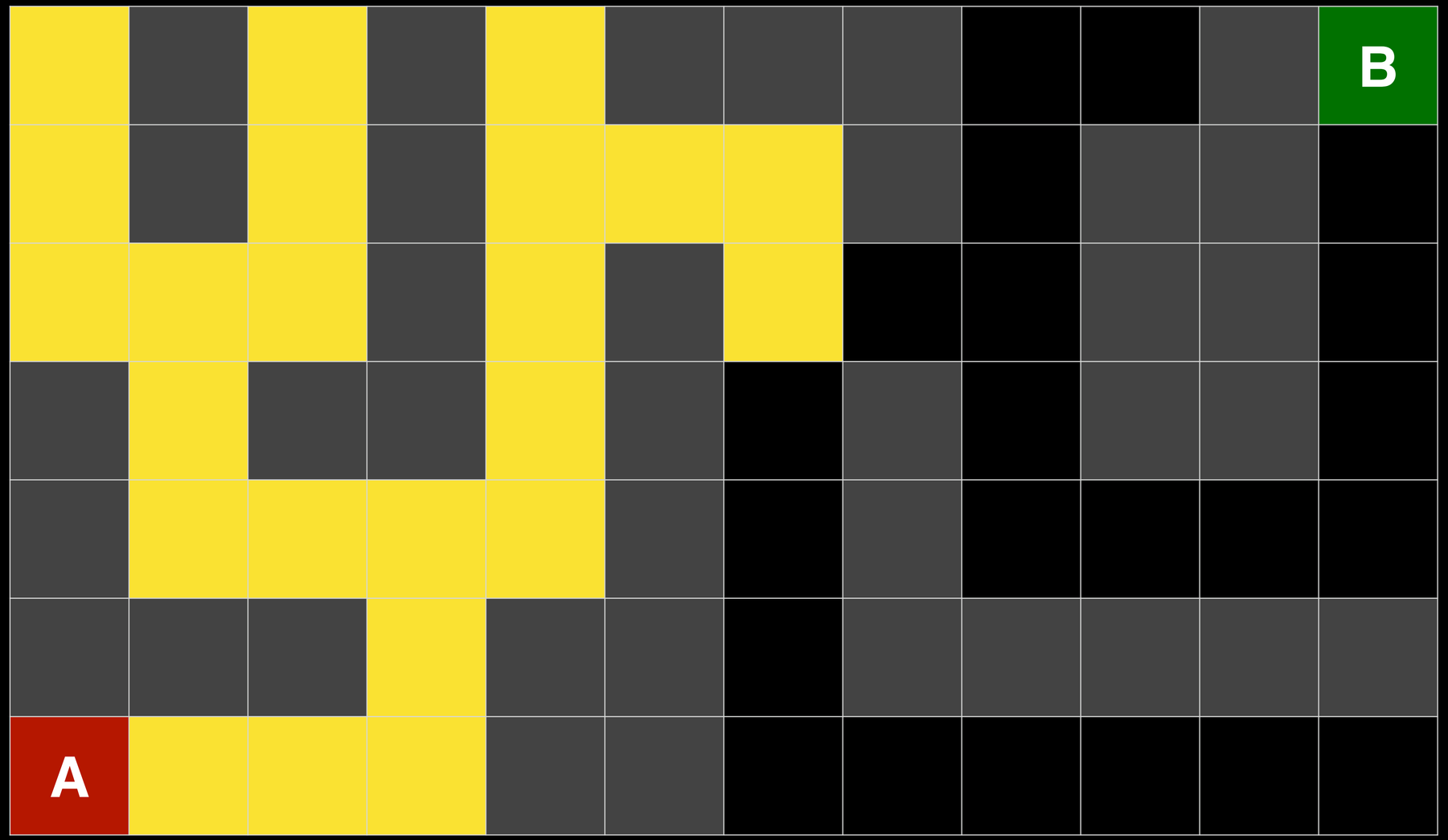
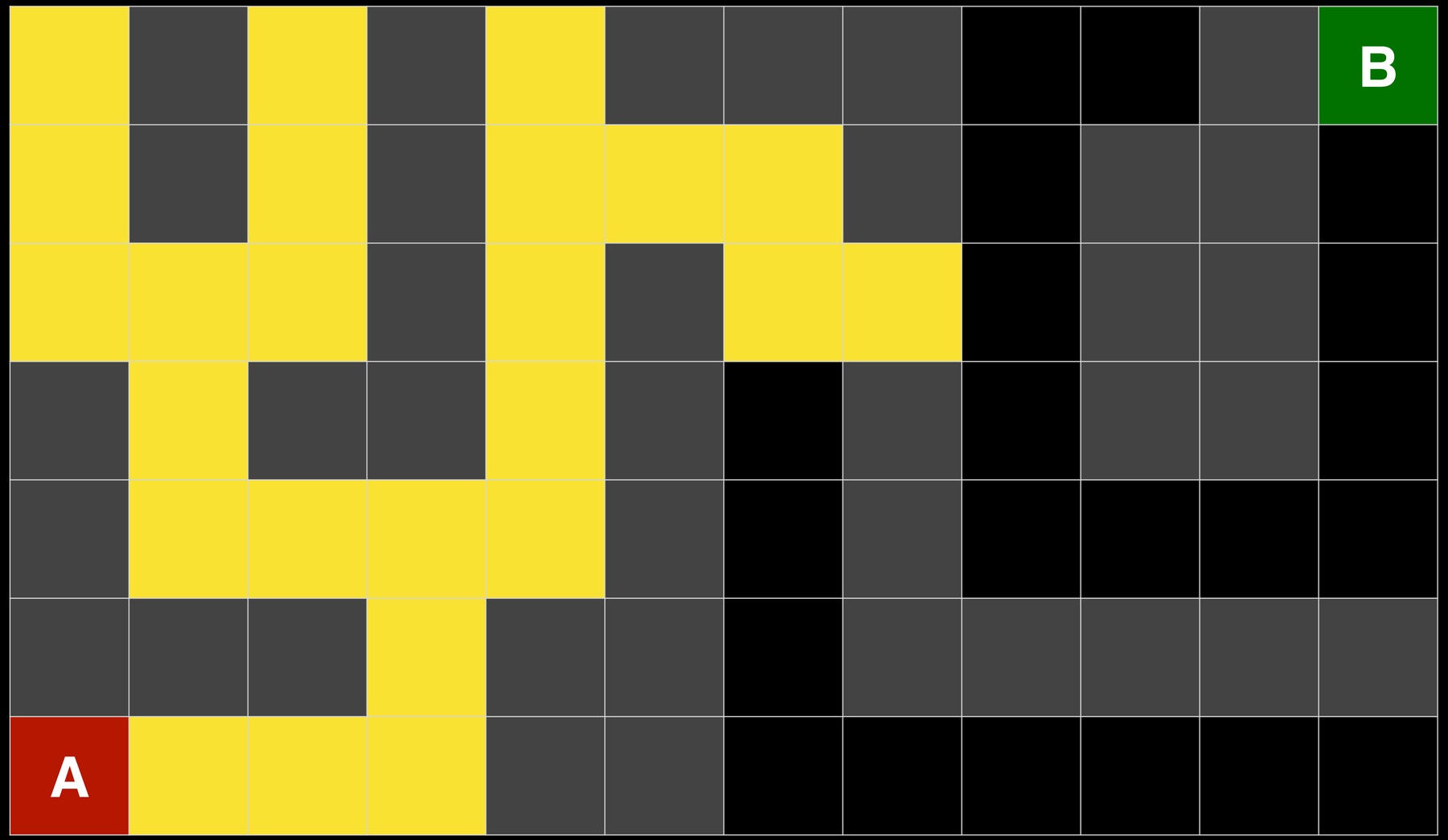


Depth-First Search



Depth-First Search



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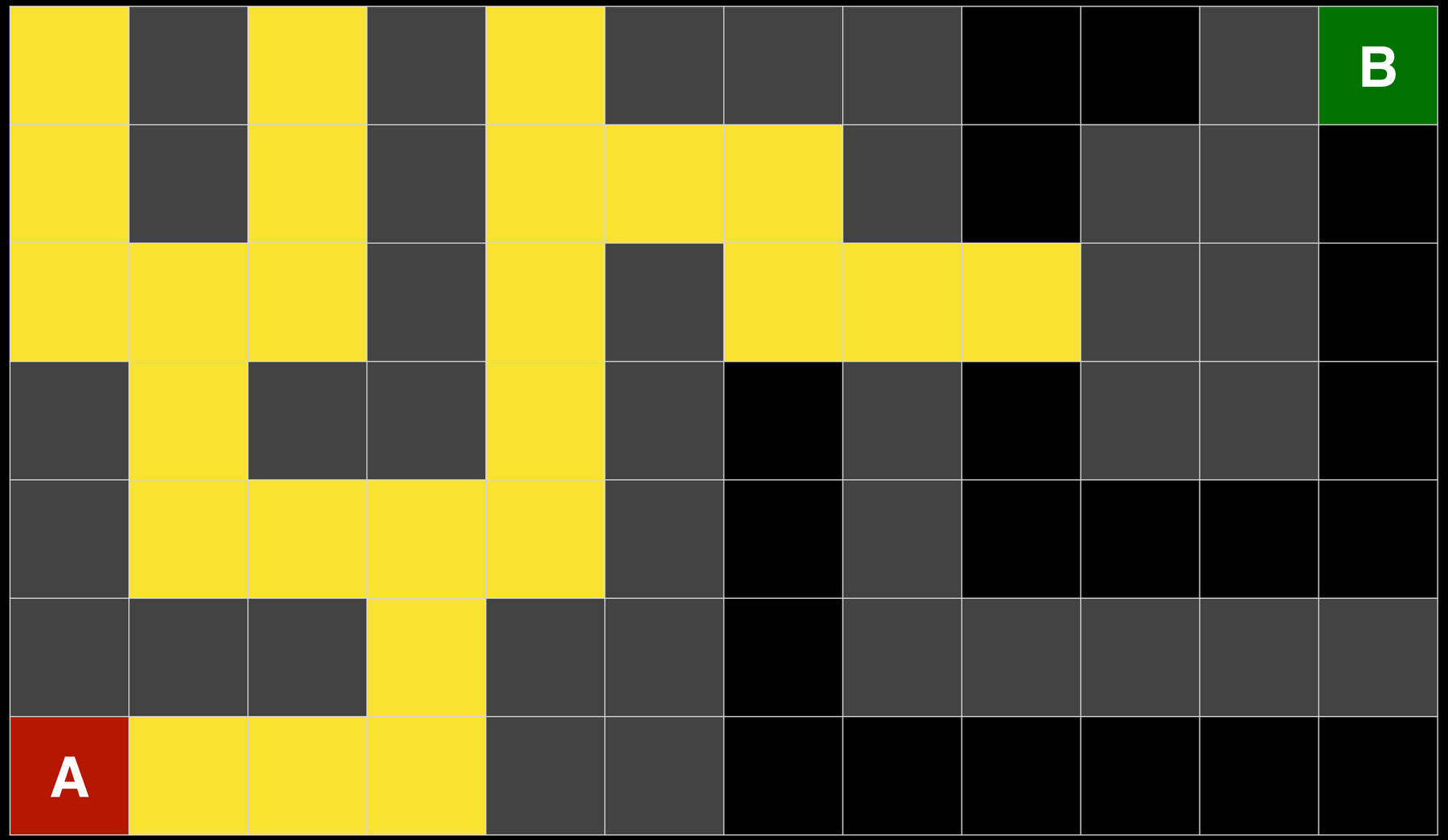
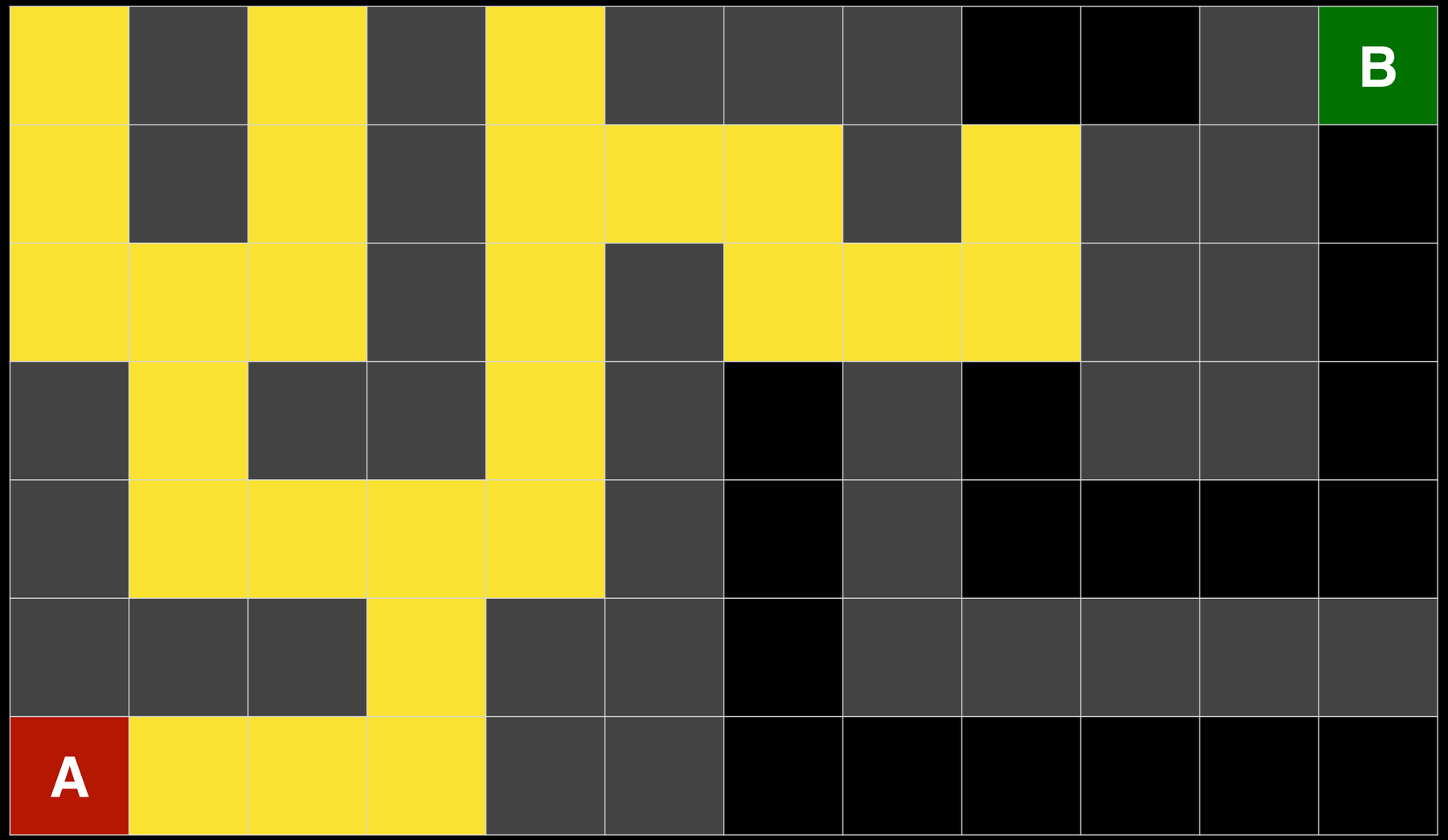
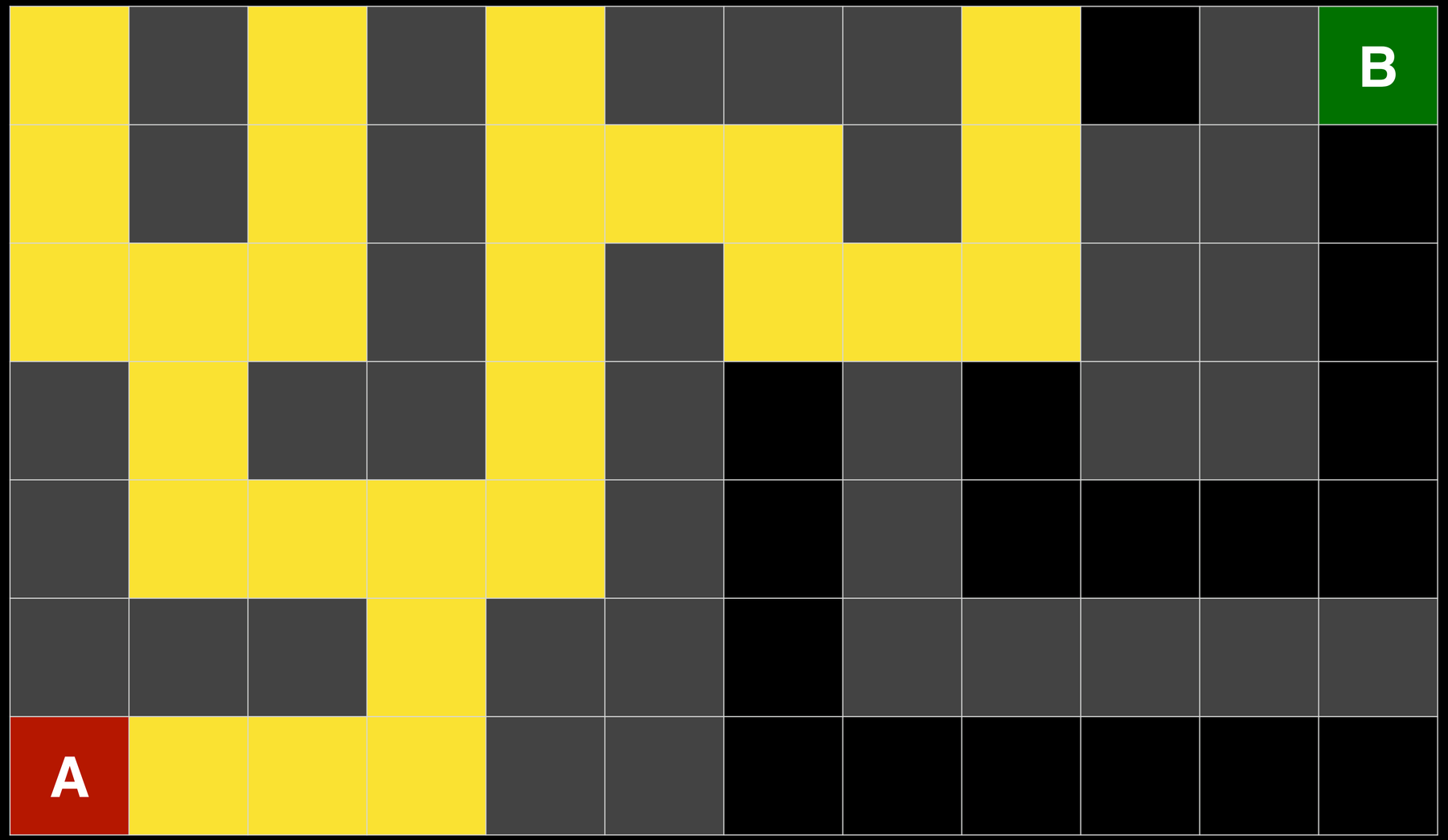


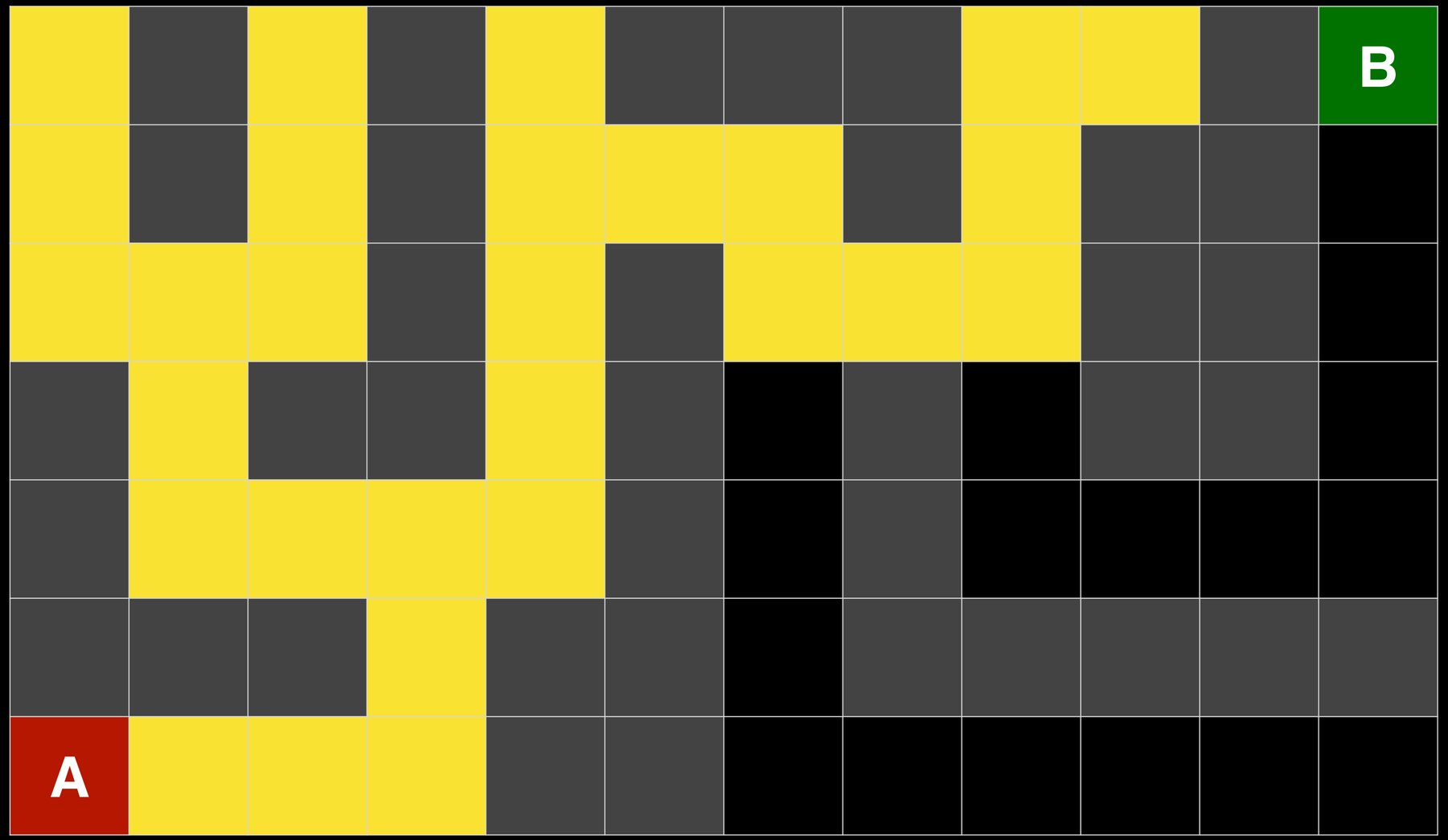
Depth-First Search



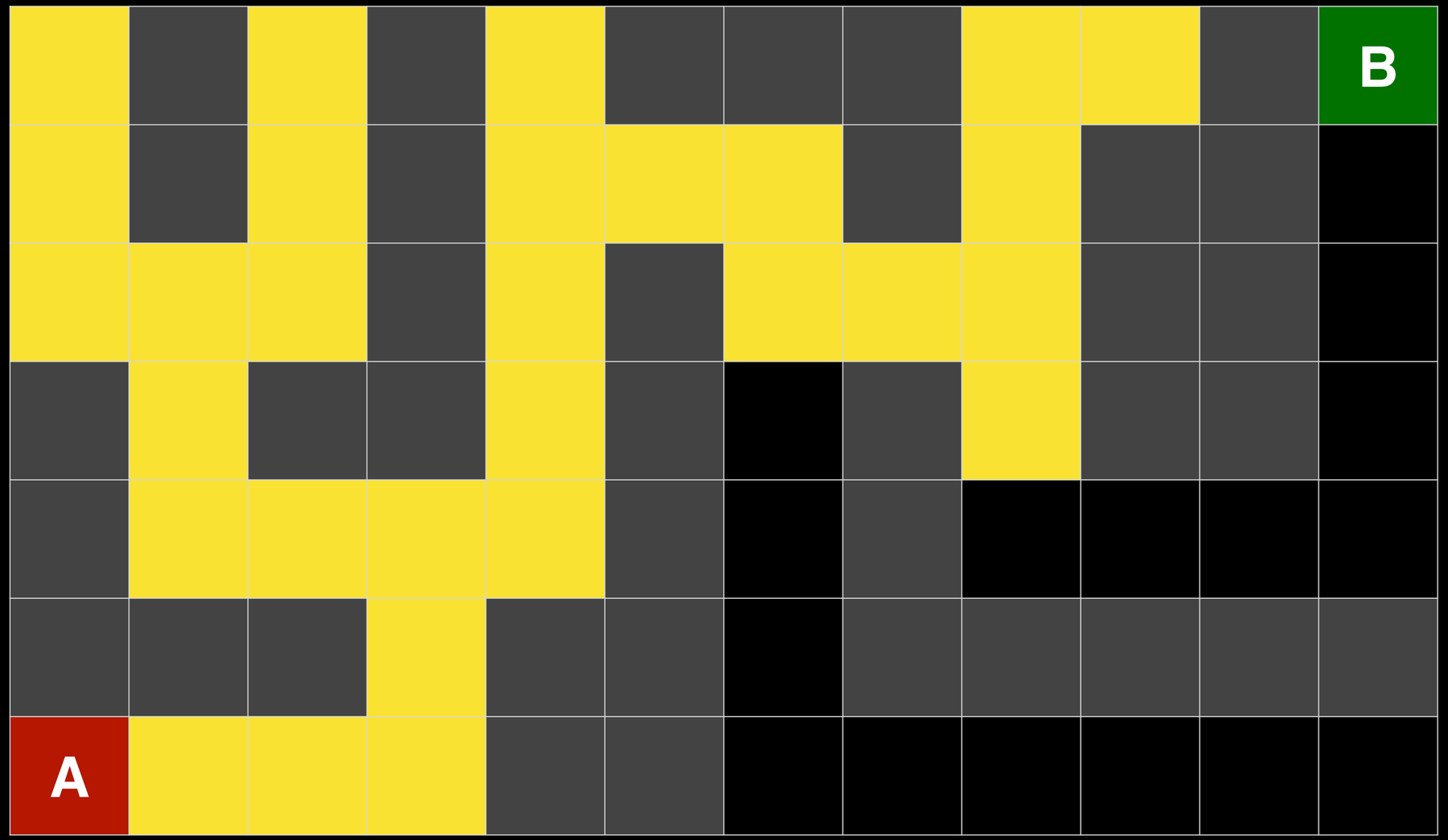
Depth-First Search



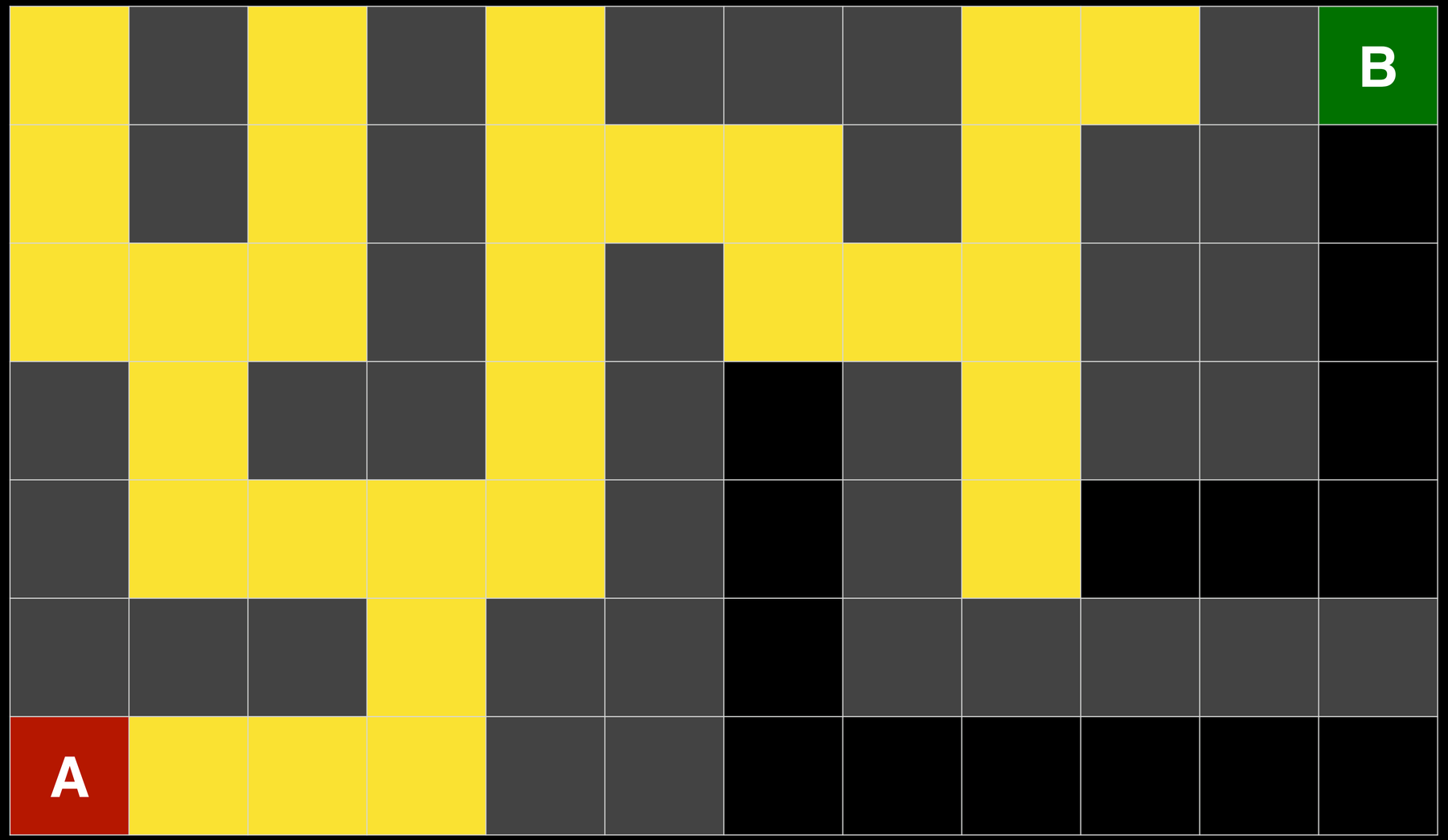
Depth-First Search



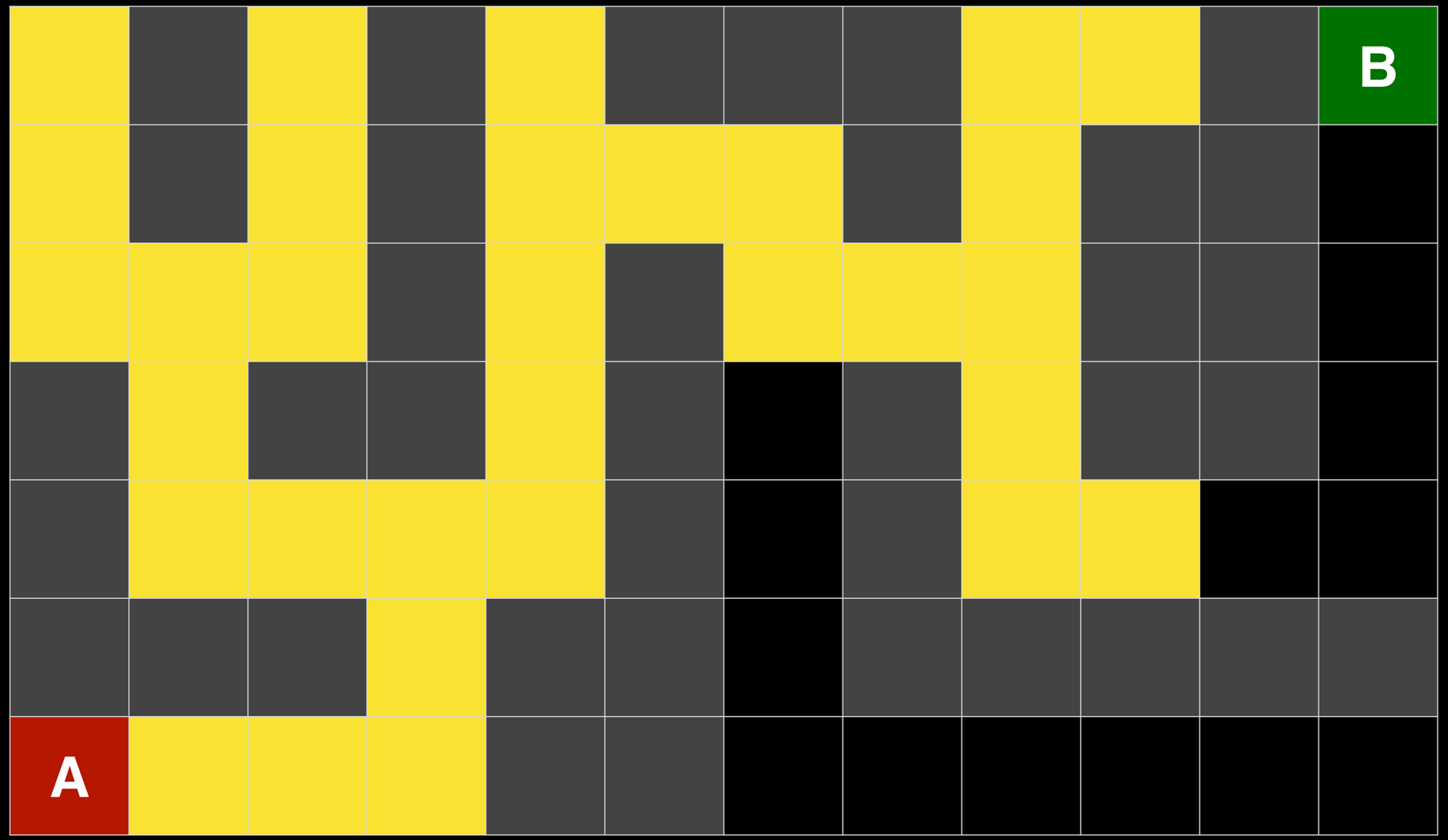
Depth-First Search



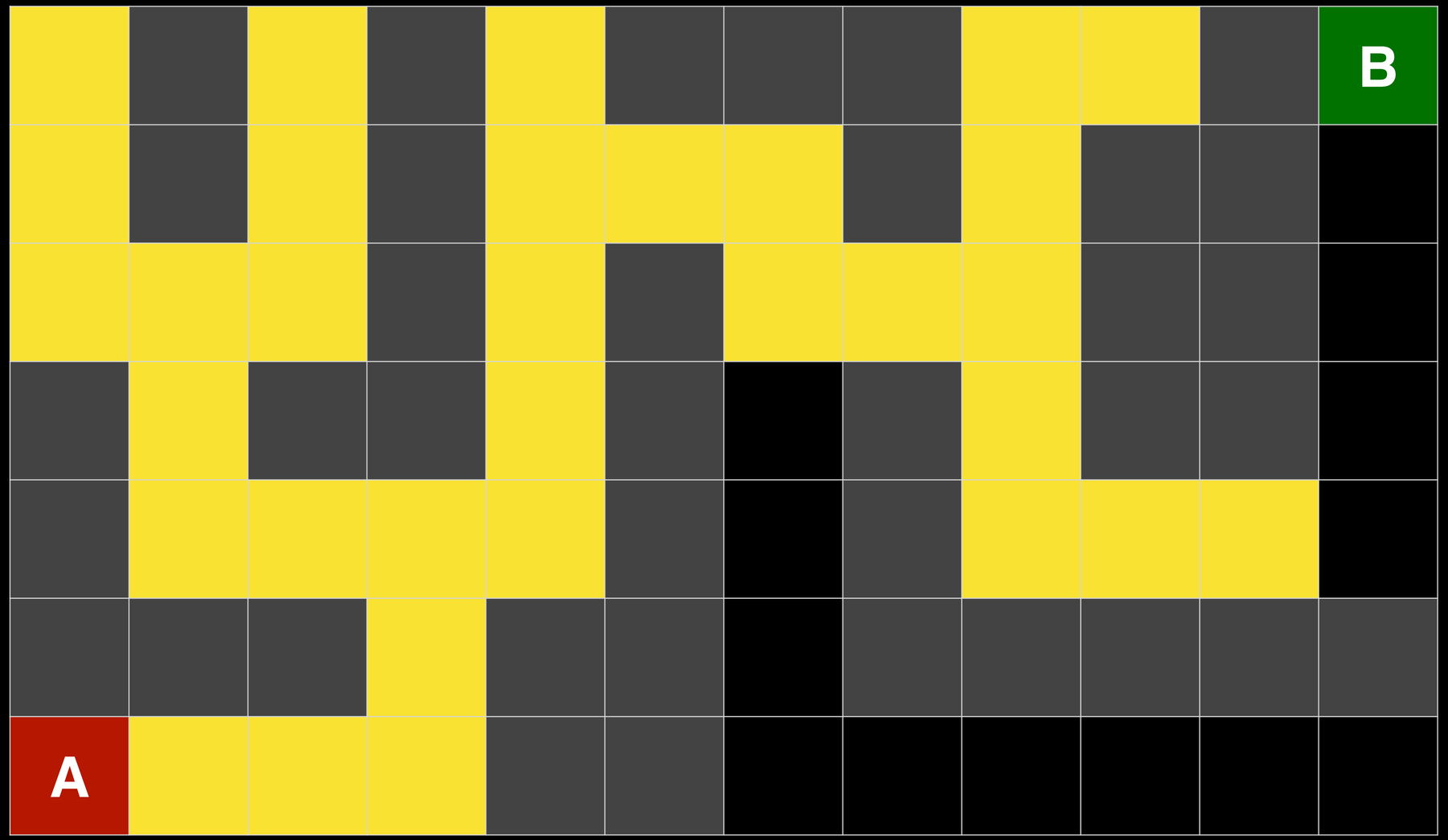
Depth-First Search



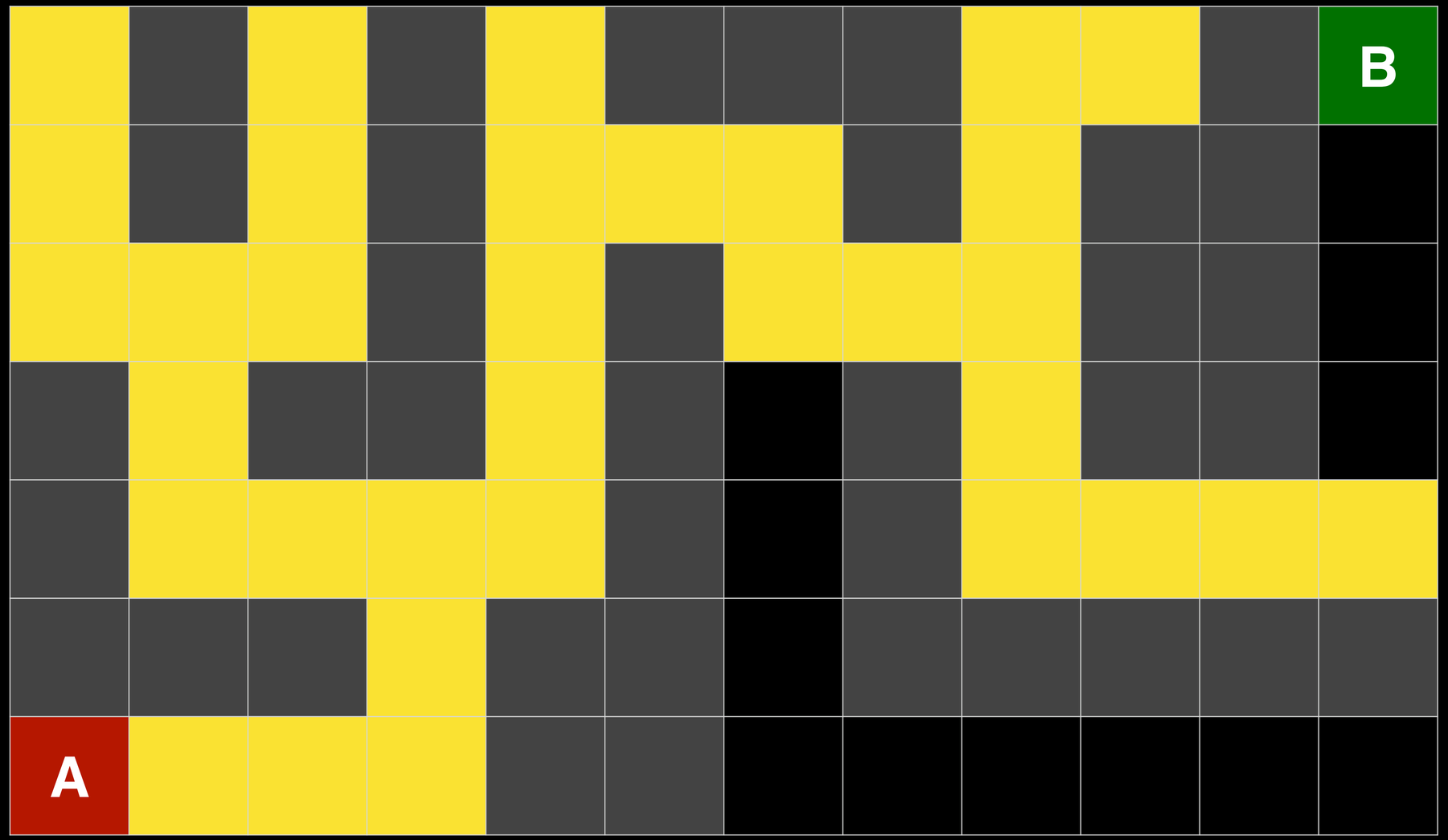
Depth-First Search



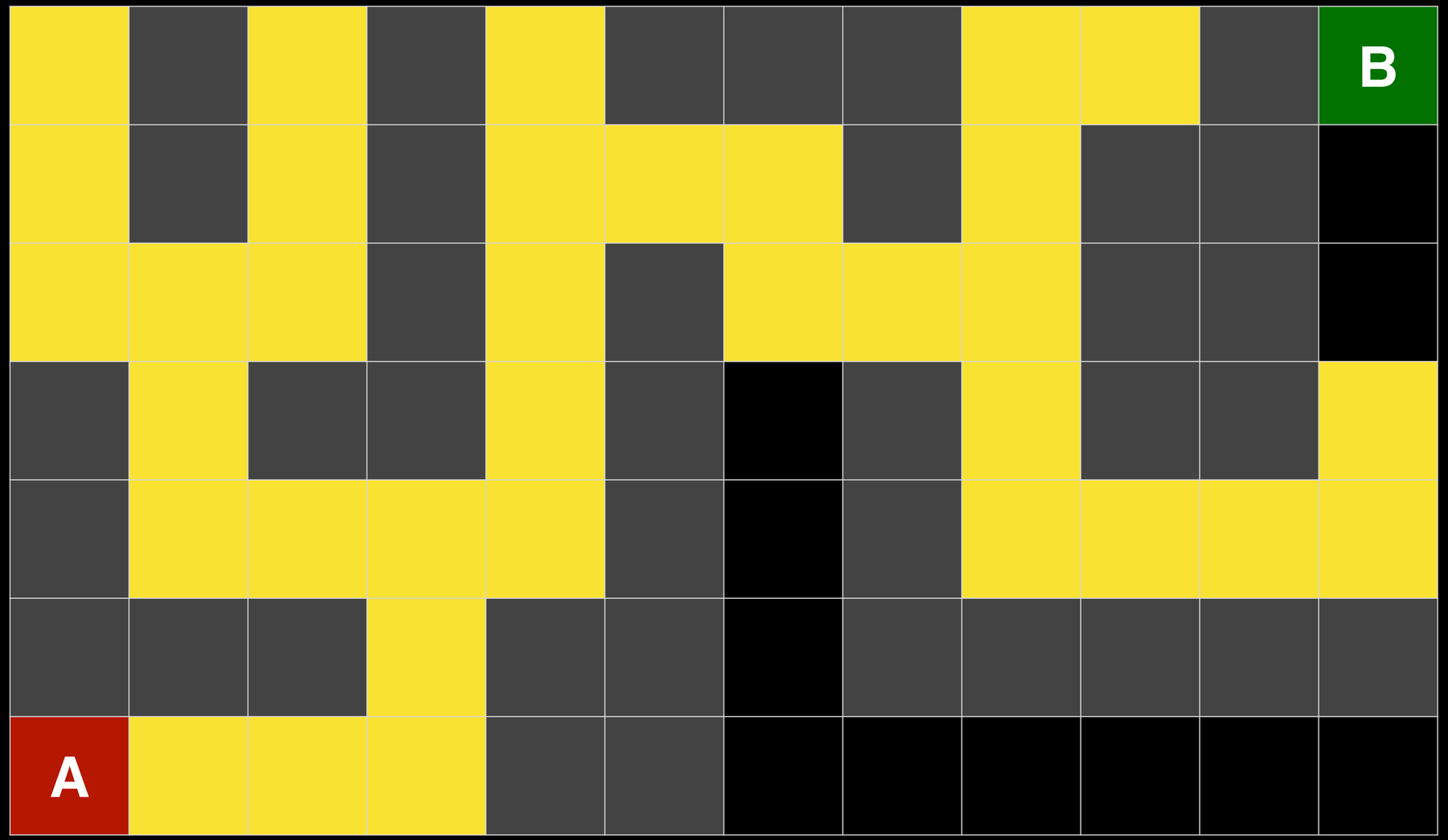
Depth-First Search



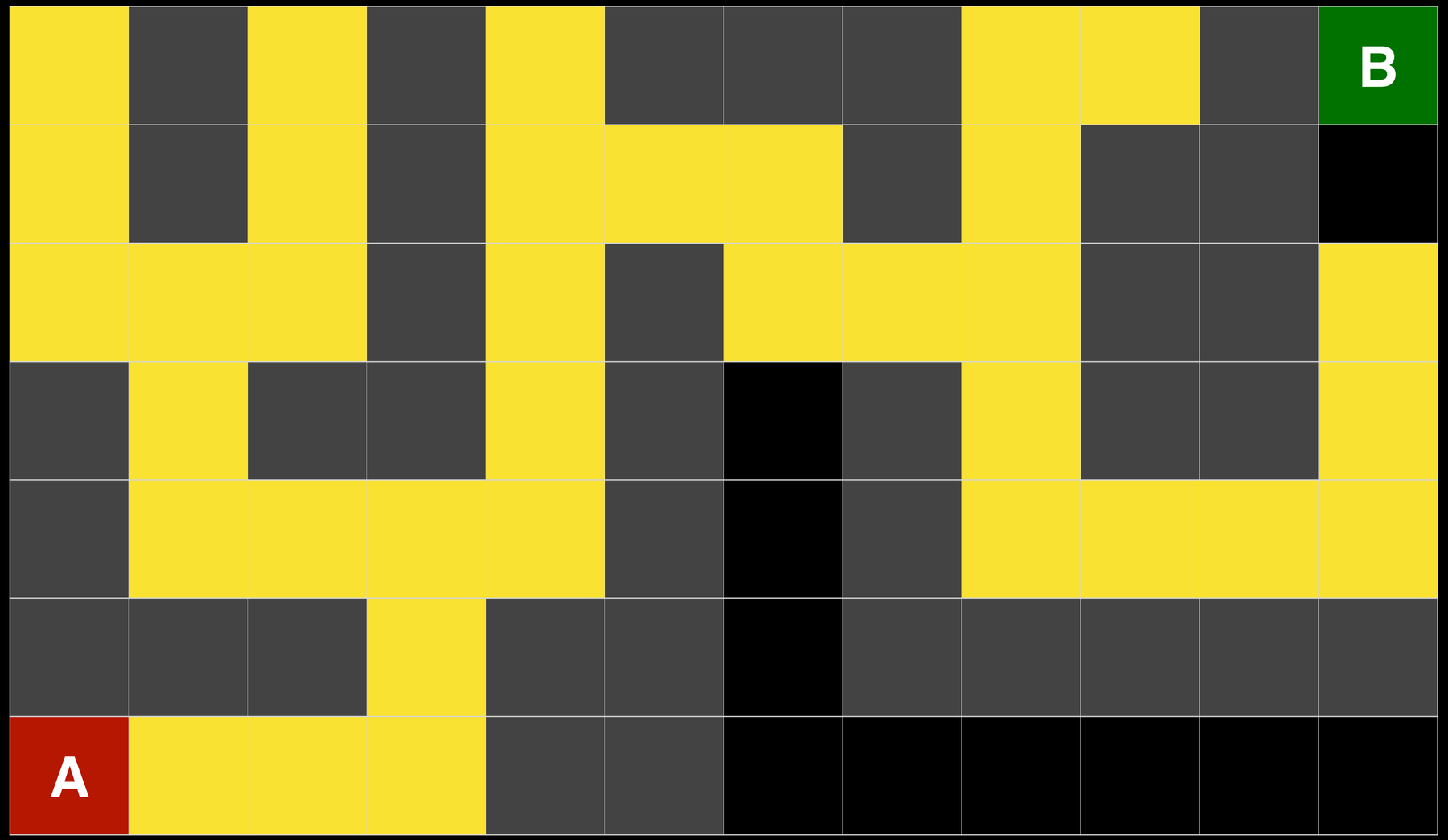
Depth-First Search



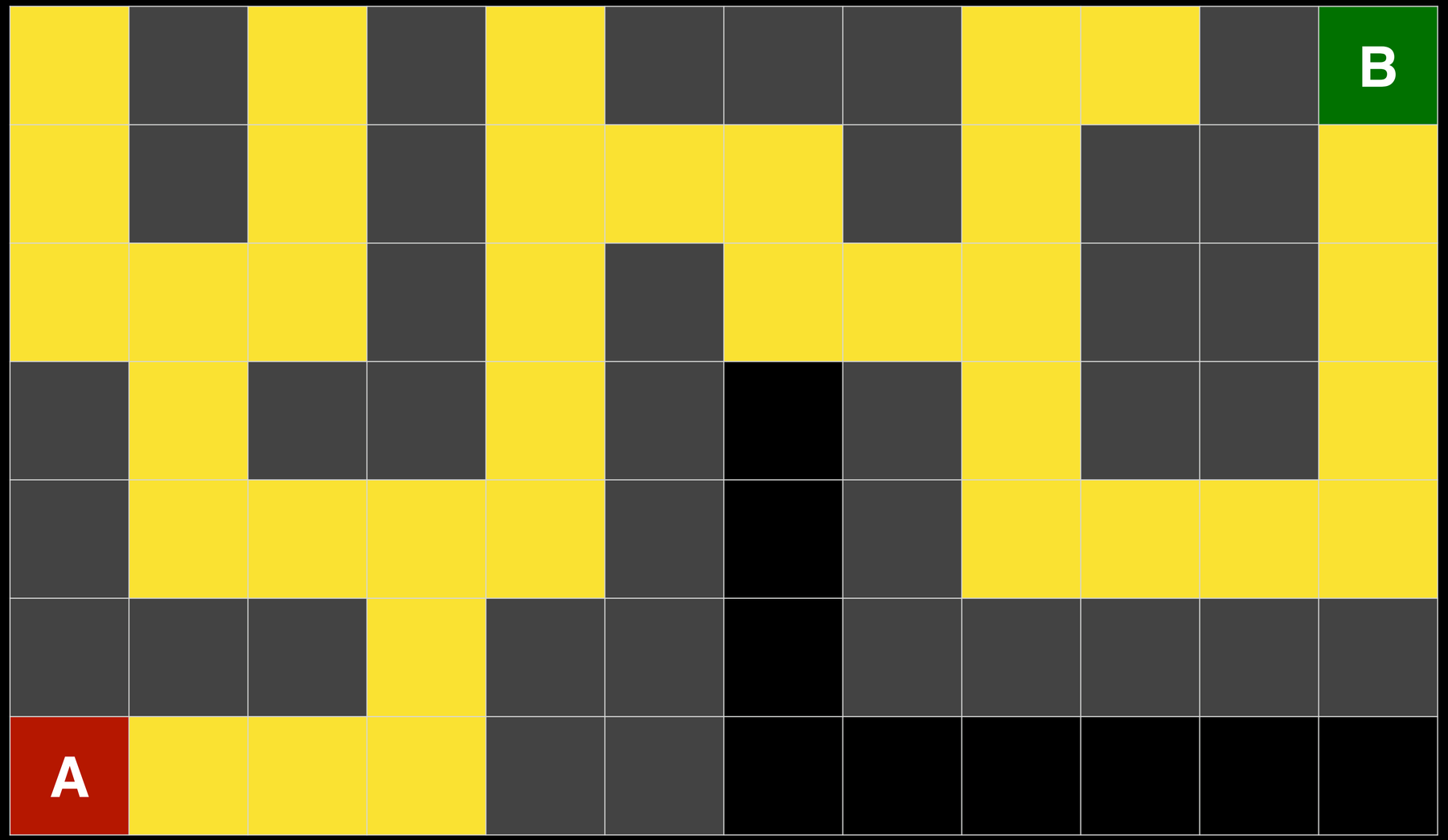
Depth-First Search



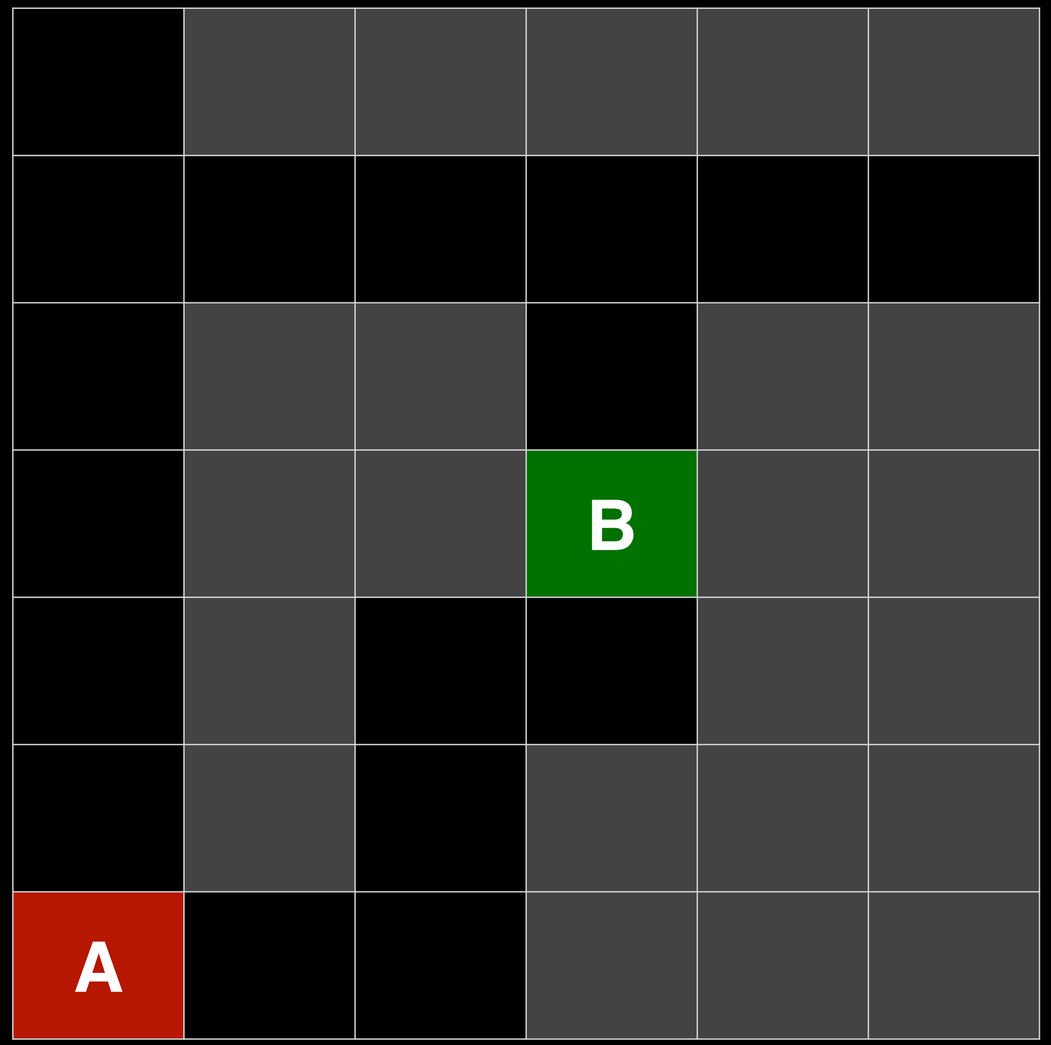
Depth-First Search



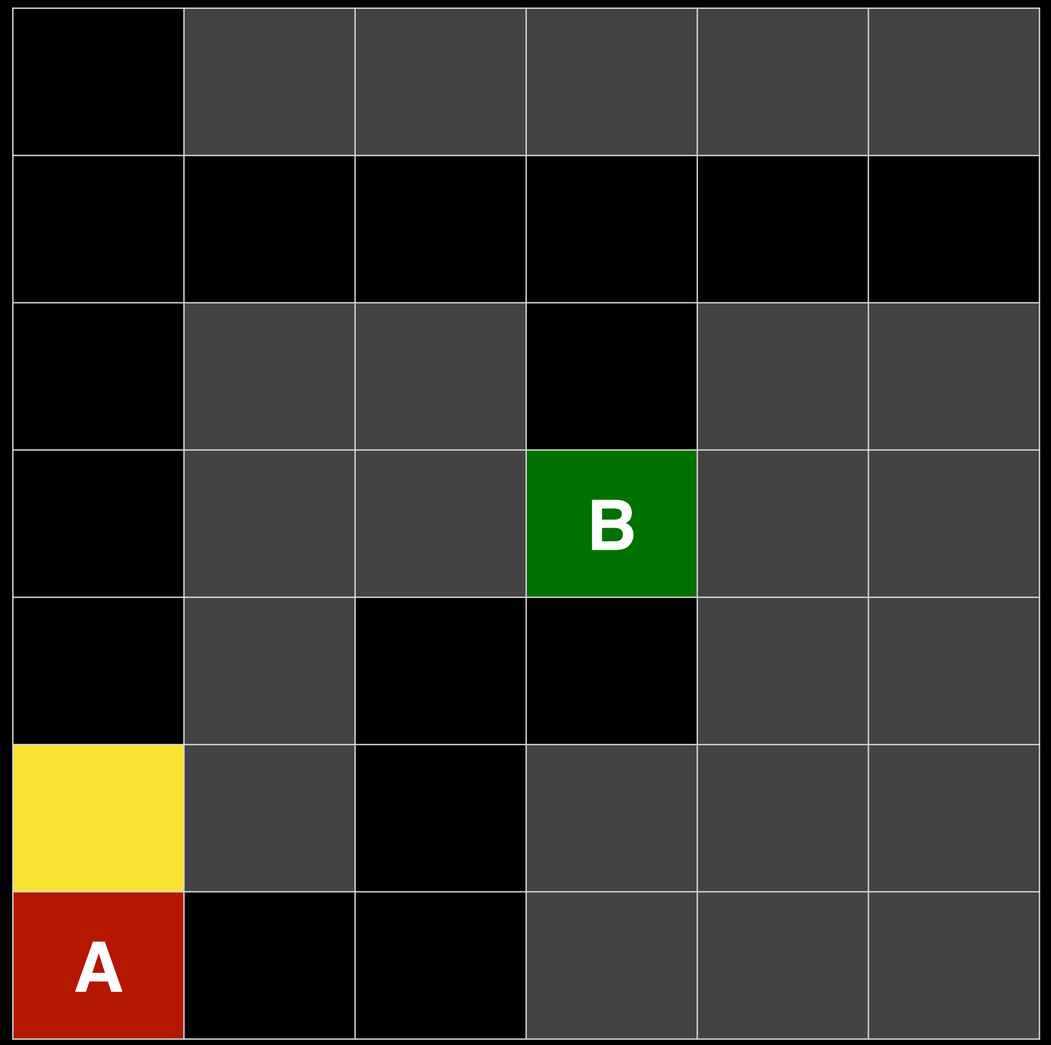
Depth-First Search



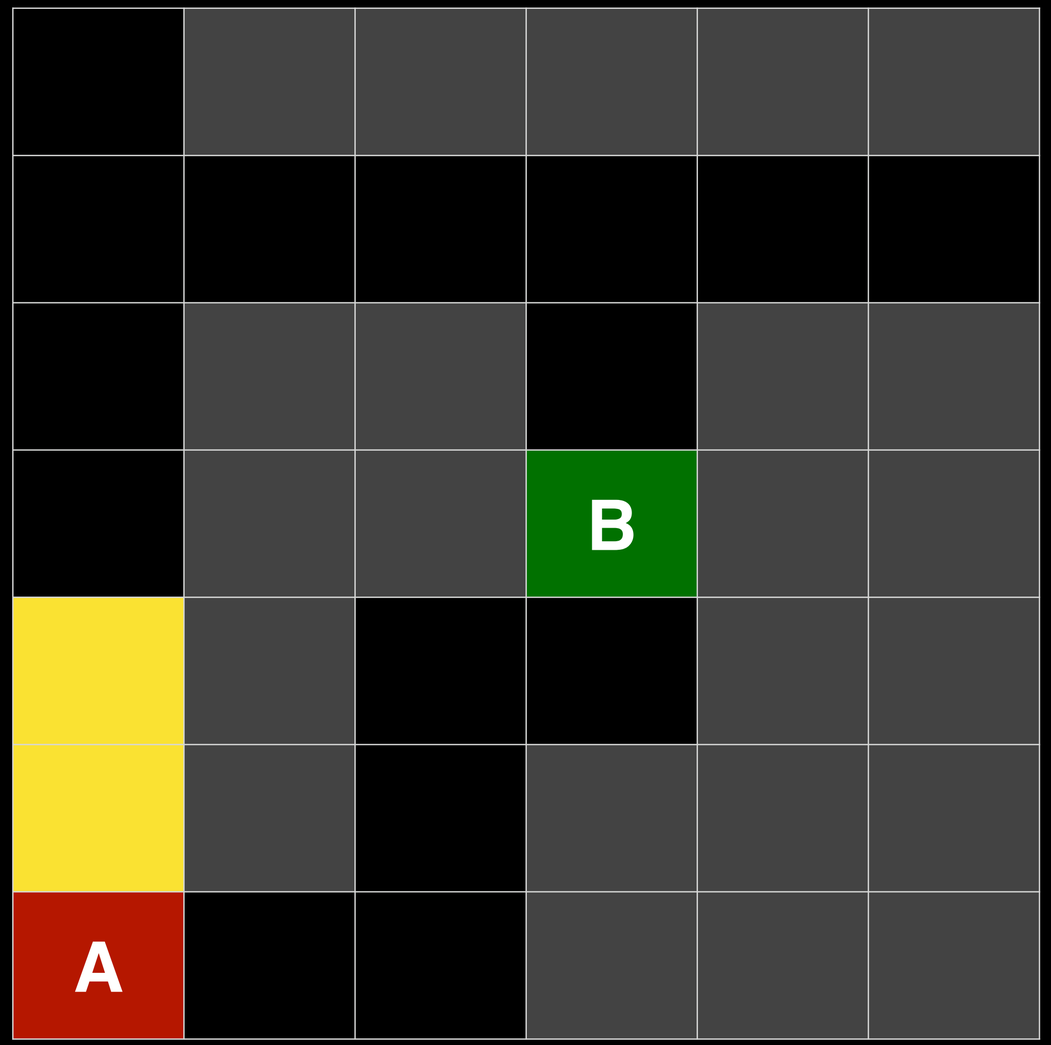
Depth-First Search



Depth-First Search



Depth-First Search



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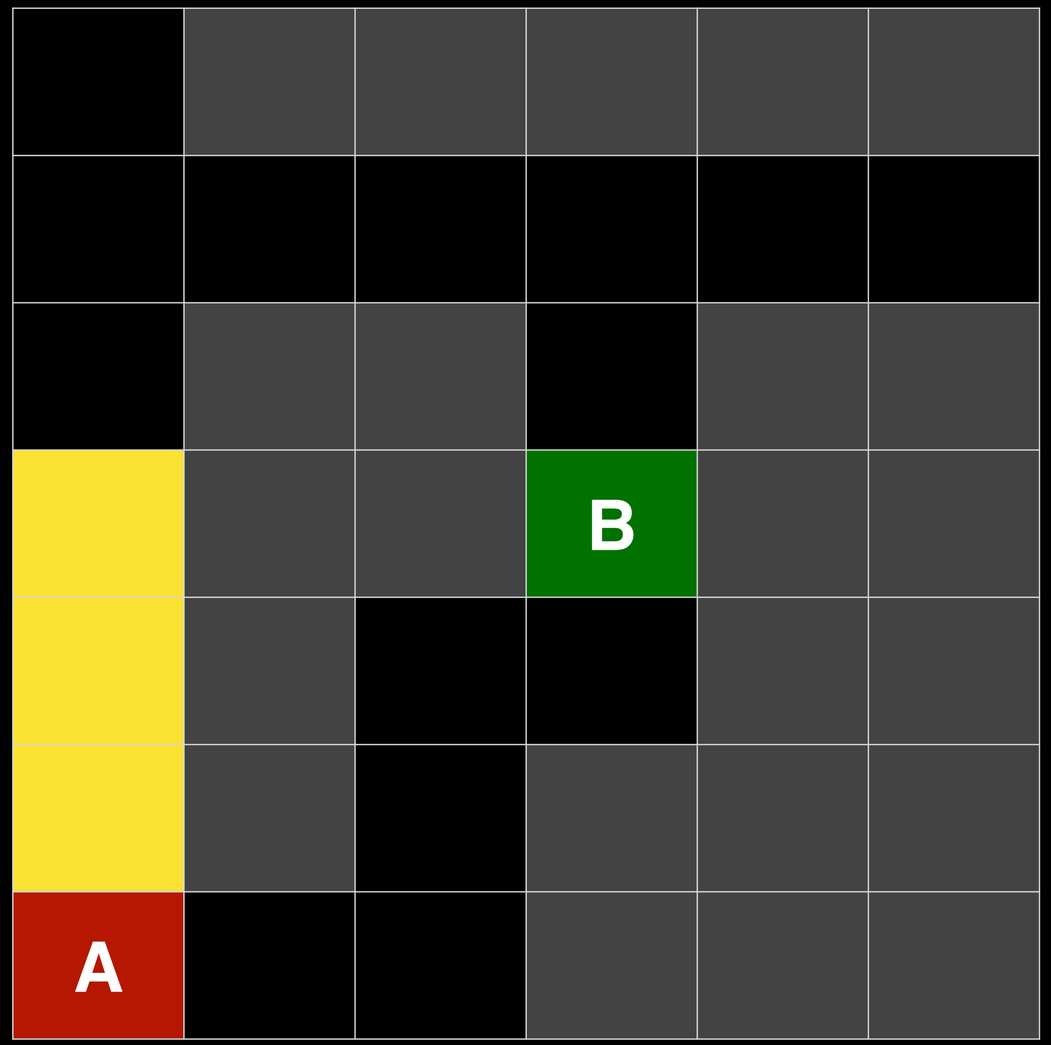

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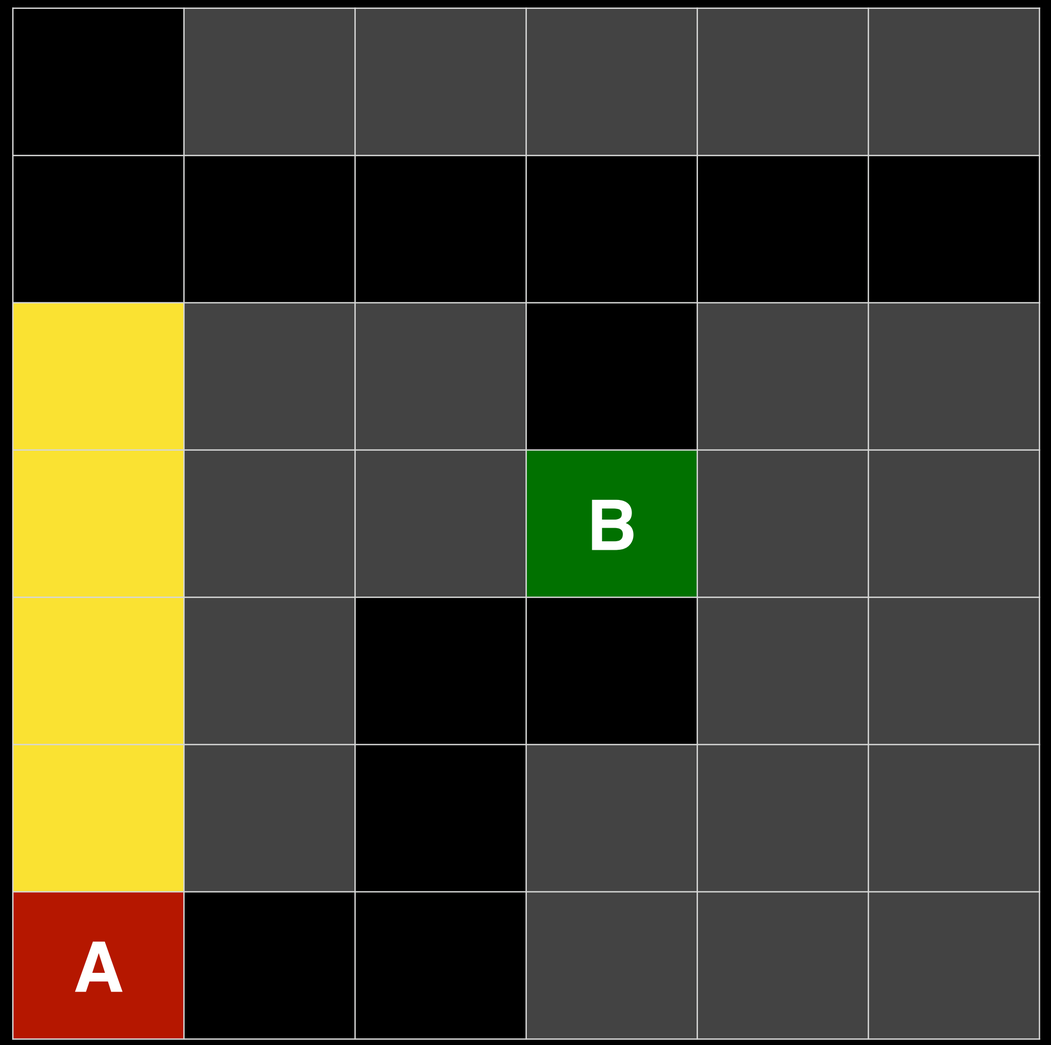

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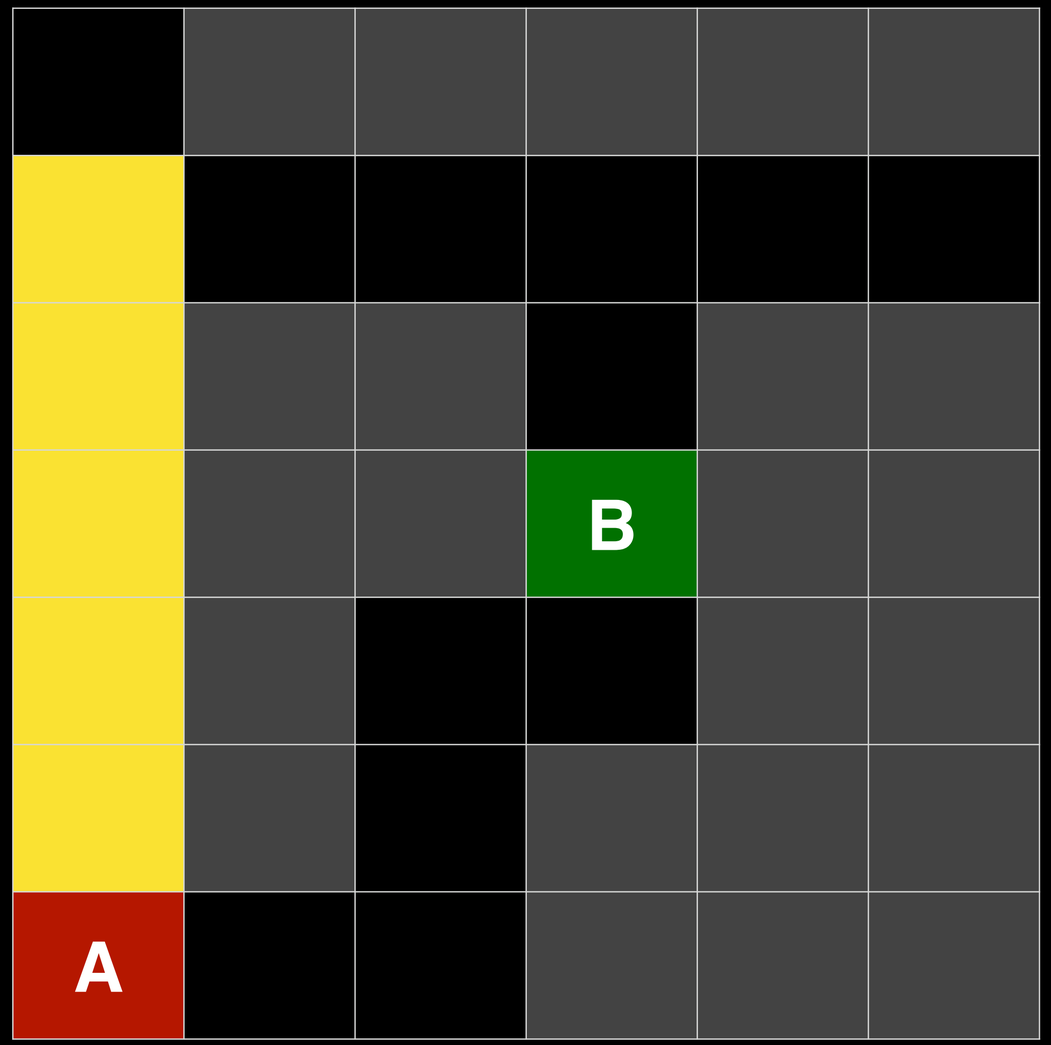

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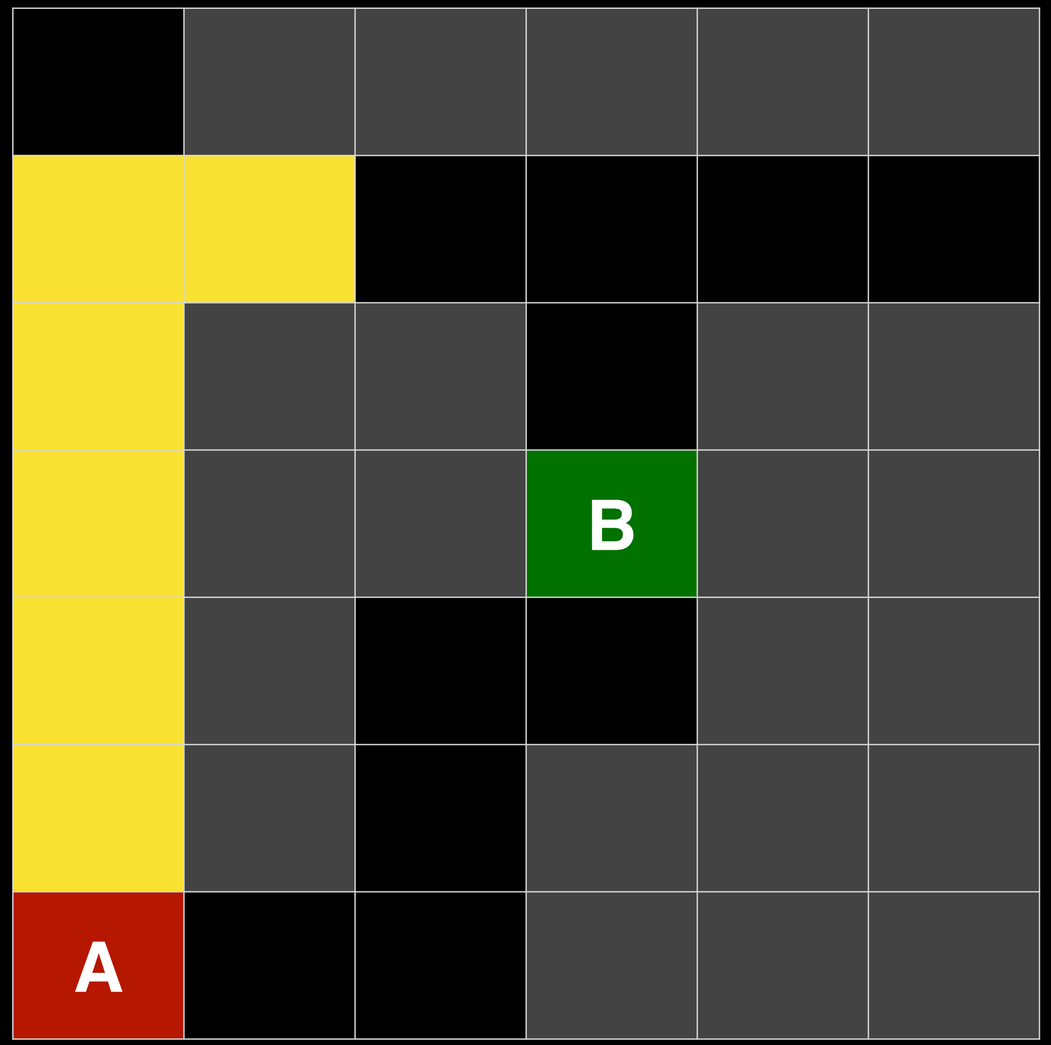

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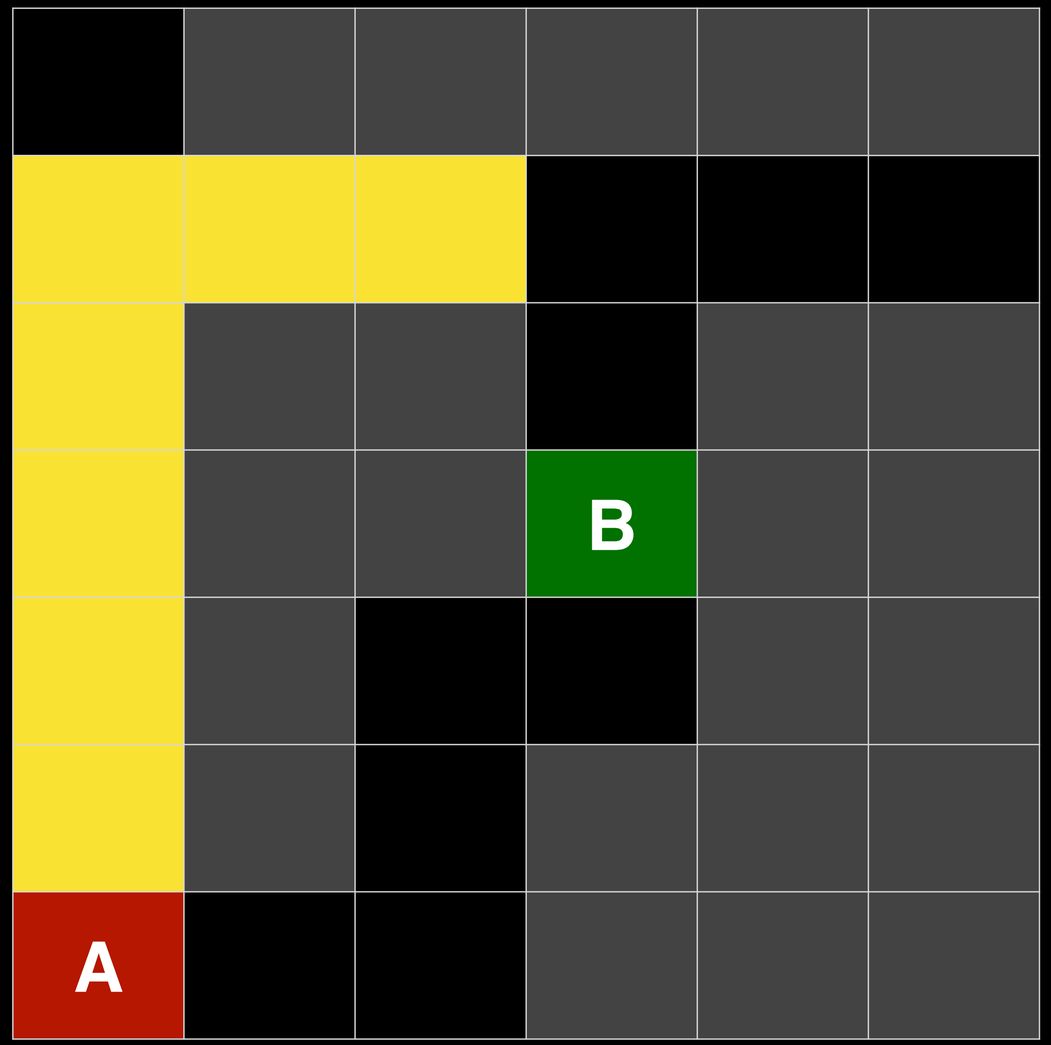

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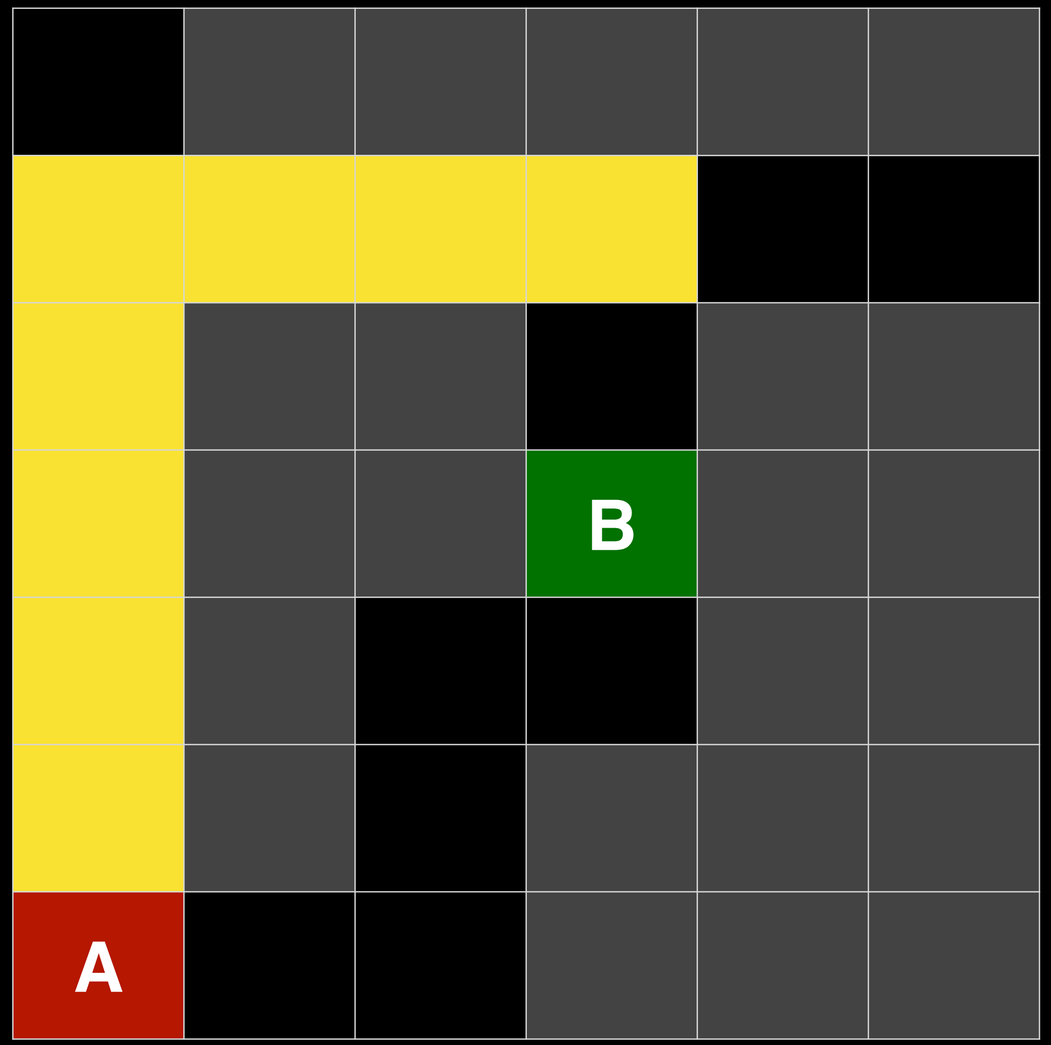

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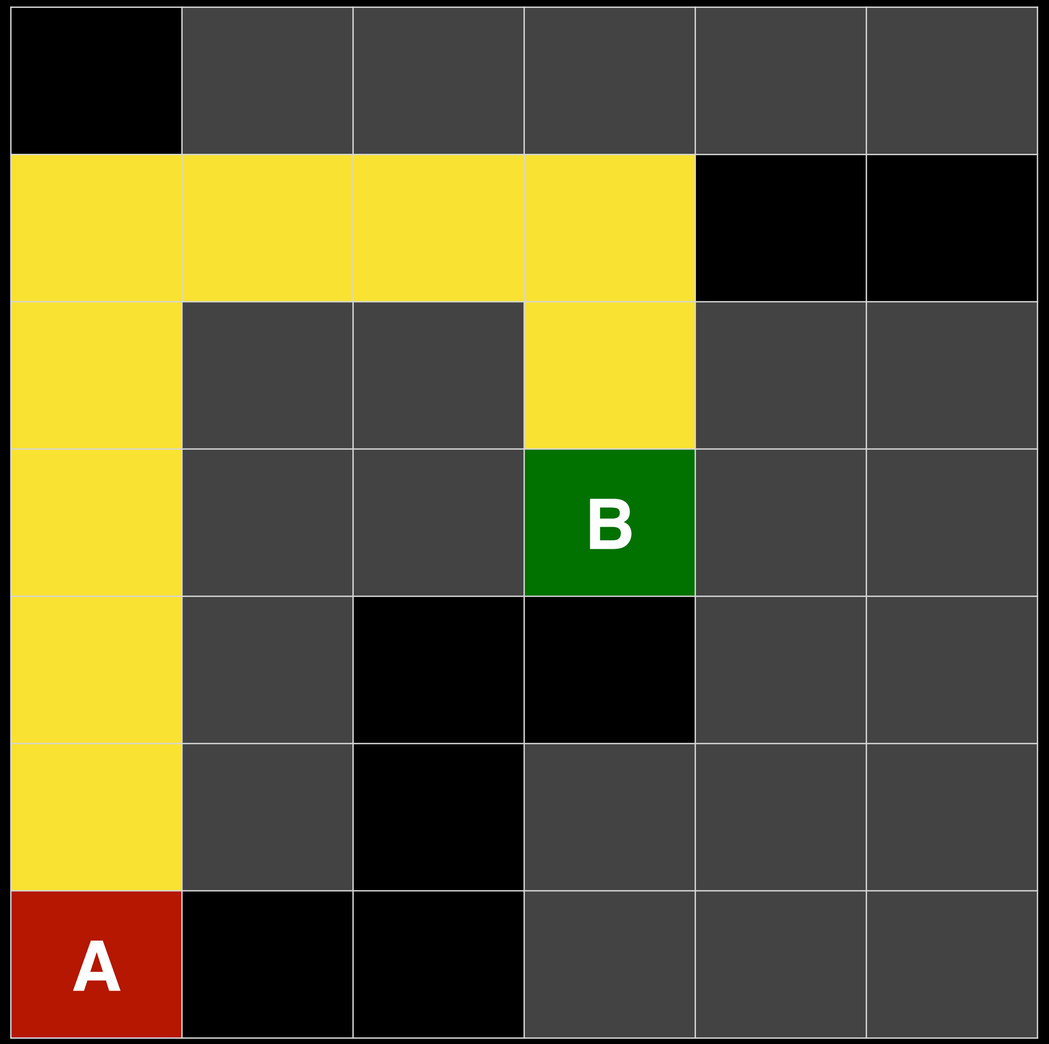

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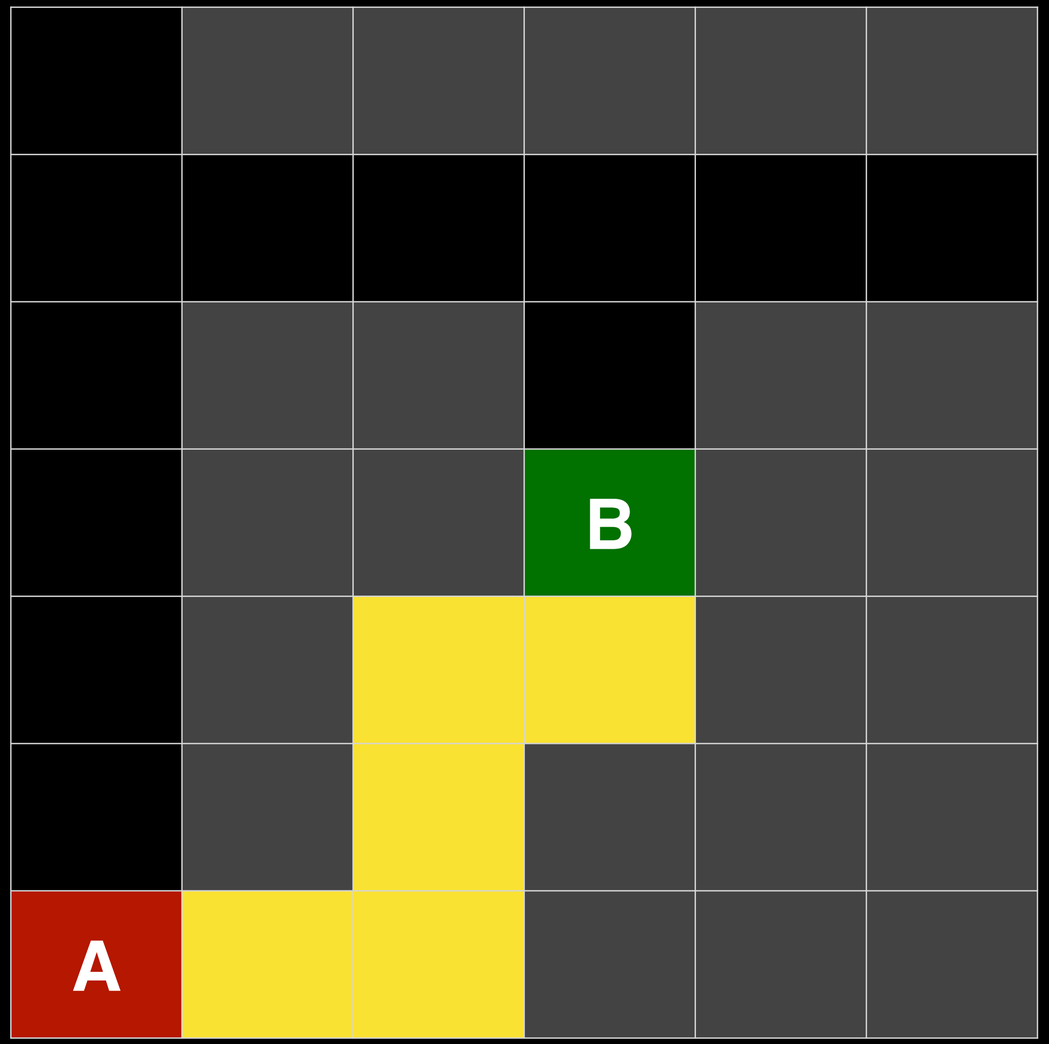

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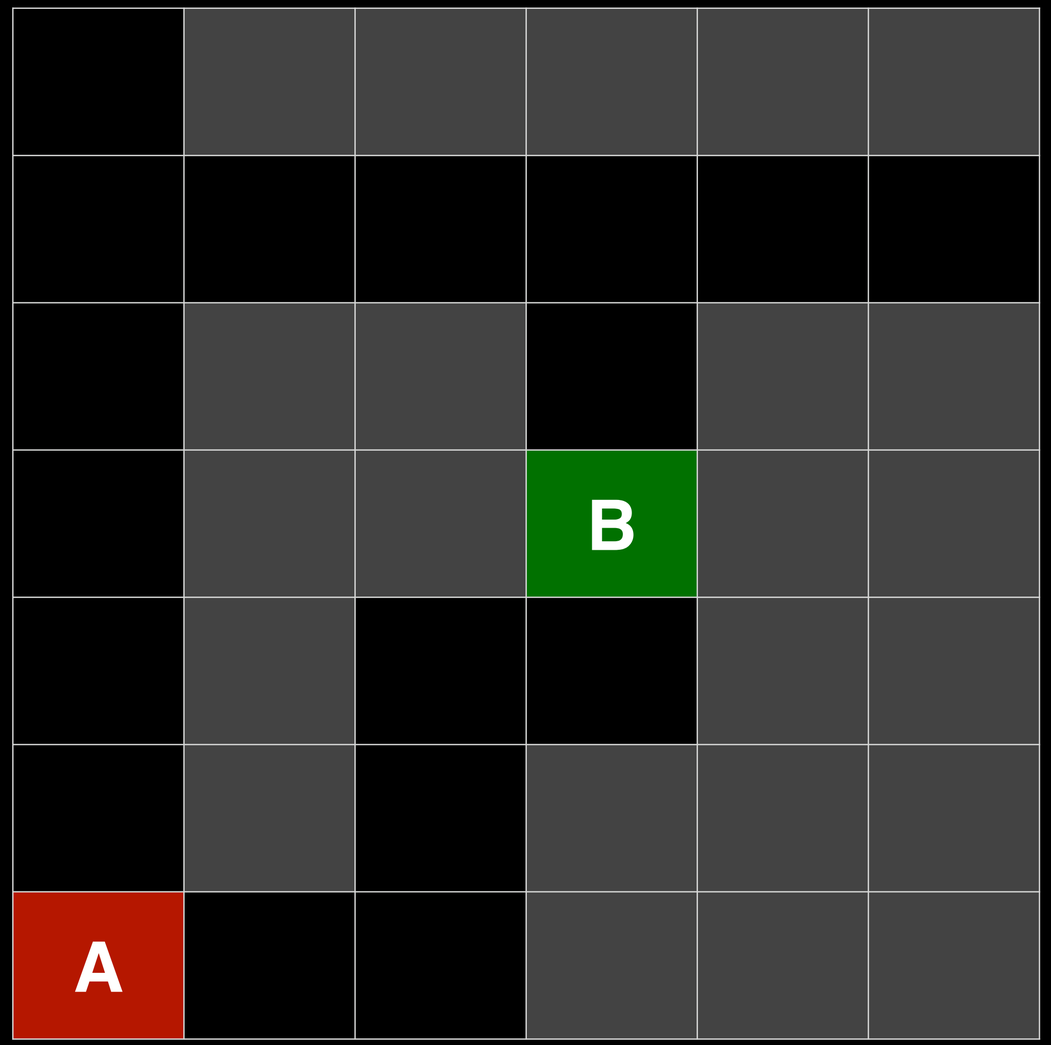

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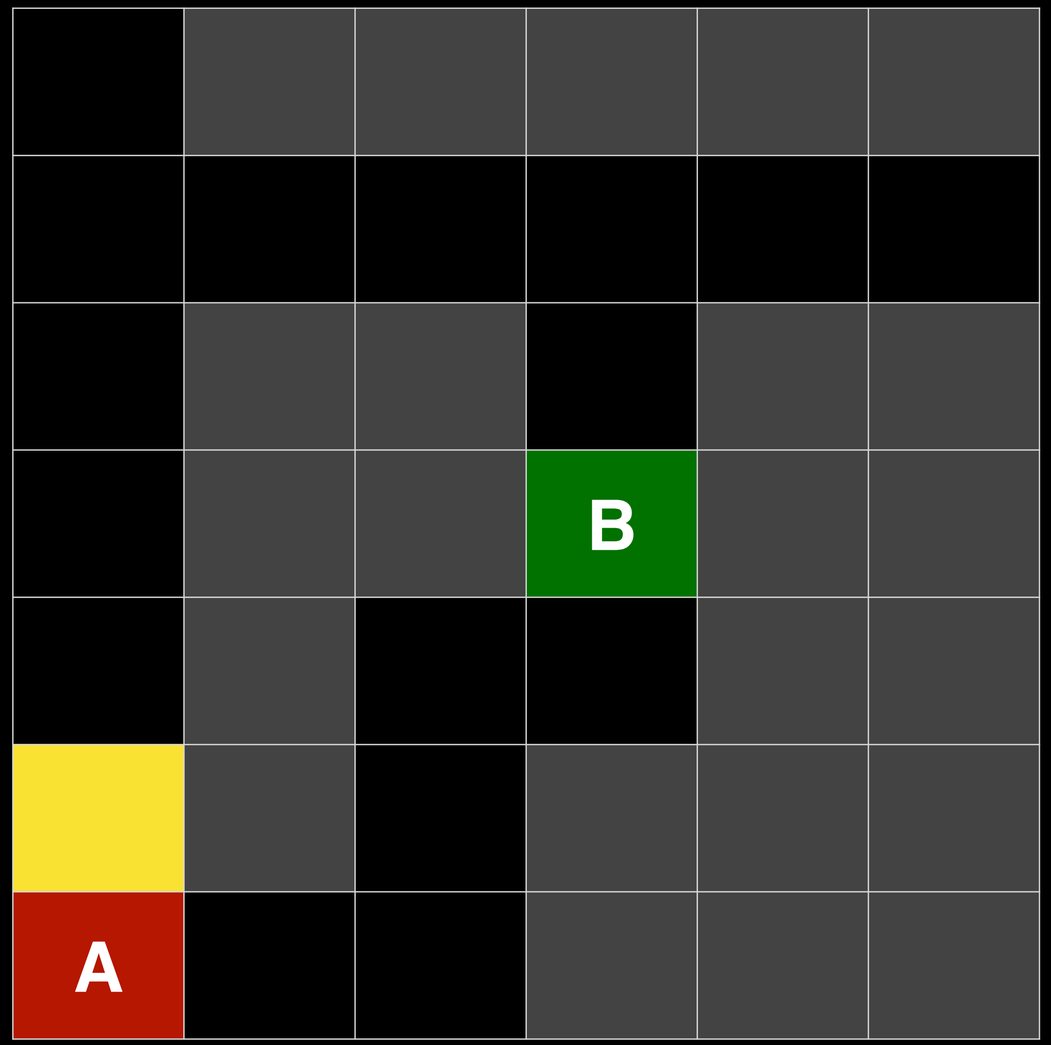

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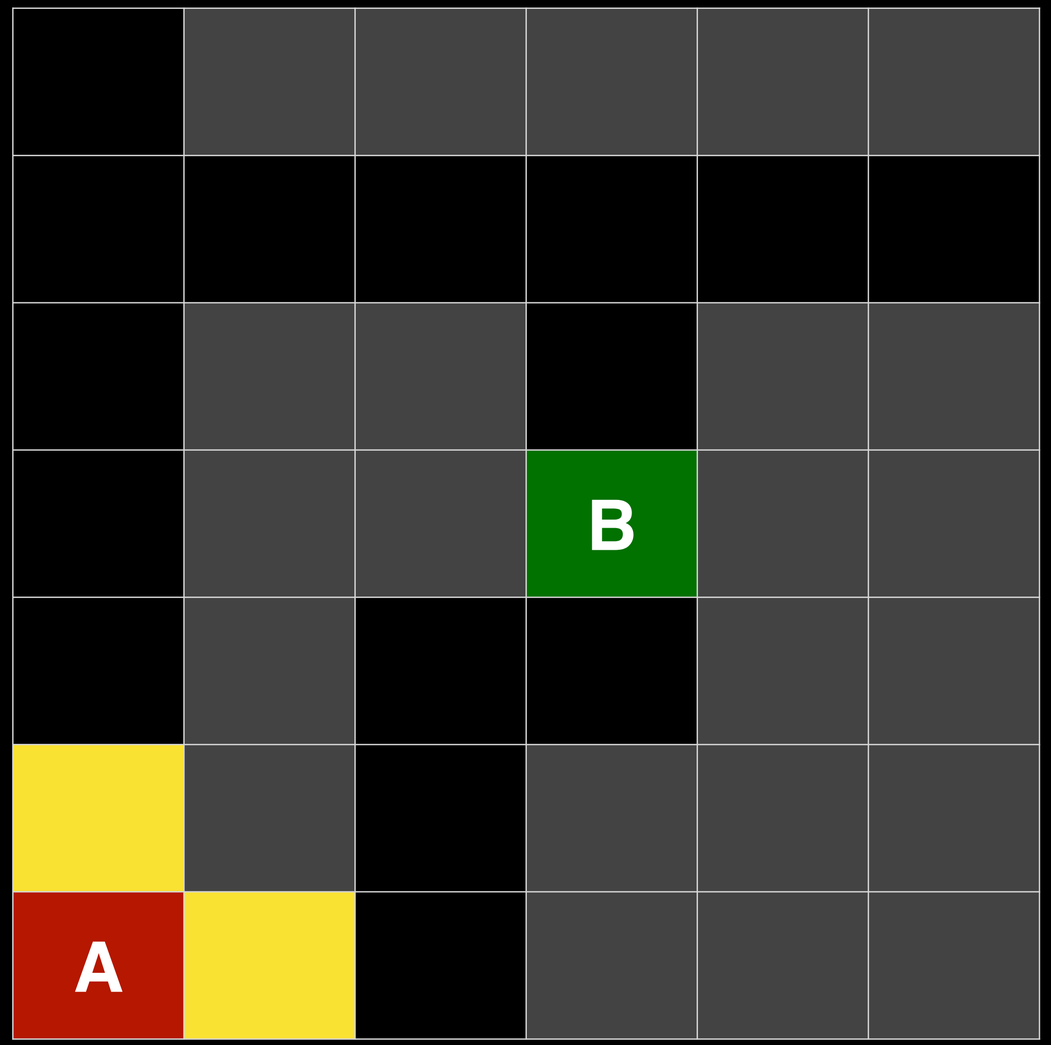

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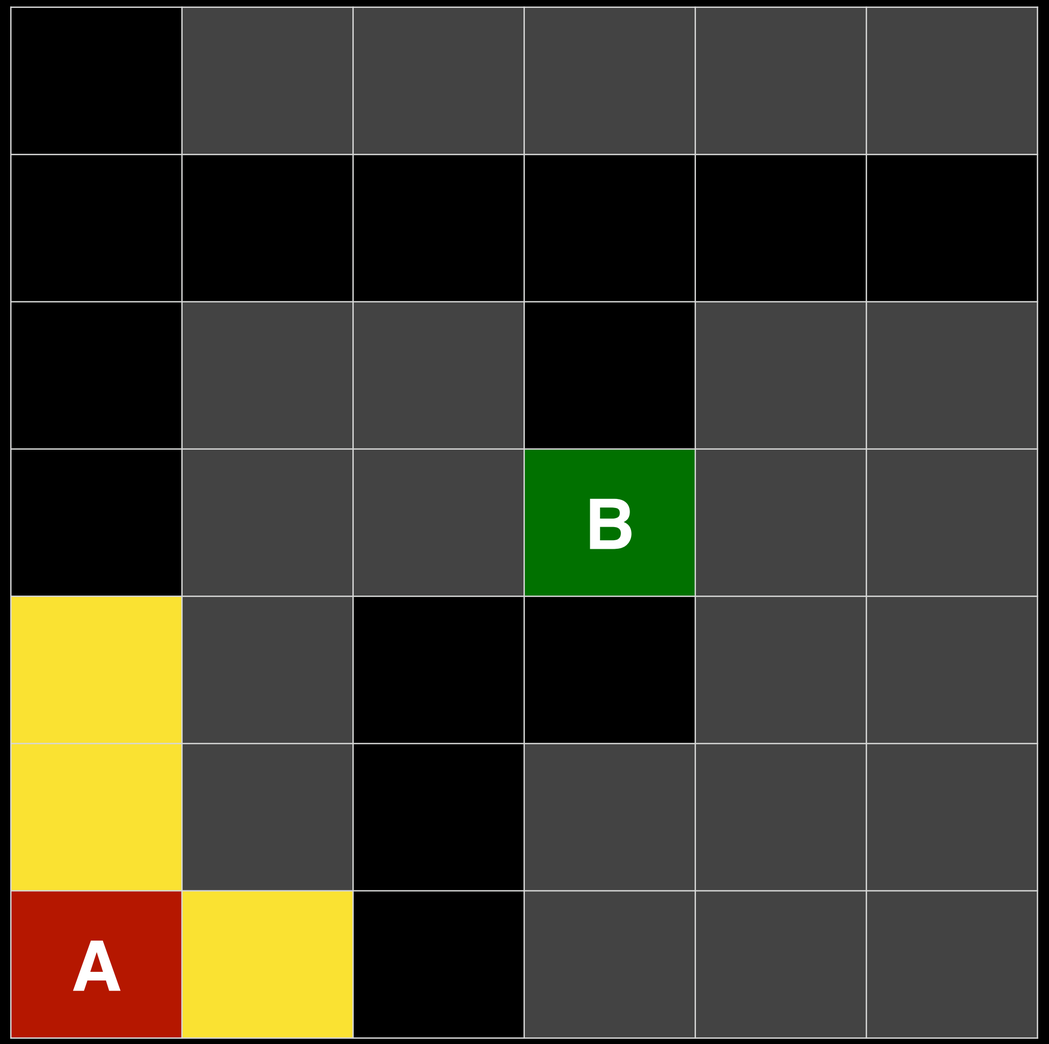

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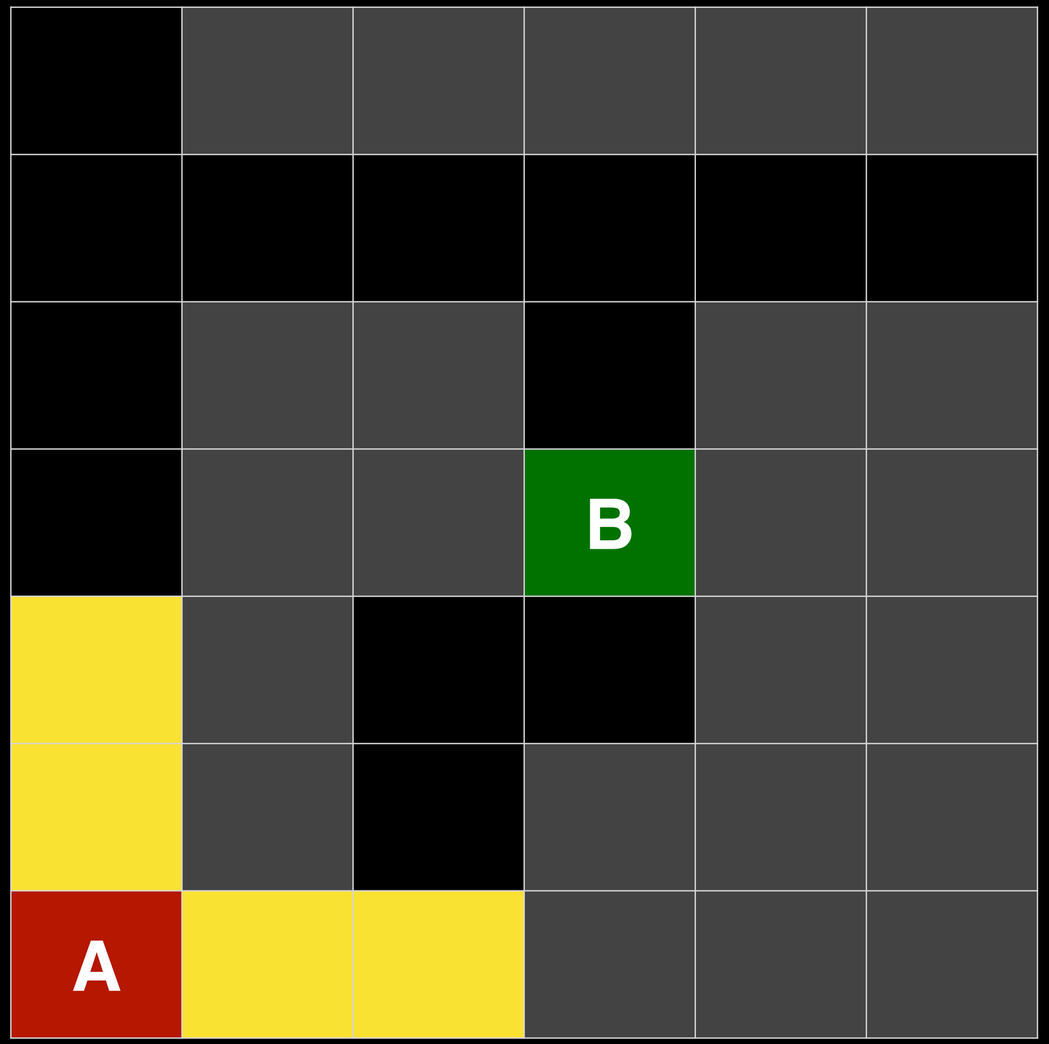

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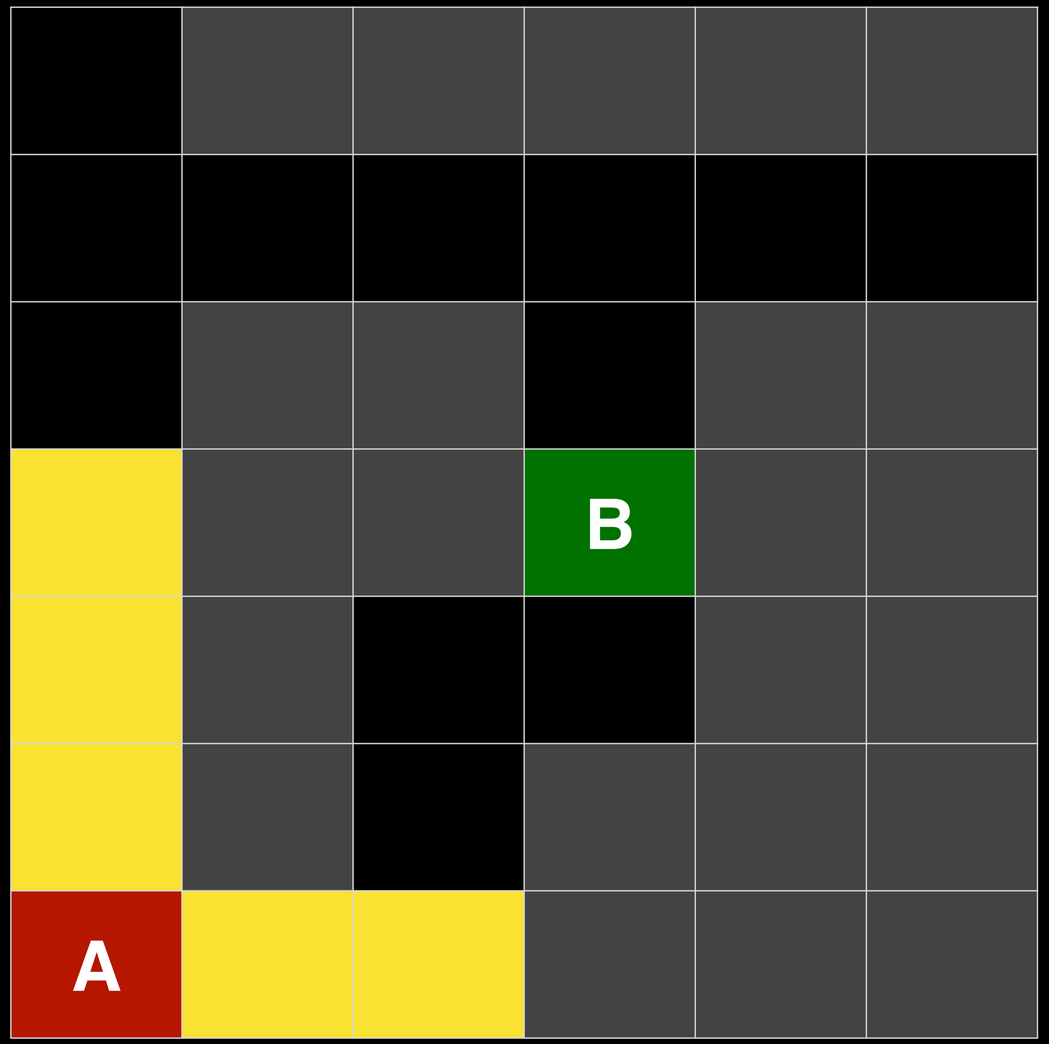

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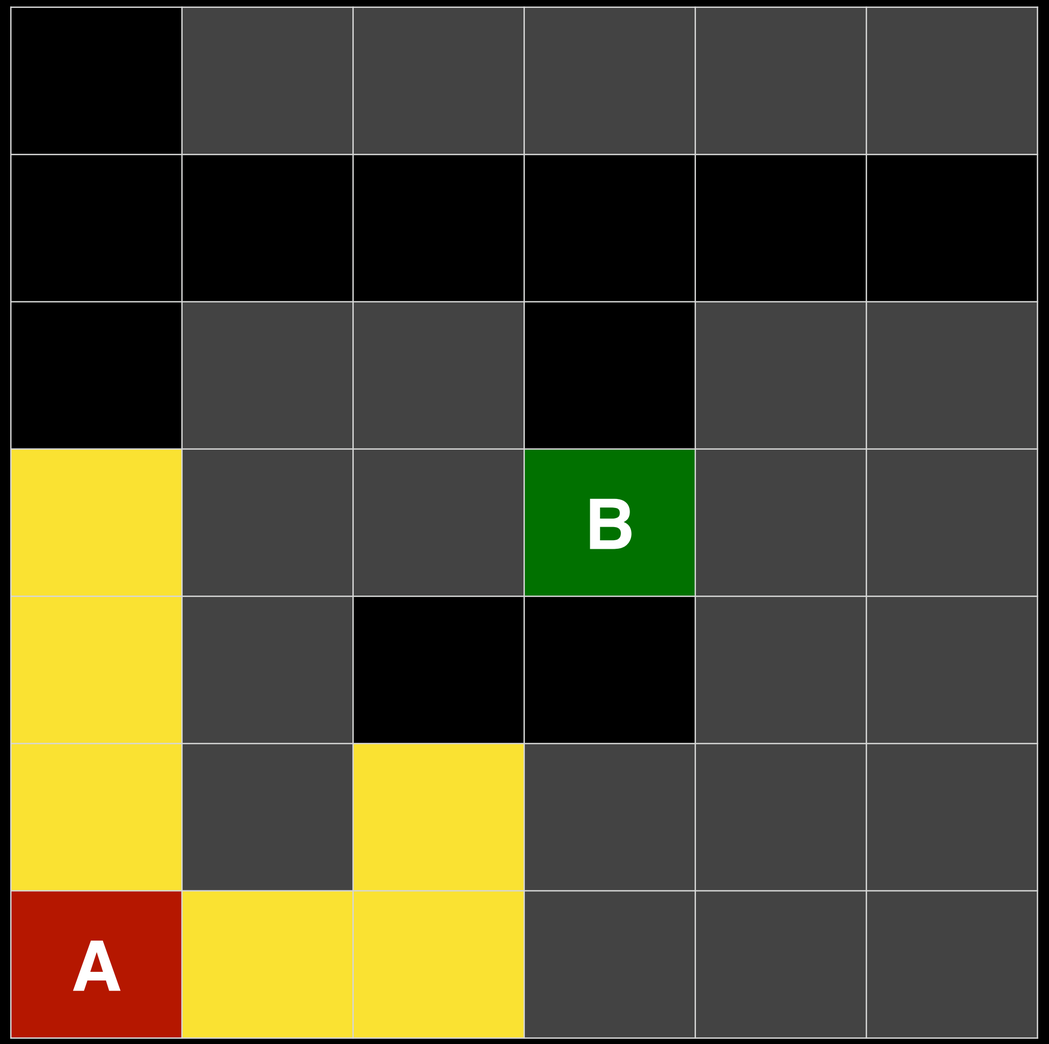

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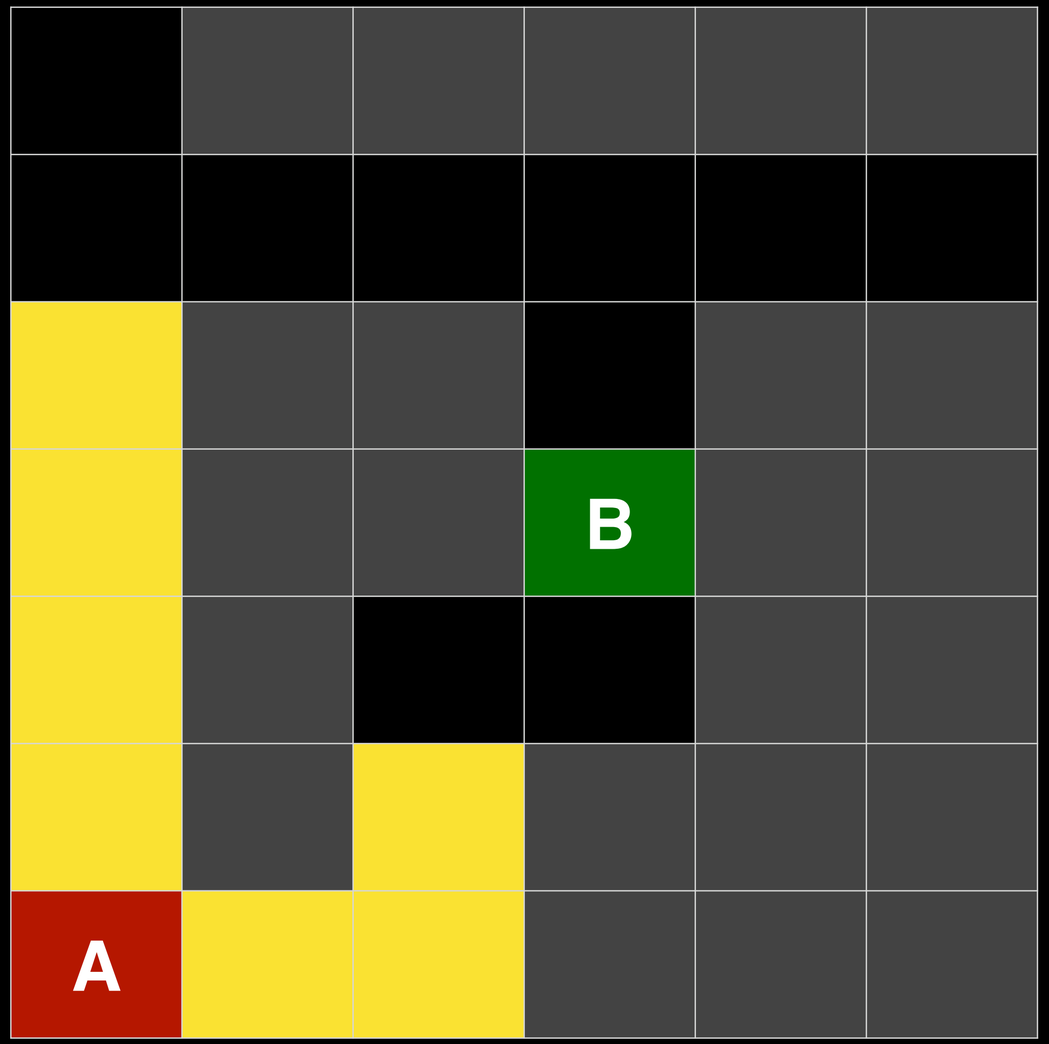

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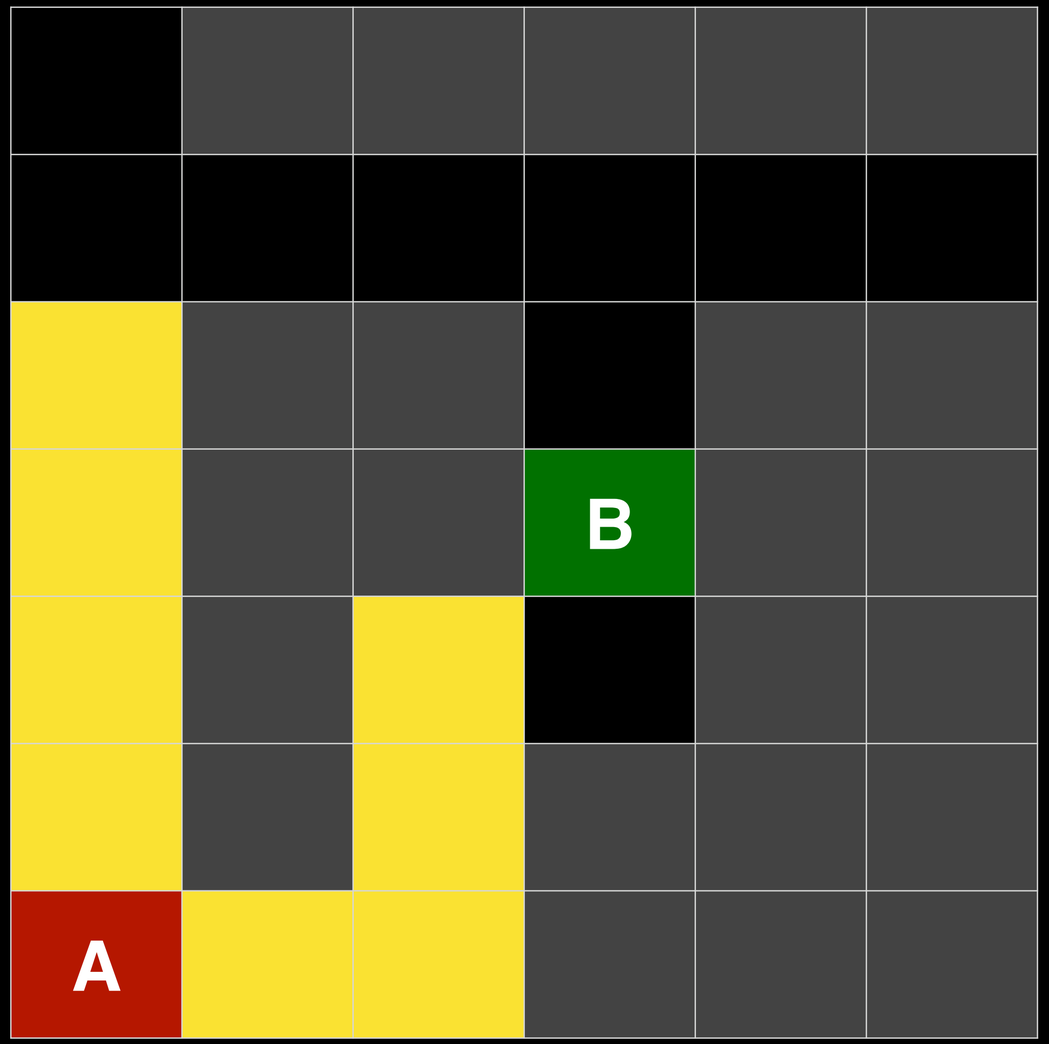

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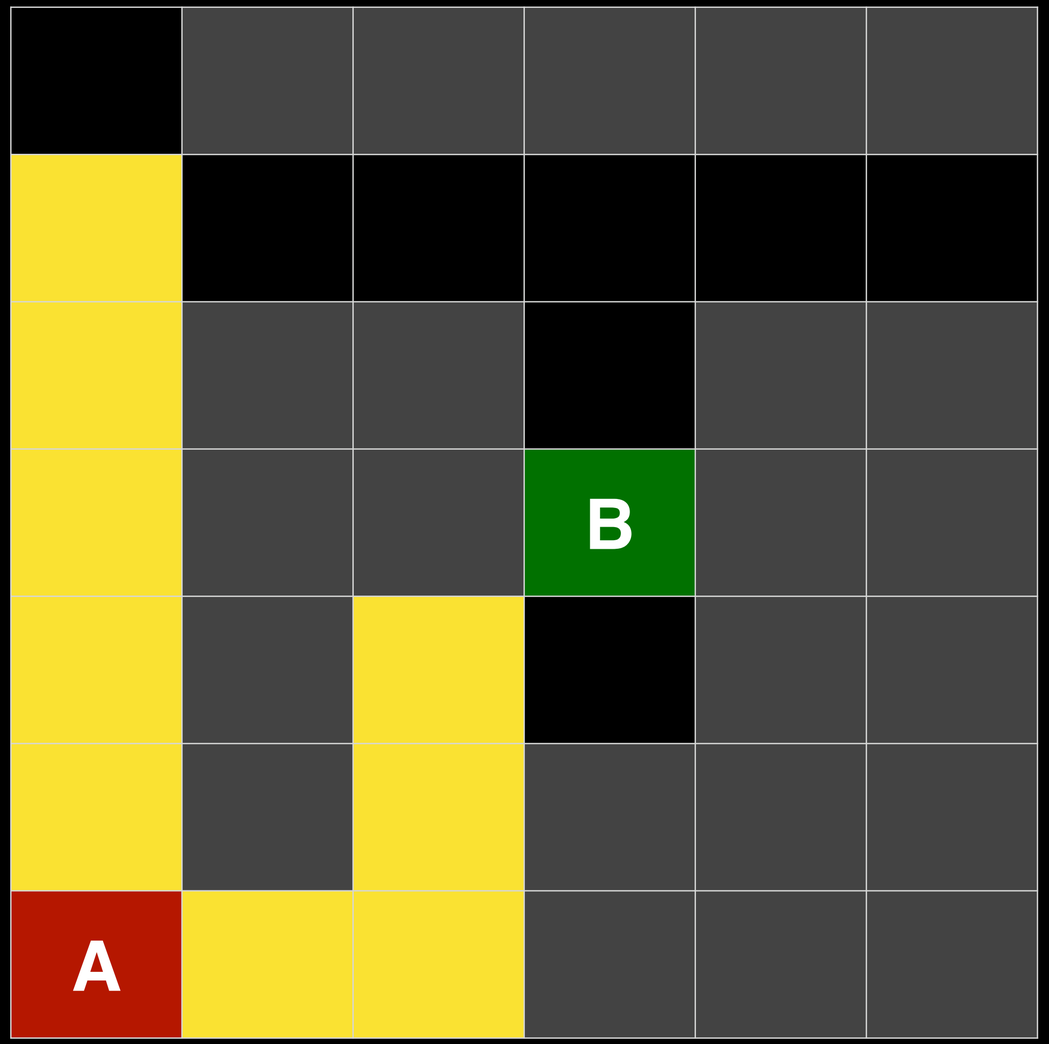

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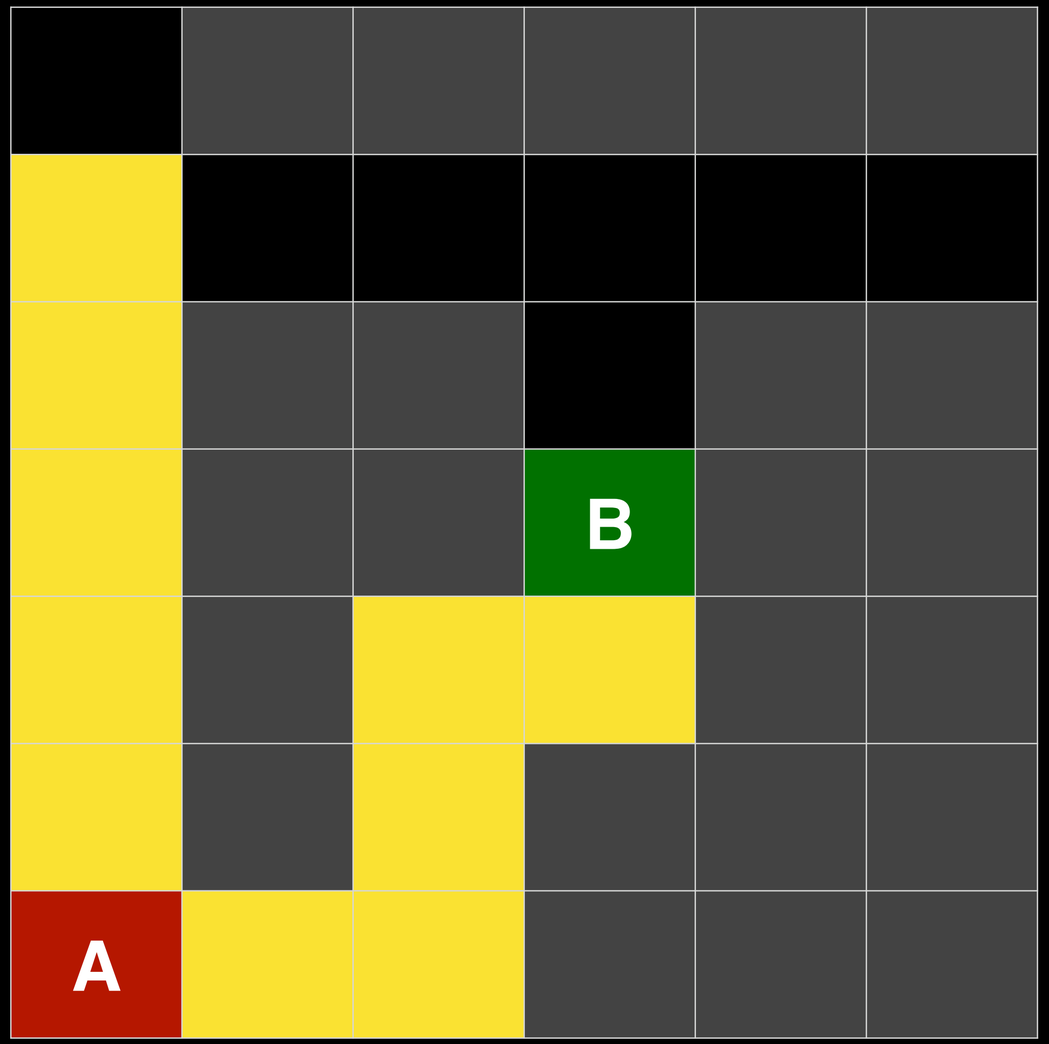

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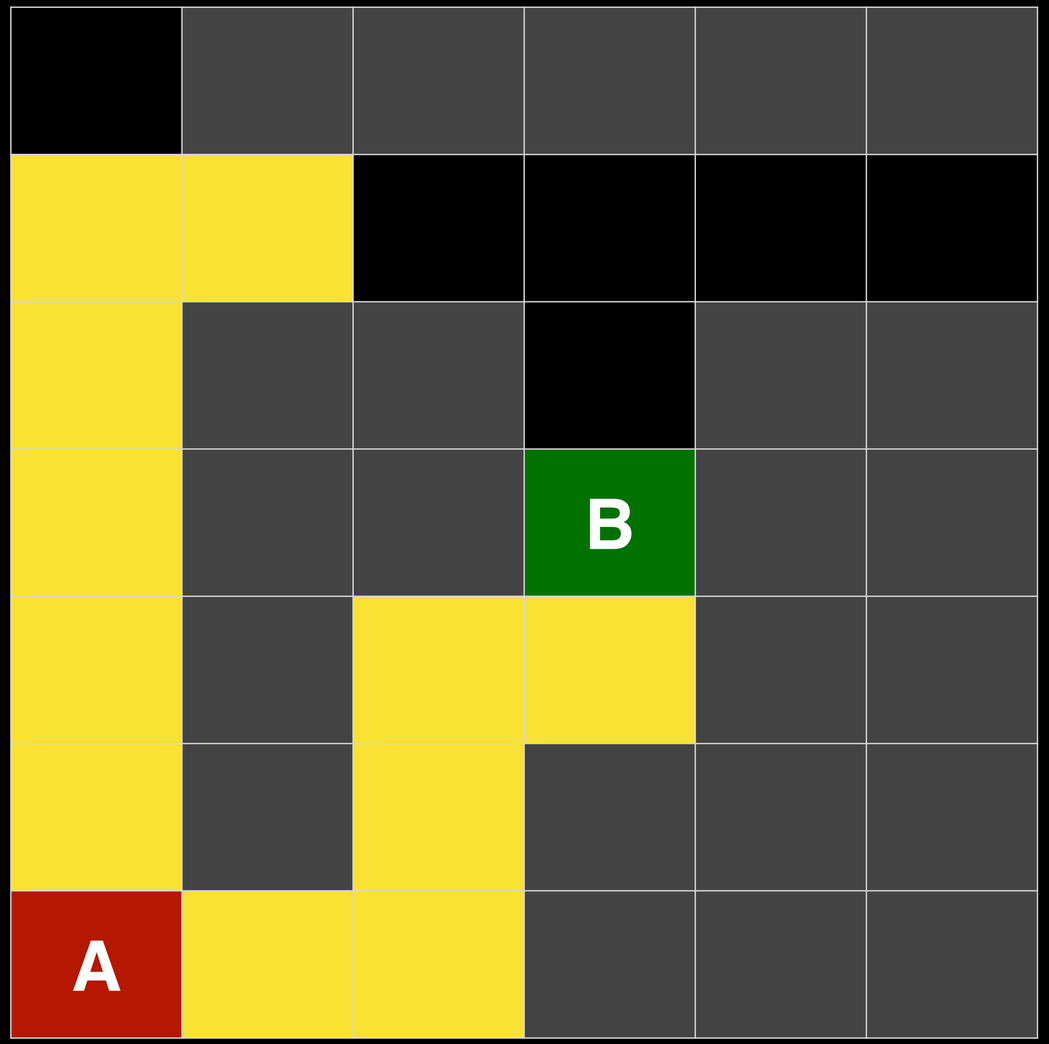

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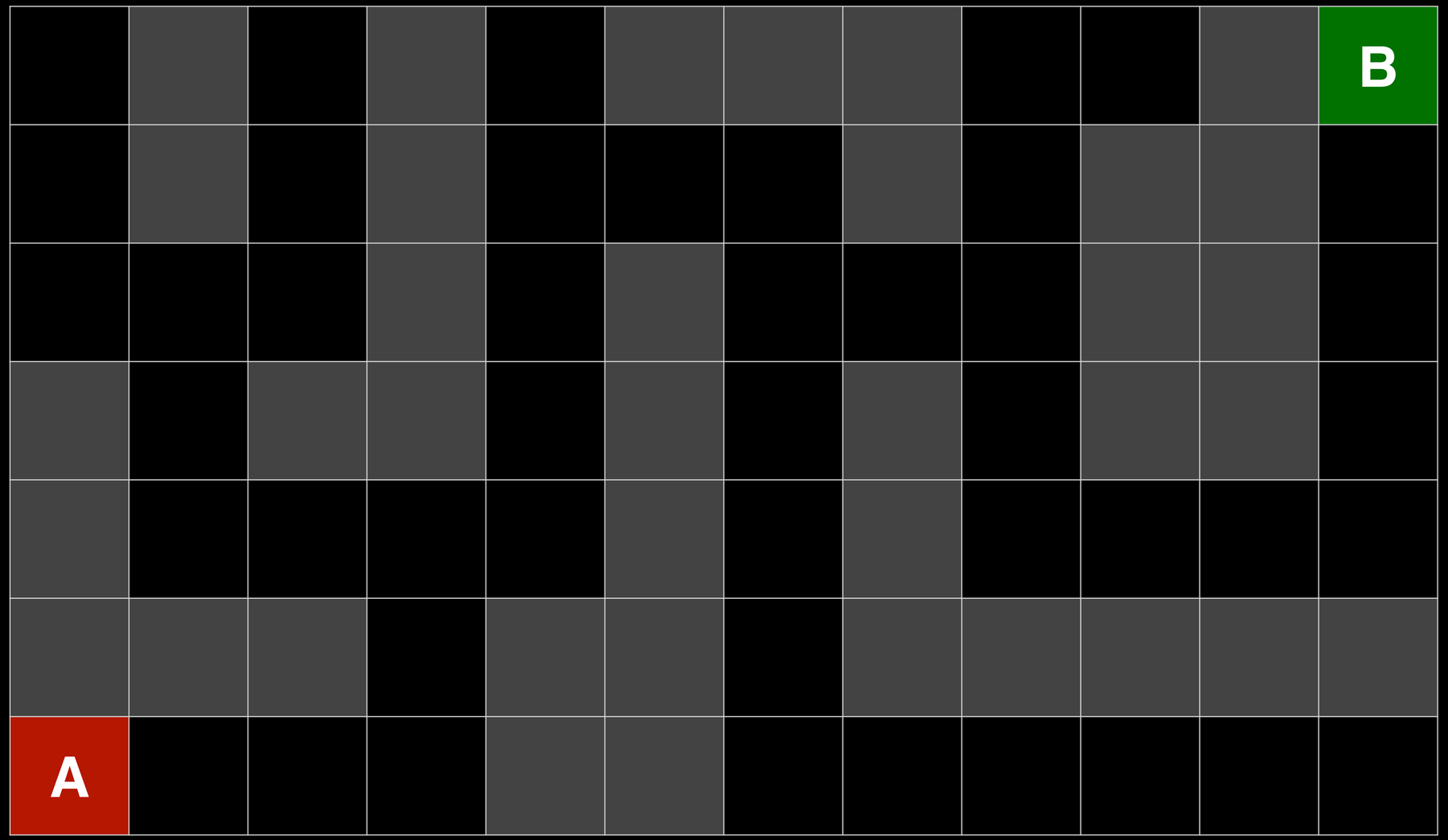

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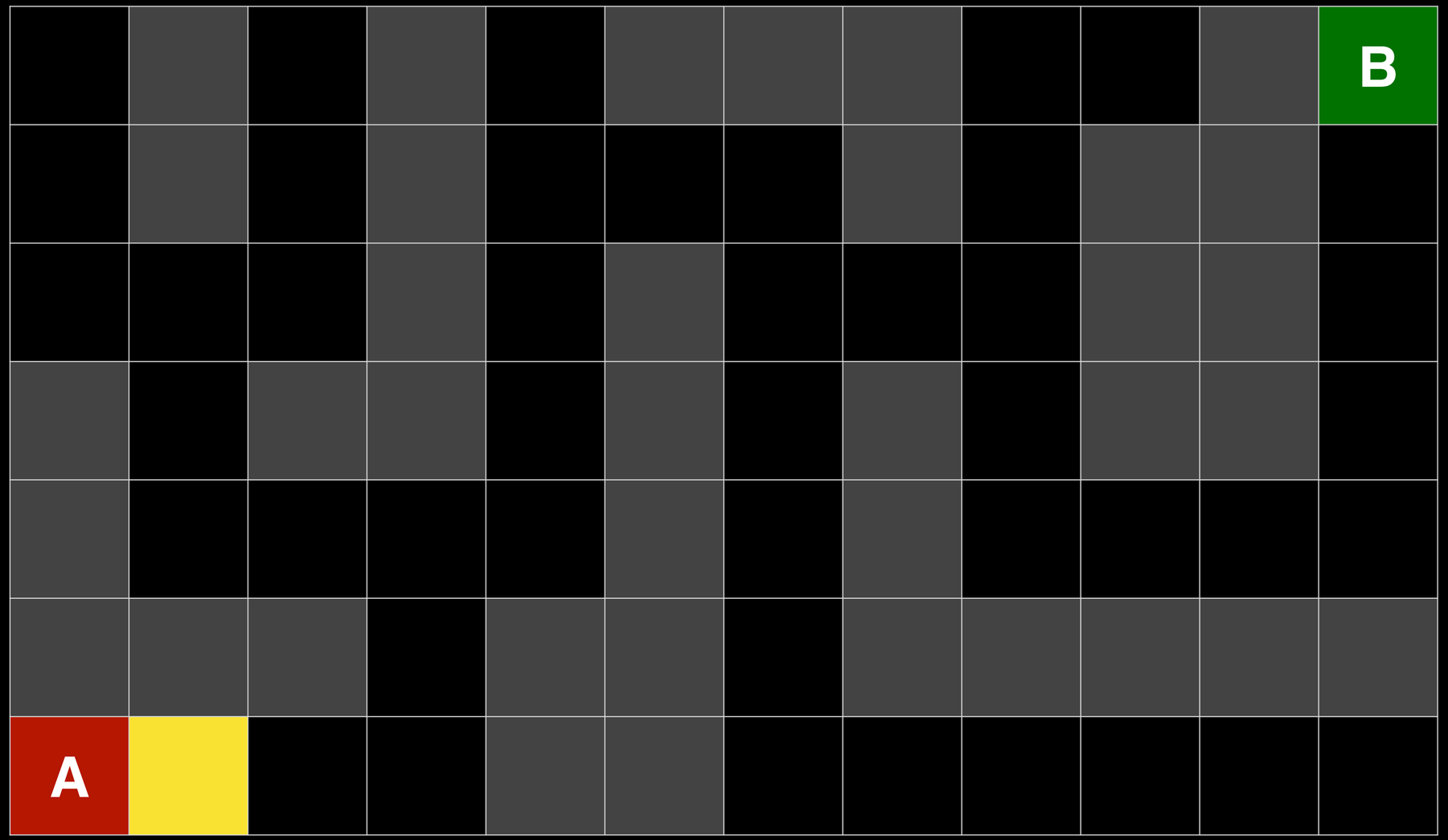

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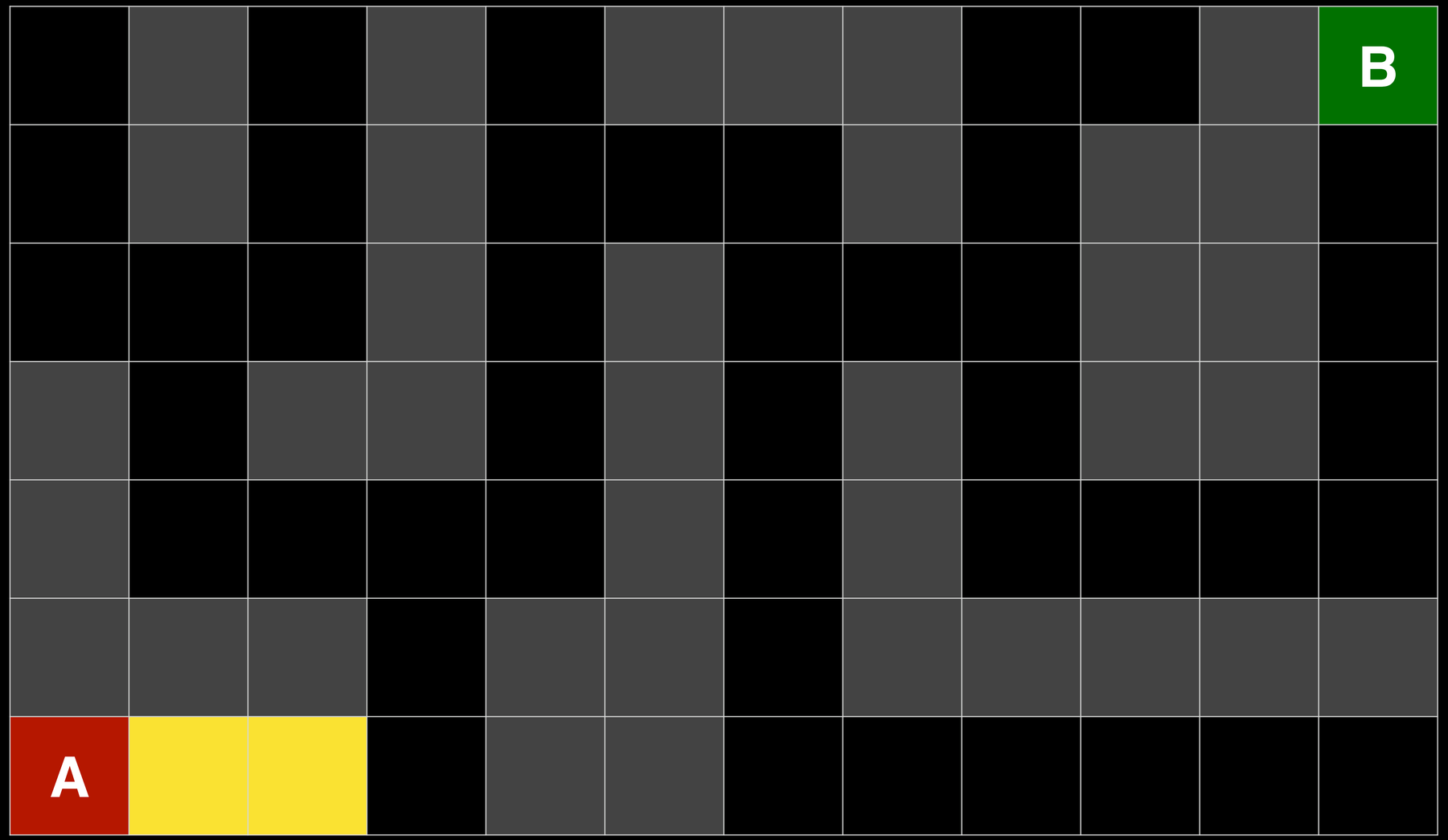

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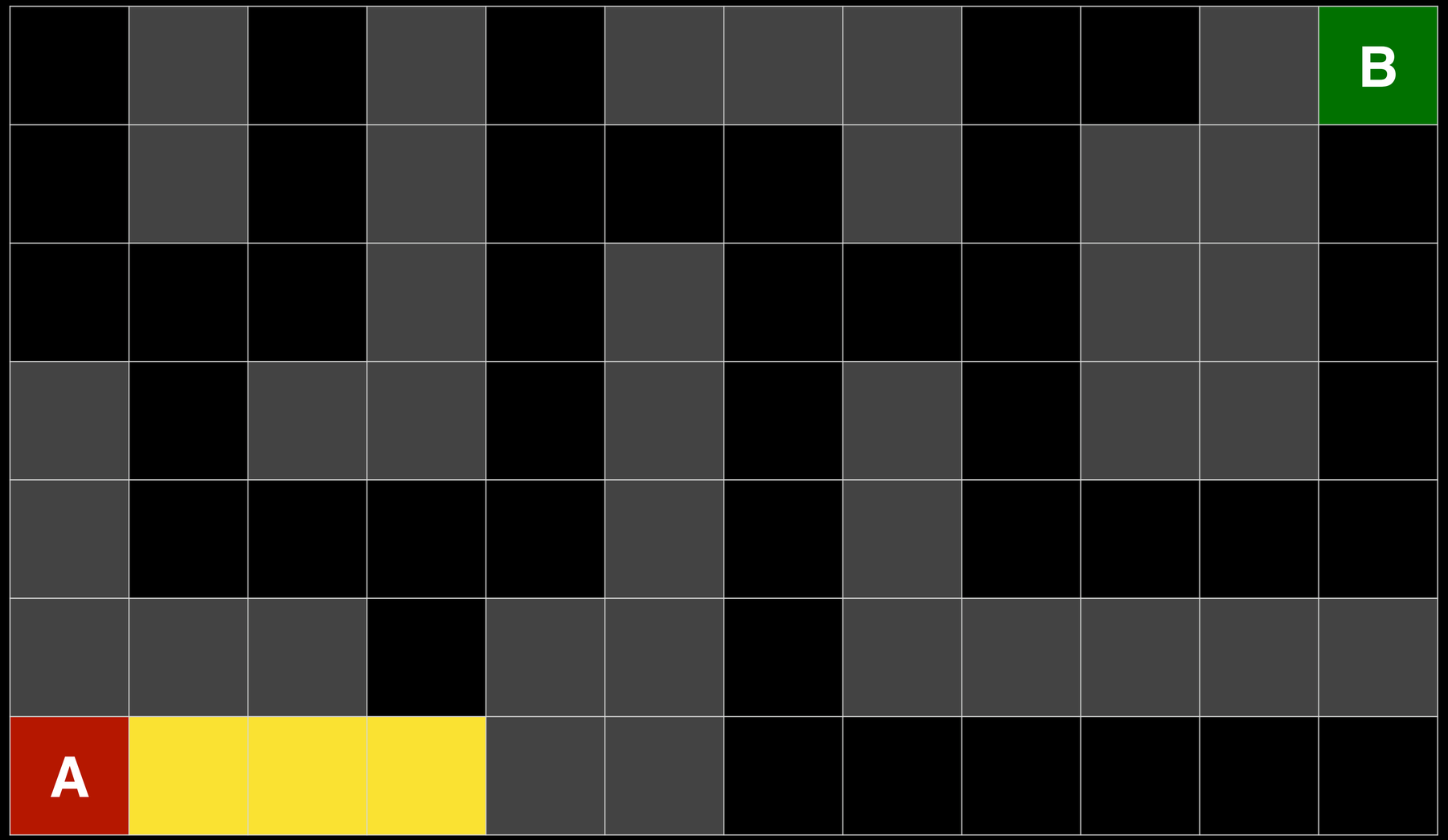

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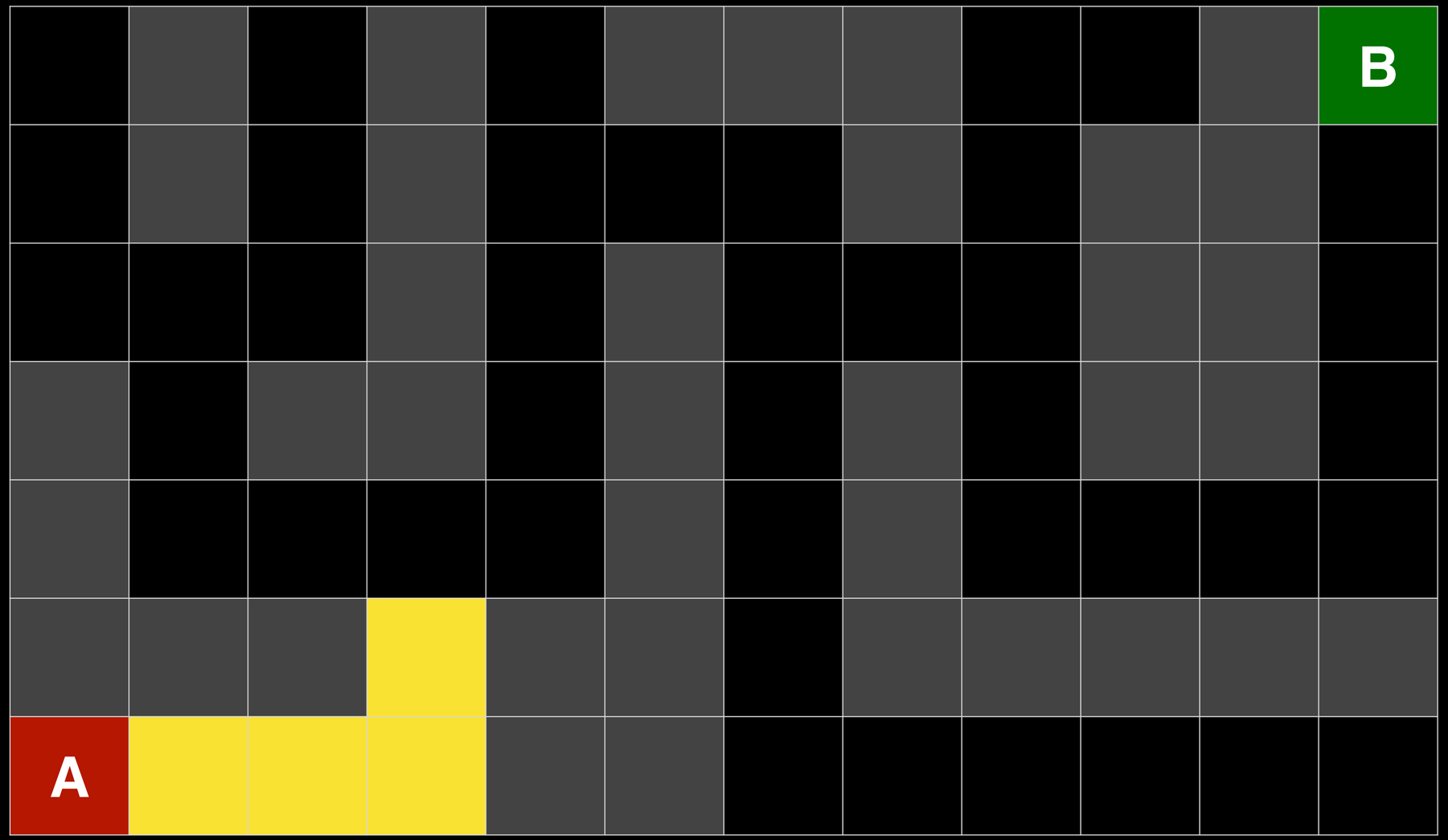

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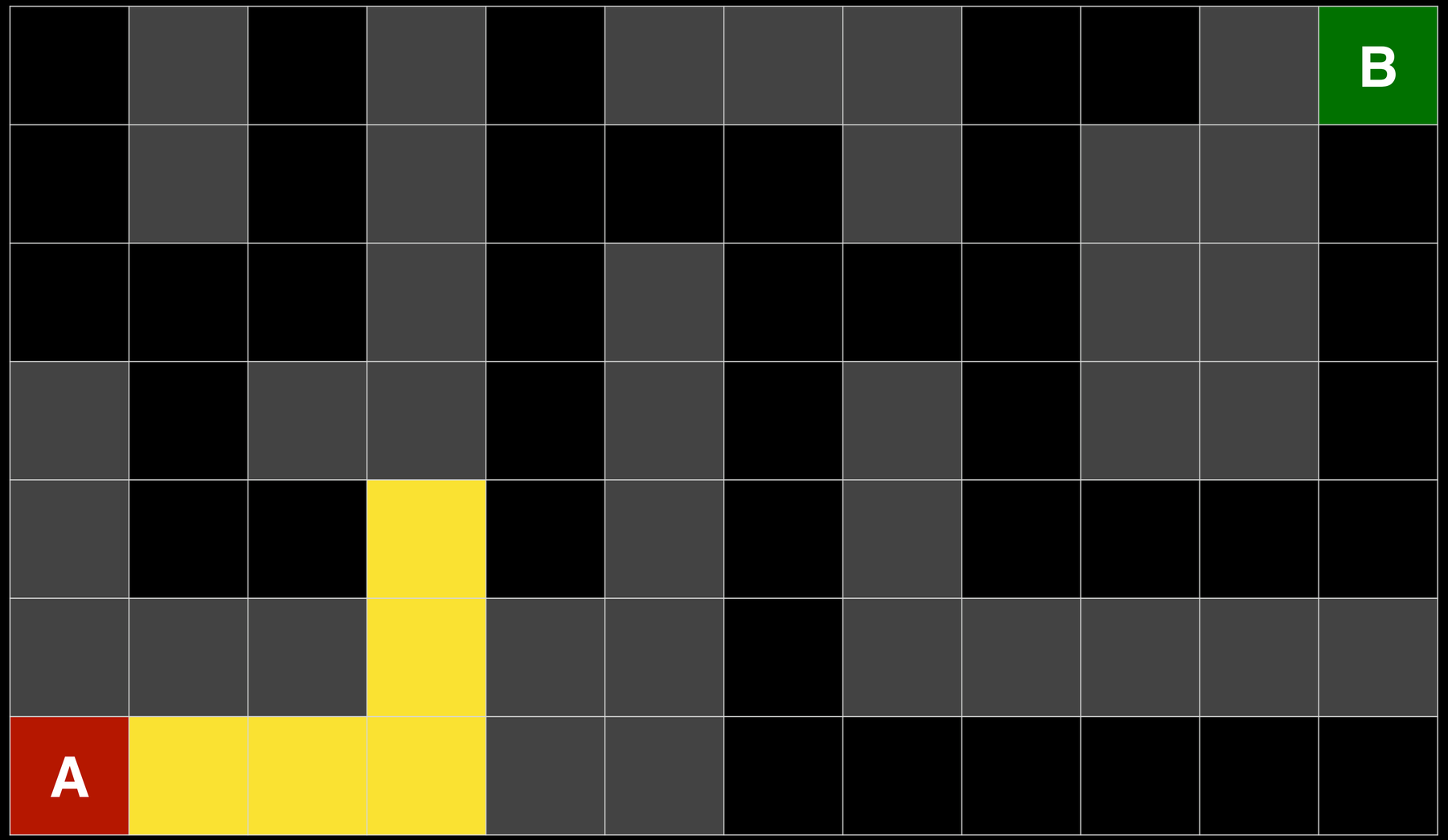

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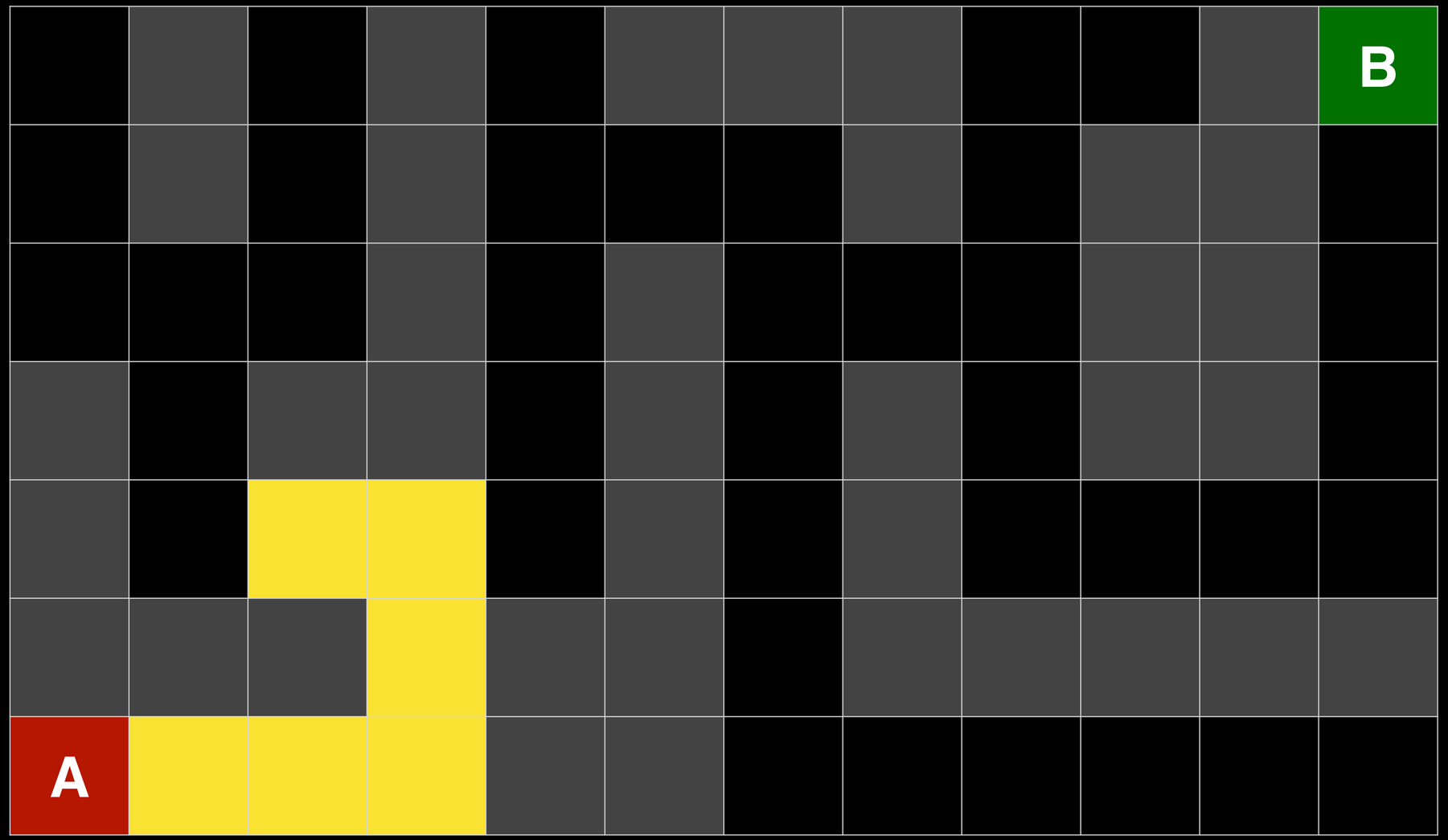

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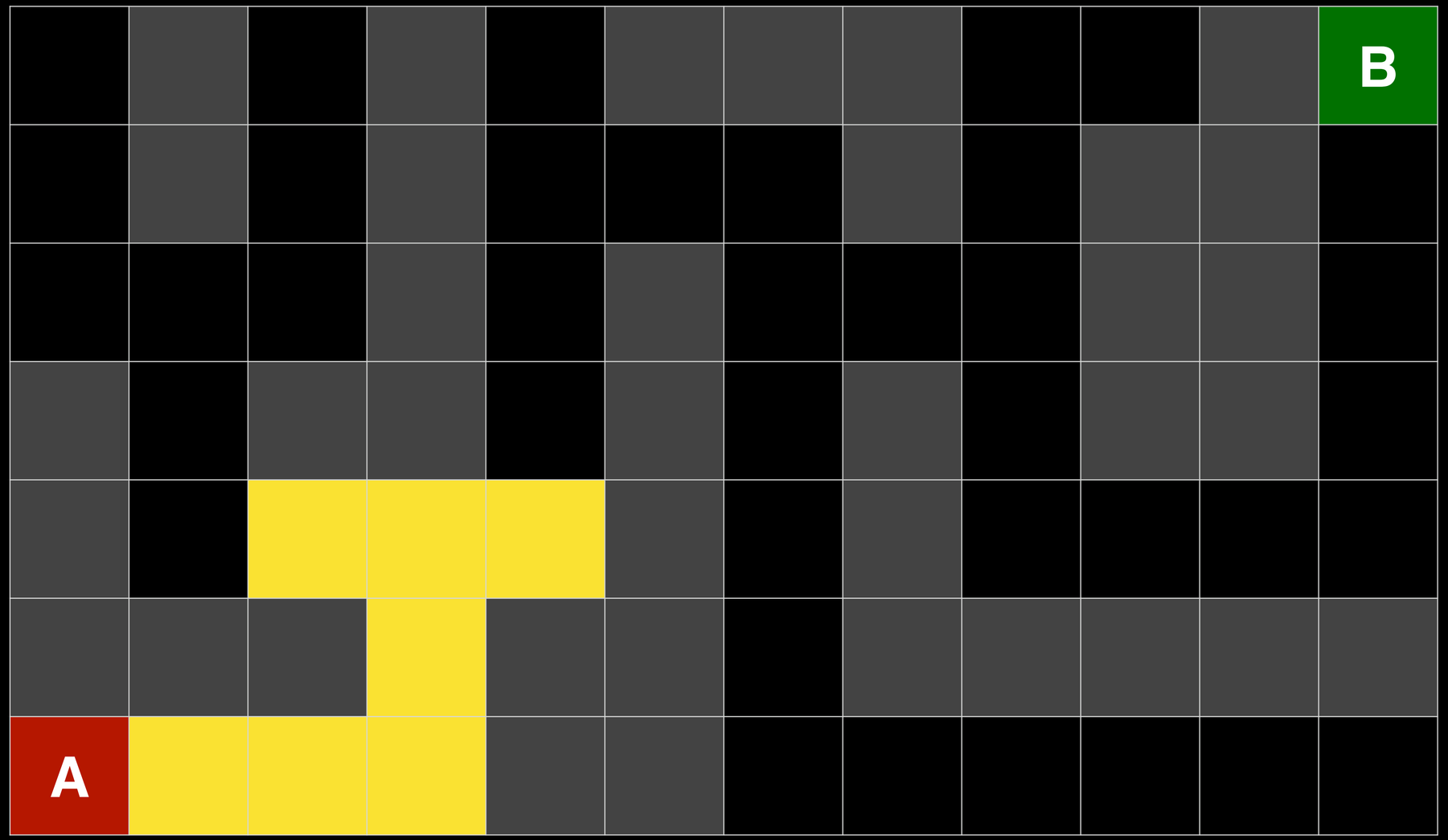

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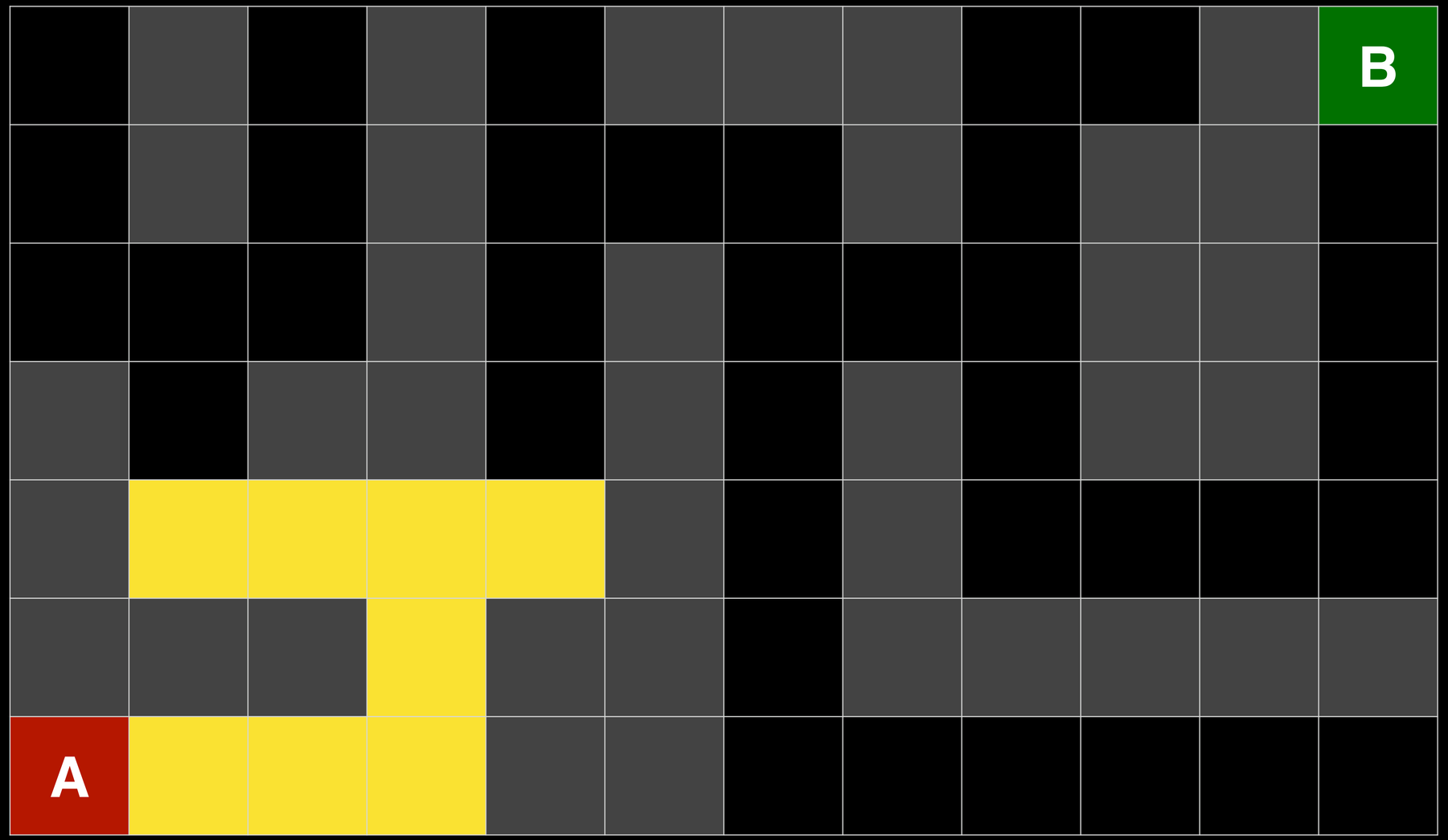

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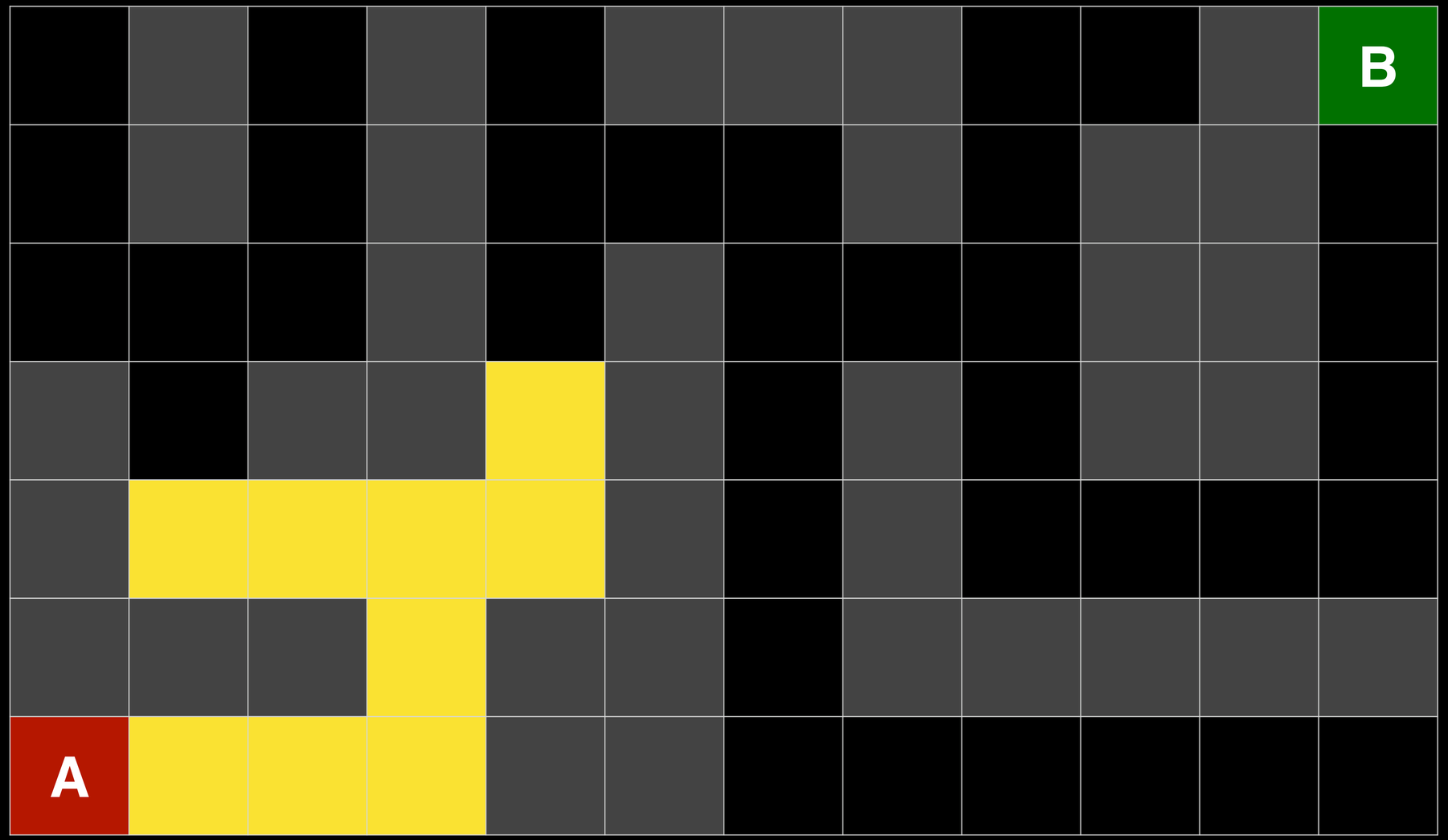

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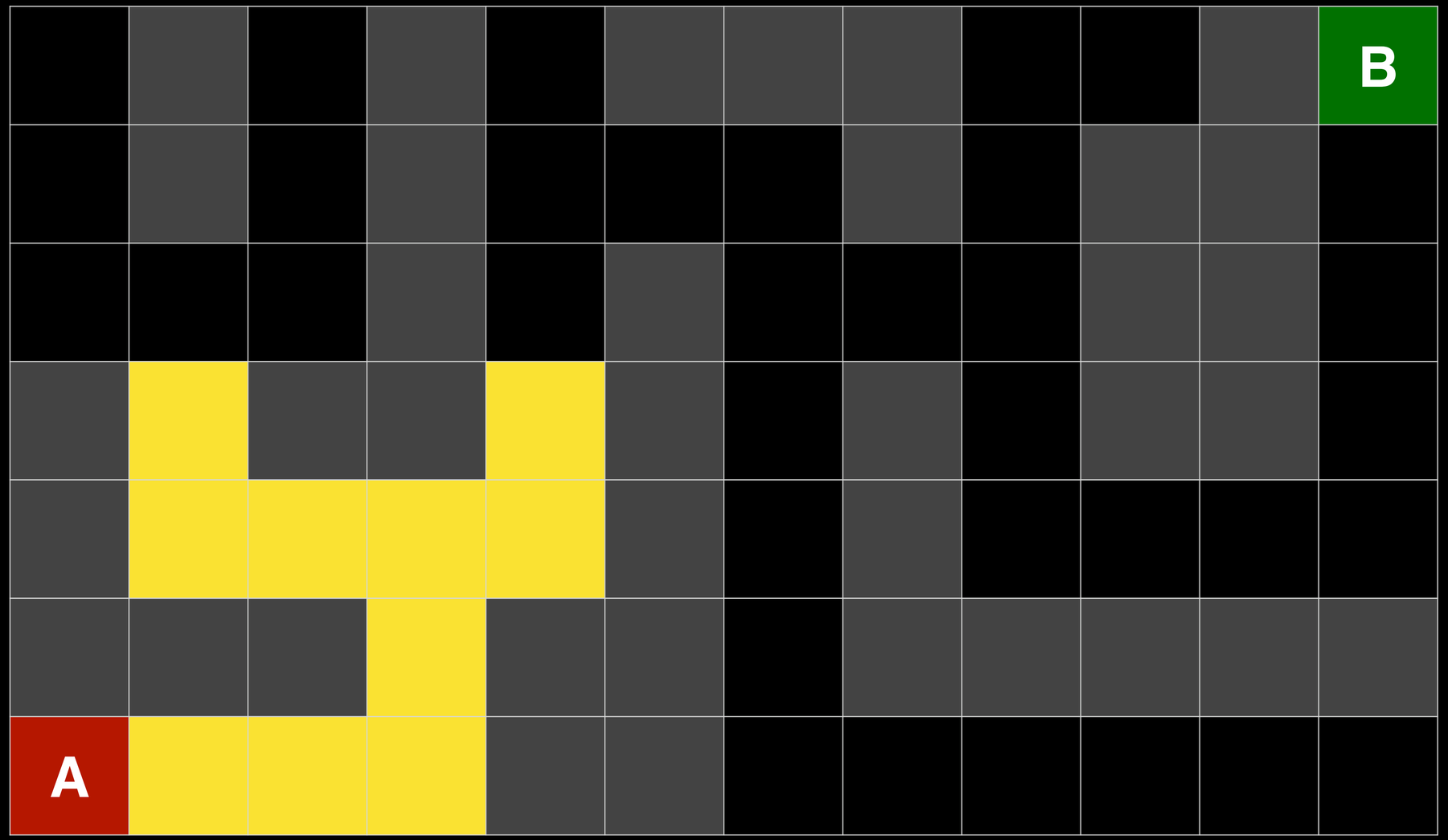

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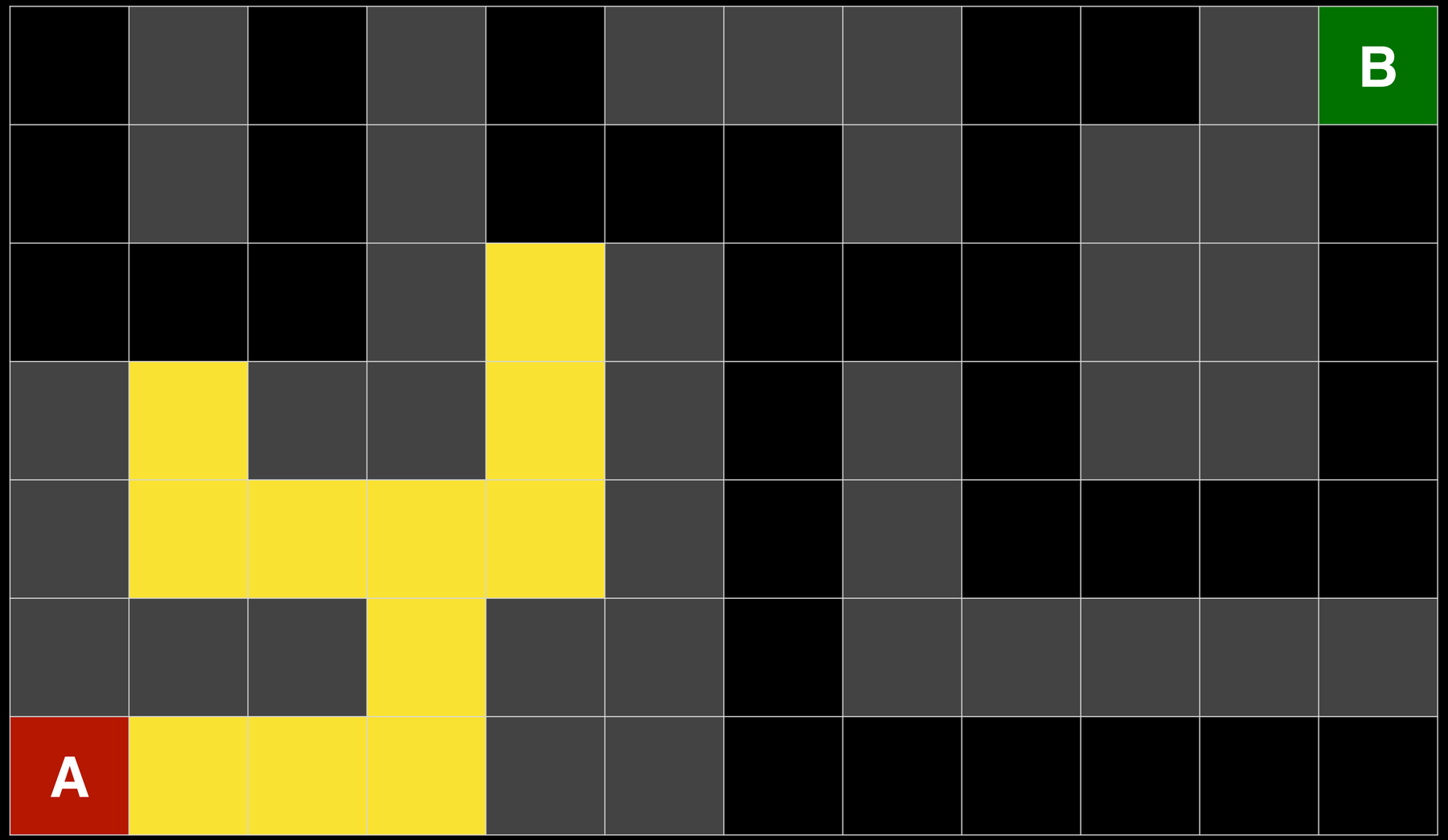

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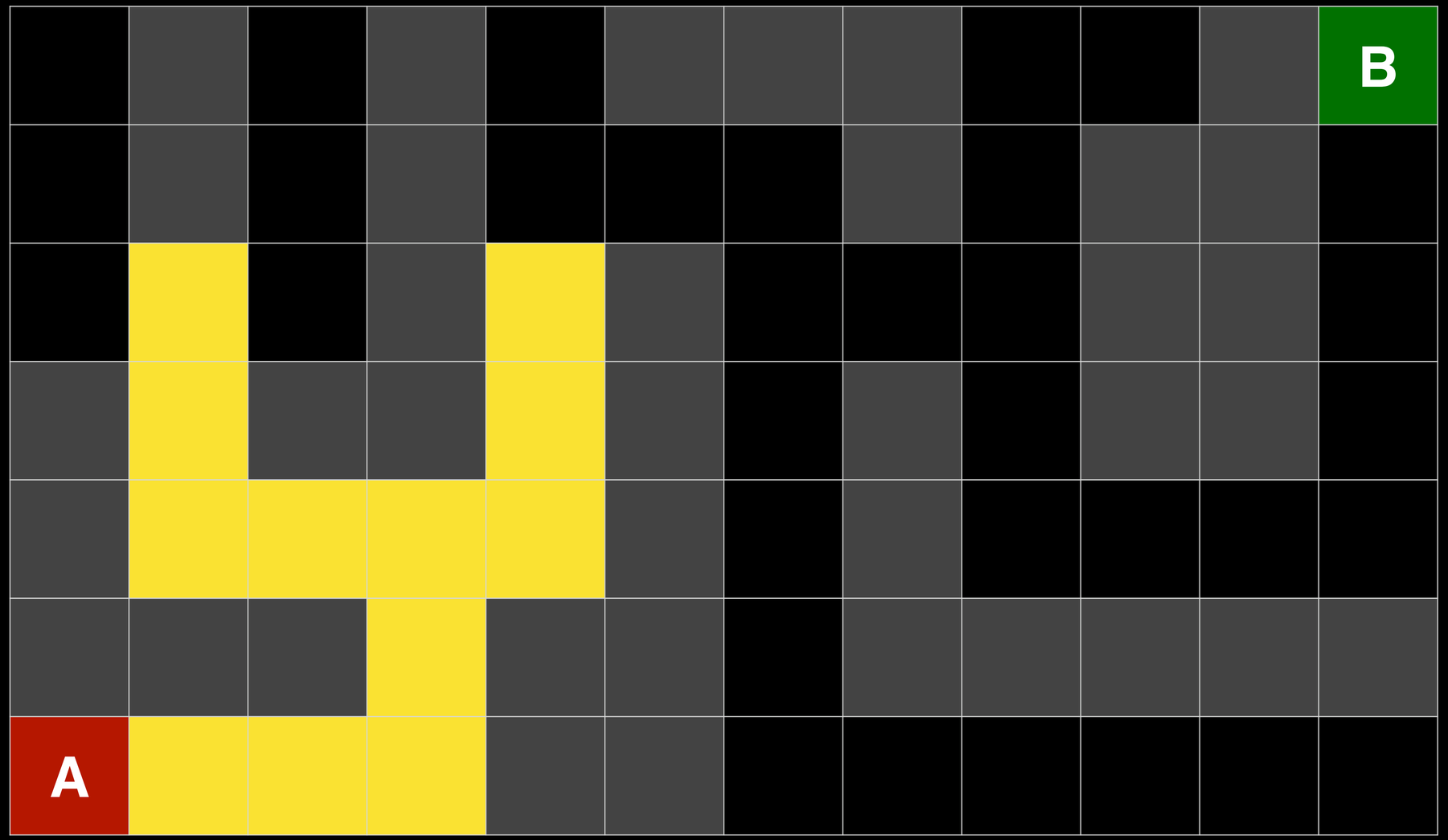

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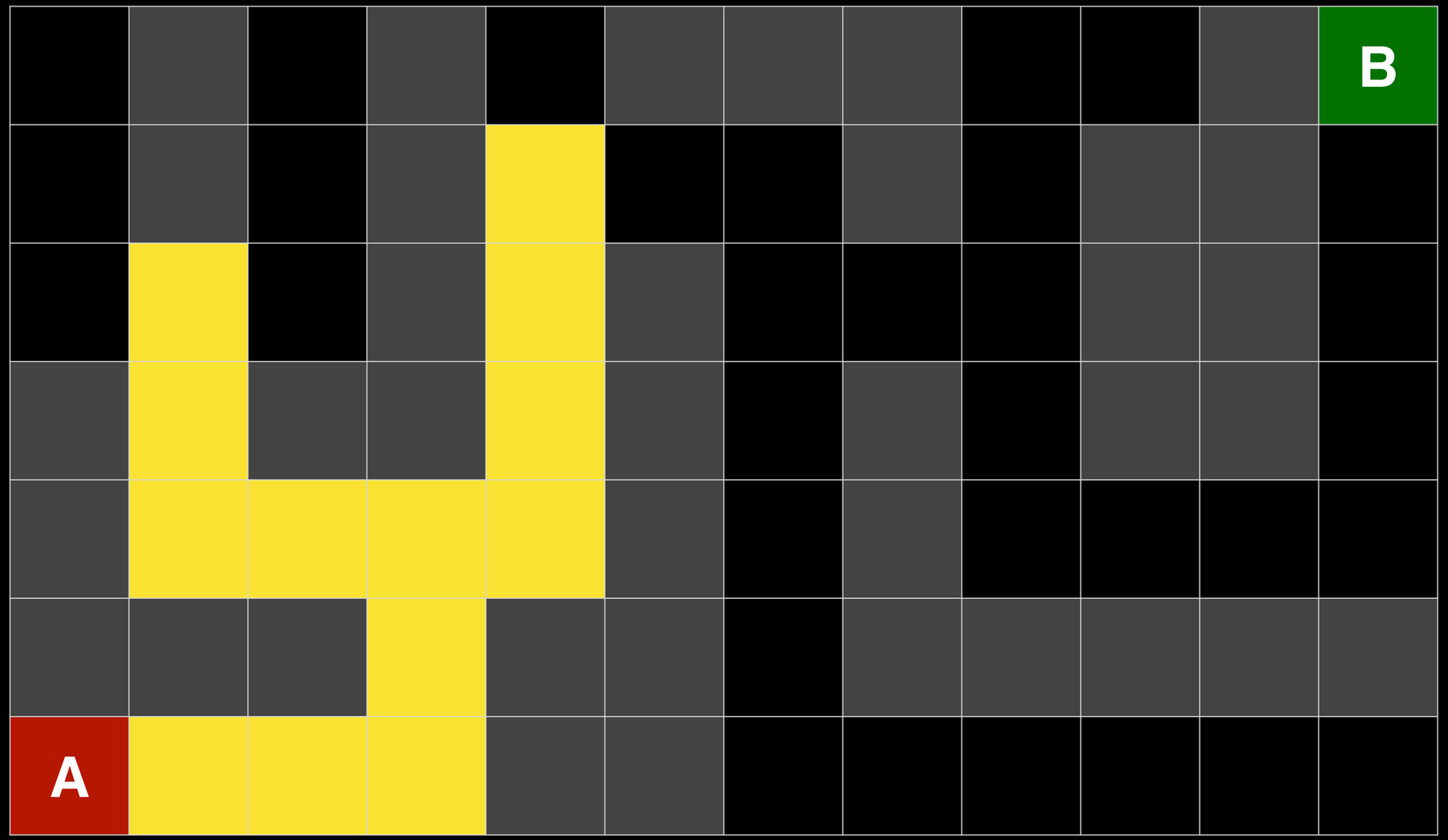

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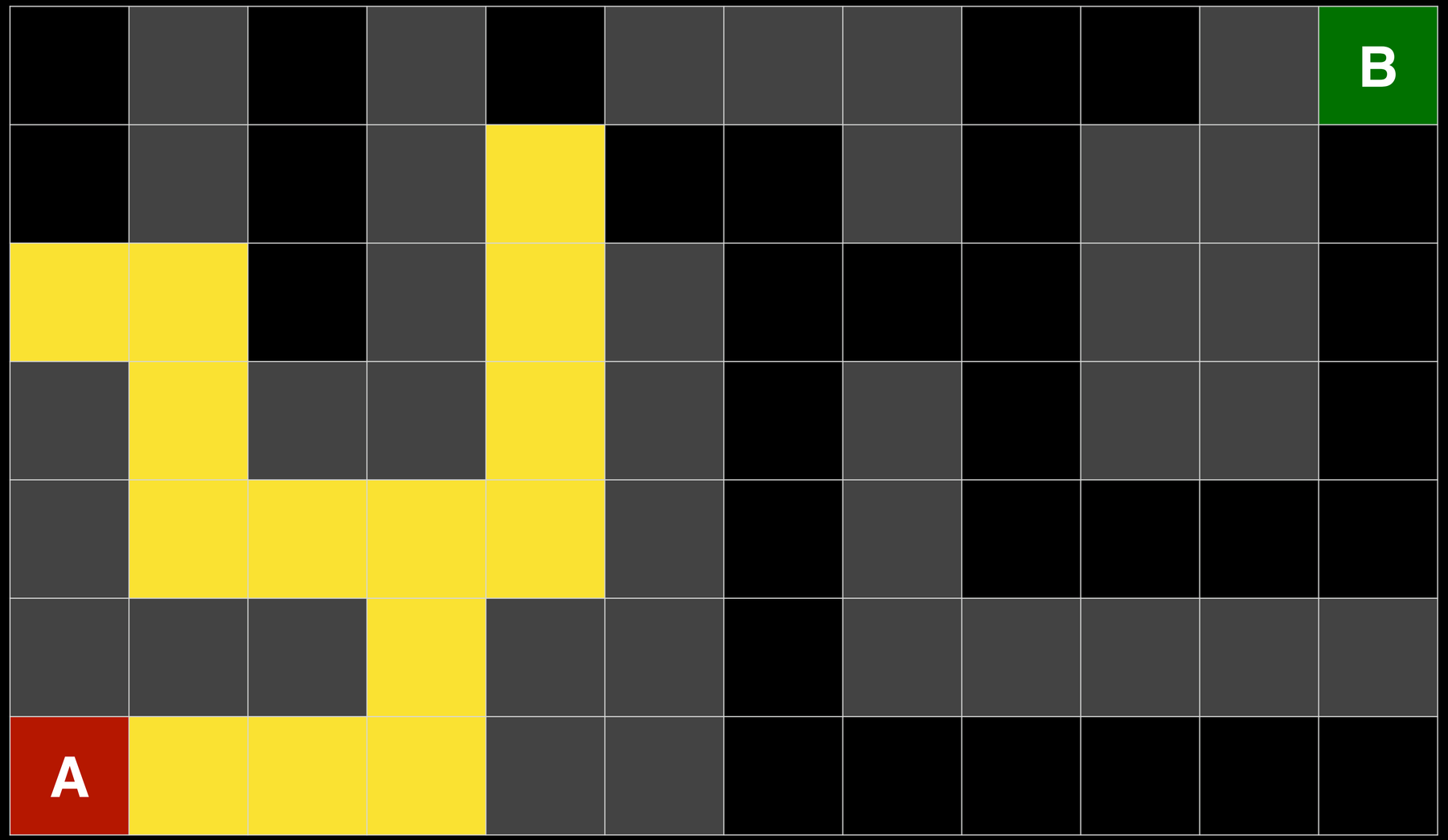

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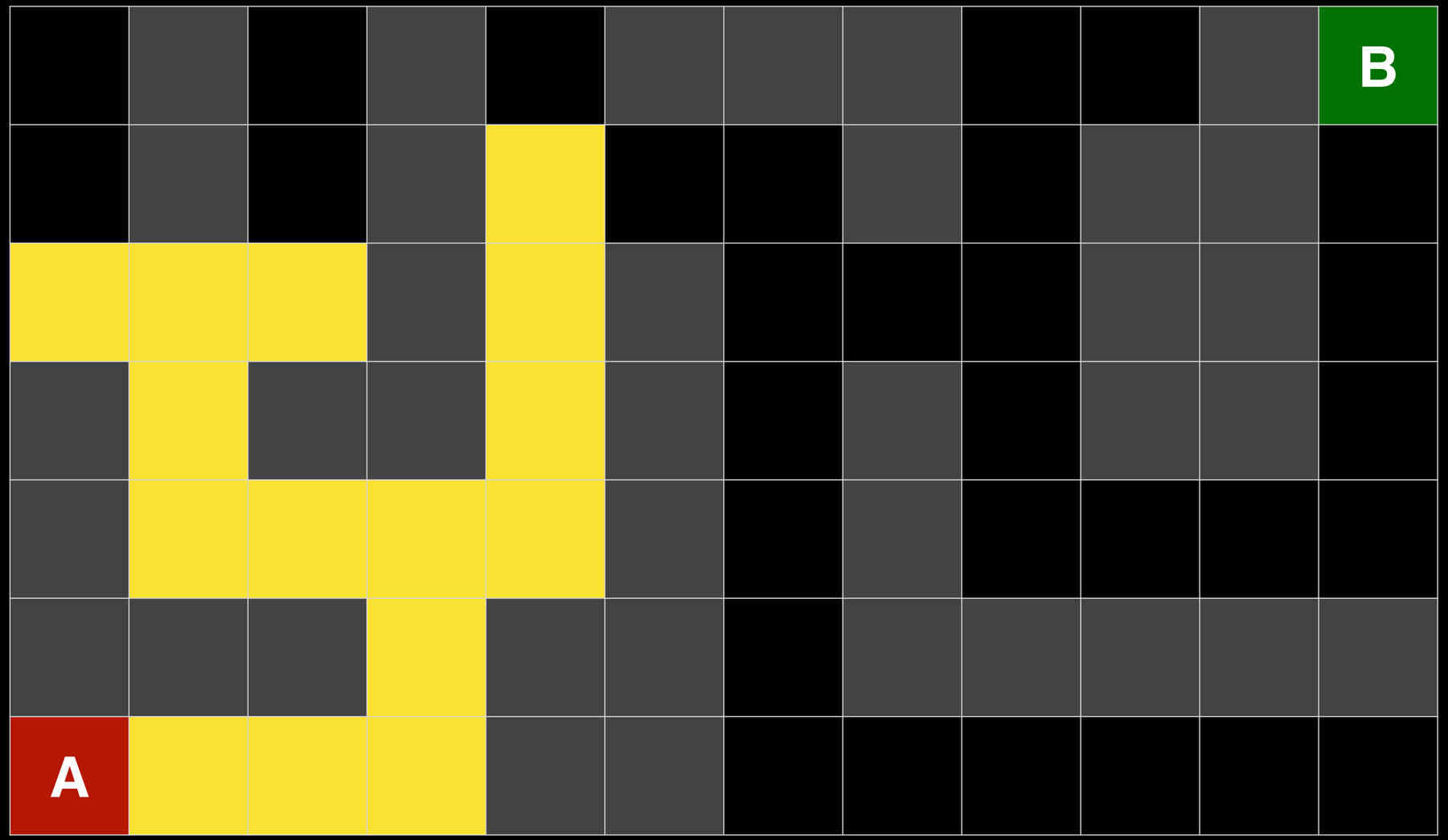

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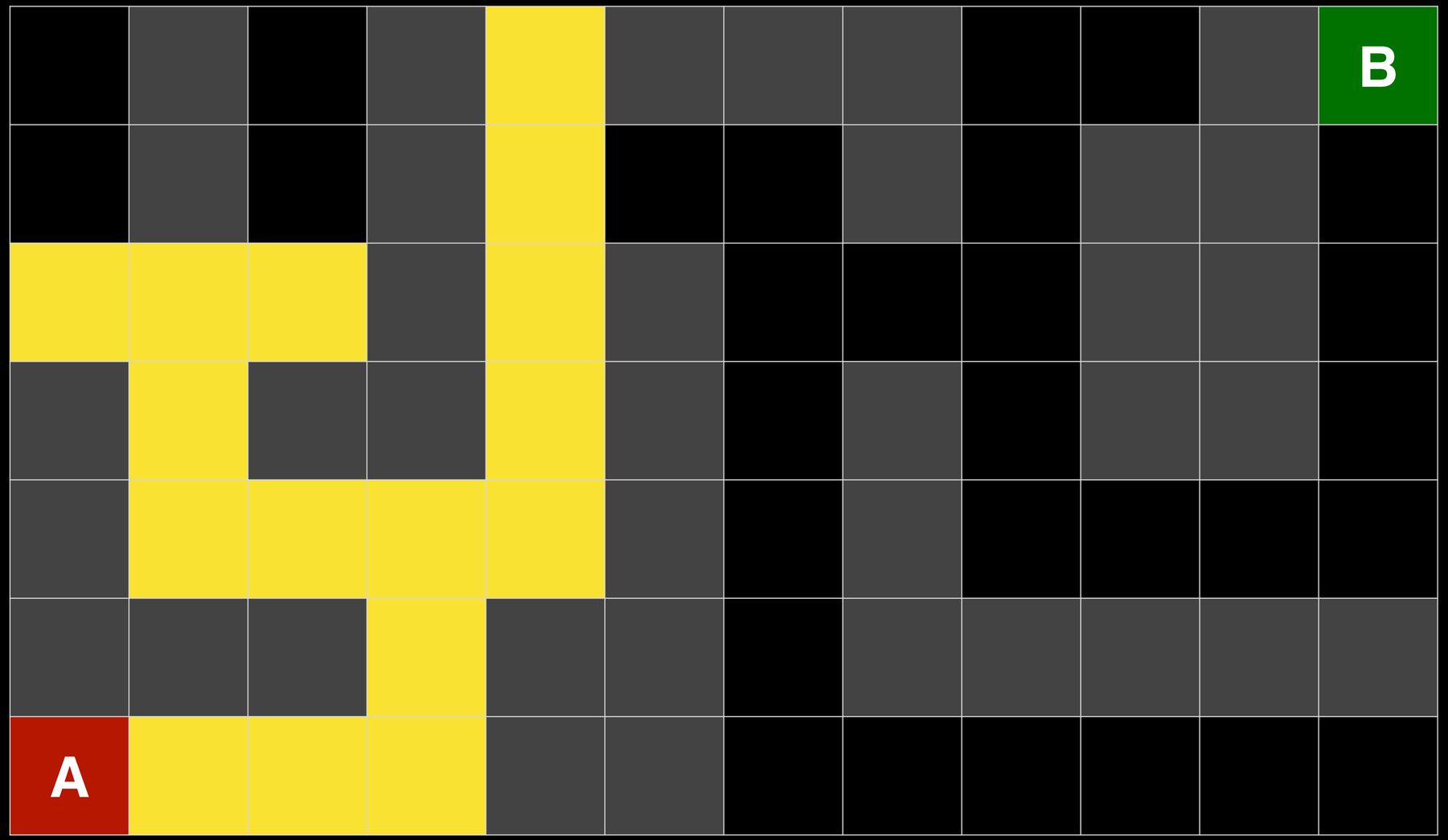

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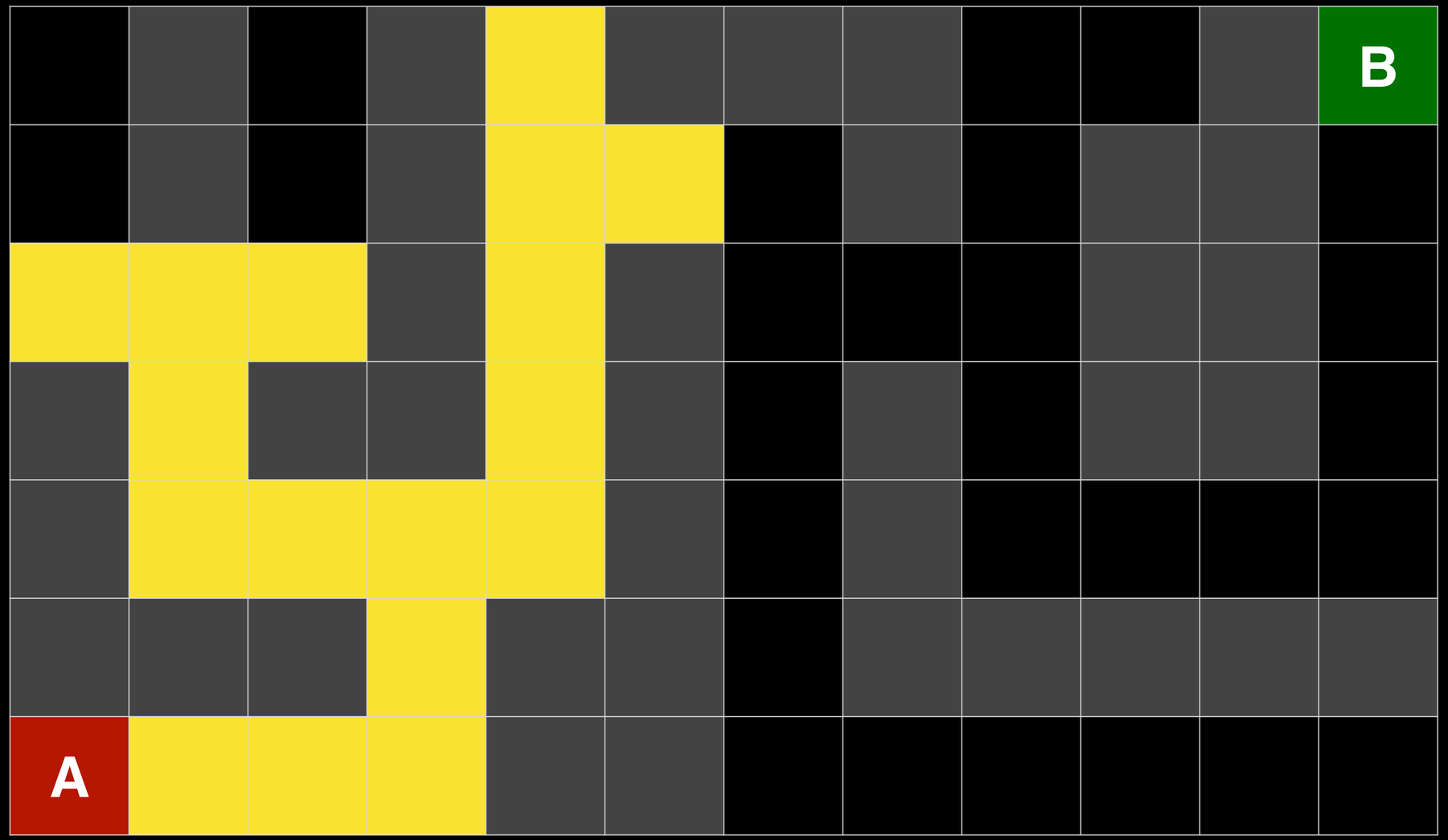

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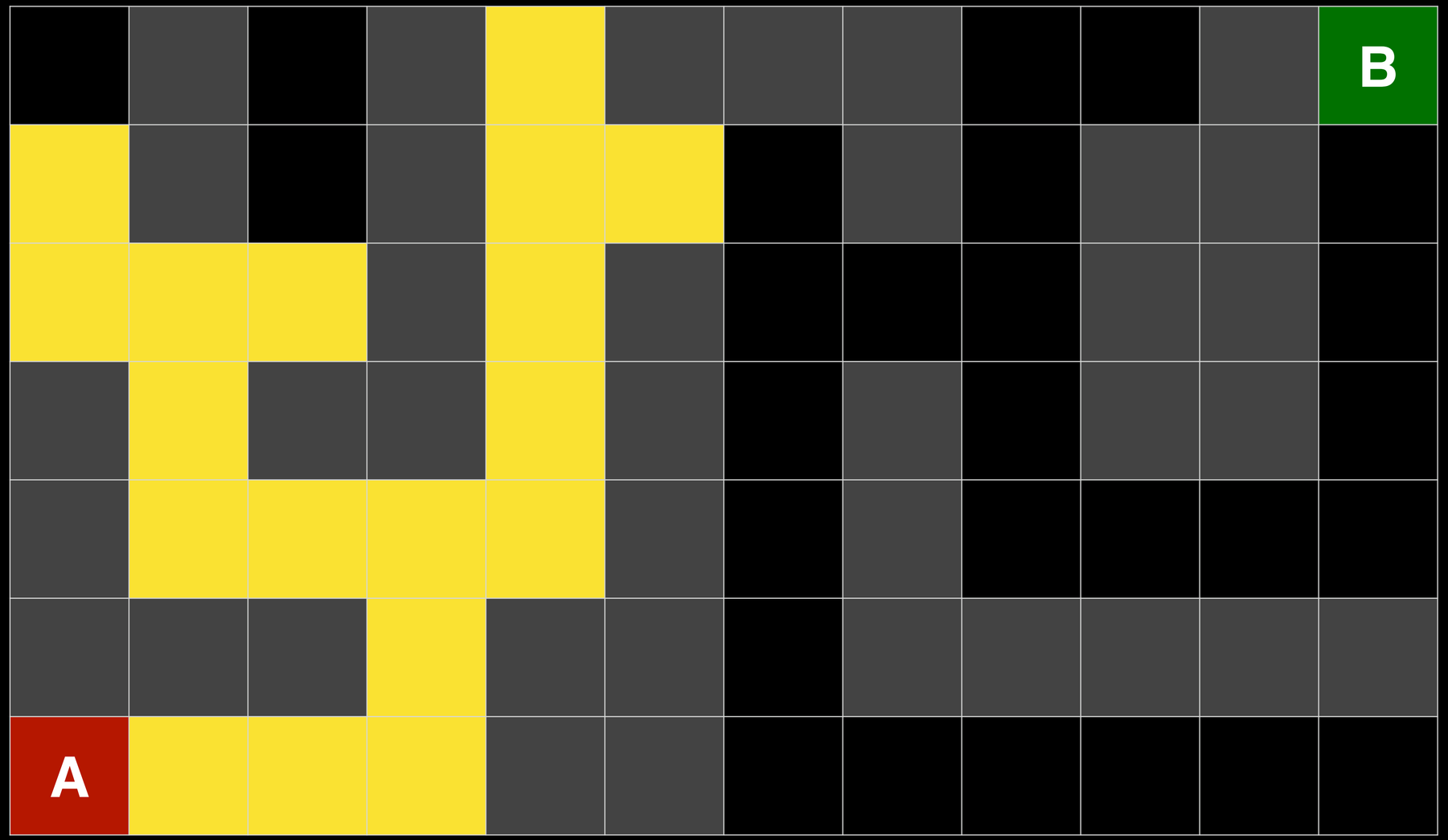

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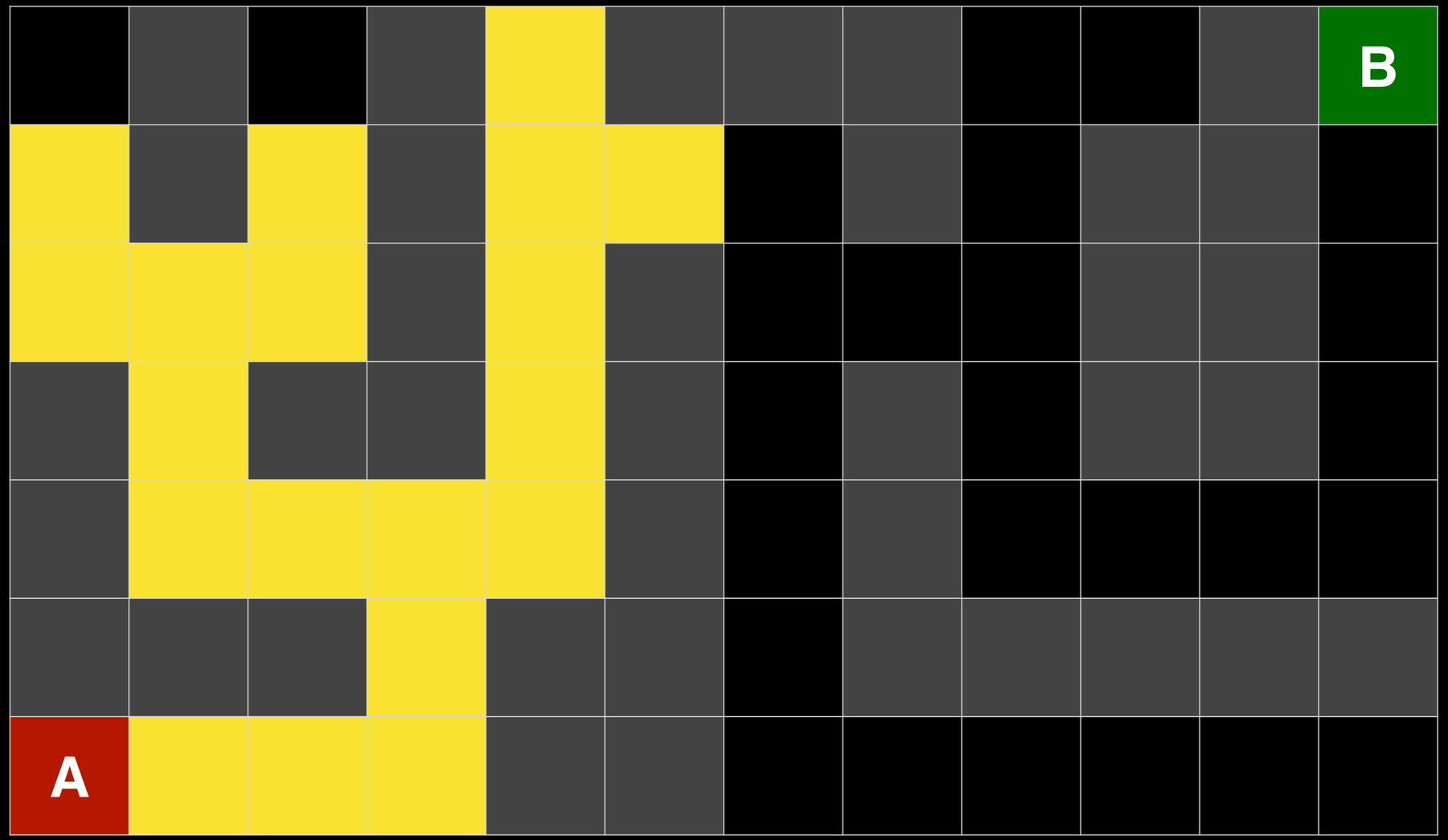

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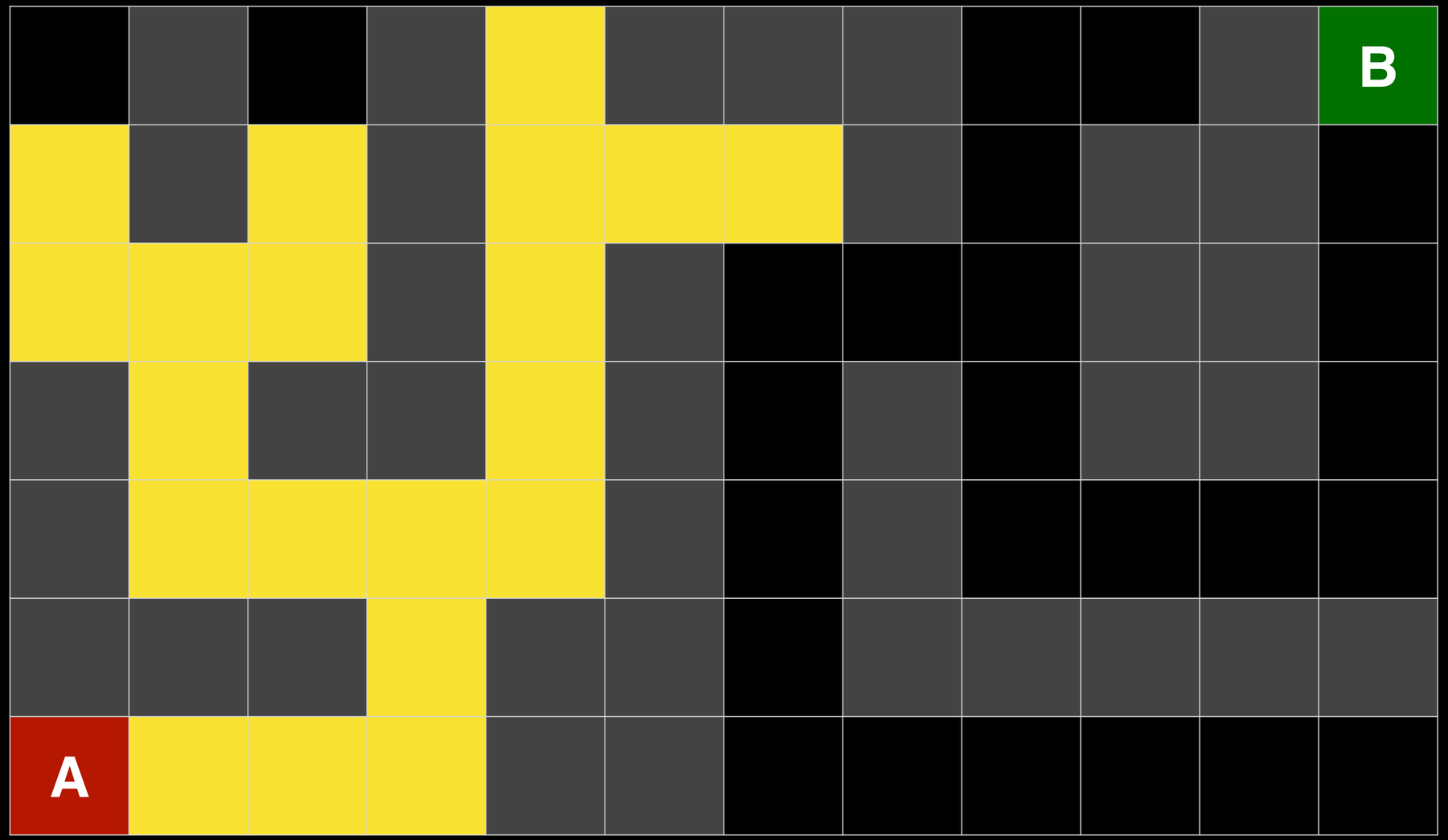

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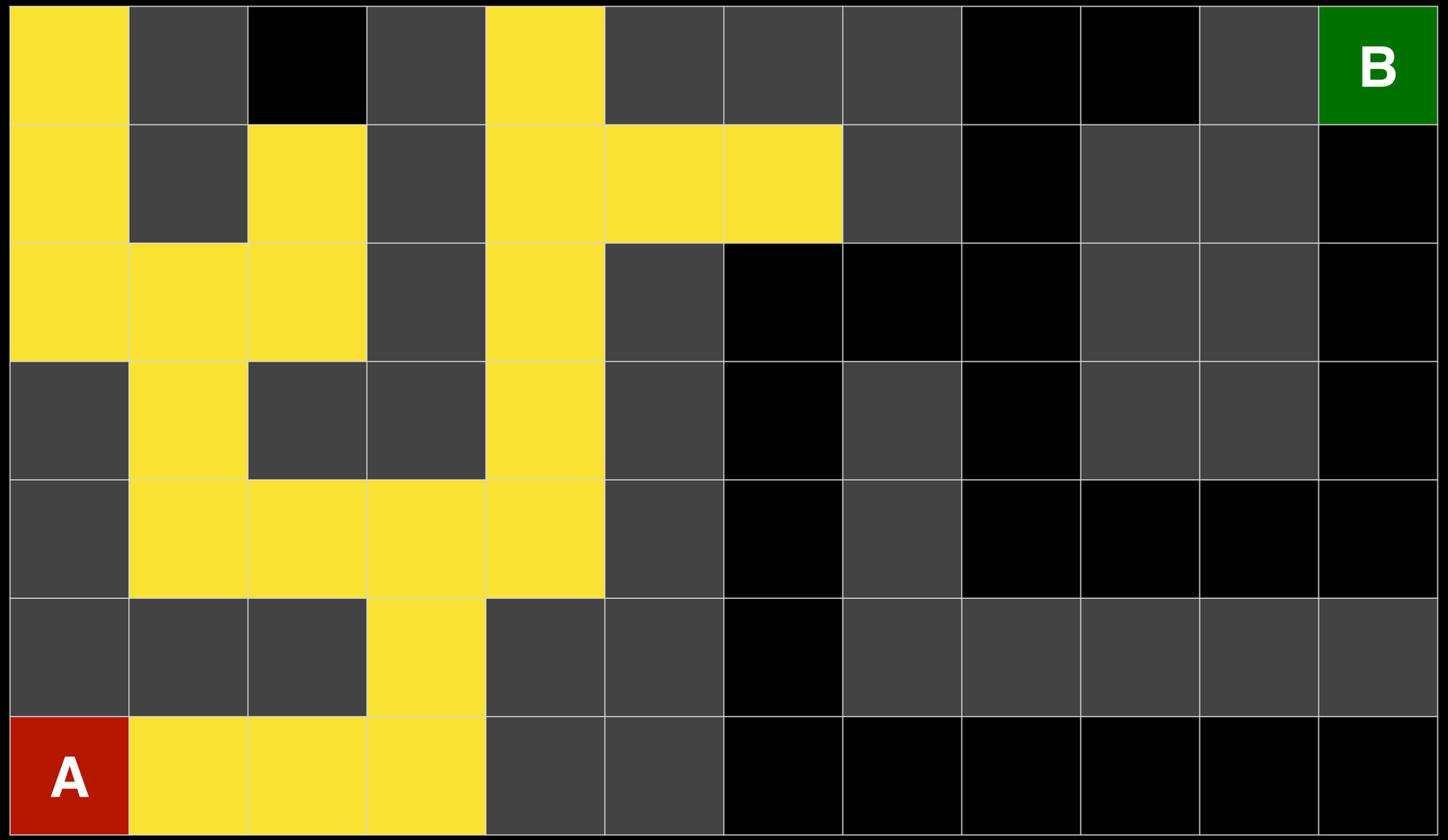

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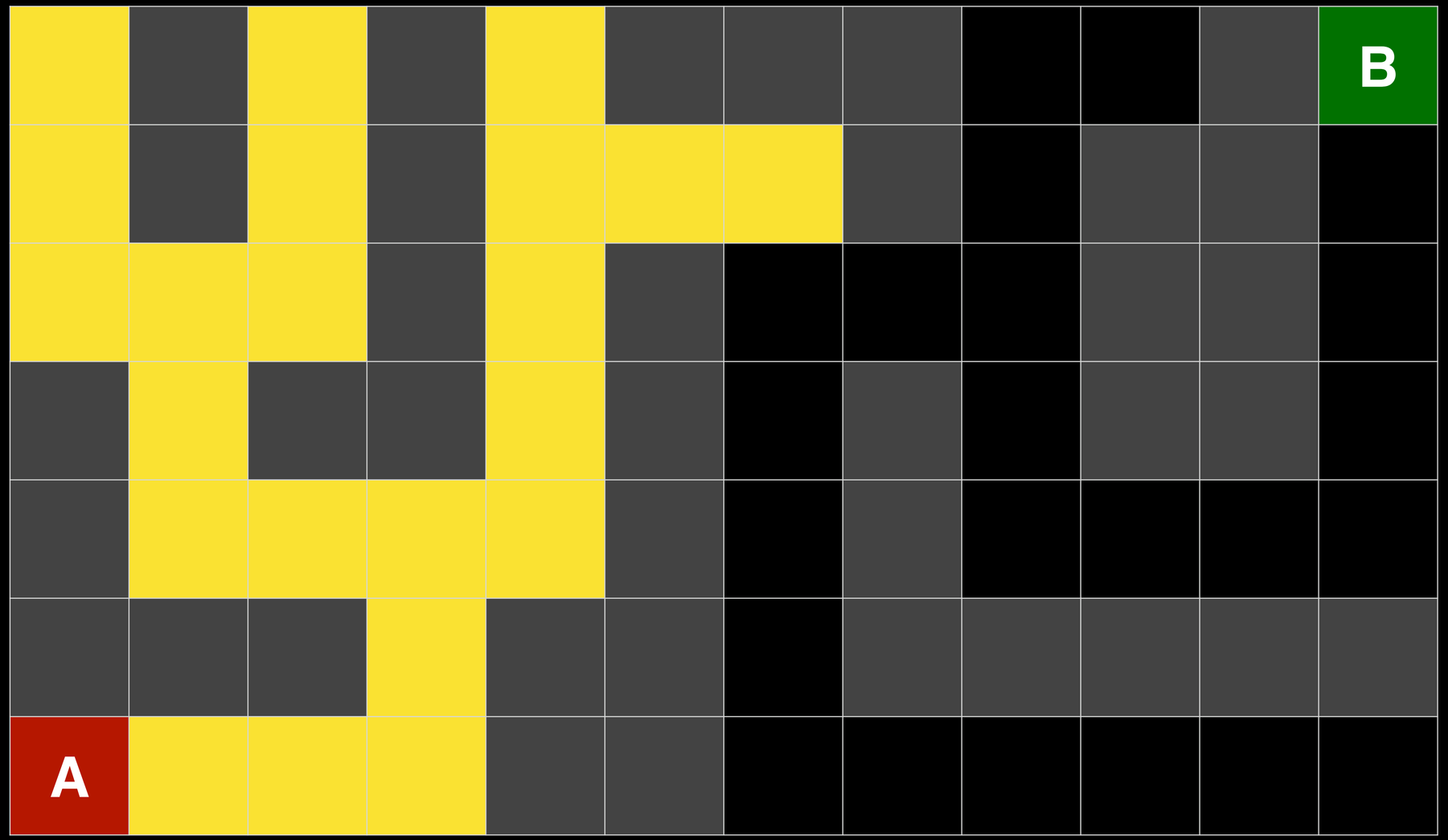

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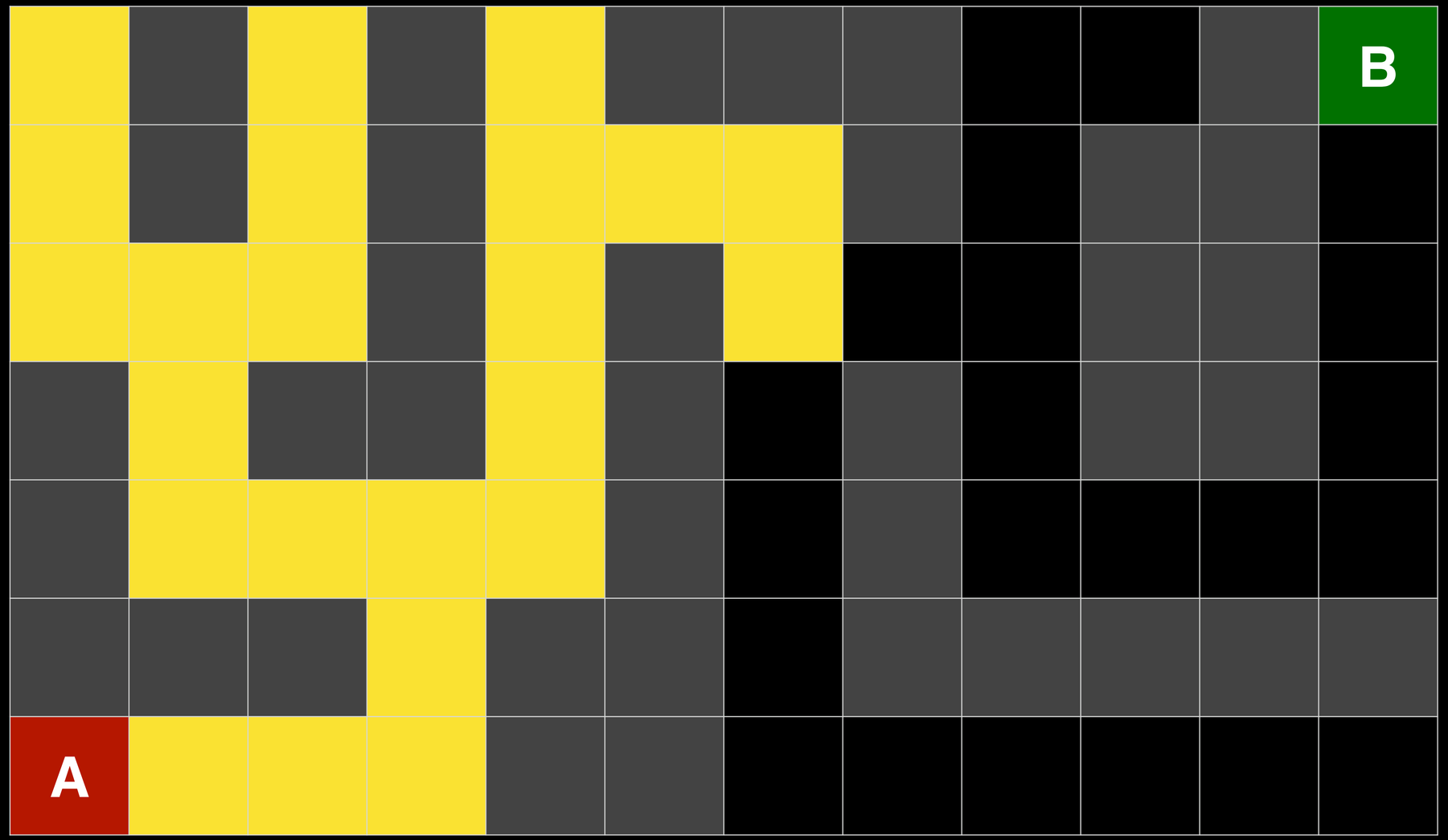

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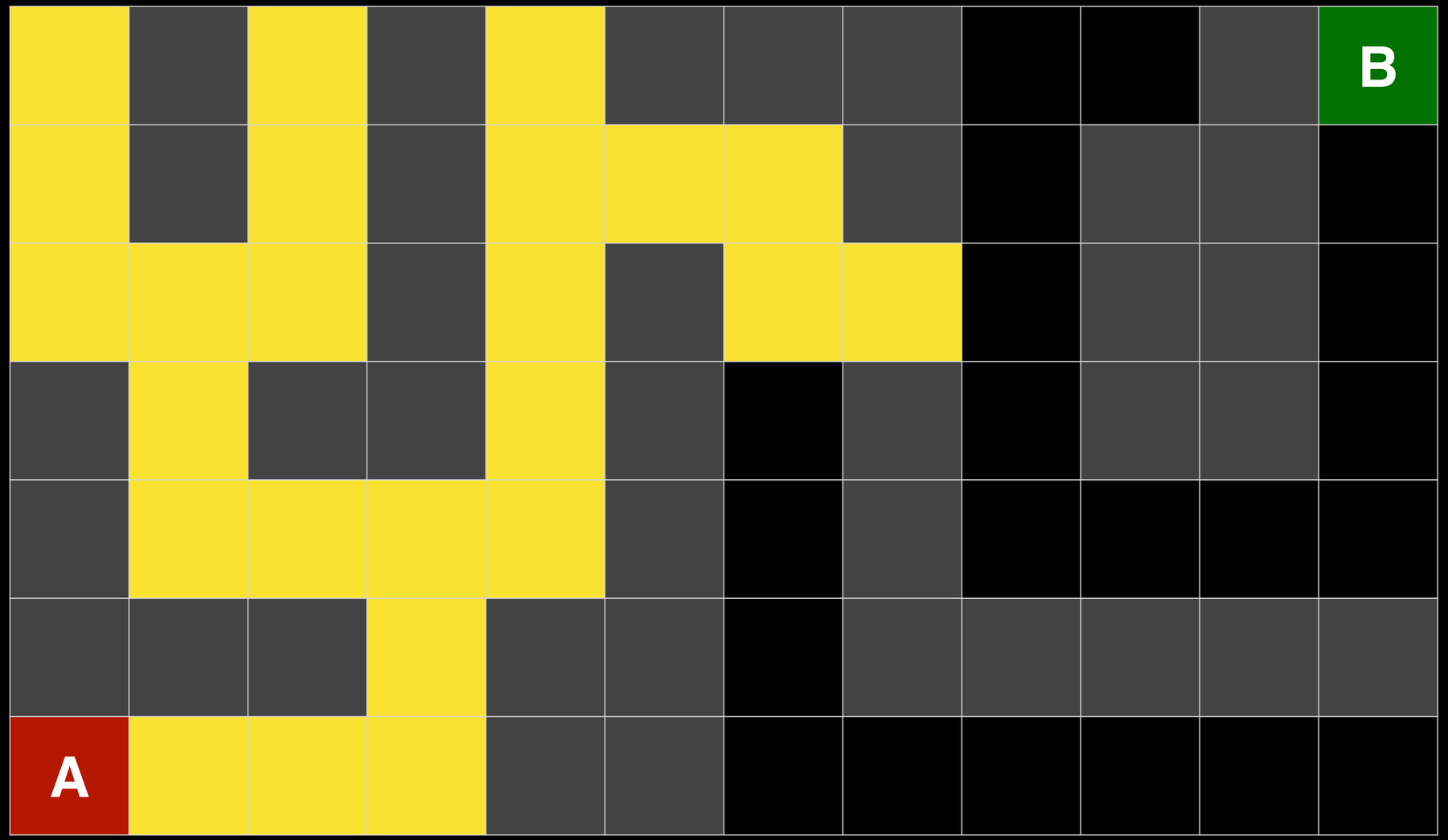

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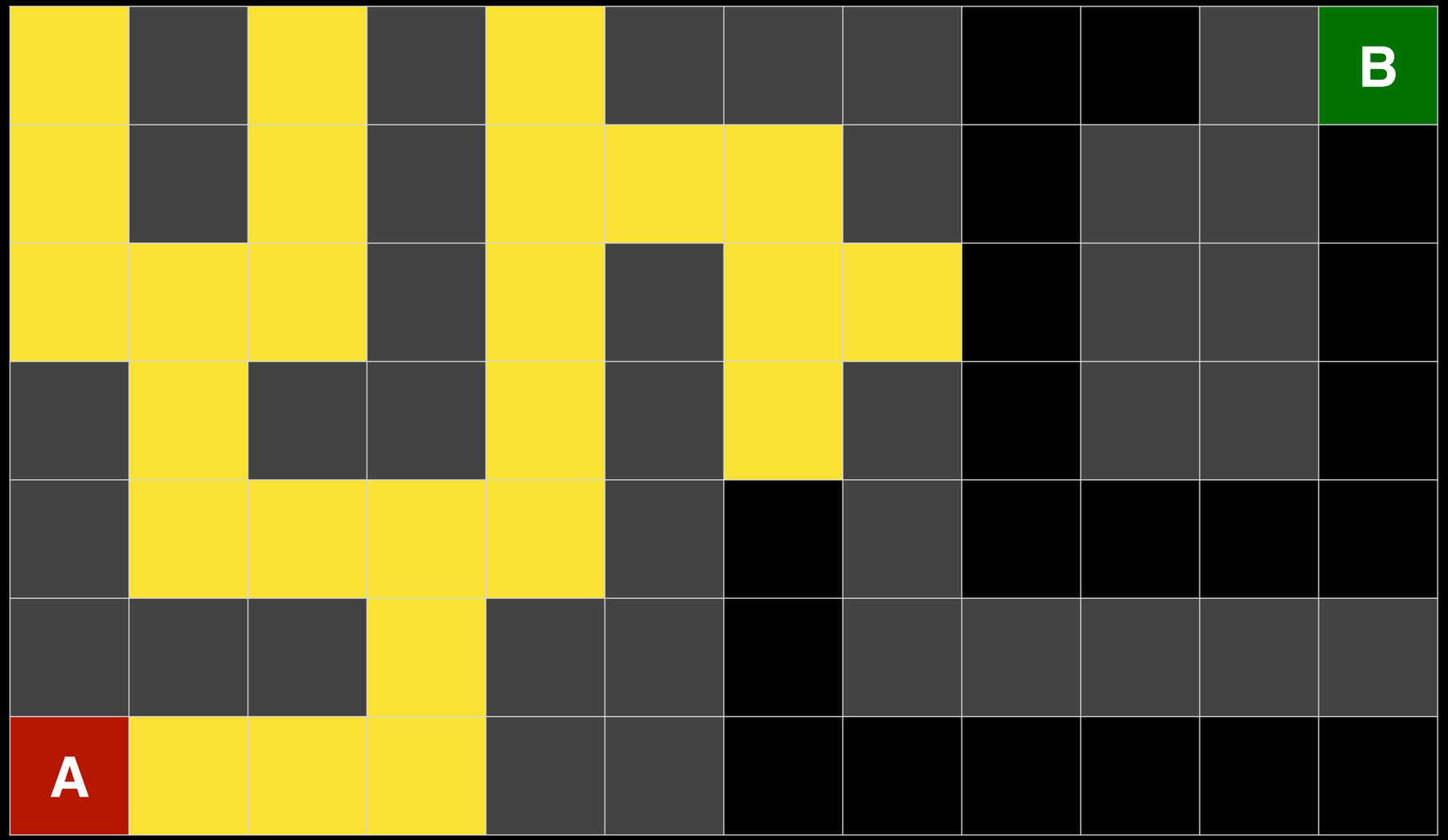

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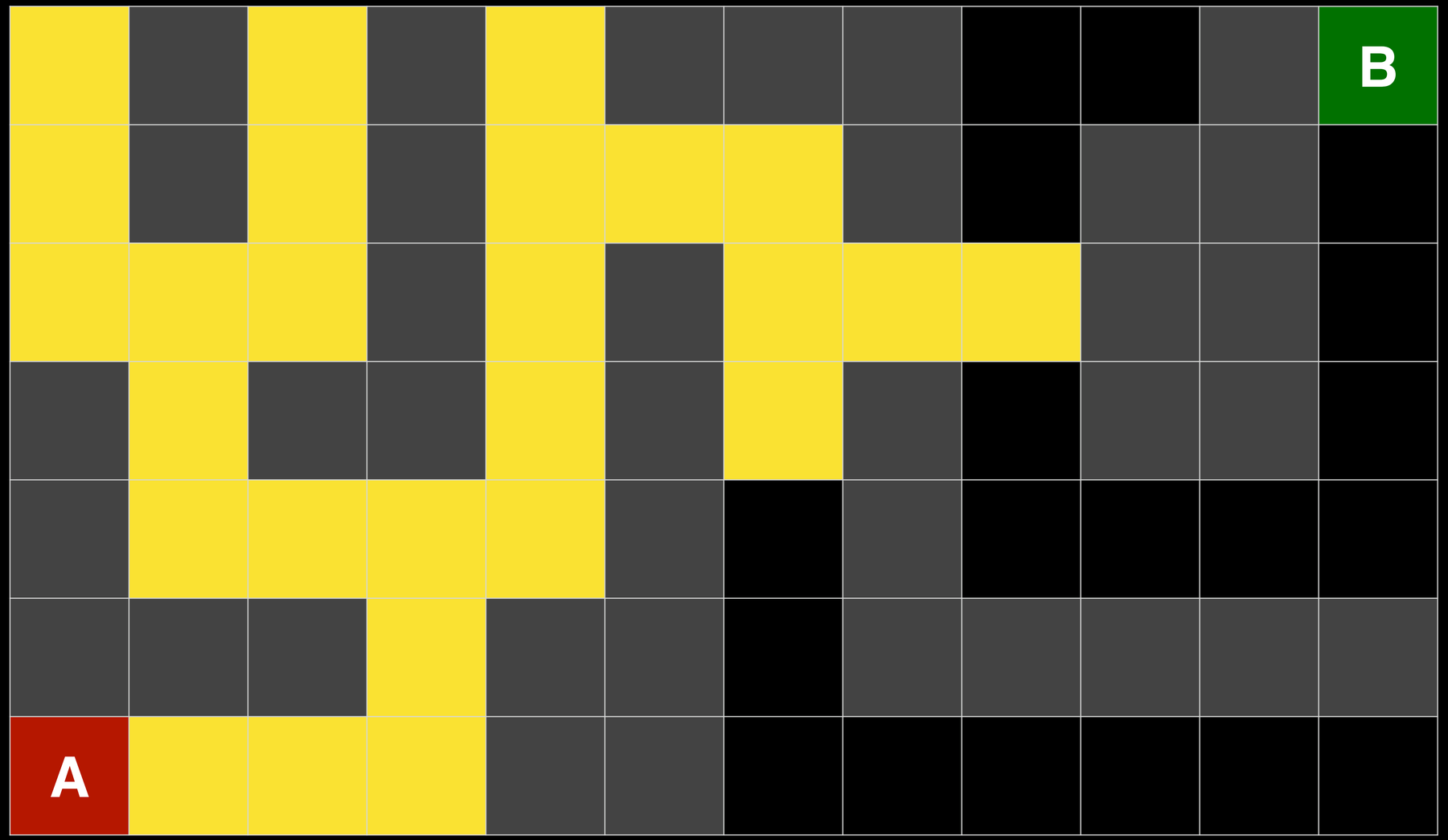

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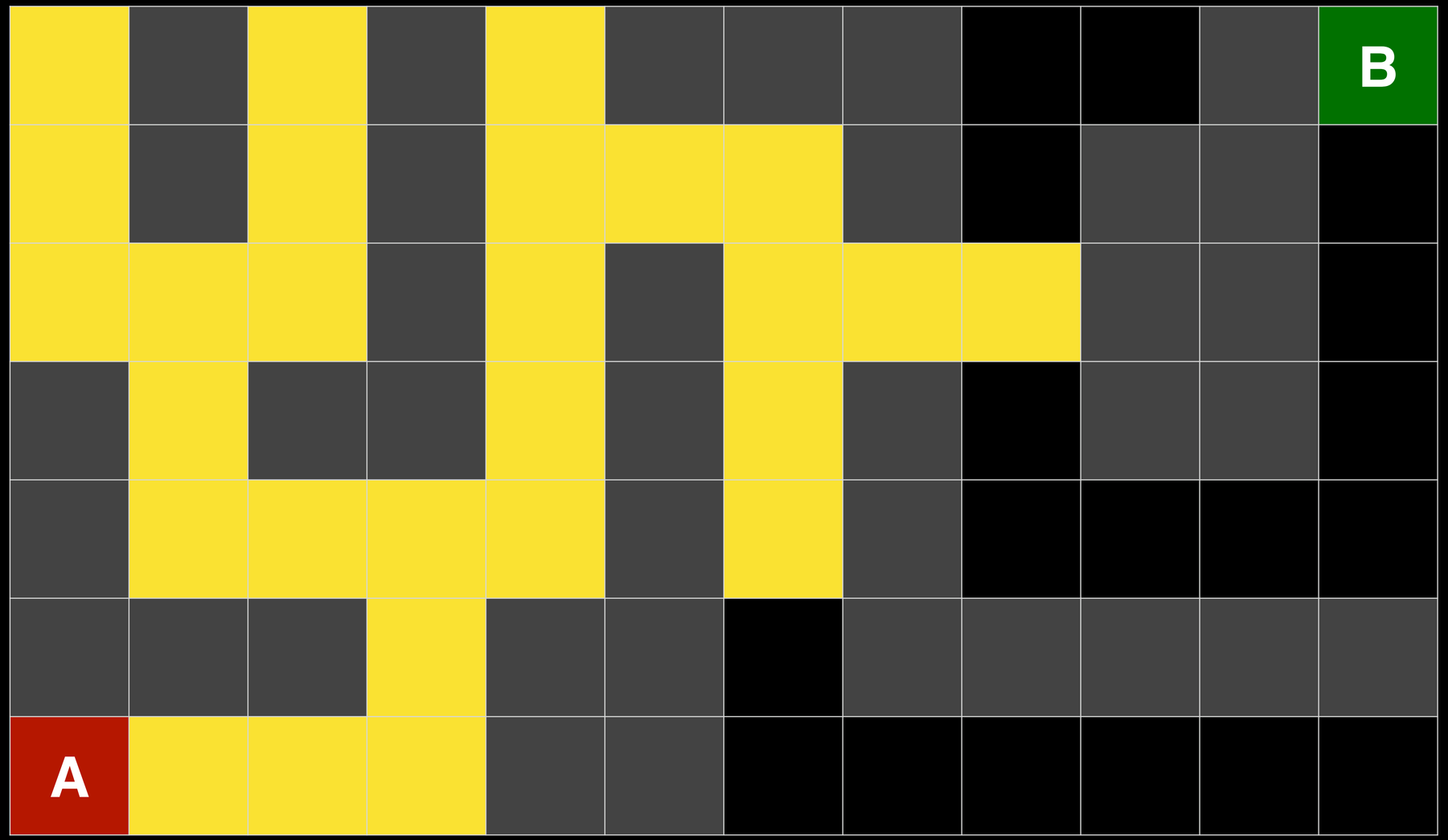

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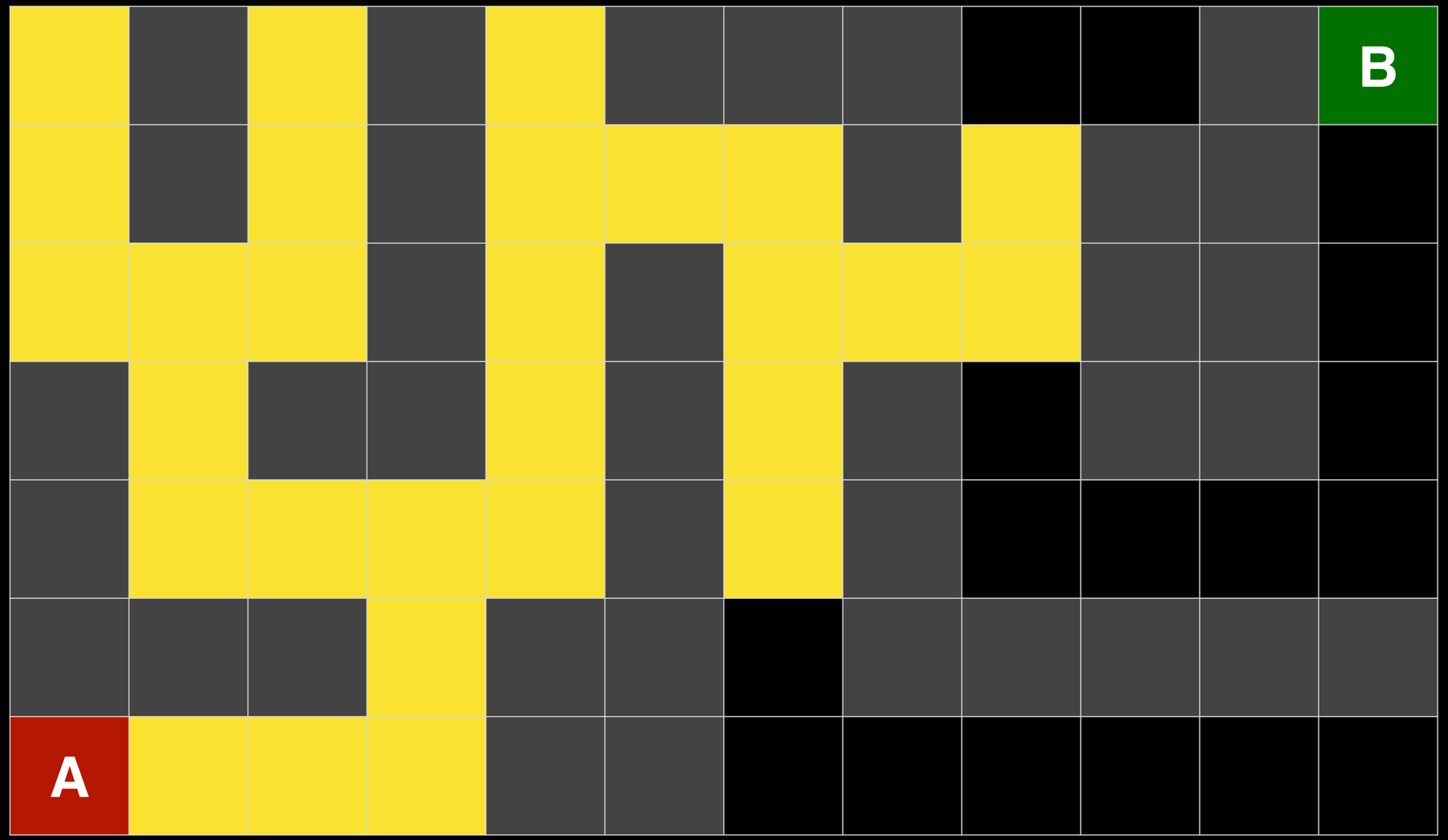

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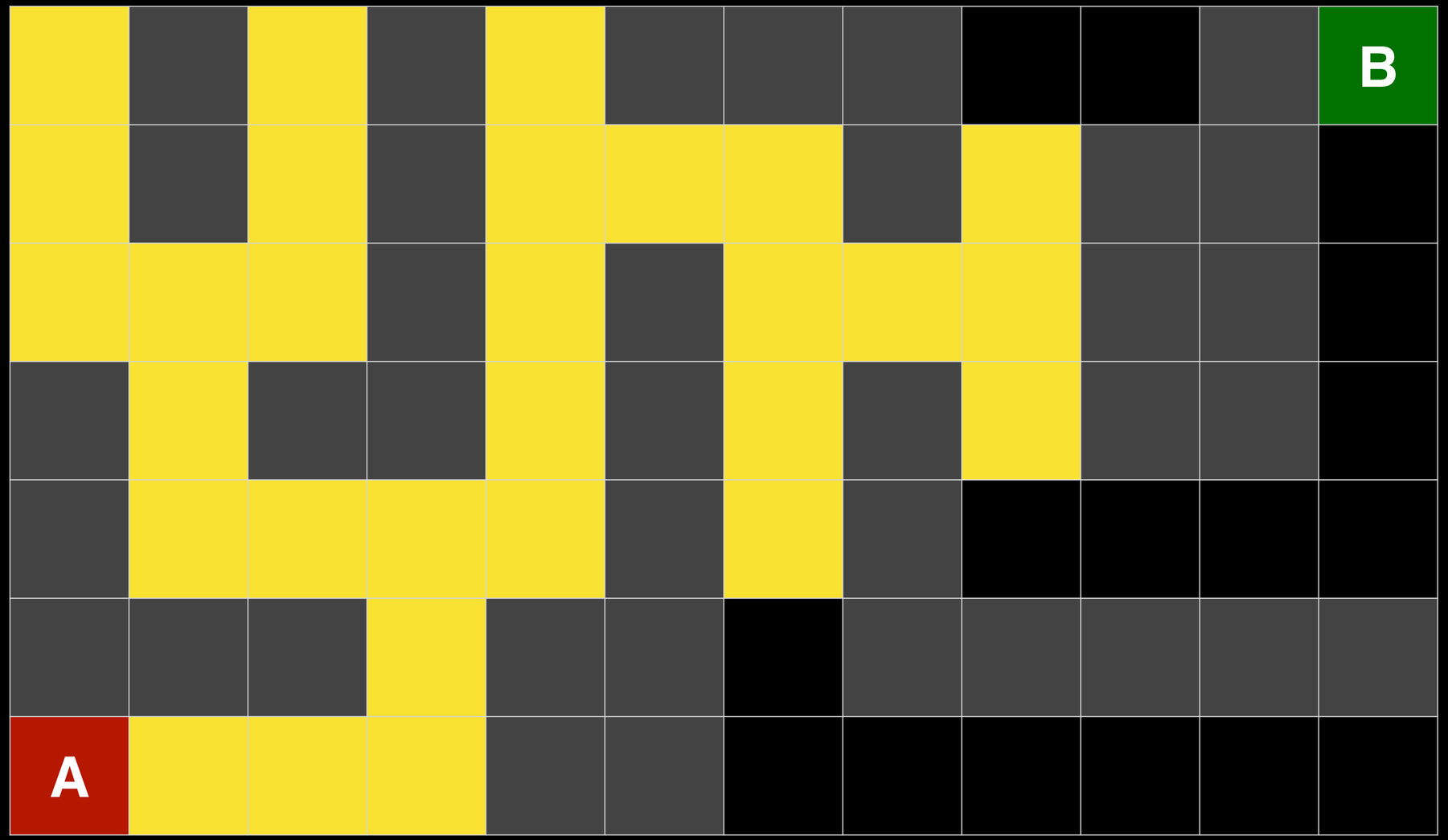

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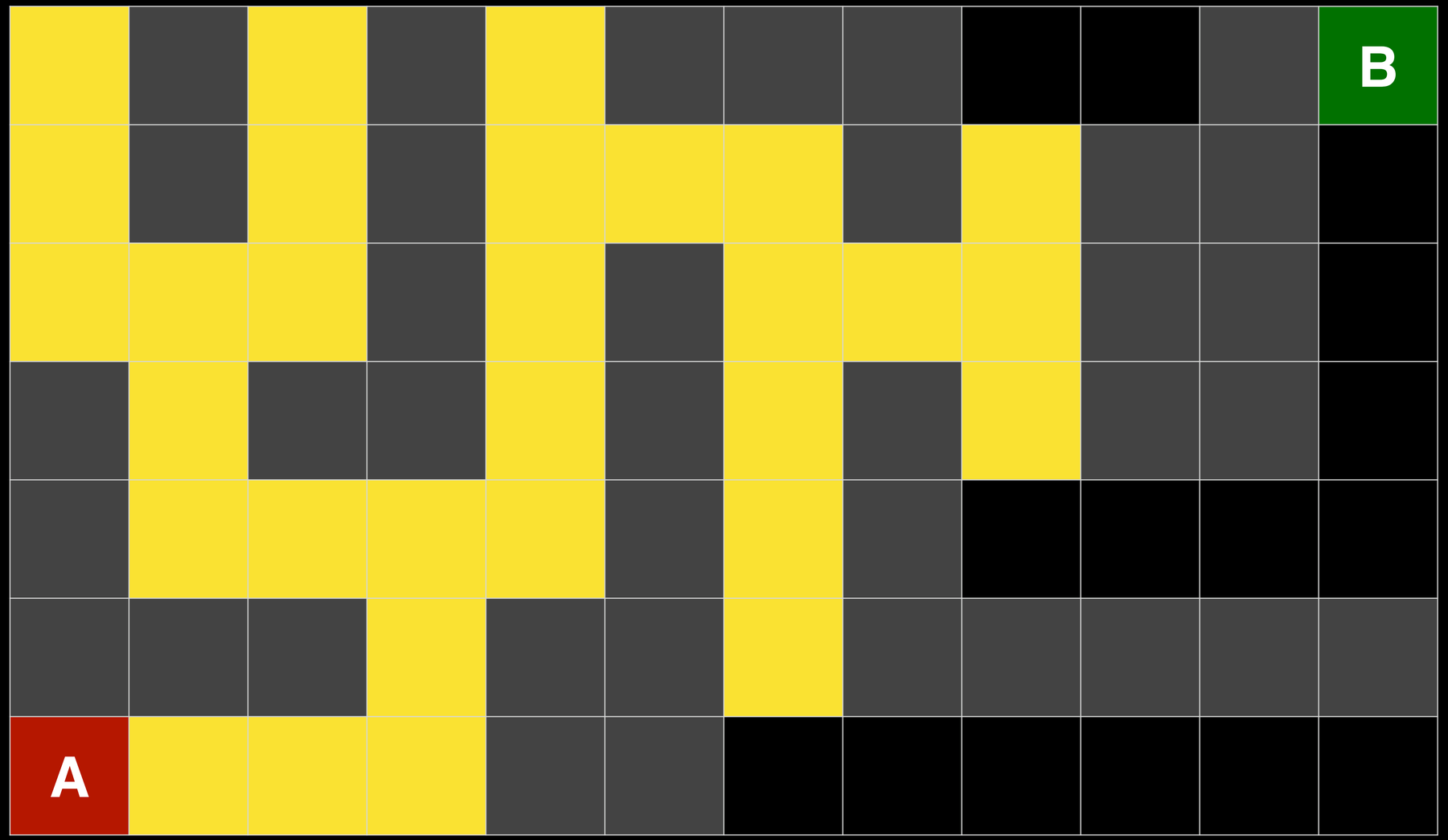

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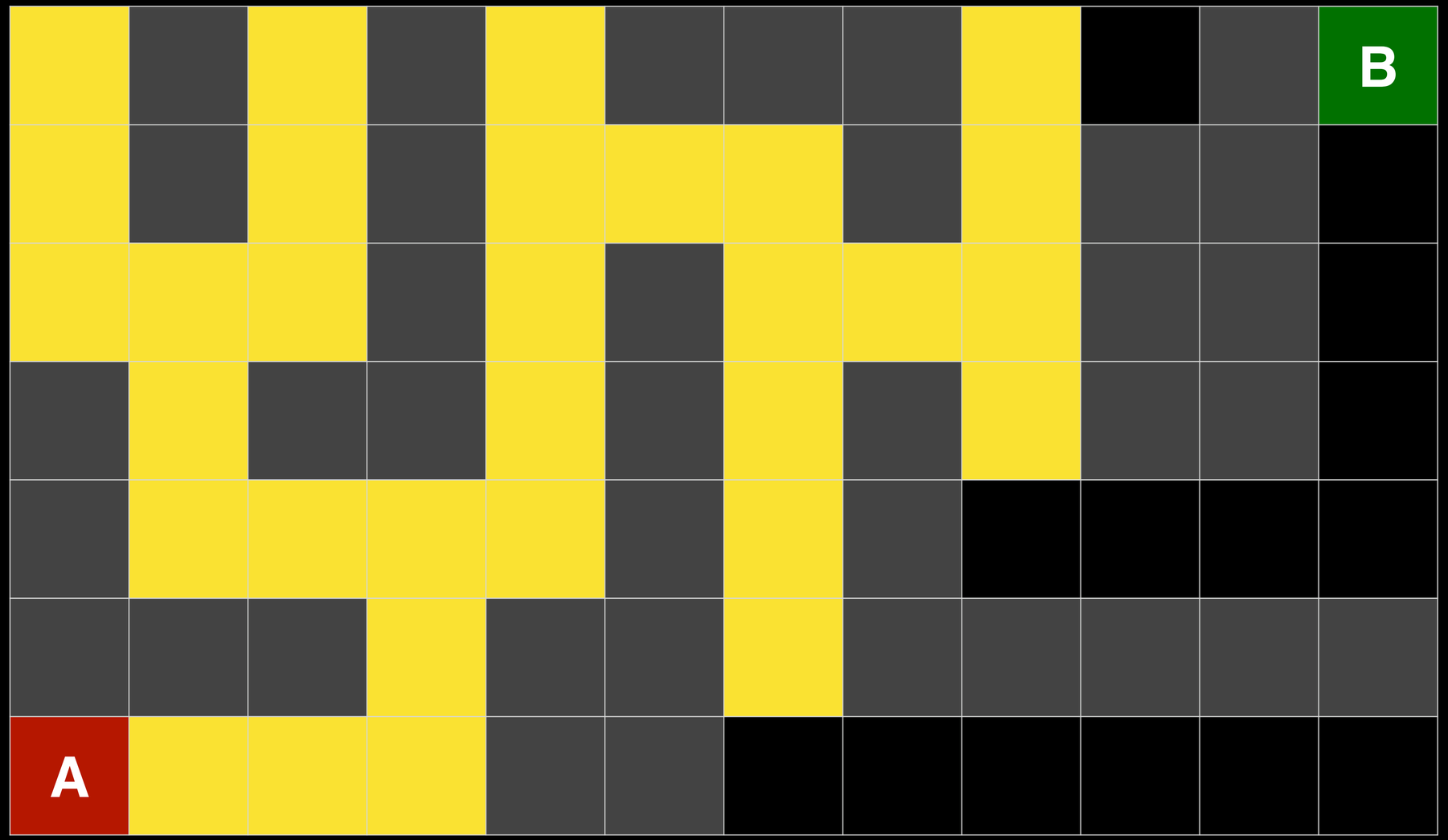

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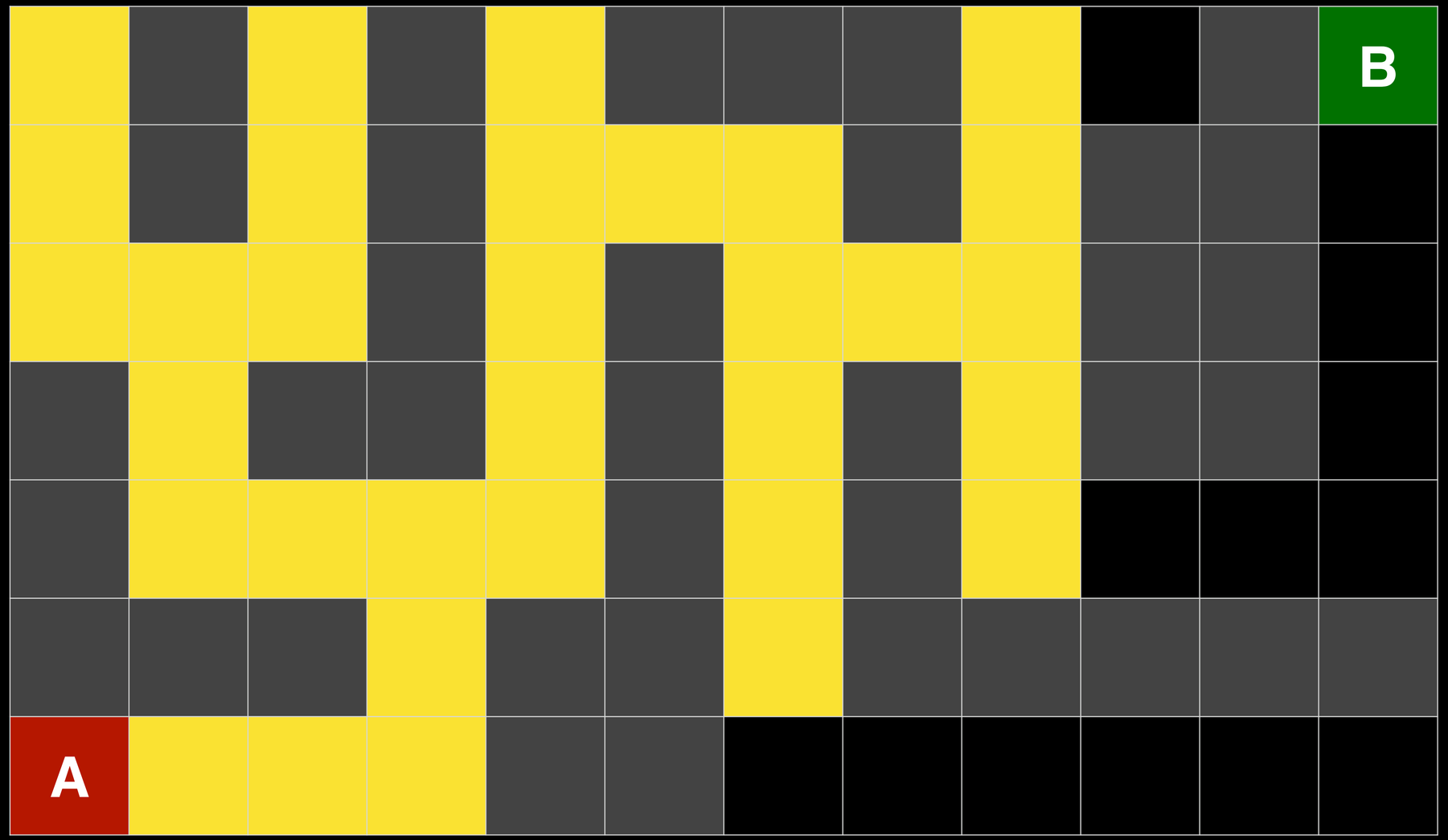

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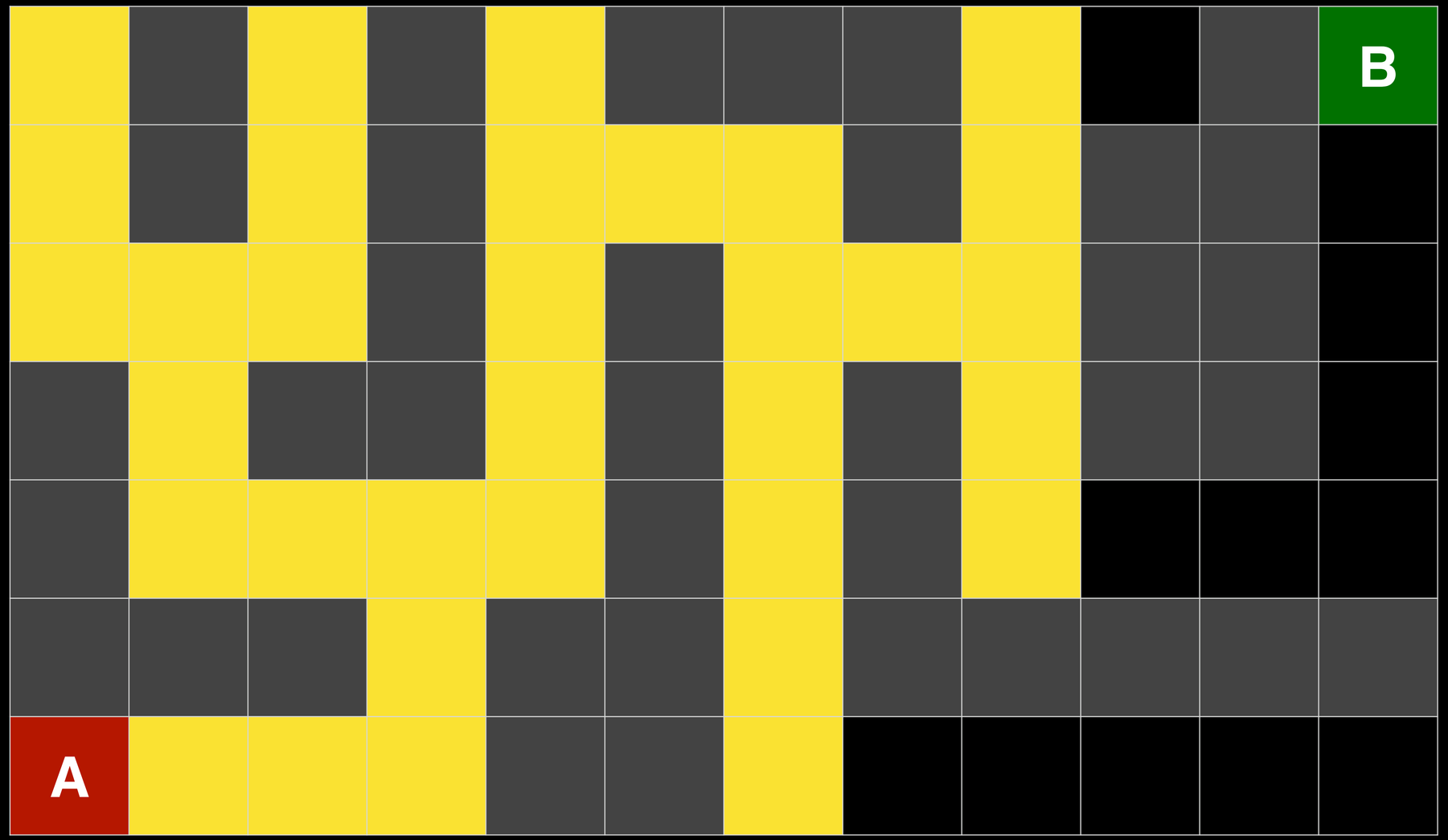

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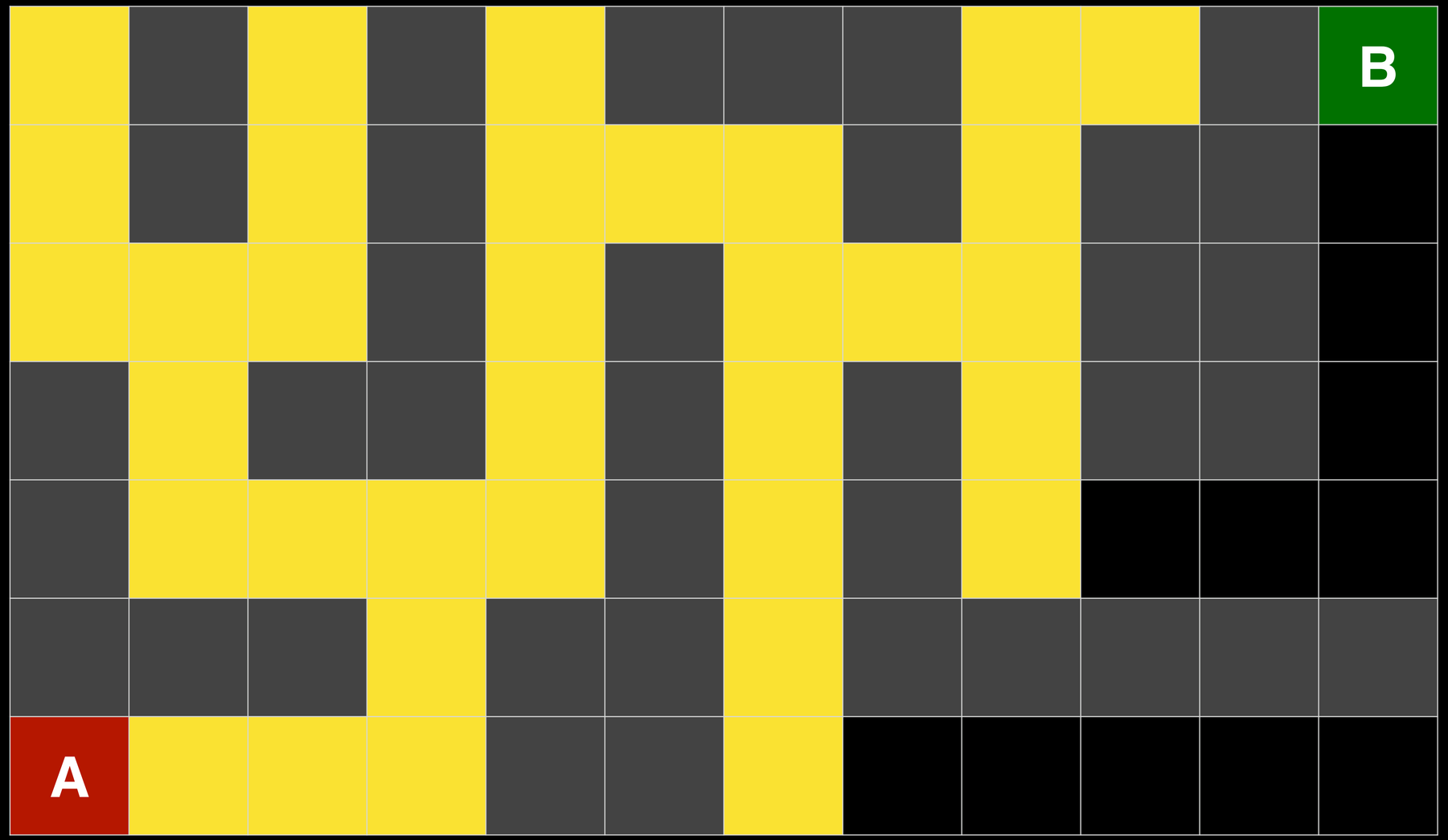

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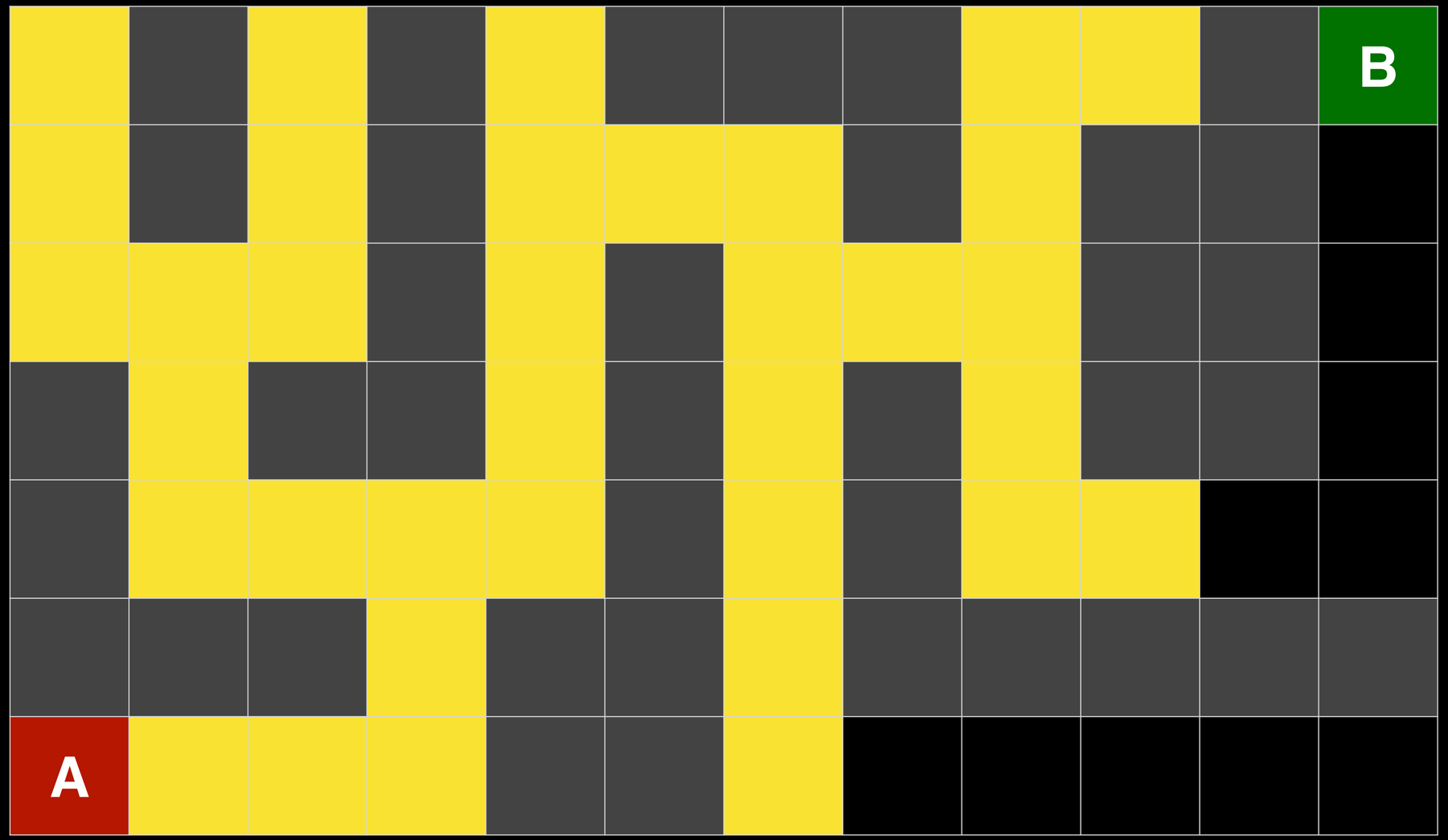

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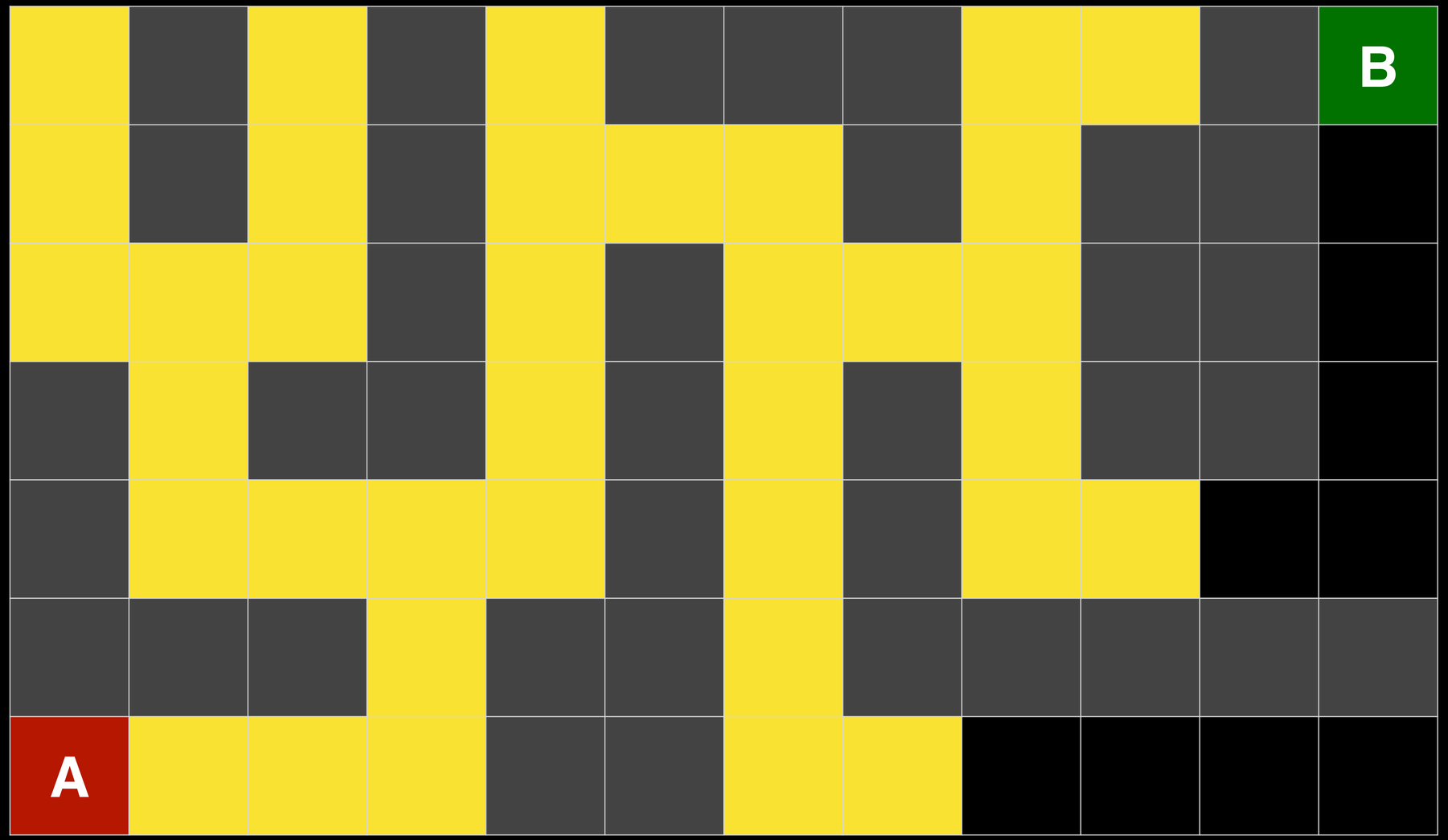

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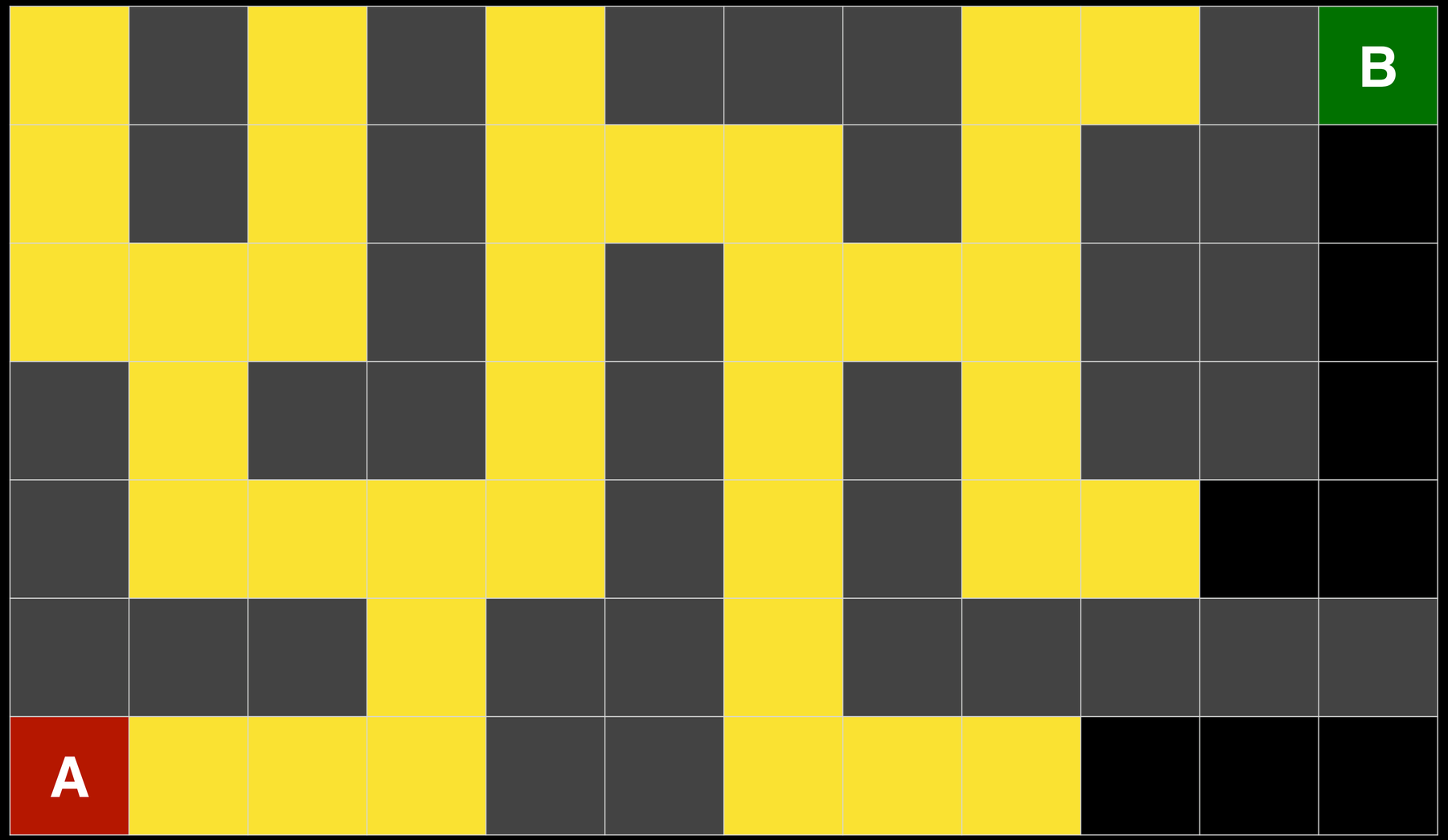

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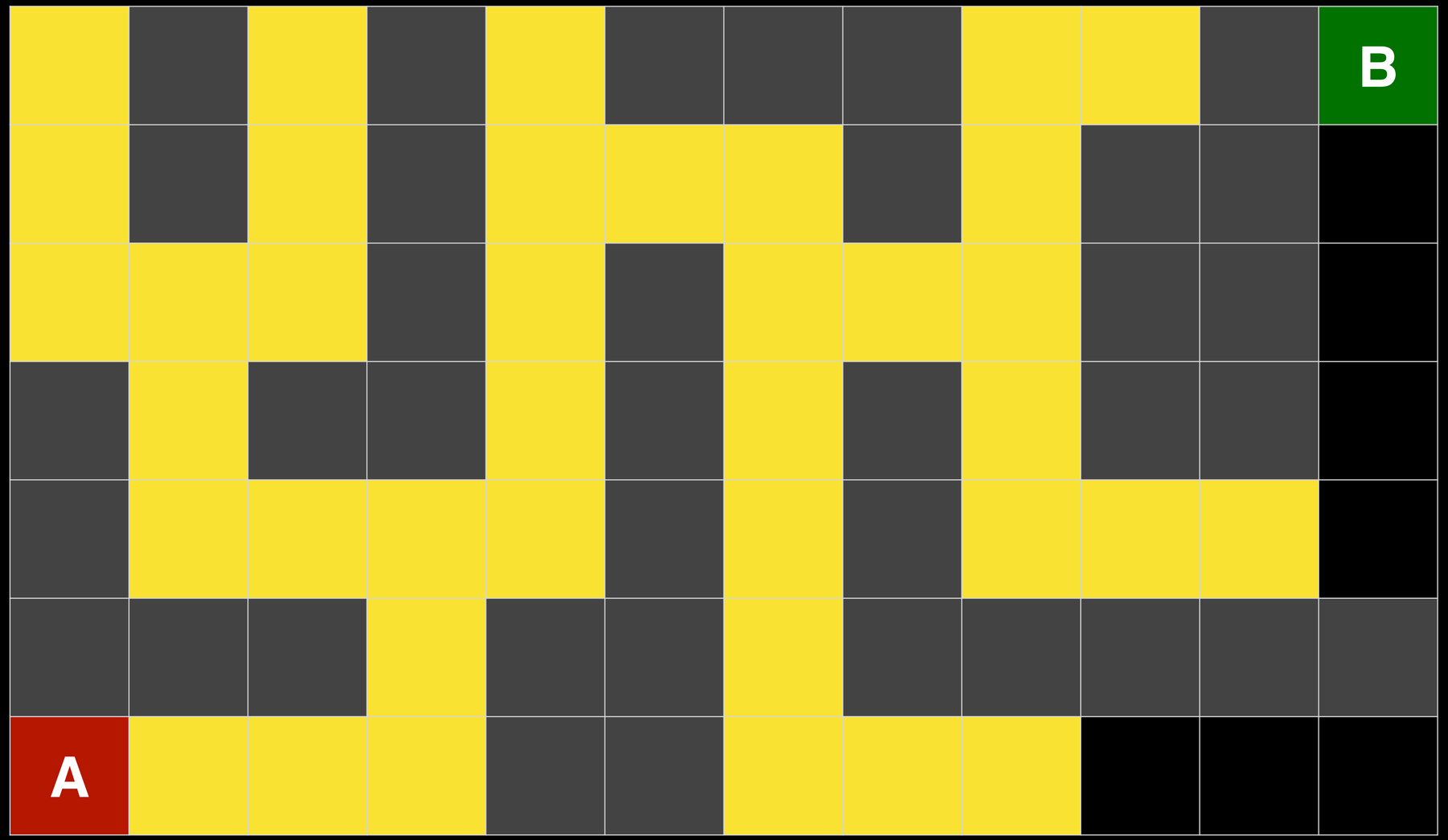

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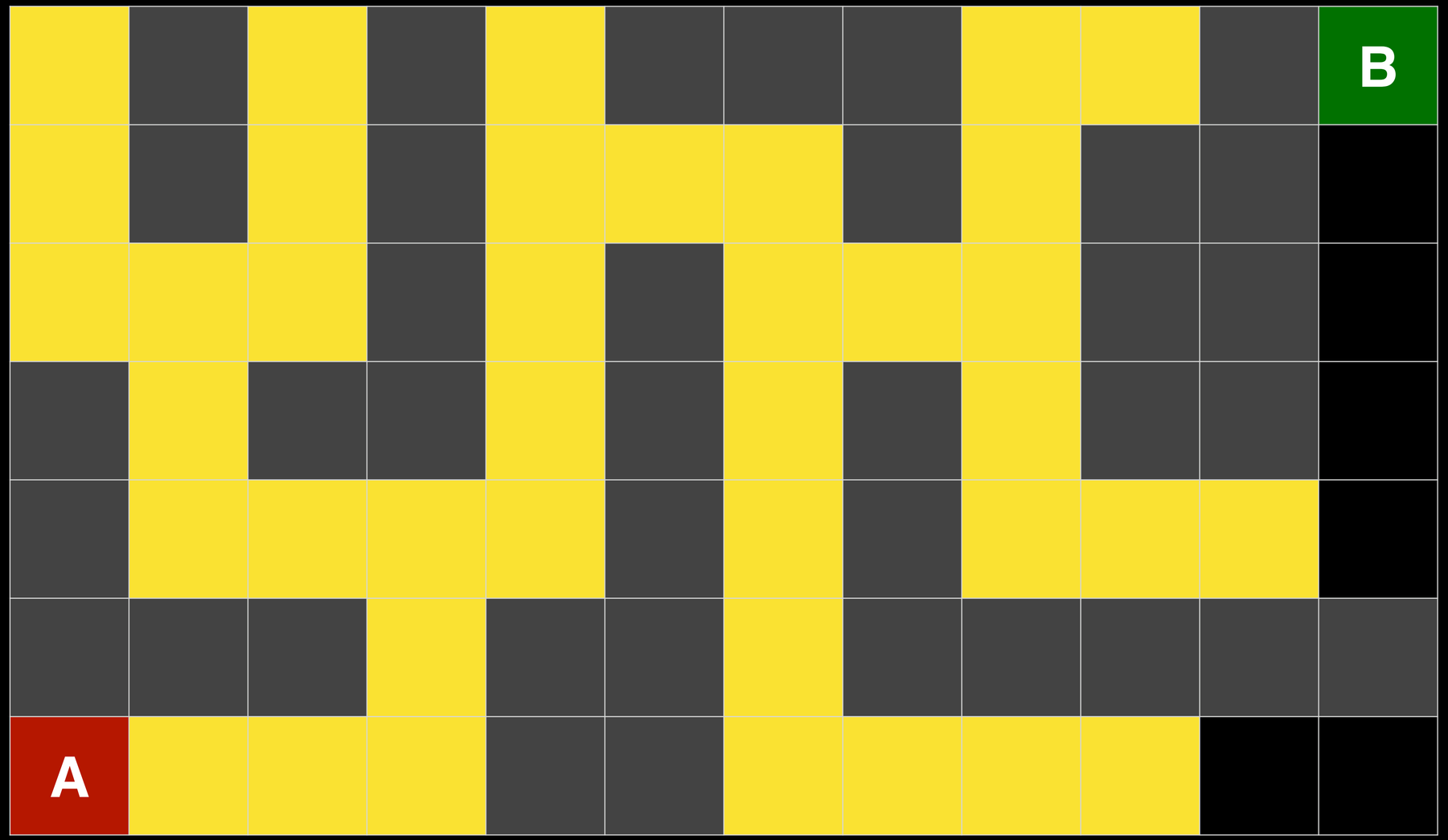

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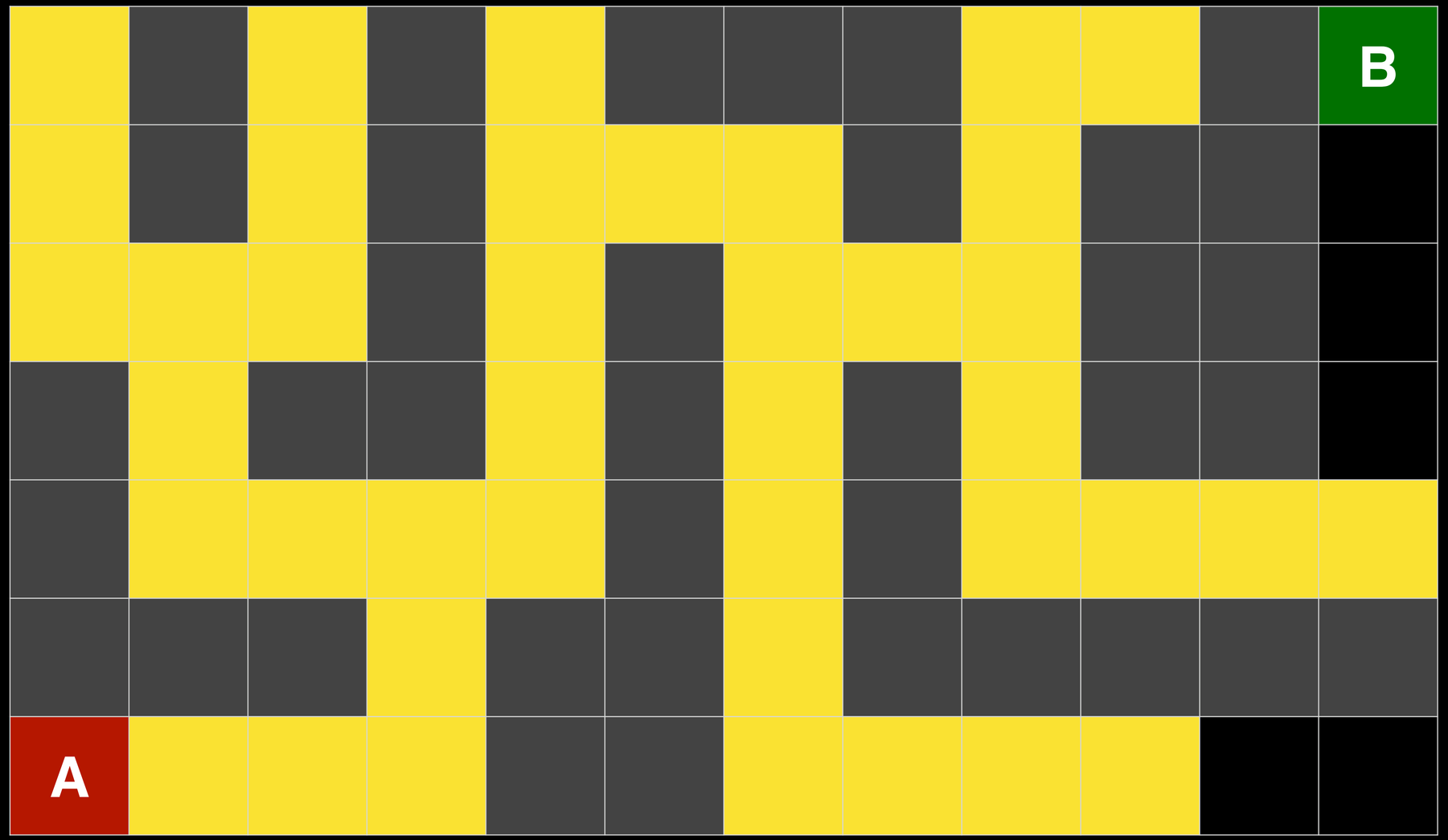

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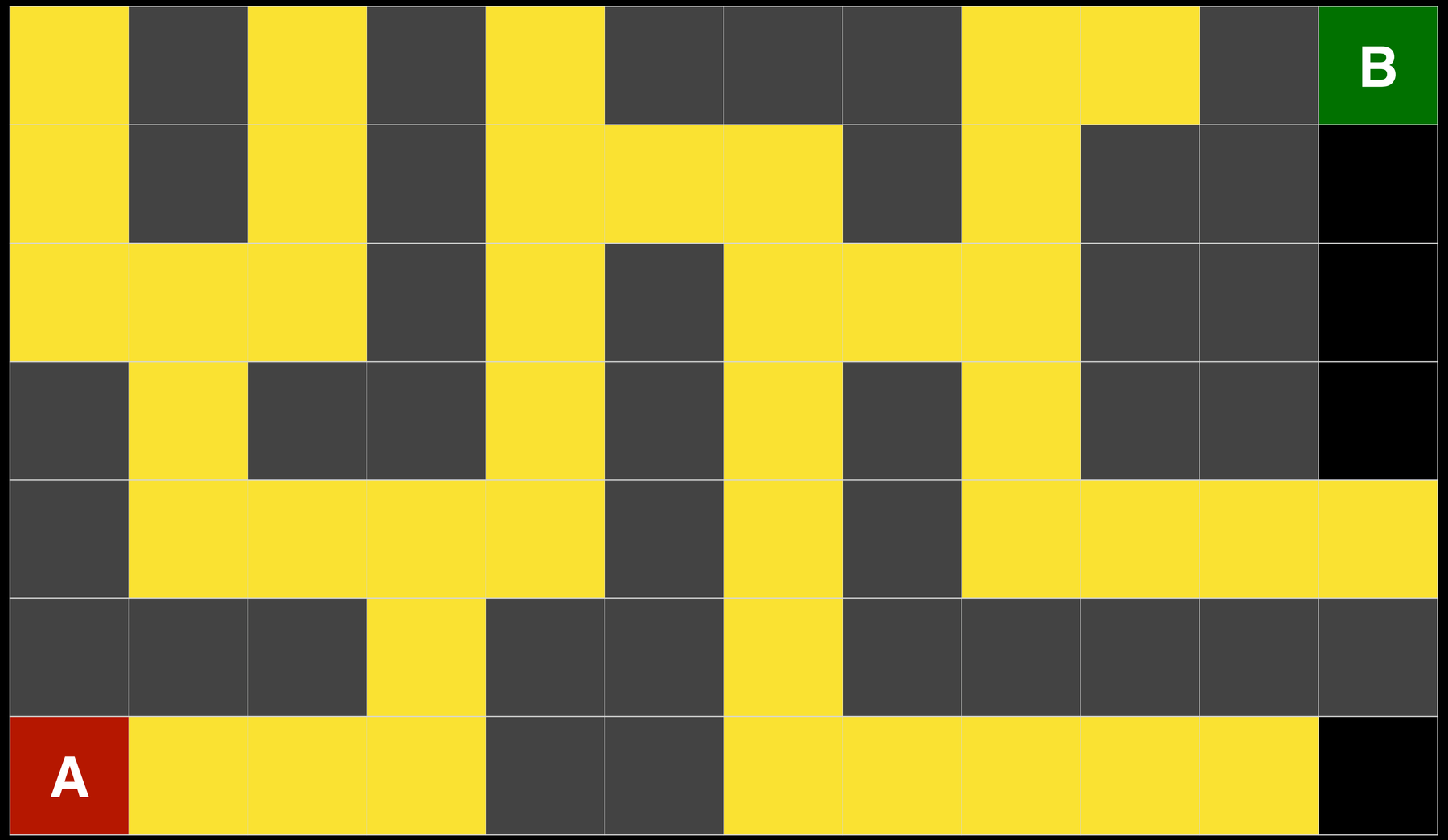

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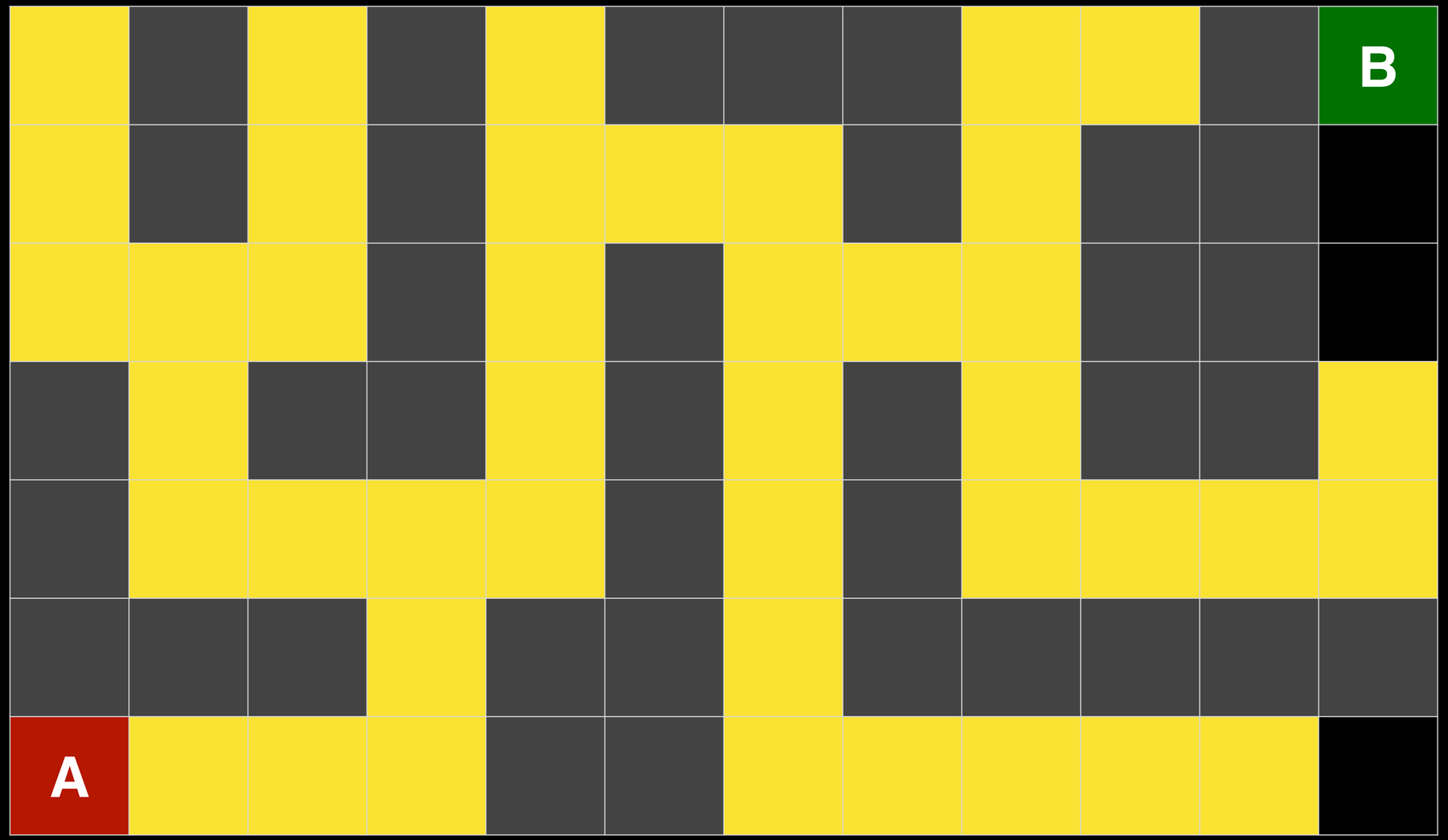

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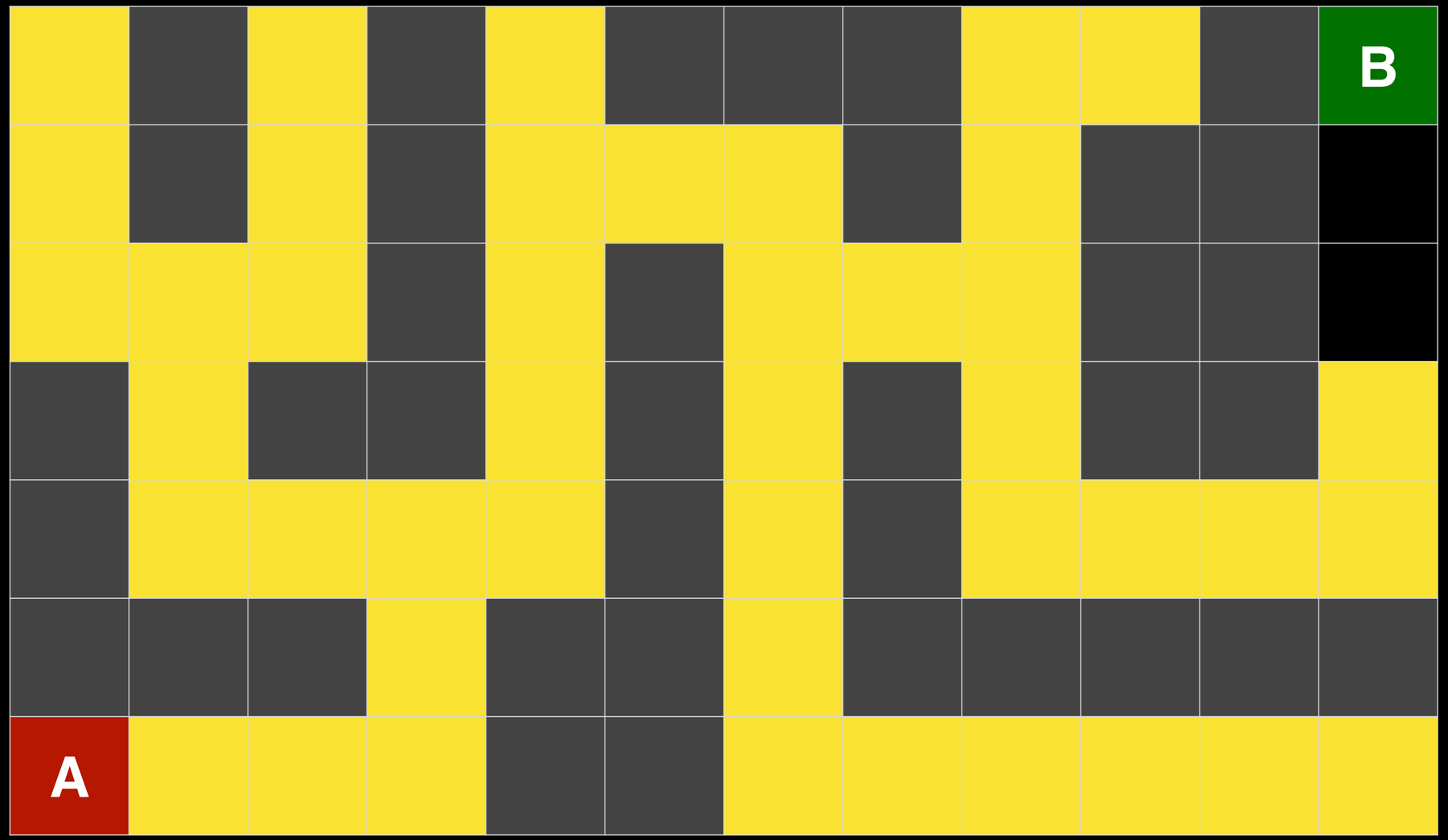

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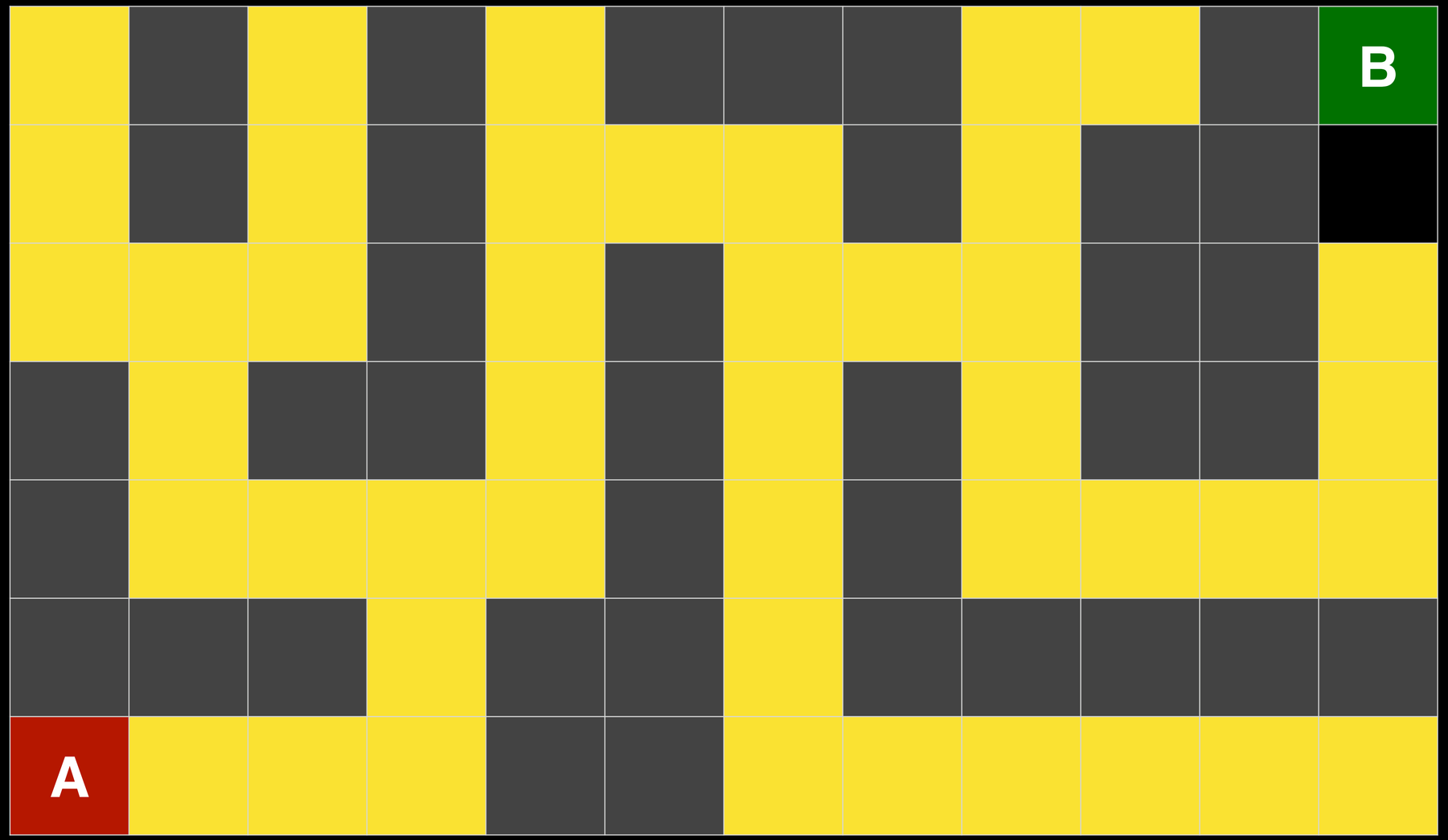

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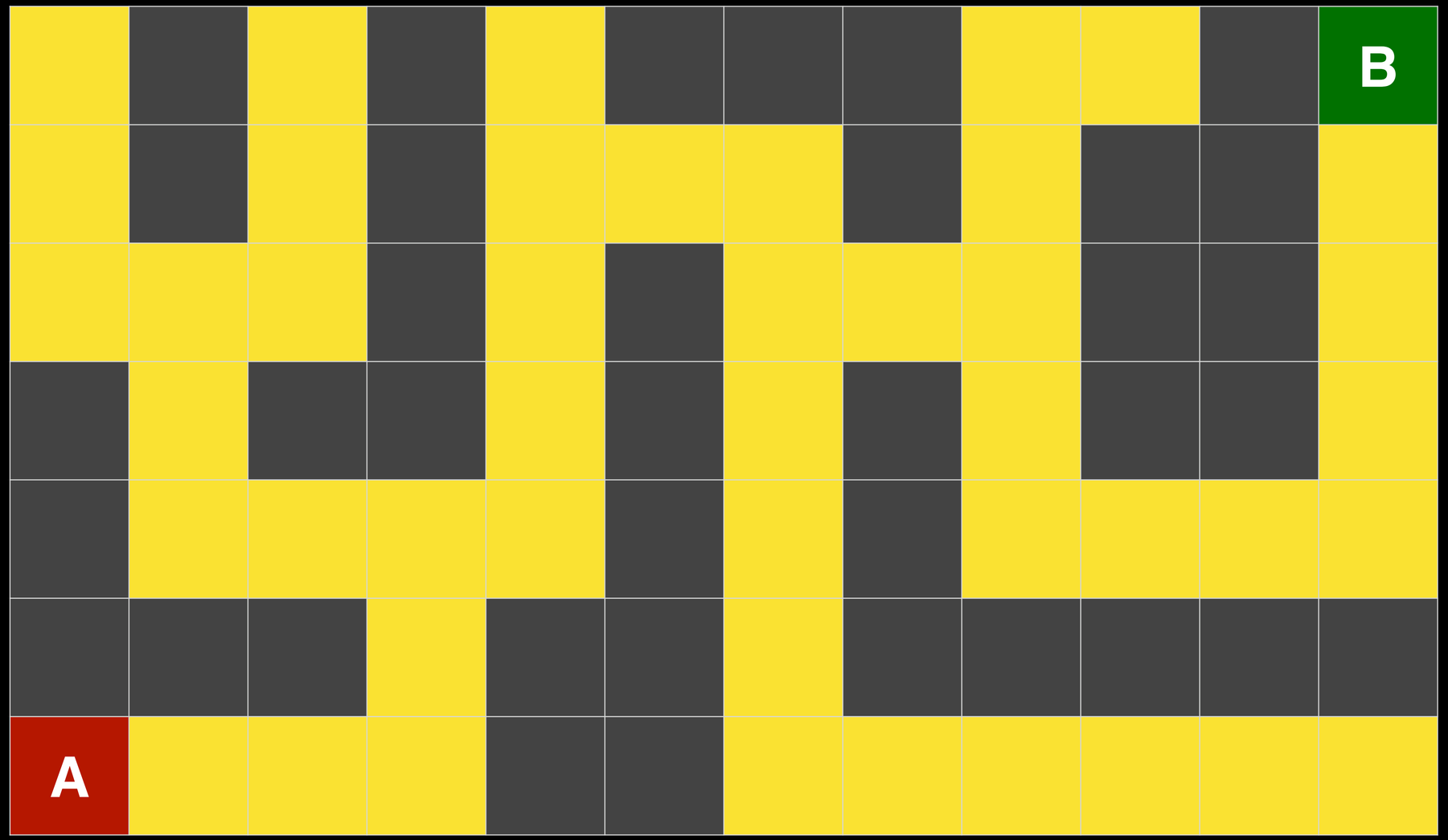

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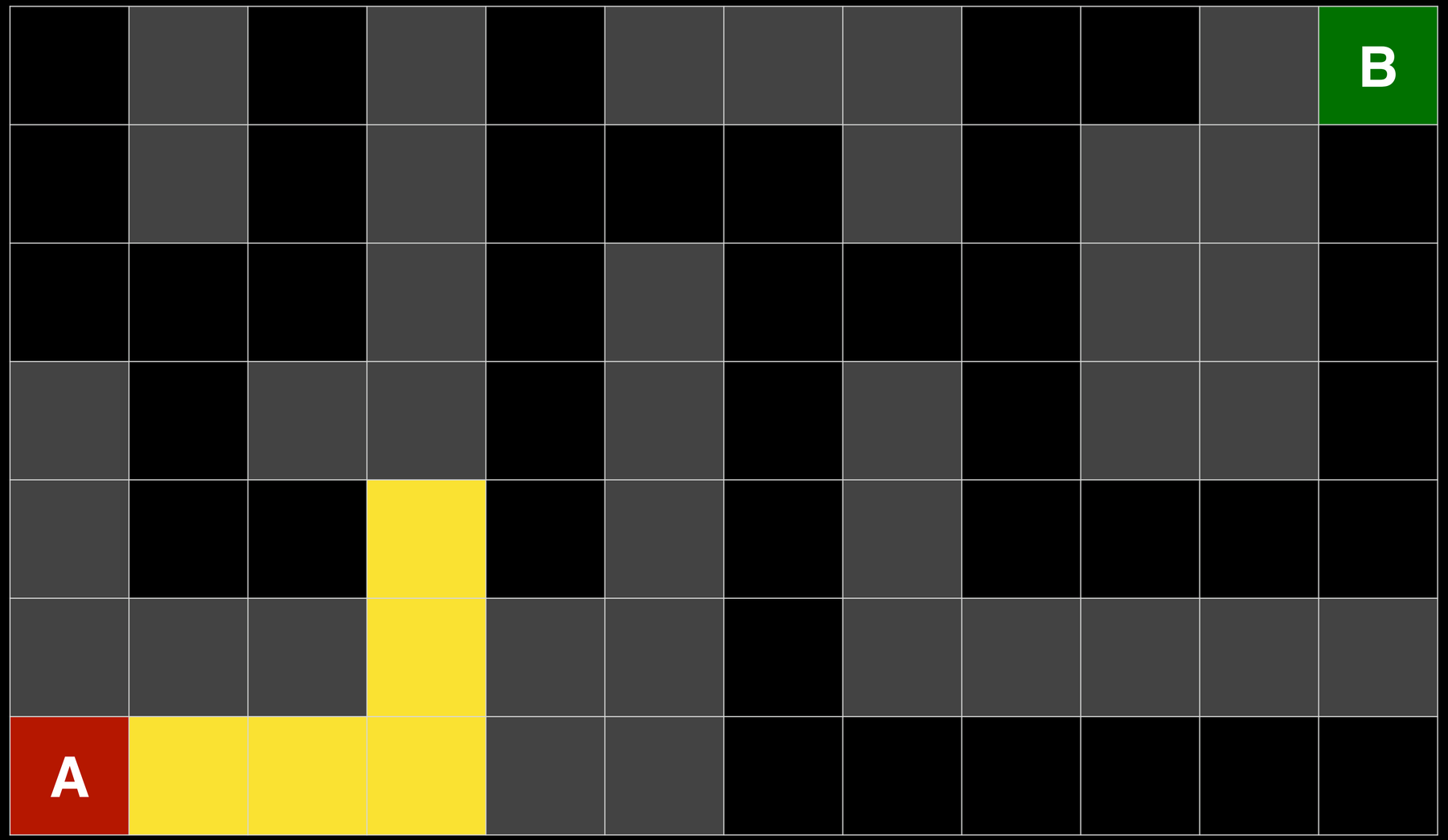

uninformed search

search strategy that uses no problem-specific knowledge